

Verification Continuum™

Verdi® Coverage User Guide and Tutorial

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Preface

The Verdi Coverage User Guide and Tutorial discusses Verdi Coverage that provides visualization for coverage data. Visualization for coverage data in Verdi Coverage is a comprehensive visualization environment that integrates with the design hierarchy display and can be used to get a summary of the coverage results or details of various types of coverage. Verdi Coverage can display code coverage, covergroup coverage, and assertion coverage. Furthermore, you can use the Exclusion Manager of Verdi Coverage to show all exclusions and unresolved exclusions and edit exclusions. You can also use Verdi Coverage to create a verification plan or verification plan components (.hvp files) and link specifications (.pdf files). The interaction between the Verdi platform and Verdi Coverage is also provided to debug with designs and the coverage report.

Customer Support

For any online access to the self-help resources, you can refer to the documentation and searchable knowledge base available in SolvNetPlus.

To obtain support for your Verdi product, choose one of the following:

- Open a case through SolvNetPlus.

Go to <https://solvnetplus.synopsys.com/s/contactsupport> and provide the requested information, including:

- Product L1 as Verdi
- Case Type

Fill in the remaining fields according to your environment and issue.

- Send an e-mail message to verdi_support@synopsys.com.

Include product name (L1), sub-product name/technology (L2), and product version in your e-mail, so it can be routed correctly.

Your e-mail will be acknowledged by automatic reply and assigned a Case number along with Case reference ID in the subject (ref:____:ref).

For any further communication on this Case via e-mail, send e-mail to verdi_support@synopsys.com and ensure to have the same Case ref ID in the subject header or else it will open duplicate cases.

Note:

In general, we need to be able to reproduce the problem in order to fix it, so a simple model demonstrating the error is the most effective way for us to

identify the bug. If that is not possible, then provide a detailed explanation of the problem along with complete error and corresponding code, if any/permissible.

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Introducing Verdi Coverage

The Verdi Coverage function in the Verdi® Automated Debug Platform provides visualization for coverage data. After you have loaded the coverage of the simulation results for a Verilog or VHDL design, you can easily use the Verdi Coverage capability to check the coverage report in the Verdi platform.

Visualization for coverage data in Verdi Coverage is a comprehensive visualization environment that integrates with the design hierarchy display and can be used to get a summary of the coverage results or details of various types of coverage. Verdi Coverage can display the following coverage information:

- **Code coverage** including:
 - **Line Coverage:** The statements that were executed during simulation.
 - **FSM Coverage:** The blocks of code that make up FSMs can be identified. FSM coverage reports on the states, transitions, and sequences of states during simulation.
 - **Toggle Coverage:** Whether the signals in the design toggle from 0 to 1 and 1 to 0 during simulation.
 - **Condition Coverage:** Conditions are expressions and sub-expressions that control the execution of code or the assignment of values to signals. Condition coverage tests whether both true and false states of these conditions were covered during simulation.
 - **Branch Coverage:** Analyzes how if, case statements, and the ternary operator (?:) establish branches of execution in your Verilog design. It shows vectors of signal or expression values that enable or prevent simulation events.
- **Covergroup Coverage:** Coverage of SystemVerilog testbench coverage groups.
- **Assertion Coverage:** Coverage of SystemVerilog cover properties or assert properties. Low Power Coverage: Coverage of UPF low power coverage groups.

Furthermore, you can use the **Exclusion Manager** of Verdi Coverage to show all exclusions and unresolved exclusions and edit exclusions. You can also use Verdi Coverage to create a verification plan or verification plan components (.hvp files) and link specifications (.pdf files). The interaction between the Verdi platform and Verdi Coverage is also provided to debug with designs and the coverage report.

Note:

Currently, the supported simulator is VCS with version 2013.06 and later.

2

Prerequisites

Before you can begin using the Verdi Coverage capability, the following criteria must be met:

- Your coverage database is generated by a VCS simulator with version 2013.06 or later.
- The coverage features must be enabled at compile-time and simulation time to include them in the coverage database and view them in the Verdi platform by using the `-cm1 line+tgl+fsm+cond+branch+assert` compile option and runtime option. For example:

```
> vcs -P $VERDI_HOME/share/PLI/VCS/LINUX/novas.tab \
    $VERDI_HOME/share/PLI/VCS/LINUX/pli.a \
    -f file_list.vc -cm line+tgl+fsm+cond+branch+assert -sverilog

> ./simv -cm line+tgl+branch+fsm+cond+assert
```

If your design includes UPF file for power simulation, the `-power=coverage+dump_hvp2` compile option is needed to enable VCS power coverage features. For example:

```
> vcs -P $VERDI_HOME/share/PLI/VCS/LINUX/novas.tab \
    $VERDI_HOME/share/PLI/VCS/LINUX/pli.a \
    -sverilog <your design> -upf <your UPF power design> \
    -cm line+tgl+fsm+cond+branch+assert -power=coverage+dump_hvp
```

- You have the Verdi license and the Verdi Coverage add-on licenses.

Note:

If the Verdi platform is invoked through Verdi Coverage, there is no need to check another Verdi license.

After the above criteria are met, the following Verdi environment variable must be set before invoking Verdi Coverage:

```
> setenv VERDI_HOME <VERDI_INST_DIR>
```

1. For details on the compilation options, see the VCS User Guide.
2. For details on the compilation options, see the VCS User Guide.

3

Invoking Verdi Coverage

To invoke Verdi Coverage, you need to add the `-cov` option to the `verdi` command line. You can also specify the directory name in the command line while invoking Verdi Coverage as follows:

```
> verdi -cov &
> verdi -cov -covdir simv.vdb &
```

Note:

It is recommended to merge or map multiple .vdb's via URG before invoking Verdi. For details on this, see Chapter, "Merging Coverage Metrics" of the *Coverage Technology User Guide*.

Verdi Coverage provides four work modes, namely Planner, Coverage, Coverage and Planner, and User-Defined Mode. To choose a work mode for Verdi Coverage, use the `-workMode` command with `verdi -cov` as follows:

```
> verdi -cov -workMode planner           //Planner Mode
> verdi -cov -workMode coverageAnalysis //Coverage Mode
> verdi -cov -workMode coveragePlanner  //Coverage & Planner mode
> verdi -cov -workMode userDefinedMode  //User-defined mode
```

To save a user-defined mode, use the **Windows -> Save/Restore User Layout** menu command from the main window.

To see the command line options for Verdi Coverage, use the following command:

```
> verdi -cov -help
```

Alternatively, you can use the **Tools -> Coverage** command in the main window to invoke Verdi Coverage from a previously opened Verdi platform.

Note:

When the `-cov` option is mixed with non-coverage options for the Verdi platform (for example, `verdi -cov -lib`), Verdi Coverage is invoked. The Verdi platform is not invoked until the **Tools -> Debug** command is used in Verdi Coverage. Unaffected options are shown in the **Message** pane as a warning message.

If you change the location of a source file, you can map the path to the relocated file using the `-pathmap <file>` option. The rule specified in the file relocates source files. For example,

```
> verdi -cov -covdir simv.vdb -pathmap <file> &
```

The syntax of the path mapping file is as follows:

```
<old_directory_path_name>:<new_directory_path_name>
```

For example,

```
## the content of the path mapping file  
/home/work/src: ~local/files/source
```

The subsequent chapters discuss how you can use Verdi Coverage.

Showing Instance-Based Scores for Coverpoints and Crosses

An Instcov mode score (different from default or union mode) is computed by calculating the average score of instances. This score applies only to the definition-level covergroup, coverpoint, cross, or bin hvp regions, with instance coverage enabled as `option.per_instance = 1..`

How to Display the Instance-Based Scores Feature

To ensure the instcov mode score is displayed, you must enable the `-group instcov_for_cp_score` option explicitly. This option has the effect of `-group instcov_for_score` showing the instcov mode score of covergroup. Refer to the **Instance Coverage Score Option** section of the *Verification Continuum™ Coverage Technology Reference Manual* to learn more.

Enabling the `-group instcov_for_cp_score` option ensures the display of the instcov mode score of covergroups through the following:

- The Summary and Coverage Detail panes) display the instcov mode score for definition-level coverpoint/cross.
- The HVP pane displays the instcov mode score for definition-level covergroup, coverpoint/cross and single-bin regions.

Usage of the `-group instcov_for_cp_score` Option

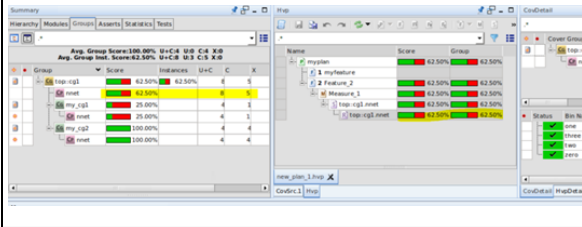
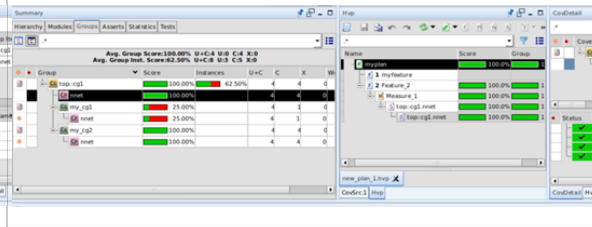
Add the `-group instcov_for_cp_score` option in the Verdi Coverage command line as given below:

```
%> verdi -cov -covdir <simv.vdb> -group instcov_for_cp_score
```

Effect of Enabling -group instcov_for_cp_score

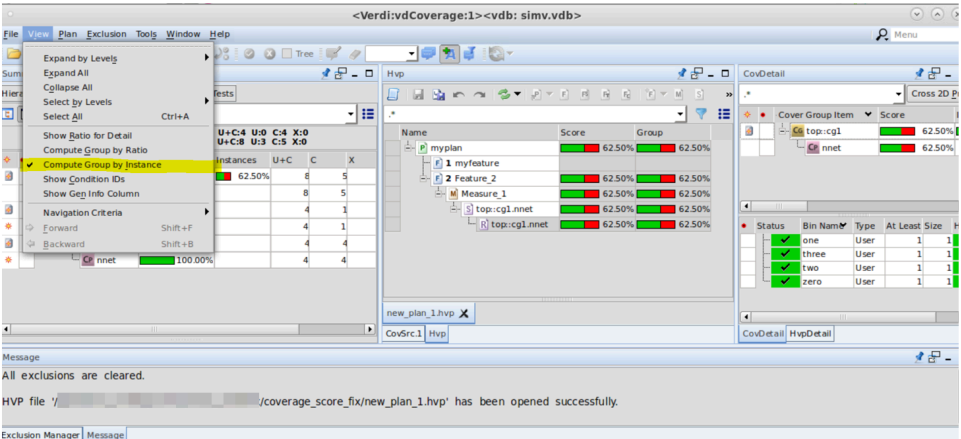
The following screenshots depict the display in the Summary, HVP, and Coverage Detail panes with and without -group instcov_for_cp_score enabled:

Table 1 Effect of Enabling -group instcov_for_cp_score

With the option enabled vdcov -covdir simv.vdb -group instcov_for_cp_score	Without the option enabled vdcov -covdir simv.vdb
	

Switching the Coverage Score between Default Mode and Instcov Mode

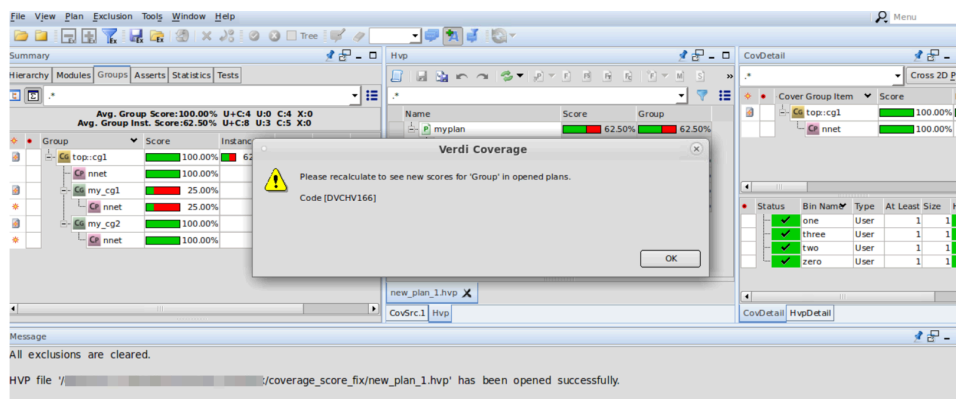
With the -group instcov_for_cp_score option, you can toggle the preference item **View -> Compute Group by Instance** to switch the coverage score between the default mode and instcov mode in the Summary and Coverage Detail panes.



If you toggle the preference item for HVP regions scores, a dialog box pops up to enable you to recalculate and view the new plan scores.

Chapter 3: Invoking Verdi Coverage

Showing Instance-Based Scores for Coverpoints and Crosses



4

Using Verdi Coverage

This chapter provides information on how you can use the capabilities of Verdi Coverage with its various coverage modes. It also describes each of the Coverage GUI tabs in the following sections:

- [Using the Summary Pane](#)
- [Using the Hierarchy Tab](#)
- [Configuring Computation Modes](#)
- [Using the Modules Tab](#)
- [Using the Groups Tab](#)
- [Using the Asserts Tab](#)
- [Using the Statistics Tab](#)
- [Using the Tests Tab](#)
- [Using the UPF Low-Power Coverage](#)
- [Using Application Preferences](#)
- [Shipped Example for User Reference](#)

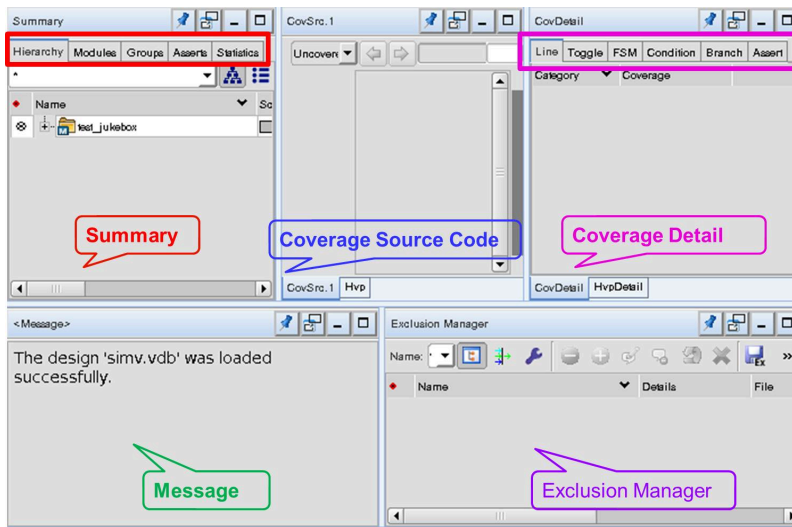
Verdi Coverage provides the following working modes with predefined window layouts. You can use the commands listed in the following table to switch to these modes.

Mode	Command	Explanation
Coverage Mode	Windows -> Coverage Mode	Focuses on the coverage report and its details that exist in the coverage database.
Plan Mode	Windows -> Plan Mode	Focuses on the hierarchical verification plan and manages the specification for higher coverage.

Mode	Command	Explanation
Coverage + Plan Mode	Windows -> Coverage + Plan Mode	Shows both coverage and planner information.

As shown in Figure [Coverage Mode](#), when Verdi Coverage is invoked, the *Coverage* mode is launched by default. The *Summary*, *Coverage Source Code (CovSrc)*, *Coverage Detail (CovDetail)*, *Message*, and *Exclusion Manager* panes are displayed in the main window.

Figure 1 Coverage Mode

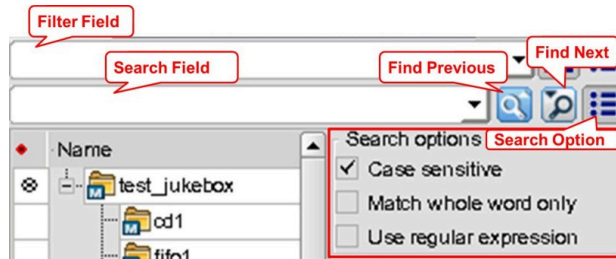


The *Summary* pane gives you the summary information for each type of coverage and navigates to the required database. It consists of the *Hierarchy*, *Module*, *Group*, *Asserts*, and *Statistics* tabs to show the summary of your design and coverage with different forms of presentation. The *Coverage Source Code* pane shows the source code and covered and uncovered highlights for specific coverage types in the *Coverage Detail* pane.

Verdi Coverage is a multi-window docking application with a flexible and easy-to-use graphical user interface. Thus, the following general functions are introduced for easier usage before describing the Verdi Coverage features:

- The layout of Verdi Coverage is flexible and you can resize, dock, and undock each pane in Verdi Coverage.
- As shown in Figure [Search and Filter](#), you can filter the content of each pages in the *Summary* pane based on the string entered in the **Filter** field.

Figure 2 Search and Filter



- You can search the content of each pane in Verdi Coverage based on the string entered in the **Search** field. You can also click the **Search Option** icon to add-on search options. The **Search** field is hidden by default. Use the **Show/Hide Search Widget** command in the right-click menu to show the field.
- The bindkey values and drop -down menu names and locations can be customized through the *Customize Menu/Toolbar* form. Use the **Customize Menu/Toolbar** command to invoke the form. For more information, see the *User Interface* chapter in the *Verdi User Guide and Tutorial*.

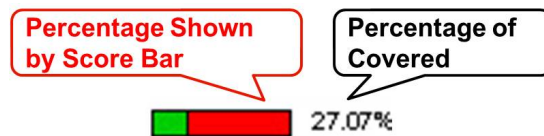
Using the Summary Pane

As shown in Figure [Summary Pane](#), you can adjust the pane size and see the line, toggle, FSM, condition, branch, and assert coverage results with their covered percentage for the instance in separate columns. After expanding the hierarchical tree by clicking the  icon, more coverage data is shown. Referring to Figures [Summary Pane](#) and [Score Bar](#), the score bar shows the covered percentage for each type of coverage. You can also see the **Exclusion** column (refer to Figure [Summary Pane](#)) to check the exclusion status for each item. An exclusion icon is displayed in the **Exclusion** column if there is an exclusion action for this item (see Chapter [Using the Exclusion Manager](#) exclusion status and details).

Figure 3 Summary Pane

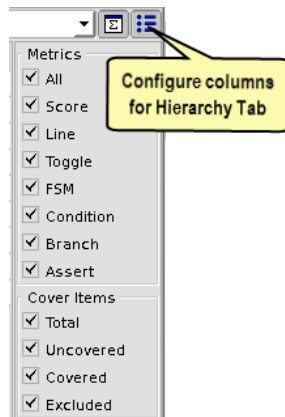
Name	Score	Line	U+C	U	C	X	Toggle	U+C	U	C	X
test_jukebox	44.52%	69.89%	641	193	448	0	50.66%	1662	820	842	
cd1	91.23%	100.00%	9	0	9	0	73.68%	38	10	28	
ffo1	76.82%	88.24%	17	2	15	0	75.56%	90	22	68	
jb1	75.90%	96.00%	25	1	24	0	53.12%	64	30	34	
st0	44.81%	72.88%	118	32	86	0	62.59%	294	110	184	
st1	73.31%	89.83%	118	12	106	0	55.10%	294	132	162	
st2	31.50%	56.78%	118	51	67	0	41.50%	294	172	122	
st3	46.44%	77.97%	118	26	92	0	52.38%	294	140	154	
st4	19.17%	41.53%	118	69	49	0	30.61%	294	204	90	

Figure 4 Score Bar



As shown in Figure [Configuring Columns for Hierarchy Tab](#), click the  icon to control the columns that need to be displayed in the *Summary* pane of each page based on the **Metrics** options. You can select the **Total** option to display the total number of coverable objects and the **Uncovered** option to display the number of uncovered objects. For more information on **Cover Items**, see table [Cover Items Description](#). All the columns can be used to sort the data by clicking the column name field.

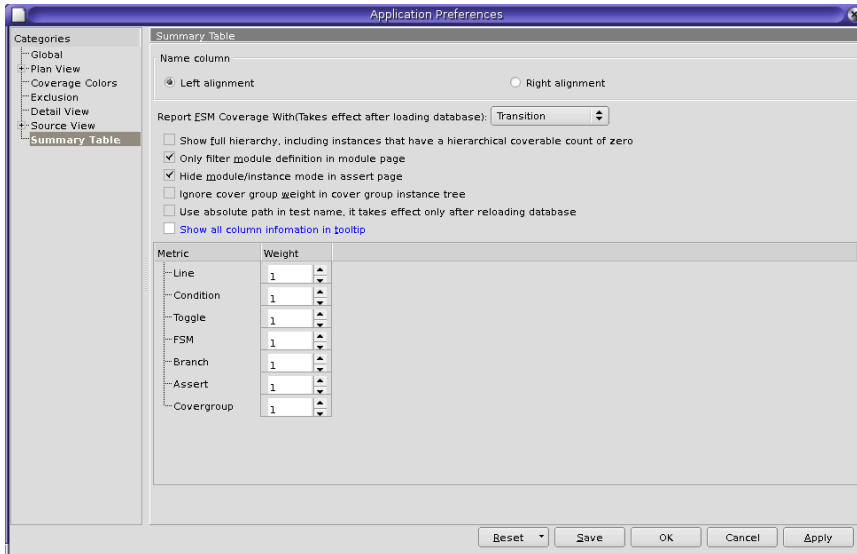
Figure 5 Configuring Columns for Hierarchy Tab



The Figure [Configuring Columns for Hierarchy Tab](#) displays the configuration options available for the *Hierarchy* tab.

Tooltip in the Summary Pane

In the *Summary* pane, the tooltip provides information related to the overall coverage score and the score for all coverage metrics, such as line, toggle, FSM, and so on. Verdi Coverage enables you to control the information in the tooltip as per your requirements. You can see complete information in the tooltip or see only column-specific information using **Tools --> Preferences**. It displays the *Application Preferences* form as shown in the following figure:



The **Show all column information in tooltip** check box is not selected by default. Therefore, Verdi shows only some information for the selected column. For example, if you move the pointer over *Score* in the following figure, it only shows the total coverage score of all coverage metrics.

Name	Score	Line	Toggle	FSM	Condition	Branch	Assert
lcd	91.23%	100.00%	73.68%			100.00%	
coin_fsm	77.93%	91.01%	83.87%	70.59%	42.59%	79.49%	100.00%
info	76.82%	88.24%	75.56%			86.67%	
jukebox	nfo (definition)						
kp_fsm	53.12%	100.00%	42.86%		54.05%	87.50%	
station	60.27%						
	74.42%						

Tooltip for 'info' row, 'Score' column: Score: 76.82%

To display the overall coverage score along with the individual score of all metrics in the tooltip of each column, select the **Show all column information in tooltip** check box in the *Application Preferences* form and click **OK**. Verdi shows the following figure if you select the check box and move the pointer over *Score*:

Name	Score	Line	Toggle	FSM	Condition	Branch	Assert
cd	91.23%	100.00%	73.68%			100.00%	
coin_fsm	77.93%	91.01%	83.87%	70.59%	42.59%	79.49%	100.00%
info	76.82%	88.24%	75.56%			66.67%	
jukebox	76.82%	88.24%	75.56%			66.67%	
kp_fsm							
station							

info (definition)

Score: 76.82%

Line : 15 / 2 / 17 (88.24%) Excluded:0

Toggle : 68 / 22 / 90 (75.56%) Excluded:0

Branch : 4 / 2 / 6 (66.67%) Excluded:0

Press 'F3' to save in Tooltip Viewer

Verdi displays the covered, uncovered, and total object counts for each metric along with the metric's score. This information is displayed in the tooltip irrespective of whether the **Show all column information in tooltip** check box is selected or not as shown in the following figure:

test_jukebox.st0

Line : 86 / 32 / 118 (72.88%)

Excluded:0

Press 'F3' to save in Tooltip Viewer

test_jukebox.st0

Score: 44.81%

Line : 86 / 32 / 118 (72.88%) Excluded:0

Toggle : 184 / 110 / 294 (62.59%) Excluded:0

FSM Transitions : 6 / 11 / 17 (35.29%) Excluded:0

Condition : 34 / 57 / 91 (37.36%) Excluded:0

Branch : 34 / 22 / 56 (60.71%) Excluded:0

Assert : 0 / 2 / 2 (0.00%) Excluded:0

Press 'F3' to save in Tooltip Viewer

In this figure, 86 is the total number of covered objects, 32 is the total number of uncovered objects, and 118 is the total number of coverable objects. Here, 72.88% is calculated by dividing 86 by 118. The total number of excluded objects is zero.

Advanced Masking

In previous releases, Verdi provided quick filters to search items based on characters, for example a covergroup name or an instance name. Now, Verdi also enables you to filter the items based on coverage scores, exclusion status, properties, and so on. Verdi provides this advanced masking feature in the *Summary* pane, the *Coverage Detail* pane, and the *HVP* pane. This advanced masking does not result in recomputation of HVP scores, which behaves differently from quick filters and modifiers.

With the advanced masking capabilities, you can filter by specifying a single condition as well as multiple conditions. For the single condition, you can use the following syntax for filtering:

```
[text filter][conditional_expression]
```

where,

- `[text filter]` is a wildcard or regular expression that matches items by their name. If the matched items also fulfill `[conditional_expression]`, Verdi retains the items. You can also retain the sub-items of the matched items, if you select the following option from **Preferences > Global**:

☒ Keep matched item's children

However, if the sub-items of the matched items do not fulfill the condition, Verdi does not show the items even if the above option is selected.

- `[conditional_expression]` is of the following format:

`[column name][operator][Number/String]`

where,

`[operator]` refers to `>`, `<`, `=`, `!=`, `>=`, `<=`, and `..`. Here, `:` refers to `=`.

`[Number]` refers to `int`, `float`, or `double`.

`[String]` is only allowed with `:` and `=`. `[String]` with empty spaces should be enclosed within `" "`. If `[String]` includes `" "`, use escaped characters.

To specify multiple conditions for filtering, use the following syntax:

```
[text filter][operator][conditional_expression1] [operator]
[conditional_expression2]
```

where,

`[text filter]` and `[operator]` are optional. Also, if no operator is specified, then it is AND by default.

For example, if you specify the filter as `*coin* Line<50% OR Toggle<80% exclude=0`, Verdi shows the search results where the region name contains `coin` and line score is less than 50% or toggle score is less than 80% and that is not excluded as shown in the following figure:

Summary						
Hierarchy Modules Groups Asserts Statistics Tests						
coin Line<50% OR Toggle<80% exclude=0						
Name	Score	Line	X	Toggle		
test_jukebox	44.98%	69.20%	89	50.38%		
st0	44.81%	72.88%	0	62.59%		
coin1	38.70%	67.42%	0	54.84%		
st2	31.50%	56.78%	0	41.50%		
coin1	22.38%	46.07%	0	29.03%		
st4	19.17%	41.53%	0	30.61%		
coin1	15.44%	38.20%	0	25.81%		

To filter only module definition in the *Modules* pane, select the following option from **Preferences > Summary:**

☒ Only filter module definition in module page

With advanced masking, if you hover the pointer over the **Filter** field, it shows the following tooltip:

Input one string to filter items.
Syntax: [text filter][condition]
text filter: wildcard/regular expression
condition: the shown items should meet the condition
i.e. 'exclude=0', 'weight=1', 'FSM<30'

Examples:

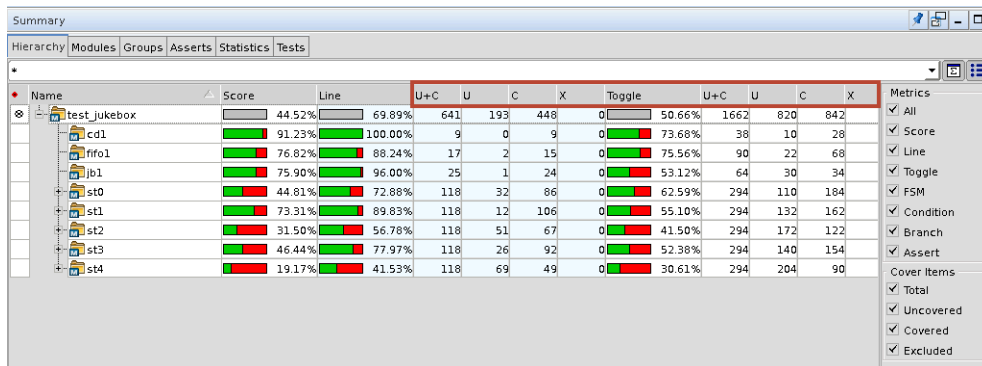
write	show items which match '*write*'
exclude=0	hide all excluded items
Toggle<=80%	show items which are toggle coverage <= 80%
	Syntax: [column name][operator] [value] operator: '=', '>', '<', '>=', '<=', ':'
write Line<50% OR Toggle<80% exclude=0	show items which match '*write*', also they are Line coverage < 50% or Toggle coverage < 80%, and not excluded Syntax: [expression] [operator] [expression] operator: 'AND', 'OR', ' ' (BLANK Space)

*[Preferences][Global][Keep matched item's children]: The children of the item matched will be kept, when they meet [condition].

Using the Hierarchy Tab

The *Hierarchy* tab displays coverage results by design hierarchy. As shown in Figure [Hierarchy Tab](#), the total covered percentage is shown in the **Score** column of the *Hierarchy* page.

Figure 6 Hierarchy Tab



On the right pane of the *Hierarchy* tab, the configuration metrics is displayed. Based on the selected metrics the columns displayed in the *Hierarchy* tab changes. In this case, all the metrics and cover items are selected, therefore, the cover items **U+C (Total)**, **C (Covered)**, **U (Uncovered)**, and **X (Excluded)** columns are visible for all the metrics in the *Hierarchy* tab as shown in Figure [Hierarchy Tab](#).

The following table explains these columns in more detail:

Table 2 Cover Items Description

Tab Column	Option Name	Explanation
U+C	Total	Total number of coverable objects <i>not counting</i> excluded and unreachable objects.
C	Covered	Number of <i>unexcluded</i> objects hit.
U	Uncovered	Number of <i>unexcluded</i> objects not hit (holes).
X	Excluded	Number of excluded objects. If a higher level object like an instance or a module is excluded, the count of all of the lower level objects within the instance and/or module are reported. Objects might be excluded using an elfile or using manual interactive exclusions. Coverage objects removed via -cm_hier are not counted in the X or any other column.

You can click the **+** icon to expand the hierarchy tree. You can check the value in the **Score** column and see the distribution for those coverage types. For example, click the

icon to expand the `test1_jukebox` and see the `jb1` sub-instance in Figure [Observe the Covered Percentage](#). The score is 75.90% but the toggle and condition coverage results are only 53.12% and 42.86%. Thus, you can first focus on the details for the low covered percentage (see Section [Show Source Code and Coverage Detail](#) for details) and how to increase these two coverage types.

Figure 7 Observe the Covered Percentage

Name	Score	Line	Toggle	FSM	Condition	Branch	Assert
jb1	75.90%	96.00%	53.12%	100.00%	42.86%	87.50%	

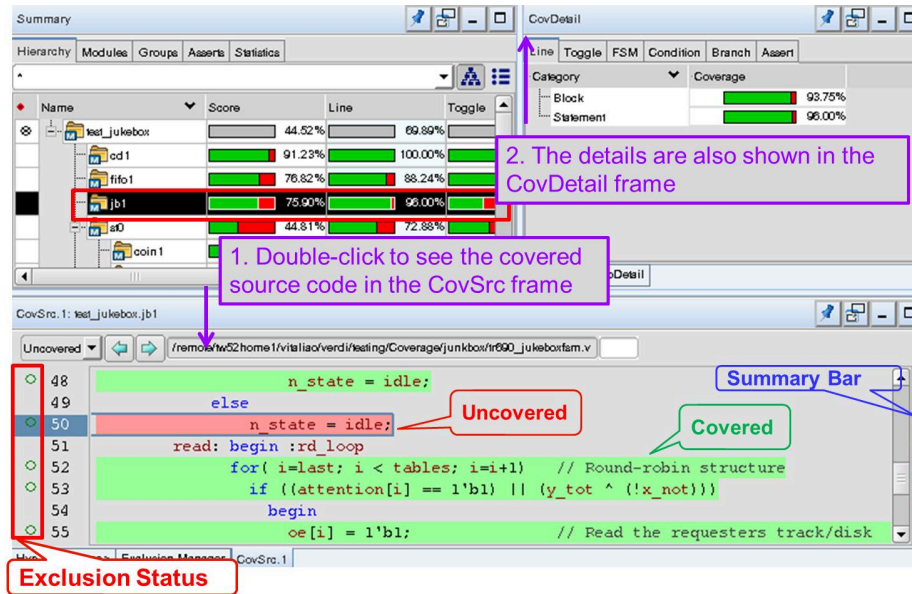
Show Source Code and Coverage Detail

You can double-click a row under a column in the *Hierarchy* page for a specified scope to see the corresponding information in the *Coverage Source Code* and *Coverage Detail* panes (refer to Figure [Check the Coverage Details](#)):

- In the *Coverage Source Code* pane, the source code with annotation indicates the covered type from the double-clicked column. For example, refer to Figure [Check the Coverage Details](#), double-click the `jb1` row under the **Line** column. The source code with annotation indicates a covered line in the `jb1` scope. You can also see the exclusion status icons shown in front of the line number. Macro instances are collapsed in the *Coverage source code* pane by default. You can click the expand/collapse icon in the line number widget to show/hide the macro content.
- In the *Coverage Detail* pane, the selected page in the *Coverage Detail* pane displays the coverage percentage for the specified coverage type. For example, refer to Figure

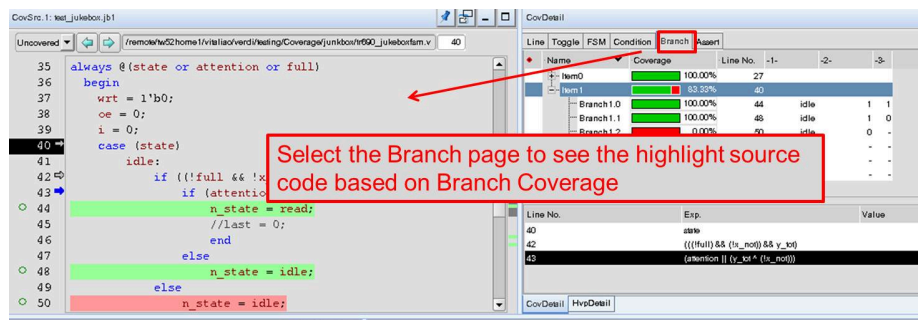
Check the Coverage Details, double-click the `jb1` row under the **Line** column. The line coverage details of `jb1` scope are shown in the *Coverage Detail* pane in the following figure:

Figure 8 Check the Coverage Details



You can also change the coverage type in the highlighted source code by selecting a different coverage page in the *Coverage Detail* pane. For example, as shown in Figure [Check Covered Source Code by Selecting Specific Coverage Type](#), the source code is shown and highlighted based on the branch coverage.

Figure 9 Check Covered Source Code by Selecting Specific Coverage Type



The following steps describe how you can show the location of the corresponding source code line automatically:

1. In the *Preferences* form (invoked using **Tools -> Preferences**), turn *on* the **Auto-scroll CovSrc when switching CovDetail metrics** option in the **Detail View** page.
2. In the *Coverage Detail* pane, select a row in one page (for example, select a row in the *Branch* page).
3. In the *Coverage Detail* pane, select another page (for example, select the *FSM* page).
4. The coverage type in the *Coverage Source Code* pane changes to the FSM coverage type.
5. In the *Coverage Detail* pane, select the previous page that contains the previously selected row (for example, select the *Branch* page).

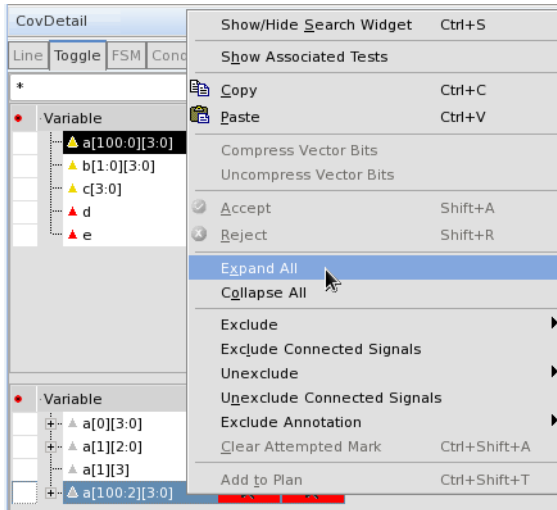
The coverage type in the *Coverage Source Code* pane changes to the branch coverage type. In addition, the location of the corresponding selected item in the page is automatically displayed in the *Coverage Source Code* pane (that is, the scroll is moved to the corresponding location).

Expanding MDAs in the Coverage Detail Pane

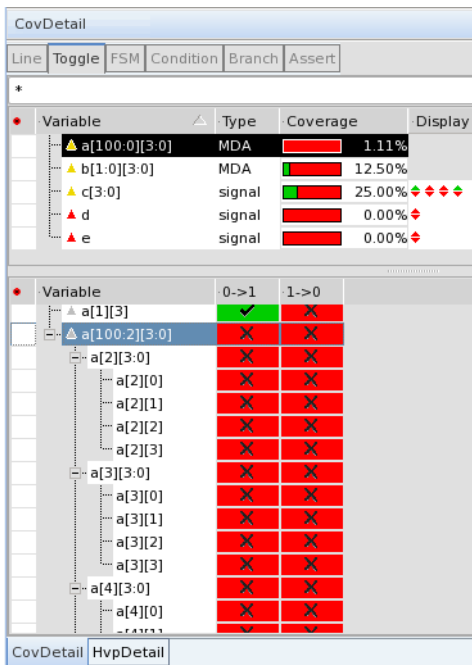
Verdi enables you to expand all variables of type Multi-Dimensional Array (MDA) on a single click using the **Expand All** option. The option allows you to see all details of MDA variables with a single click. Verdi also provides the **Collapse All** option to hide the details of MDA using a single click.

To expand MDAs in the Coverage Detail (CovDetail) pane, perform the following steps:

1. Right-click a variable and select **Expand All**. For example, right-click the `a[100:2][3:0]` variable and click **Expand All** as follows:

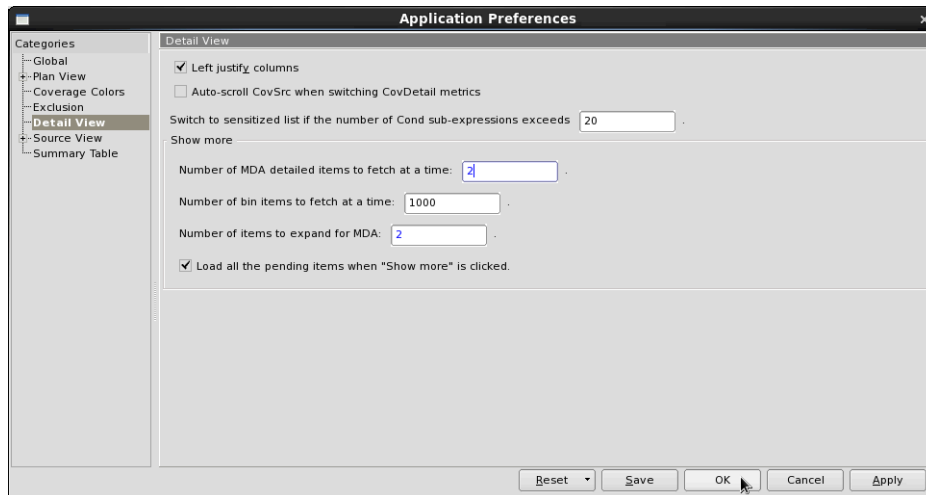


2. The **Expand All** option expands all the items of the selected variable. For example, the figure below shows the details of the items, such as `a[2][3:0]`, `a[3][3:0]`, and so on of the `a[100:2][3:0]` variable.



You can restrict the number of items to expand for MDA. For this, perform the following steps:

1. Select **Tools > Preferences** to open the **Application Preferences** dialog box as shown in the following figure:



2. Specify the number of items to expand for MDA. For example, specify 2 in the **Number of items to expand for MDA** field to expand only two items of `a[100:2][3:0]` (see the figure below):

Variable	0->1	1->0
a[1][3]	✓	✗
a[100:2][3:0]	✗	✗
a[2][3:0]	✗	✗
a[2][0]	✗	✗
a[2][1]	✗	✗
a[2][2]	✗	✗
a[2][3]	✗	✗
a[3][3:0]	✗	✗
a[3][0]	✗	✗
a[3][1]	✗	✗
a[3][2]	✗	✗
a[3][3]	✗	✗
a[4][3:0]	✗	✗
a[5][3:0]	✗	✗
a[6][3:0]	✗	✗

Note:

The remaining items are not expanded and show the + sign.

You can further restrict the number of items that need to be displayed using the **Number of MDA detailed items to fetch at a time** option in the **Application Preferences** form. The option shows **Show more** for the remaining items as shown in the following figure:

Variable	0->1	1->0
▲ a[0][3:0]	✓	✓
▲ a[1][2:0]	✗	✗
a[1][0]	✗	✗
a[1][1]	✗	✗
Show more of a[1][2:0] ...		
Show more of a[100:0][3:0] ...		

You can control whether the remaining items should be displayed on clicking **Show more**. To load the remaining items, select the **Load all the pending items when "Show more" is clicked** option in the **Application Preferences** dialog box.

Filters in Cross Tables in the Coverage Detail Pane

Verdi has an advanced filter on bins in cross table for ease of analysis when there are a large number of bins in a cross. With the filter capability, you can easily search the bin items as per your requirements. It also enables you to sort the items. This capability is available in the *CovDetail* pane of Verdi Coverage.

For example, consider a cross table, namely `src_addr_vs_dst_addr`, which is a cross between two coverpoints, namely `src_addr` and `dst_addr`.

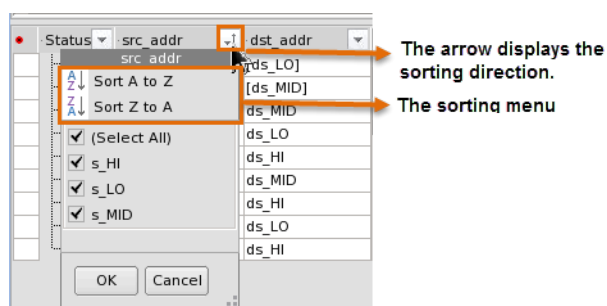
CovDetail							
* Cover Group Item	Score	Instances	U+C	U	C	X	
Cr ine_done	50.00%		2	1	1	0	
Cr ine_err	50.00%		2	1	1	0	
Cr mode	100.00%		2	0	2	0	
Cr src_addr	100.00%		3	0	3	0	
Cr src_addr_mask	34.38%		64	42	22	0	
Cr src_addr_vs_dst_addr	77.78%		9	2	7	0	

Status	src_addr	dst_addr	Typ	At Least	Size	Hit Count
✗	[s_LO]	[ds_LO]	Auto	1	1	0
✗	[s_MID]	[ds_MID]	Auto	1	1	0
✓	s_HI	ds_MID	Auto	1	1	2
✓	s_HI	ds_LO	Auto	1	1	4
✓	s_HI	ds_HI	Auto	1	1	7
✓	s_LO	ds_MID	Auto	1	1	2
✓	s_LO	ds_HI	Auto	1	1	5
✓	s_MID	ds_LO	Auto	1	1	1
✓	s_MID	ds_HI	Auto	1	1	4

For filters, Verdi provides a drop-down menu with text-based columns, such as **Status**, **src_addr**, **dst_addr**, and **Type**. You can sort or filter based on the values of these columns. For example, if you click the drop-down menu of **src_addr**, it shows the options to sort and to filter based on `s_HI`, `s_LO`, and `s_MID`.

To sort the bins two options are available, that is **Sort A to Z** and **Sort Z to A**. When you select any of these options, the arrow on the column gets changed and displays the sorting direction as shown in the following figure:

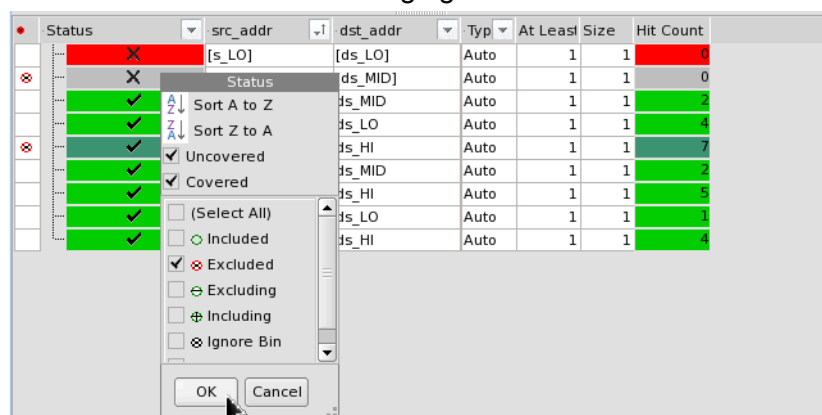
Chapter 4: Using Verdi Coverage Using the Hierarchy Tab



If you filter the bins based on `s_HI`, Verdi shows the following results:

Status	src_addr	dst_addr	Type	At Least	Size	Hit Count
✓	s_HI	ds_MID	Auto	1	1	2
✓	s_HI	ds_LO	Auto	1	1	4
✓	s_HI	ds_HI	Auto	1	1	7

Verdi also enables you to filter based on the status of exclusions. For example, to view only excluded covered and uncovered items, select **Excluded** from the **Status** drop-down menu as shown in the following figure:.

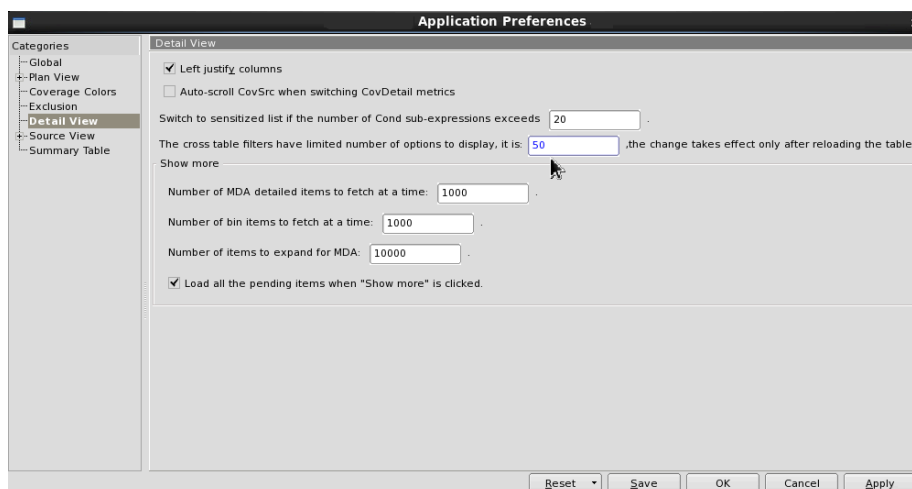


When **Excluded** is selected from the menu, Verdi shows the following results:

Status	src_addr	dst_addr	Type	At Least	Size	Hit Count
X	[s_MID]	[ds_MID]	Auto	1	1	0
✓	s_HI	ds_HI	Auto	1	1	7

Verdi provides you the capability to control the number of options to display in the drop-down menu. For this, perform the following steps:

1. Select **Tools > Preferences** to open the **Application Preferences** form and then select **Detail View** as shown in the following figure:



2. In the **Application Preferences** form, specify the number of options to display, say 50, in the following field and click **OK**:

The cross table filters have limited number of options to display, it is 50, the change takes effect only after reloading the table.

Verdi displays 1000 bin items by default. If more bin items are available than the displayed options, Verdi shows the tooltip when you hover over a column to inform you that more bin items are also available. For example, Verdi shows the following tooltip when the number of options to display is selected as 50:

The Filter's options exceed the limit number, only first 50 options are displayed. You can change it at pull down menu: Tools->Preference...->Detail View.

If you specify 0, it means no limit and Verdi displays all items.

Configuring Computation Modes

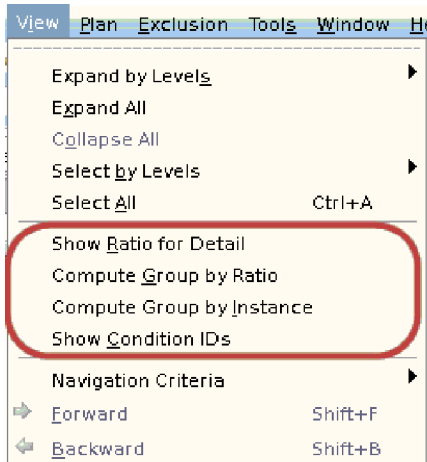
You can configure the display of the Coverage report data based on the following computation modes:

- [Show Ratio for Detail](#)
- [Compute Group by Ratio](#)

- [Compute Group by Instance](#)
- [Show Condition IDs](#)

These modes are available in the **View** menu and can be set/unset as shown in the following figure:

Figure 10 Computation Options in the View Menu

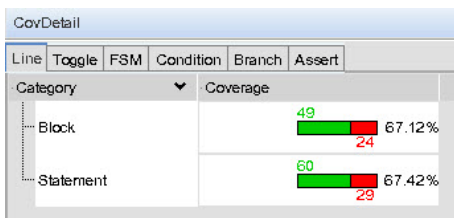


These modes are discussed in detail in the subsequent sections.

Show Ratio for Detail

As shown in Figure [Show Ratio for Detail](#), you can use the **View -> Show Ratio for Detail** command to show the actual counts in the *Coverage Detail* panes.

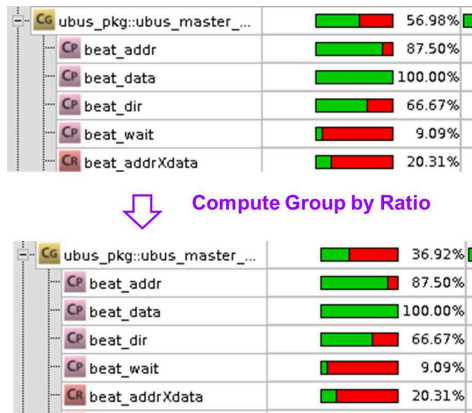
Figure 11 Show Ratio for Detail



Compute Group by Ratio

The score for a covergroup is calculated as the average of the percentages of individual cover point scores by default. You can use the **View -> Compute Group by Ratio** command to calculate the covergroup scores and overall group scores as a ratio between the number of covered bins and the total number of coverable bins. [Compute Group by Ratio](#) shows the difference between the score calculation with and without the ratio.

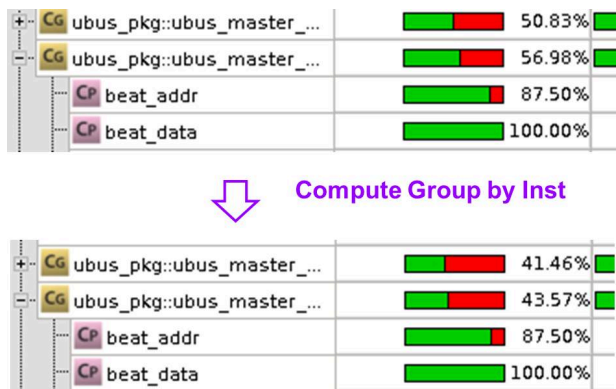
Figure 12 Compute Group by Ratio



Compute Group by Instance

You can use **View -> Compute Group by Instance** to calculate the covergroup scores using the coverage of each instance for the covergroups with instance coverage. Figure [Compute Group by Instance](#) shows the difference between the score calculation with and without enabling this option.

Figure 13 Compute Group by Instance



Show Condition IDs

You can use **View -> Show Condition IDs** to view the condition IDs for an expression in the *Condition* tab as shown in the [Showing Condition IDs](#).

Figure 14 Showing Condition IDs

CovDetail					
Line	Toggle	FSM	Condition	Branch	Assert
*					
Expression	ID	Coverage	Line No.		
☐ ((go > 1'b0) && (press x_not) &...	5	28.57%	70		
☐ ((go > 1'b0) && (press x_not) &...	10	42.86%	21		
☐ ((trko_n == 8'd32) && (dsko_n =...	9	0.00%	20		
☐ (go && ((!x_not) && press && y...	4	50.00%	65		
☐ (go && (press x_not) && (!oe_...	1	71.43%	45		
☐ (oe_s && (y_tot > 1'b0))	3	66.67%	51		
☐ (oe_s ? dsko_n : 8'bzzzzzz)	8	100.00%	18		
☐ (oe_s ? trko_n : 8'bzzzzzz)	7	100.00%	17		

Line Coverage

The *Line* page in the *Coverage Details* pane displays the coverage percentage for total block and statement coverage. The named objects or lines are shown in the lower part of the *Line* page. Also, the source code with annotation indicates covered and uncovered lines or objects in the *Coverage source code* pane.

Figure 15 Show Line Coverage

Line Coverage of Coverage Detail

Block and Statement Coverage

Category	Coverage
Block	66.67%
Statement	87.50%

Named Objects

Status	Name	Line No.	Source
✓	testline1	14	if (a && b) c = 1'b0; // exclud...
✓	test_line2	18	x <= w & z;
✓	test_line3	21	y <= w[3:0] & z[3:0];
✓	bob	24	bob <= w & 3'b1;

Click for Next Uncovered Line

Current Line

Annotated line coverage in Source Code

Covered Line

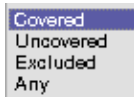
Uncovered

Summary Bar

The covered lines are highlighted in green and uncovered lines are highlighted in red. You can use the **Uncovered** **Navigation** menu to select the type of line for the   searching

icons. For example, selecting the **Covered** type in the search menu and clicking the  icon moves forward to the next covered line in the *Coverage Source Code* pane.

Figure 16 Navigation Menu



As shown in Figure [Show Ratio for Detail](#), you can use the **View -> Show Ratio for Detail** command to show the actual counts in the *Coverage Detail* panes.

Figure 17 Show Ratio for Detail

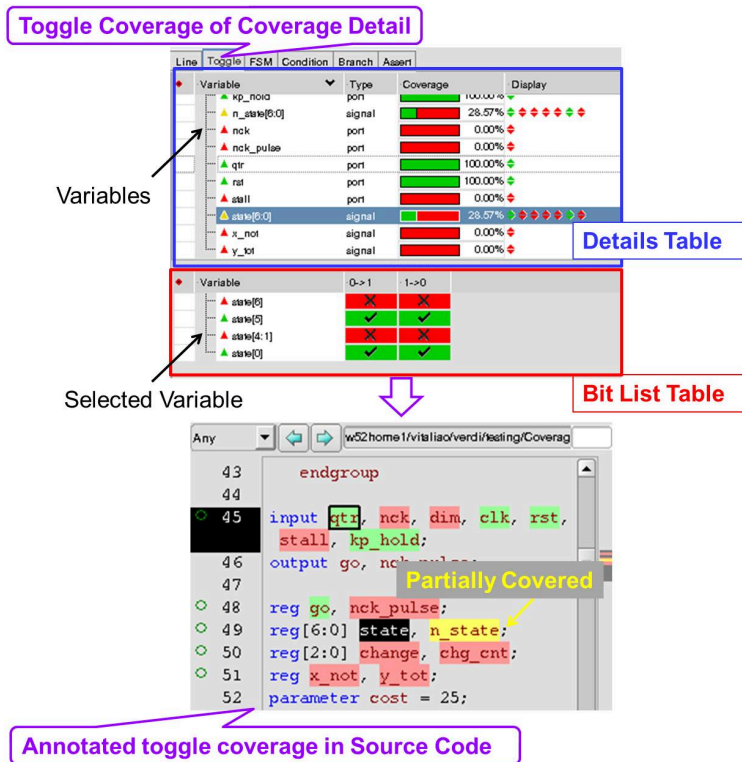
CovDetail		
Line	Toggle	FSM Condition Branch Assert
Category	Coverage	
Block	49	67.12%
	24	
Statement	60	67.42%
	29	

Toggle Coverage

The *Toggle* page in the *Coverage Details* pane displays the value transitions from 0 to 1 and 1 to 0 for each net and register. Refer to Figure [Check Toggle Coverage Details](#), for the following actions:

- Double-click a signal in the **Details** table and the signal is highlighted with its coverage range in the *Coverage source code* pane.
- Select a signal in the **Details** table, and the bit list of the signal is displayed in the **Bit List** table.
- The bits in the **Bit List** table can be compressed or uncompressed by selecting the **Compress Vector Bits** and **Uncompress Vector Bits** commands in the right-click menu or by double-clicking the bit.

Figure 18 Check Toggle Coverage Details

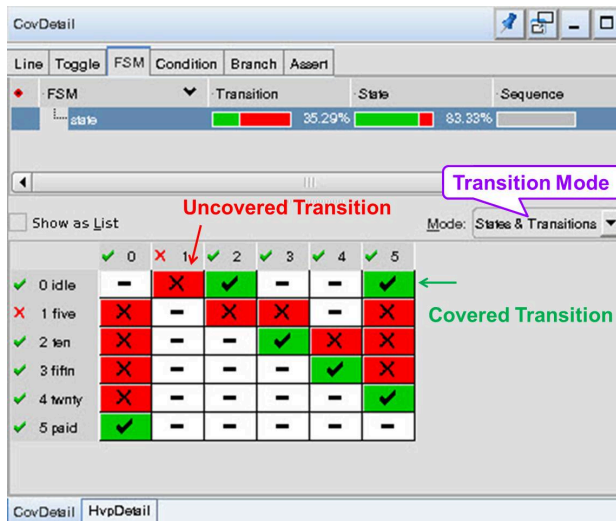


Using FSM Coverage

The default *FSM* page has two panes (see Figure [Check FSM Coverage Details](#))

- The upper pane displays the state, transition, and sequence coverage for each FSM with a percentage bar annotated with the total number of covered and uncovered objects and the percentage of each.
- The lower pane displays transition or sequence results for all possibilities in a table or list format.

Figure 19 Check FSM Coverage Details



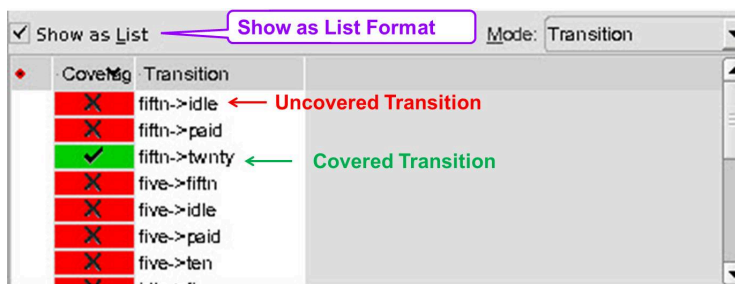
Displaying Transition Details

Transition mode for the selected FSM displays all the possible transitions and indicates whether the current state of the FSM made any of these transitions. You can display the results in a table or list format. In the table format (opened by default), select the **States & Transitions** mode in the **Mode** field. In the list format, select the **Transition** mode in the **Mode** field.

Figure [Check FSM Coverage Details](#) shows the table view with states and transitions displayed with covered transitions in green and uncovered transitions in red. Transitions that are not possible are white with a dash.

To view the transition details in list format with coverage ranges, enable the **Show as List** option. Figure [FSM Transition List](#) shows transitions with the covered transitions displayed in green and uncovered in red. Transitions that are not possible are not listed in the table.

Figure 20 FSM Transition List



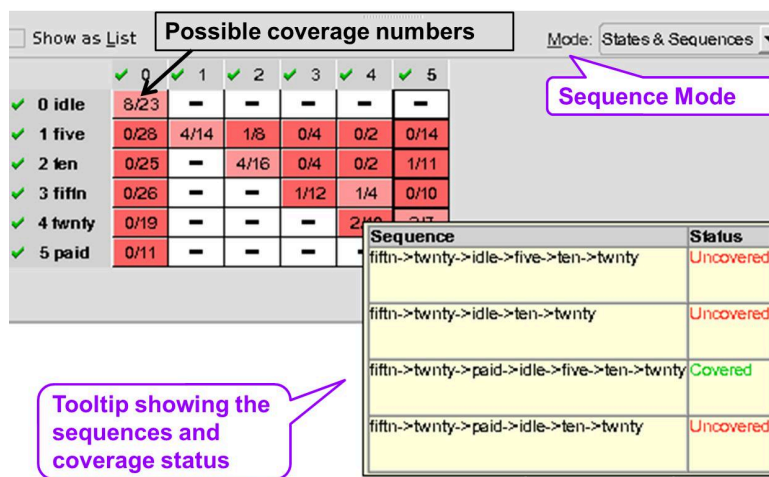
Displaying Sequences

Select the **States & Sequences** mode in the **Mode** field. As shown in Figure [FSM Sequence Table](#), the Sequences mode shows states and sequences and their possible coverage numbers that an FSM can follow to go from one state to another state. Holding the cursor on a cell displays a tooltip showing the sequences and their coverage.

Note:

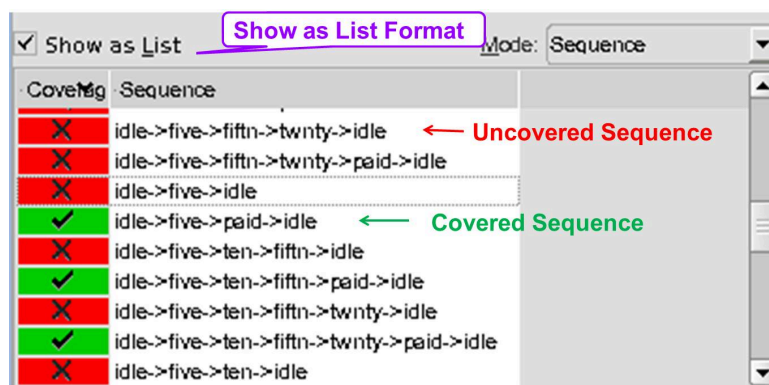
You must enable the sequences at compile-time with the **-cm_fsmopt sequence** option to view sequences in the Verdi platform.

Figure 21 FSM Sequence Table



To view the state details in list format, enable the **Show as List** option. Figure [FSM Sequence List](#) shows sequences with covered sequences displayed in green and uncovered in red.

Figure 22 FSM Sequence List

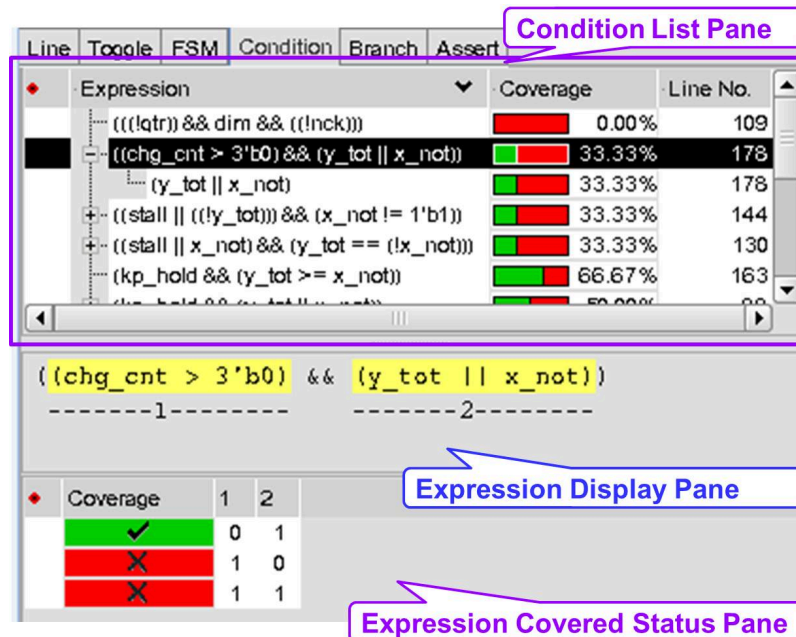


Using Condition Coverage

The *Condition* page helps you monitor whether both true and false states of the conditional expressions and sub-expressions in your code are covered. See Figure [Check Condition Coverage Details](#) for the following actions:

- The **Condition List** pane expands to show coverage by possible vectors or sub-conditions, percent coverage, and the line number of the expression.
- Double-click an expression in the **Condition List** pane, and the expression is highlighted with its coverage range in the *Coverage source code* pane.
- Select an expression in the **Condition List** pane, and the **Expression Display** pane displays the selected expression above with possible conditions and their coverage status below.

Figure 23 Check Condition Coverage Details

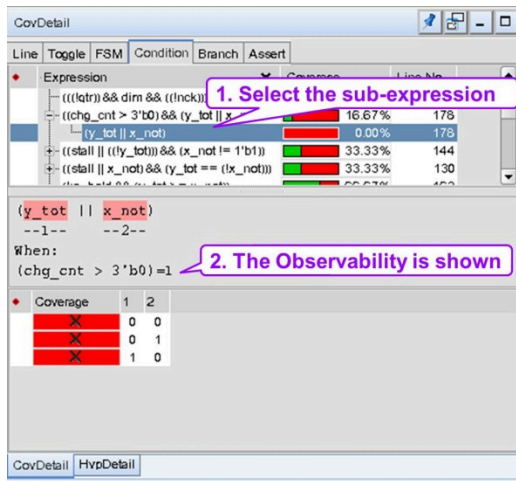


The *Condition* page also provides an observability feature to locate situations where a certain part of the combinational circuit is controlling (and hence observable) the primary output of the combinational circuit. Certain vector combinations of a sub-expression is marked as covered only if that combination is controlling the primary output in the context of the whole expression. As shown in Figure [Condition Coverage Observability](#), you can view the observability expression in the **Expression Display** pane.

Note:

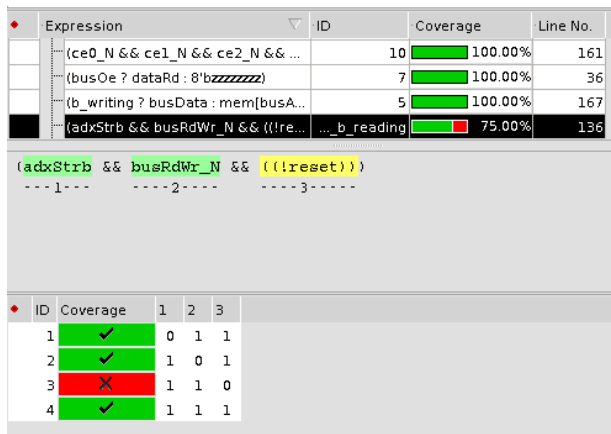
You must use the **-cm cond** and **-cm_cond obs** options at compile-time to enable the observability based condition coverage and view the observability in the Verdi platform.

Figure 24 Condition Coverage Observability



You can use the **View -> Show Condition IDs** command to show the condition names in the *Condition* page of the *CovDetail* pane.

Figure 25 Showing Condition Names

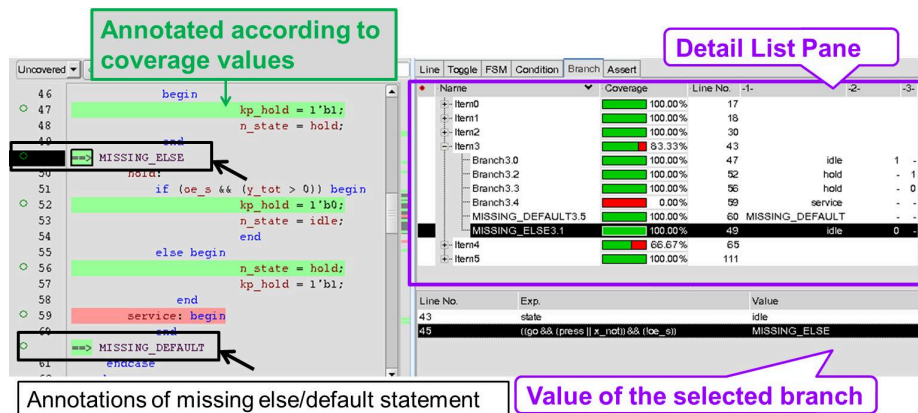


Using Branch Coverage

The *Branch* page displays the coverage percentage for branch coverage. See Section [Check Branch Coverage Details](#) for the following actions:

- The **Detail List** pane shows the items and branches of the selected scope. Expand an item and select a branch, and then the value of the selected branch is shown in the **Value** column of the lower pane.
- Double-click a branch to view the annotation in the *Coverage Source Code* pane for the branch. The corresponding statement in the source file is annotated to identify code that is covered and that is not covered. The source code lines are color-annotated according to their coverage values. You can see only the line of branches as color-annotated and not the line of terms.
- The implied branches are shown in the *Coverage source code* pane only when you view branch coverage or are analyzing the branch metric. The implied branches of case and if statements are indicated by *MISSING_ELSE* and *MISSING_DEFAULT* tags.

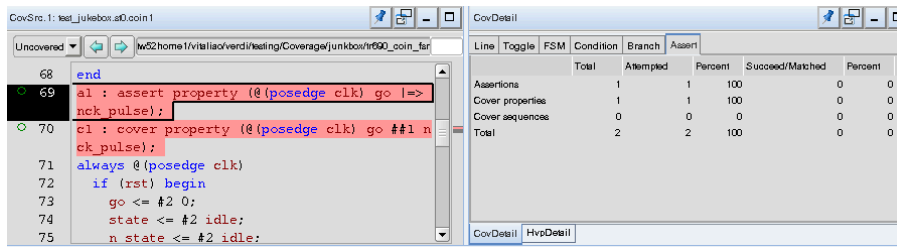
Figure 26 Check Branch Coverage Details



Using Assert Coverage

The *Assert* page displays the coverage counts and percentage for SystemVerilog Assertions (SVA) **assert** and **cover** statements.

Figure 27 Check the Assert Coverage



See Section [Using the Asserts Tab](#) for more details about the Assert coverage.

Using the Modules Tab

The *Module* tab displays coverage results by listing module definitions, variants for parameterized modules and module instances and their total coverage score and individual scores for line, toggle, FSM, condition, branch, and assert coverage. Click the icon to the left of the module name to see the instances under it (see Figure [Modules Page](#)).

Figure 28 Modules Page

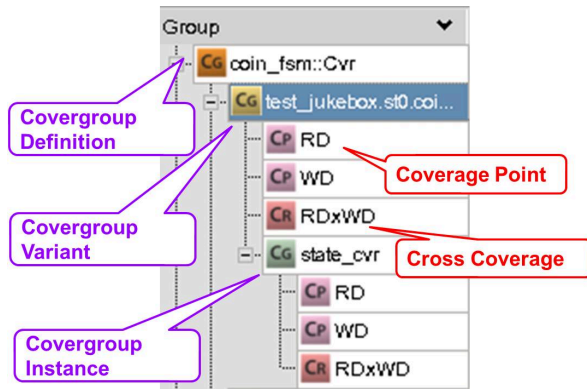
Name	Score	Line	Toggle	FSM	Condition	Branch	Assert
coin1	91.23%	100.00%	73.68%			100.00%	
coin_fern	77.93%	91.01%	83.87%	70.59%	42.59%	79.49%	100.00%
fifo	76.82%	88.24%	75.56%			66.67%	
jukebox	75.90%	96.00%	53.12%	100.00%	42.86%	87.50%	
kp_fsm	70.35%	89.66%	60.27%		43.24%	88.24%	
station	74.42%		74.42%				

The *Modules* page usage is the same as the *Hierarchy* page usage. Refer to the [Using the Hierarchy Tab](#) section for details.

Using the Groups Tab

The *Groups* page shows the function groups report. As shown in Figure [Covergroup Icons](#), the icons for covergroup represent the definition, variant, and instance of a covergroup. In addition, coverage points and cross coverage have their own icons.

Figure 29 Covergroup Icons



You can change the covergroup display by selecting or deselecting the  Tree icon. The figures [Groups Tab View With Tree Icon Selected](#), [Groups Tab With Definitions and Instances Trees Expanded](#) and [Groups Tab View With Tree Icon Deselected](#) show the display when the Tree is selected and deselected respectively.

Figure 30 Groups Tab View With Tree Icon Selected

Group	Score	Definition	U+C	U	C	X	Goal	Weight	AtLeast	PerInst	Ovr
Definitions	48.82%		1052998	1052970	28	0					
Instances	39.71%		3368655	32633184	1053369	0					

Figure 31 Groups Tab With Definitions and Instances Trees Expanded

Group	Score	Definition	U+C	U	C
Definitions	48.82%		1052998	1052970	
top	48.82%		1052998	1052970	
t	48.82%		1052998	1052970	
ral_cvr_block_OUT_BLK	100.00%		3	0	
addr_map	100.00%		3	0	
DUMMY_REG	100.00%		1	0	
STATUS_REG	100.00%		1	0	
TEMP_REG	100.00%		1	0	
ral_cvr_reg_TEMP_REG	59.38%		32	13	
ral_reg_DUMMY_REG	33.33%		3	2	
ral_reg_TEMP_REG	2.58%		1052960	1052955	
Instances	39.71%		3368655	32633184	10
OUT_BLK1	30.91%		16843272	5790036	10
DUMMY_REG	33.33%		9	6	
TEMP_REG	59.38%		512	208	
TEMP_REG_1	0.01%		2.58%	6842752	5789824
OUT_BLK2	8.33%		16843272	6843144	
DUMMY_REG	0.00%		9	9	
DMY_value	0.00%		3	3	
TEMP_REG	25.00%		512	384	
TEMP_REG_1	0.00%		2.58%	6842752	6842752
global_group_item	100.00%		6	0	

Figure 32 Groups Tab View With Tree Icon Deselected

Group	Score	Instances	U+C	U	C	X	Goal	Weight	AtLeast	PerInst	Ov
top.t:ral_cvr_block_DUT...	100.00%	100.00%	3	0	3	0	100%	1	1	1	1
top.t:ral_cvr_reg_TEMP_R...	59.38%	42.19%	32	13	19	0	100%	1	1	1	1
top.t:ral_reg_DUMMY_RE...	33.33%	16.67%	3	2	1	0	100%	1	1	1	1
top.t:ral_reg_TEMP_REG:f...	2.58%	0.01%	1052960	1052955	5	0	100%	1	1	1	1

You can also configure the **Attributes** that you want to see displayed in the *Groups* tab by selected them in the **Attributes** pane on the right-hand side of the *Groups* tab as shown in the Figure [Attributes and Cover Items in the Groups Tab](#).

The values of the cover items **U+C (Total)**, **C (Covered)**, **U (Uncovered)**, and **X (Excluded)** are also displayed at the top of the *Groups* tab as shown in the Figure [Attributes and Cover Items in the Groups Tab](#).

Figure 33 Attributes and Cover Items in the Groups Tab

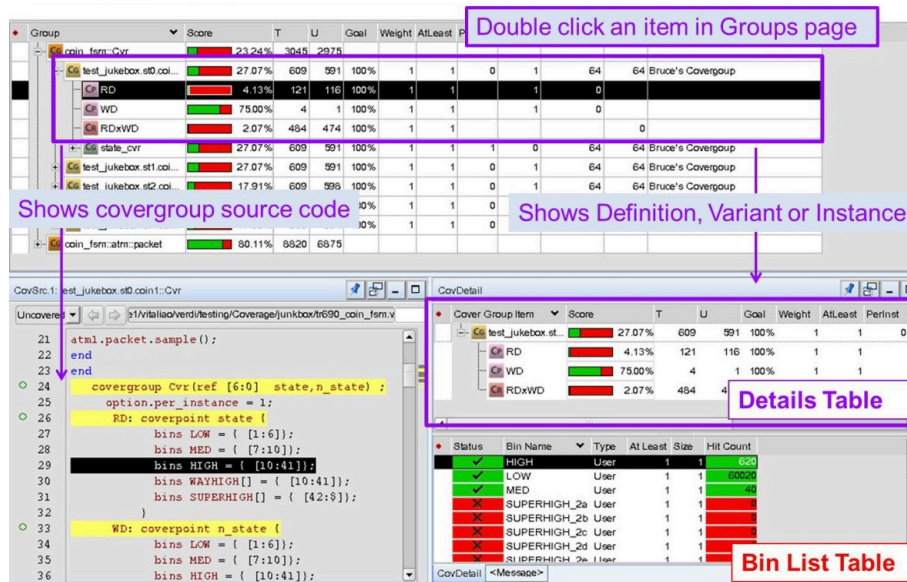
Group	Score	Instances	U+C	U	C	X	Goal	Weight	AtLeast	PerInst	Overlap	AutoBin	Missing	Con
coin_fsm:Cvr	23.24%	23.24%	3045	2975	70	0								
test_juke...	27.07%	27.07%	609	591	18	0	100%	1	1	1	1	64	64	Br
test_juke...	27.07%	27.07%	609	591	18	0	100%	1	1	1	1	64	64	Br
test_juke...	17.91%	17.91%	609	598	11	0	100%	1	1	1	1	64	64	Br
test_juke...	27.07%	27.07%	609	591	18	0	100%	1	1	1	1	64	64	Br
test_juke...	17.08%	17.08%	609	604	5	0	100%	1	1	1	1	64	64	Br
coin_fsm:at...	80.11%		8820	6875	1945	0								
test_juke...	80.11%		1764	1375	389	0	100%	1	1	0	1	64	64	
test_juke...	80.11%		1764	1375	389	0	100%	1	1	0	1	64	64	
test_juke...	80.11%		1764	1375	389	0	100%	1	1	0	1	64	64	
test_juke...	80.11%		1764	1375	389	0	100%	1	1	0	1	64	64	

For more information on cover items, see table [Cover Items Description](#).

The *Groups* page lists covergroup items with coverage score, goal, weight, and a coverage map view. See Figure [Groups Page](#) for the following actions:

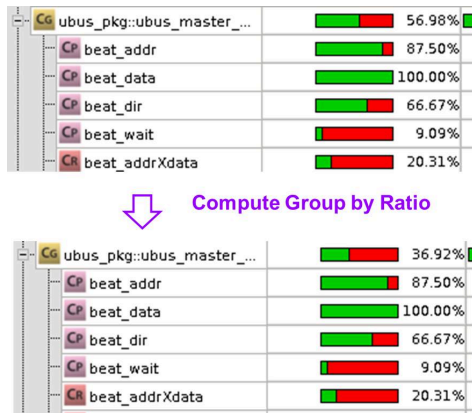
- Expand the covergroup definition to display the variant, coverage points, crosses, and instances.
- Double-click a covergroup item to view the source code in the *Coverage source code* pane.
- The double-clicked definition, variant, or instance can also be viewed in the **Details** table and the cover bins can be viewed in the **Bin** table of the *CovDetail* pane.

Figure 34 *Groups Page*



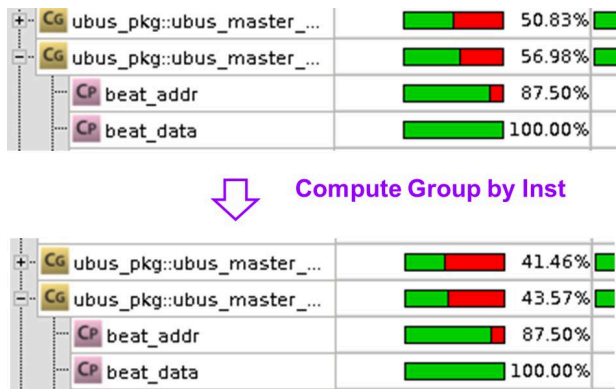
The score for a covergroup is calculated as the average of the percentages of individual cover point scores by default. You can use the **View -> Compute Group by Ratio** command to calculate the covergroup scores and overall group scores as a ratio between the number of covered bins and the total number of coverable bins. Figure [Compute Group by Ratio](#) shows the difference between the score calculation with and without the ratio.

Figure 35 Compute Group by Ratio



You can use the **View -> Compute Group by Instance** command to calculate the covergroup scores using the coverage of each instance for the covergroups with instance coverage. Figure [Compute Group by Instance](#) shows the difference between the score calculation with and without enabling this option.

Figure 36 Compute Group by Instance



The summary of the covergroup is shown in the **Summary Bar**. You can click the  icon to show or hide the **Summary Bar** in the **Groups** page.

Figure 37 Covergroup Summary Bar



You can search and filter the **Group** and **Comment** columns in the *Groups* page accordingly. The matching expression format is `[Column Name]:[Expression]`. For example, only show the groups which contain the `st0*` string in the **Group** column or the groups which contain the `Bruce*` string in the **Comment** column:

Figure 38 Filtering Groups by the Group Column

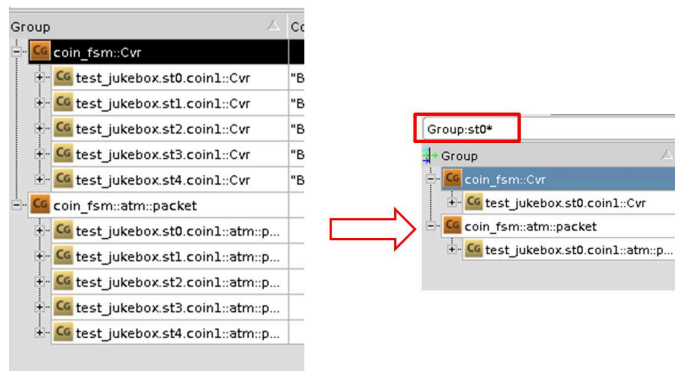
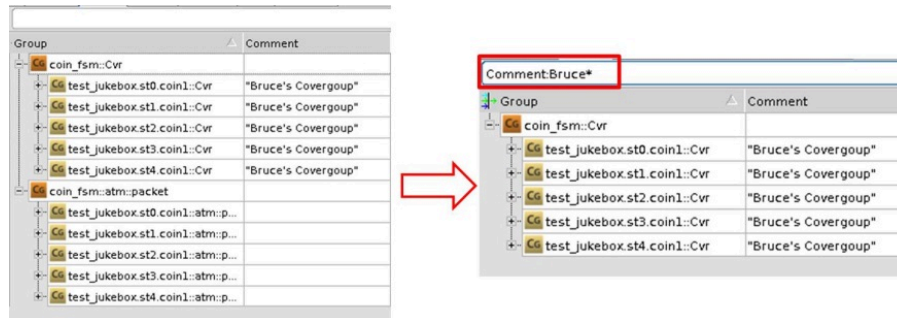


Figure 39 Filtering Groups by the Comment Column



You can specify multiple matching expressions by separating them using a space. If the expression contains a space, bracketing the expression with `" "` is required. For example,

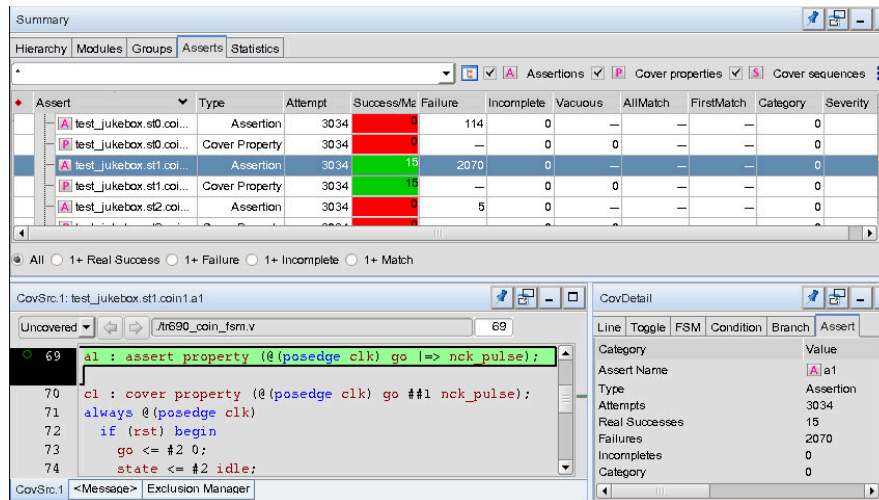
```
*ohio*
Comment:*ohio*
Comment:"ohio" Group:*coin*
Comment:"coin ohio"
```

Using the Asserts Tab

The *Asserts* tab shows the SystemVerilog Assertions (SVA) report about the coverage counts and percentage for SVA *assert* and *cover* statements (see Figure [Asserts Coverage](#)). For example, the highlighted row in Figure [Asserts Coverage](#) shows the simulator checked 3034 times (in the **Attempts** column) whether the *assert* statement or directive succeeded or the *cover* statement or directive matched. The *assert* statement

succeeded (in the **Successes/Match** column) 15 times and the *assert* statement did not succeed (in the **Failure** column) 2070 times. A real success is when the entire expression succeeds or matches without vacuous successes. The *cover* statement did not have any failure, so the entry for the *cover* statement in this column is always 0.

Figure 40 Asserts Coverage



The ☒ **A** Assertions, ☒ **P** Cover properties, and ☒ **S** Cover sequences options can be used to select the asserts to show in the *Summary* pane. You can also use the **All**, **1+Real Success**, **1+Real Success**, **1+Failure**, **1+Incomplete** and **1+Match** toggle options to show all asserts, at least one real success, at least one failure, at least one incomplete or at least one match statement in the *Asserts* tab of the *Summary* pane.

Double-click a statement in the *Asserts* page of the *Summary* pane to display the source code in the *Coverage source code* pane and the static report in the *Assert* page of the *CovDetail* pane.

You can switch Assert view using the mode selection menu shown in Figure [Switch Asserts View Mode](#). There are three options you can choose from: *Flat*, *Module/Instance* and *Hierarchy*.

Figure 41 Switch Asserts View Mode

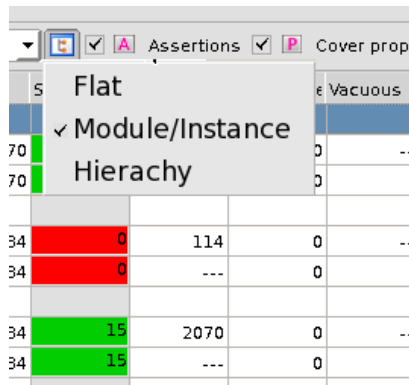


Figure [Flat View of Assertions](#) shows the *Flat* view of the assertions. In this mode, all the instance assertions are shown in a flat list. Note that only instance level assertions are shown in this mode, module level ones are not shown.

Figure 42 Flat View of Assertions

Hierarchy Modules Groups Asserts Statistics Tests										
* [Flat] [A] Assertions [P] Cover properties [S] Cover sequences [List Icon]										
Assert	Type	Attempt	Success/F	Failure	Incomplete	Vacuous	AllMatch	FirstMatch	Category	Severity
[A] arb.arb_sva_inst.cover_all_reqs_same_time	... Property	0	0	---	0	0	---	---	0	0
[A] arb.arb_sva_inst.cover_no_gnts	... Property	0	0	---	0	0	---	---	0	0
[A] arb.arb_sva_inst.gnt_oh	Assertion	0	0	0	0	---	---	---	0	0
[A] arb.arb_sva_inst.gnt_oh0	Assertion	0	0	0	0	---	---	---	0	0
[A] arb.arb_sva_inst.never_gnt_wo_req	Assertion	0	0	0	0	---	---	---	0	0
[A] arb.gnt_oh	Assertion	0	0	0	0	---	---	---	0	0
[A] test_jukebox.st0.coin1.a1	Assertion	3034	0	114	0	---	---	---	0	0
[A] test_jukebox.st0.coin1.c1	... Property	3034	0	---	0	0	---	---	0	0
[A] test_jukebox.st1.coin1.a1	Assertion	3034	15	2070	0	---	---	---	0	0
[A] test_jukebox.st1.coin1.c1	... Property	3034	15	---	0	0	---	---	0	0
[A] test_jukebox.st2.coin1.a1	Assertion	3034	0	5	0	---	---	---	0	0
[A] test_jukebox.st2.coin1.c1	... Property	3034	0	---	0	0	---	---	0	0
[A] test_jukebox.st3.coin1.a1	Assertion	3034	0	12	0	---	---	---	0	0
[A] test_jukebox.st3.coin1.c1	... Property	3034	0	---	0	0	---	---	0	0
[A] test_jukebox.st4.coin1.a1	Assertion	3034	0	0	0	---	---	---	0	0
[A] test_jukebox.st4.coin1.c1	... Property	3034	0	---	0	0	---	---	0	0

Figures [Module/Instance View of Assertions](#) and [Hierarchy View of Assertions](#) show the module/instance and hierarchy mode for assertions respectively.

Figure 43 Module/Instance View of Assertions

Hierarchy Modules Groups Asserts Statistics Tests									
* Assert									
Type	Attempt	Success/M	Failure	Incomplete	Vacuous	AllMatch	FirstMatch	Category	
arb_svas									
arb_svas.never_gnt_wo_req	Assertion	0	0	0	0	---	---	0	
arb_svas.gnt_oh0	Assertion	0	0	0	0	---	---	0	
arb_svas.gnt_oh	Assertion	0	0	0	0	---	---	0	
arb_svas.cover_no_gnts	... Property	0	0	---	0	0	---	0	
arb_svas.cover_all_reqs_same_time	... Property	0	0	---	0	0	---	0	
arb.arb_sva_inst									
arb.arb_sva_inst.never_gnt_wo_req	Assertion	0	0	0	0	---	---	0	
arb.arb_sva_inst.gnt_oh0	Assertion	0	0	0	0	---	---	0	
arb.arb_sva_inst.gnt_oh	Assertion	0	0	0	0	---	---	0	
arb.arb_sva_inst.cover_no_gnts	... Property	0	0	---	0	0	---	0	
arb.arb_sva_inst.cover_all_reqs_same_time	... Property	0	0	---	0	0	---	0	
arb									
arb.gnt_oh	Assertion	0	0	0	0	---	---	0	
arb									
arb.gnt_oh	Assertion	0	0	0	0	---	---	0	
coin_fsm									
coin_fsm.a1	Assertion	15170	15	2201	0	---	---	0	
coin_fsm.c1	... Property	15170	15	---	0	0	---	0	
test_jukebox.st0.coin1									
test_jukebox.st0.coin1.a1	Assertion	3034	0	114	0	---	---	0	
test_jukebox.st0.coin1.c1	... Property	3034	0	---	0	0	---	0	
test_jukebox.st1.coin1									
test_jukebox.st1.coin1.a1	Assertion	3034	15	2070	0	---	---	0	
test_jukebox.st1.coin1.c1	... Property	3034	15	---	0	0	---	0	
test_jukebox.st2.coin1									
test_jukebox.st2.coin1.a1	Assertion	3034	0	5	0	---	---	0	
test_jukebox.st2.coin1.c1	... Property	3034	0	---	0	0	---	0	
test_jukebox.st3.coin1									
test_jukebox.st3.coin1.a1	Assertion	3034	0	12	0	---	---	0	
test_jukebox.st3.coin1.c1	... Property	3034	0	---	0	0	---	0	
test_jukebox.st4.coin1									
test_jukebox.st4.coin1.a1	Assertion	3034	0	0	0	---	---	0	
test_jukebox.st4.coin1.c1	... Property	3034	0	---	0	0	---	0	

Figure 44 Hierarchy View of Assertions

Hierarchy Modules Groups Asserts Statistics Tests									
* Assert									
Type	Attempt	Success/M	Failure	Incomplete	Vacuous	AllMatch			
arb									
arb.gnt_oh	Assertion	0	0	0	0	---	---		
arb.arb_sva_inst									
arb.arb_sva_inst.cover_all_reqs_same_time	... Property	0	0	---	0	0	---		
arb.arb_sva_inst.cover_no_gnts	Property	0	0	---	0	0	---		
arb.arb_sva_inst.gnt_oh	Assertion	0	0	0	0	---	---		
arb.arb_sva_inst.gnt_oh0	Assertion	0	0	0	0	---	---		
arb.arb_sva_inst.never_gnt_wo_req	Assertion	0	0	0	0	---	---		
test_jukebox									
st0									
coin1									
test_jukebox.st0.coin1.a1	Assertion	3034	0	114	0	---	---		
test_jukebox.st0.coin1.c1	... Property	3034	0	---	0	0	---		
st1									
coin1									
test_jukebox.st1.coin1.a1	Assertion	3034	15	2070	0	---	---		
test_jukebox.st1.coin1.c1	... Property	3034	15	---	0	0	---		
st2									
coin1									
test_jukebox.st2.coin1.a1	Assertion	3034	0	5	0	---	---		
test_jukebox.st2.coin1.c1	... Property	3034	0	---	0	0	---		
st3									
coin1									
test_jukebox.st3.coin1.a1	Assertion	3034	0	12	0	---	---		
test_jukebox.st3.coin1.c1	... Property	3034	0	---	0	0	---		

Note:

Coverage information does not exist in the database for **assume** statements.

Using the Statistics Tab

The *Statistics* tab shows the overall coverage scores and statistics analysis report. It summarizes the types of coverage elements.

Figure 45 Coverage Statistics

Hierarchy	Modules	Groups	Asserts	Statistics	Tests										
Design Hierarchy				Module List				Assertions				Cover Groups			
Metric	Total	Covered	Score	Metric	Total	Covered	Score	Category	Total	Covered	Category	Total	Covered	Score	Uncovered
Assert	4	3	75.00	Assert	6	3	50	Assertion	6		Total Group Coverage Score			31.15%	
Condition	10	10	100.00	Condition	10	10	100	Cover Property			Total Group Coverage Instance Score			31.15%	
FSM	4	3	75.00	FSM	4	3	75	Cover Sequence			Cover Point	8	0	0.00%	
Line	50	33	66.00	Line	51	33	64	Total	6		Cover Point Bins	91	23	25.27%	
Total	68	49	72.06	Total	71	49	69				Cover Cross				
Average			79.00	Average			72				Cover Cross Bins				
											All Bins	91	23	25.27%	
											Group	3	0	0.00%	
											Group Instance Point	8	0	0.00%	
											Group Instance Point Bins	91	23	25.27%	
											Group Instance Cross				
											Group Instance Cross Bins				
											Group Instance All Bins	91	23	25.27%	
											Group Instance	6	0	0.00%	

The *Statistics* page contains four sections: *Design Hierarchy*, *Module List*, *Assertions*, and *Covergroups*. Code. All the tabs provide the Total, Covered and Score of each coverage metric. The *Design Hierarchy* and *Module List* tabs displays coverage results based on the design hierarchy and the module list respectively. The *Assertions* tab displays the coverage results for *Assertions*, *Cover Properties*, and *Cover Sequences*. The *Covergroups* tab displays the coverage scores of *Groups*, *Cover Points*, and *Cover Crosses* are also displayed.

The following table describes how the scores are calculated for the **Total Group Coverage Score** and **Total Group Coverage Instance Score**. The remaining fields mentioned under *Covergroups* are descriptive. For example, *Cover Point* represents the number of cover points in the covergroup.

Covergroup Category	Definition	How the Score Value is Calculated
Total Group Coverage Score	Average score of all Covergroups present in a group calculated based on the Score column in <i>Groups</i> tab.	The value of Total Group Coverage Score changes based on the computation modes used. When the Compute Group By Ratio mode is off, the score is calculated as follows: Total Group Coverage Score: SUM (every group variant' score* its weight) / SUM (every group variant' weight) When the Compute Group By Ratio option is on, the score is calculated as follows: Total Group Coverage Score: SUM (every groups instance's covered number* its weight) / SUM (every groups instance's coverable number* its weight), If a groups variant has no instance (per_instance==0), the variant is used for calculation

Total Group Coverage Instance Score	Average score of all Covergroup Instances within a group calculated based on the Instance column in <i>Groups</i> tab.	The value of Total Group Coverage Instance Score changes based on the computation modes used. When the Compute Group By Ratio mode is off, the score is calculated as follows: Total Group Coverage Instance Score: SUM ((every group instance's score* its weight) / SUM (every group instance's score* its weight)) If a groups variant has no instance (per_instance==0), use variant score/weight for calculation. When the Compute Group By Ratio option is on, the score is calculated as follows: Total Group Coverage Instance Score: SUM (every groups instance's covered number* its weight)/SUM (every groups instance's coverable number* its weight) If a groups variant has no instance (per_instance==0), use the variant for calculation.
--	--	---

Using the Tests Tab

The *Tests* tab shows test details, such as test name, test location, type, status, start time, finish time, and simulation time. It also provides summary of overall test reports. The following figure shows the *Tests* tab:

Figure 46 The Tests Tab

Hierarchy	Modules	Groups	Asserts	Statistics	Tests	User Data
Data from 11 tests was loaded.						
Test Name/Test Location	Type	Status	Started	Finished	Simulation Time	
TEST1						
simv1/test11	Test Record	pass	2017/05/02 02:16:20	2017/05/02 02:16:21	41095000 ms	
userdata.txt:2	Test Record	pass				
TEST2						
TEST3						
simv2/test21	Test Record	pass	2017/05/02 02:16:26	2017/05/02 02:16:27	23395000 ms	
simv3/test31	Test Record	pass	2017/05/02 02:16:32	2017/05/02 02:16:33	37575000 ms	
userdata.txt:4	Test Record	fail				
TEST4						
TEST5						
Category	Value					
Status	pass					
Simulation Time	41095000 ms					
workdir	/slowfs/client_TD/unit_COVNEW/hvp_TestMetric/local_tests_testNameStatus					
Peak Memory	98660 kbytes					
Test Name	TEST1					
Finished	2017/05/02 02:16:21					
Host Name	host1					
Started	2017/05/02 02:16:20					
Test Location	simv1/test11					
Command	simv1 -cm line+cond+tgl+assert+fsm+branch +ntb_random_seed=1 -cm_name test11					
CPU Time	0.620 seconds					
User Name	usr1					

The *Tests* tab groups test records with the same name in the same folder. For example, all the test records with name as `TEST3` are grouped together in the above figure. If a test record has a unique name, then it is displayed as a single line with the test name as `[Test Name]--(test location)`.

```
[file name]:[Line number]
```

Using the UPF Low-Power Coverage

The *UPF LP* page shows the UPF report about the coverage counts and percentage by low power design hierarchy (refer to Figure [Power Coverage](#)). Power domain with the same prefix is grouped. All the columns are used to sort the data by clicking the column name field.

Power Domain

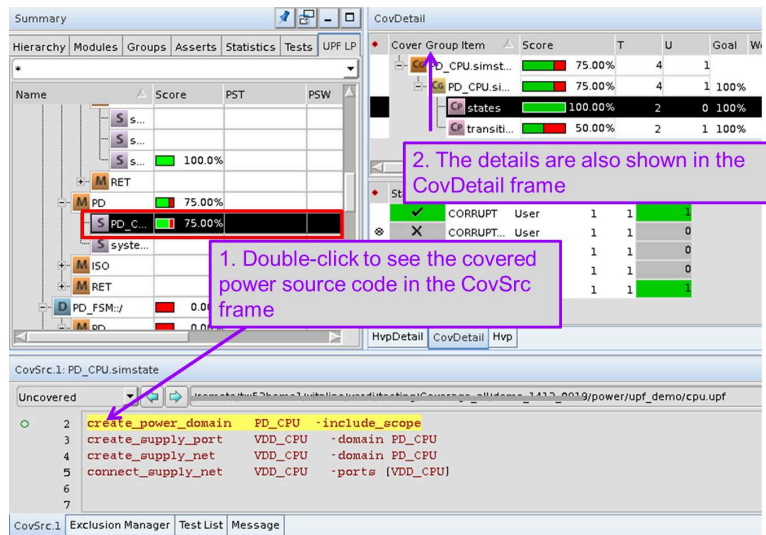
Name	Score	PST	PSW	SUPPLY_STAT1	PD	ISO
PD_TOP	72.54%	83.33%	75.00%	80.95%	23.44%	
PD_CPU::i_cpu	96.88%				93.75%	
PD_AL					100.0%	
PD_C					100.0%	
PD_PCU	100.0%				100.0%	
PD	75.00%				75.00%	
PD_C...	75.00%				75.00%	
system						
ISO						
RET						
PD_FSM::f	0.00%				0.00%	
PD_MEM::f	0.00%				0.00%	
PD	0.00%				0.00%	
PD_M...	0.00%				0.00%	

Metric

Source

You can double-click a power domain of the *Summary* pane to highlight where the **create_power_domain** command is specified in the source code in the *CovSrc* pane. You can double-click a source to highlight the source code in the *CovSrc* pane. The corresponding Covergroup is also shown in the *CovDetail* pane (see Figure [Show Details of Power Coverage](#)).

Figure 48 Show Details of Power Coverage



You can invoke the Verdi debug platform from the Verdi Coverage GUI with the **Tools -> Debug** menu and add the power domain into the *nWave* window. The `-f design.f -ssf novas.fsd` options from the command line are shown in the *Invoke Verdi Debug* form, when the following command is used:

```
% verdi -cov [coverage options] [debug options]
```

In the *UPF LP* page, select a power domain and use the **Add Power Domain to nWave** command in the right-click menu to add it into Verdi *nWave* window.

Using Application Preferences

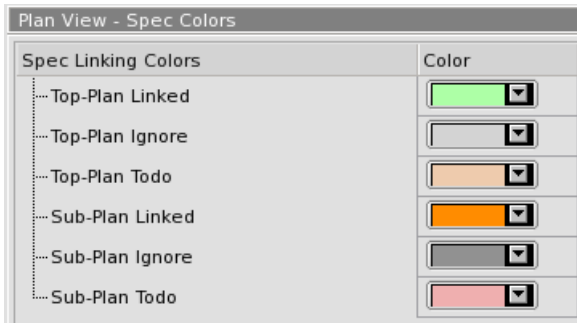
This section consists of the following subsection:

- [Color Settings of a Verification Plan](#)
- [Changing Coverage Aggregation Based on Metric Coverpoints](#)

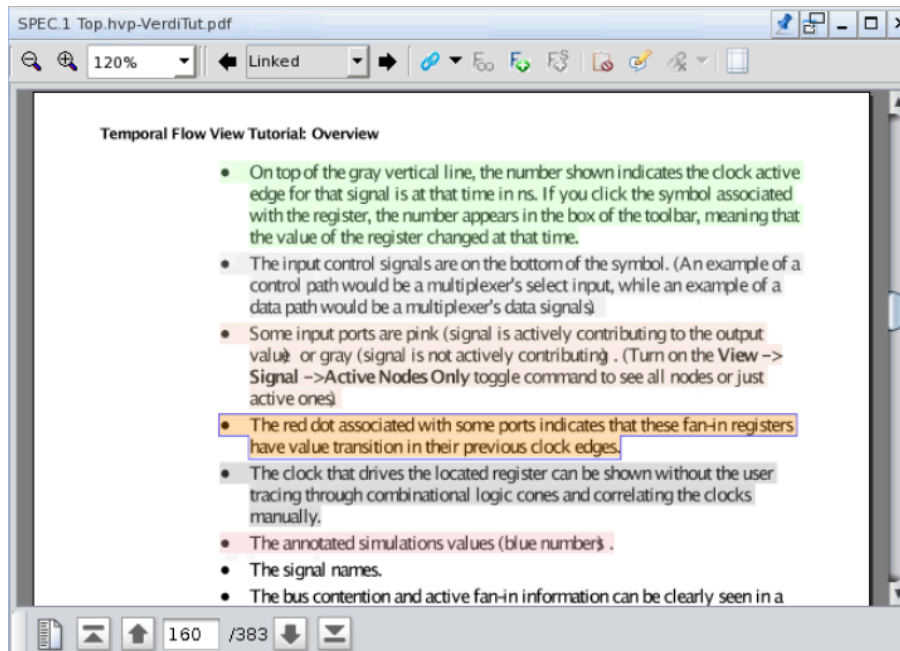
Color Settings of a Verification Plan

Verdi enables you to update the color settings of verification plans, their sub-plans, specification linking colors, and coverage colors. To update the color settings, perform the following steps:

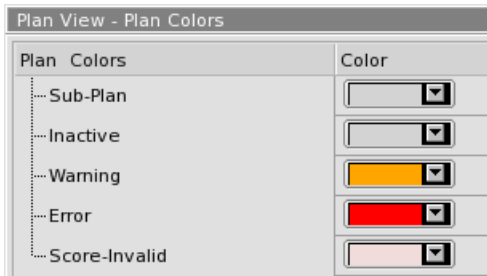
1. Click **Tools > Preferences** to open the **Application Preferences** form.
2. In the **Plan View** Category, select **Spec Colors** or **Plan Colors** as per your requirement:
 - Select **Spec Colors** to update the color settings of **Linked**, **Ignore**, and **Todo** text of top and sub plans in a PDF document.
 - Select **Plan Colors** to update the color settings of sub-plans, inactive cells, warning and error messages, and an invalid score of an item in the *Hvp* pane.
3. To change spec-linking colors, select **Spec Colors** to show the following list:



An example of the PDF document after setting its color using **Spec Colors** is shown in the following figure:



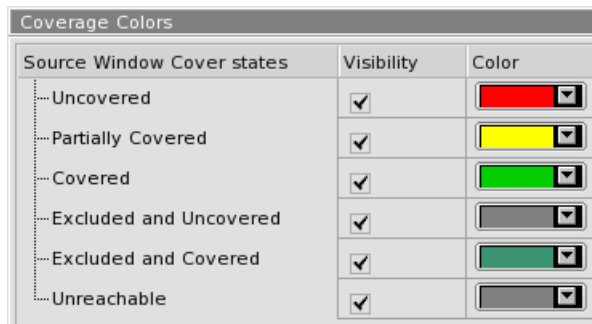
4. To change plan colors, select **Plan Colors** to show the following list:



An example of plan colors is shown in the following figure:

Name	owner	weight	Score	Line	goal.Line
myplan		1	34.25%	70.38%	
1 module	User1	1	52.39%	73.68%	(Line >= 0.9)
1.1 kp_fsm	User1	1	64.38%	89.66%	(Line >= 0.9)
1.2 coin_fsm	User1	1	69.70%	91.01%	(Line >= 0.9)
1.3 jukebox	User1	1	71.49%	96.00%	(Line >= 0.9)
1.4 station	User1	1	0.00%	0.00%	(Line >= 0.9)
1.5 test_jukebox	User1	1			(Line >= 0.9)
1.6 fifo	User1	1			(Line >= 0.9)
1.7 cd	User1	1	86.84%	100.0%	(Line >= 0.9)
2 instance1	User2	2	44.68%	69.89%	
3 instance2	User3	0	49.76%	75.72%	
4 Fun_Cov	User4	1	11.62%		
4.1 fun_cov	User4	1	11.62%		
4.1.1 cov_assertion	User4	1	0.00%		
4.1.2 cov_group	User4	1	23.24%		

You can also change coverage colors to update the color settings of uncovered, partially covered, and covered objects. Also, you can update the color settings of excluded and covered, excluded and uncovered, and unreachable objects. To change coverage colors, select the **Coverage Colors** category from the **Application Preferences** dialog box as shown in the following figure:



Changing Coverage Aggregation Based on Metric Coverpoints

By default, coverage score is calculated based on the average coverage rate of a metric. However, coverage metric with higher coverage points must impact coverage score than the one with lower coverage points. Therefore, Verdi Coverage enables you to automatically set weight according to metric coverpoints.

This section consists of the following two subsections:

- [Setting Weight Based on Metric Coverpoints](#)
- [Updates to the Verdi Coverage GUI](#)
- [Calculating the Auto-Weight Score](#)
- [Limitation](#)

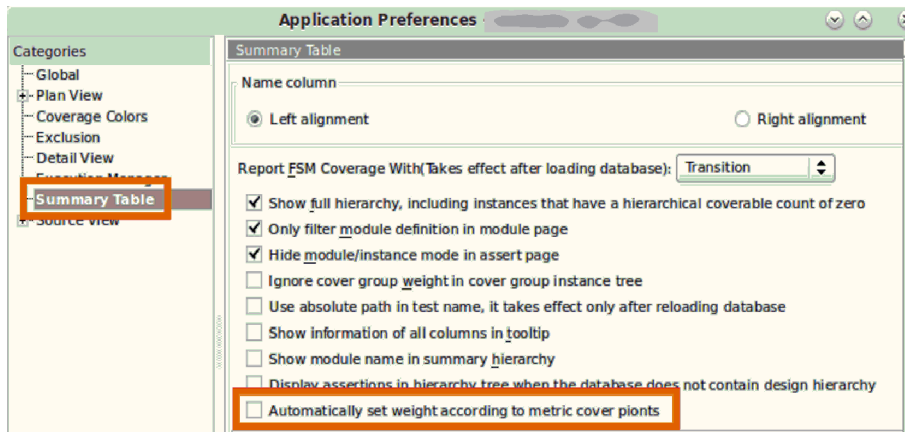
Setting Weight Based on Metric Coverpoints

To automatically set weight according to the number of coverage points, perform the following steps in the Verdi Coverage GUI:

1. Select **Tools > Preference** to display the **Application Preferences** dialog box.
2. In the dialog box figure ([Application Preferences](#)), select the following check box from the **Summary Table** category:
 - Automatically set weight according to metric coverpoints

This option is disabled by default.

Figure 49 Application Preferences



Updates to the Verdi Coverage GUI

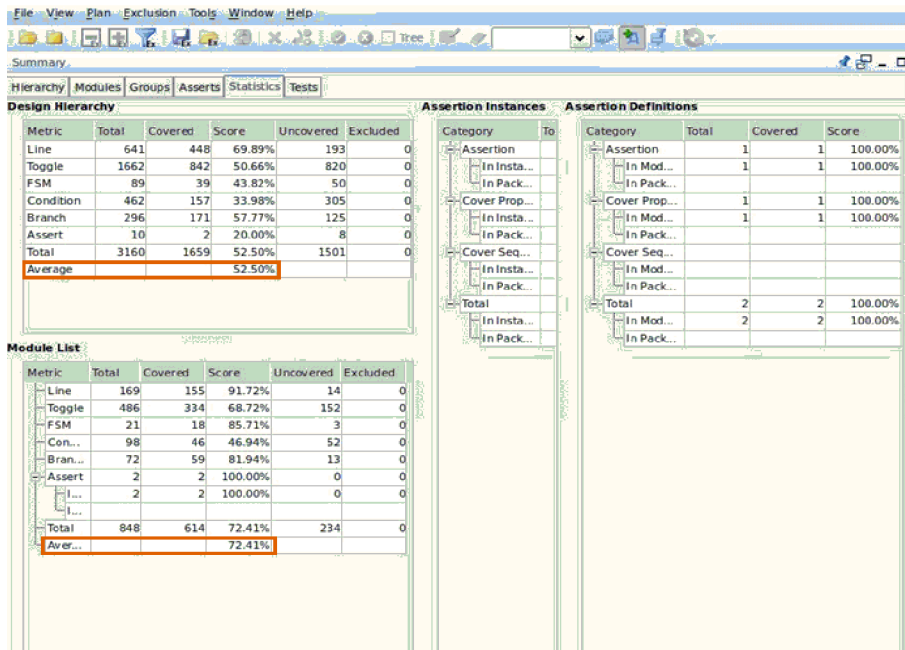
Once the *Set Weight* check box (see Figure [Application Preferences](#)) is enabled in the **Application Preferences** dialog box, the **Score (AutoWt)** column is included on the *Hierarchy* and *Modules* tabs of the *Summary* pane as shown in Figure [The Score \(AutoWt\) Column in the Summary Pane](#).

Figure 50 The Score (AutoWt) Column in the Summary Pane

Name	Score (AutoWt)	Line	Toggle	FSM	Condition
test_jukebox	52.50%	69.89%		50.66%	43.82%
cd1	79.59%	100.00%		73.68%	
fifo1	76.99%	88.24%		75.56%	
jb1	66.67%	96.00%		53.12%	100.00%

In addition, the average score in *Design Hierarchy* and *Module List* of the *Statistics* tab (see Figure [Average Score in the Statistics Tab](#)) is recalculated based on metric coverage points.

Figure 51 Average Score in the Statistics Tab



Calculating the Auto-Weight Score

Once the **Set Weight** check box Figure ([Application Preferences](#)) is enabled in the **Application Preferences** dialog box, the value of the **Score (AutoWt)** column is recalculated based on the coverage points in each metric. For calculating the coverage score, the following formula is used:

$$[\text{sum_of_metric-score_times_metric-point}] = (\text{line-score} * \text{line U+C}) + (\text{toggle-score} * \text{toggle U+C}) + (\text{FSM-score} * \text{FSM U+C}) + (\text{Condition-score} * \text{Condition U+C}) + (\text{Branch-score} * \text{Branch U+C}) + (\text{Assert-score} * \text{Assert U+C})$$

$$[\text{sum_of_metric-point}] = (\text{line U+C}) + (\text{toggle U+C}) + (\text{FSM U+C}) + (\text{Condition U+C}) + (\text{Branch U+C}) + (\text{Assert U+C})$$

$$\text{Score(Auto)} = [\text{sum_of_metric-score_times_metric-point}] / [\text{sum_of_metric-point}]$$

For example, the **Score (AutoWt)** value for test_jukebox in Figure [The Score \(AutoWt\) Column in the Summary Pane](#) is 52.50% and is calculated as:

$$\begin{aligned} [\text{sum_of_metric-score_times_metric-point}] &= (69.89\% * 641) + (50.66\% * 1662) + (43.82\% * 89) + (33.98\% * 462) + (57.77\% * 296) + (20\% * 10) \\ &= (44799.49\%) + (84196.92\%) + (3899.98\%) + (15698.76\%) + (17099.92\%) + (200\%) \\ &= 165895.07\% \end{aligned}$$

Chapter 4: Using Verdi Coverage Shipped Example for User Reference

```
[sum_of_metric-point] = (641+1662+89+462+296+10)= 3160
```

```
Score for auto-weight = 165895.07%/3160 = 52.50%
```

Limitation

- The formula to calculate auto-weight score is only applied on the *Hierarchy*, *Modules*, and *Statistics* tabs in the *Summary* pane. Other areas, such as URG, or Verdi Source and CovDetails windows, HVP Scores do not support this formula.

Shipped Example for User Reference

You can refer to a coverage example that is shipped for your reference. To do so, perform the following steps:

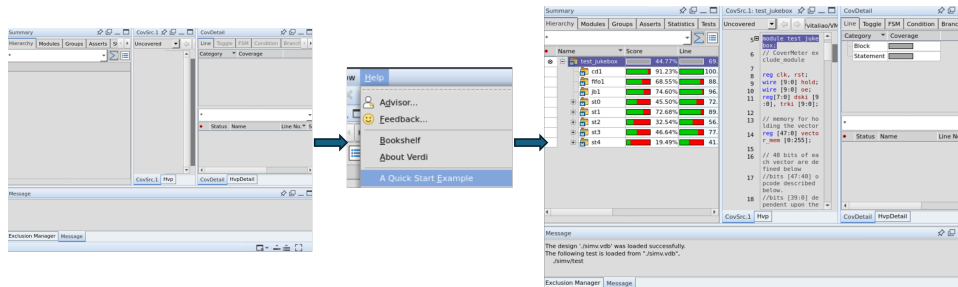
- Set up your environment with Verdi and VCS settings.
- Use the following command to open an empty coverage GUI:

```
%> verdi -cov
```

- From the **Help** menu, select **A Quick Start Example**.

Result

- The example shipped for coverage is copied to your local directory.
- The simulation results, including coverage data, are generated in your local directory through an automatic compilation and simulation process.
- The `simv.vdb` file is loaded automatically into the Verdi Coverage GUI.



5

Using Verification Planner

The Verdi platform also provides the *Verification Planner* feature to help you focus on a hierarchical verification plan and manage the specification for higher coverage.

The verification planner (HVP planner) in Verdi Coverage allows you to create a specific plan to measure what is important for each design and also allows for accountability, so that what gets measured gets done. Plans should be created as early as possible to determine the important metrics; therefore, as the design progresses, the important features in the plan are not forgotten. Verification plans can be saved into a readable format, for you to present to management, which keeps everyone informed about the coverage plan.

In this chapter, you are able to:

- Associate a plan with multiple specs.
- Link a feature to multiple sections in the spec.
- Link a section in the specification with multiple features.

This chapter encompasses the following sections:

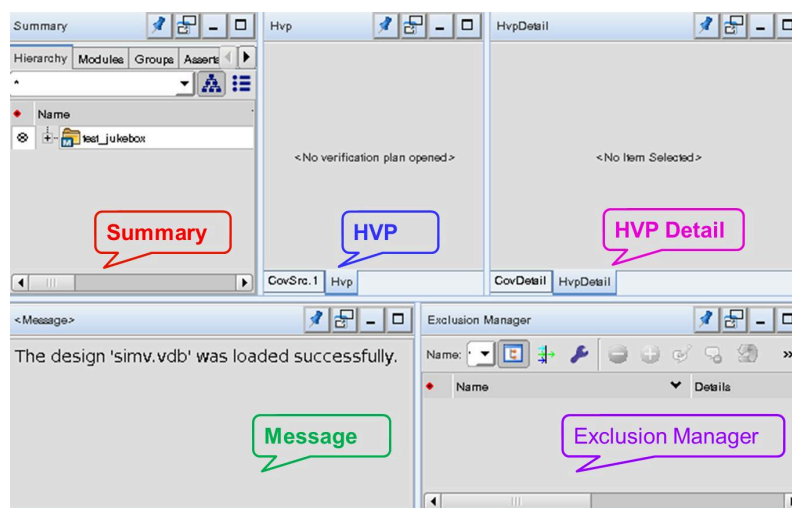
- [Using the Layout](#)
- [Using the HVP Planner](#)
- [Creating Features for Planner](#)
- [Loading a Predefined HVP File](#)
- [Saving the Plan](#)
- [User Data As a Spreadsheet Worksheet](#)
- [Loading and Saving a Verification Plan Using Environment Variables](#)
- [Showing Missing Scores](#)
- [Graphical Traceability Support](#)
- [Using Derived Test Metrics](#)
- [Using Test Name as HVP Source](#)

- [Using User-Defined Attributes and Metrics](#)
- [Using Modifiers](#)
- [Hiding Complete Features Using the Toggle Filter](#)
- [Propagation of Values](#)
- [Calculating the Count of Covergroup Bins](#)
- [Shipped Example for User Reference](#)

Using the Layout

Before starting the HVP planner, you need to change Verdi Coverage to Plan mode. Use the **Window -> Plan Mode** command to invoke the **Plan** mode. As shown in Figure [Plan Mode](#), the *Summary*, *Hierarchical Verification Plan (HVP)*, *Hierarchical Verification Plan Detail*, *Message*, and *Exclusion Manager* panes are displayed in the main window.

Figure 52 Plan Mode



Using the HVP Planner

You can create a verification plan and set the features using the HVP planner in the *Hvp* pane. The inputs to the *HVP* planner include a hierarchical verification plan in (*.hvp) format, spec-linking *.pdf files for features, and Excel (*.xml) spreadsheet format ³.

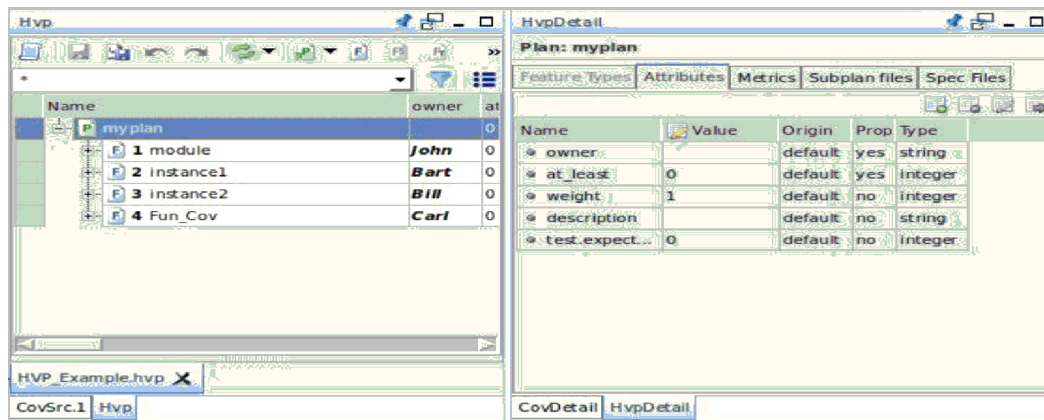
3. This document focuses on HVP and PDF files.

Creating a Plan

Use the following steps to create a new plan in Verdi Coverage:

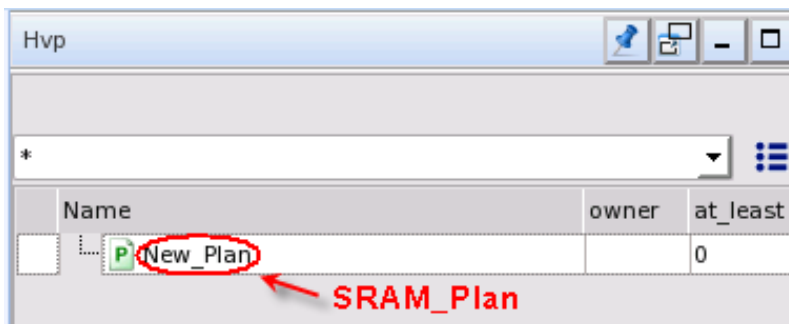
1. Select the **Plan -> New Plan** command to create a new plan in the *Hvp* pane. As shown in Figure 53, the *Attributes*, *Metrics*, *Subplan Files*, and *Spec Files* panes are shown in the *HvpDetail* pane for the new plan. You can also create attributes or metrics by clicking the buttons (see Section [Using User-Defined Attributes and Metrics](#) for more details about creating attributes and metrics).

Figure 53 *Hvp and HvpDetail Default Layout*



2. Double-click the *New_Plan* term in the **Name** column to change the plan name. For example, as shown in Figure 54, change the plan name to *SRAM_Plan*.

Figure 54 *Change Plan Name*



3. To add a description for the plan, double-click the **Description** cell and add the description in the **Edit String** dialog box that appears as shown in Figure 55. The **Edit String** dialog box displays the multi-line text that you have entered, making it easy to read. You can also right-click the string attribute and choose **Edit String** in the menu bar to enter the string as shown in Figure 56.

Figure 55 Edit String Dialog Box

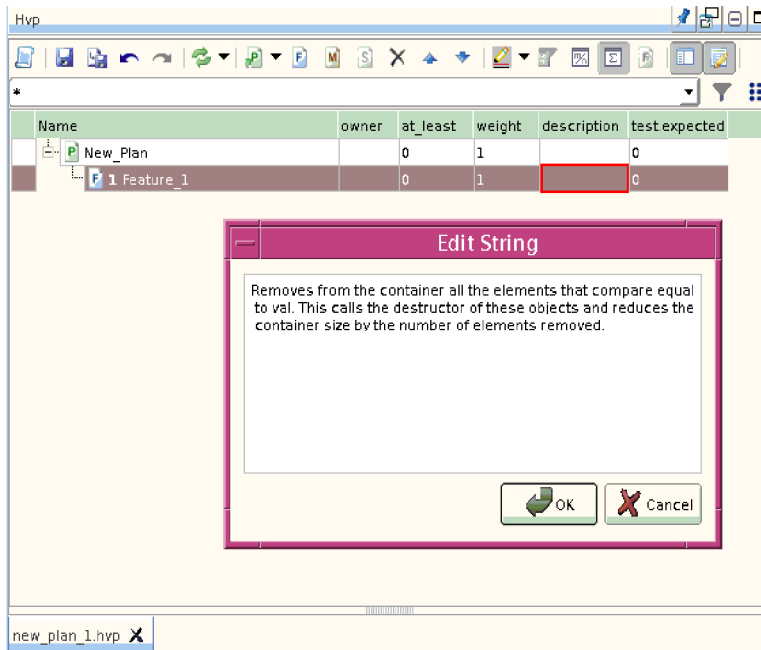
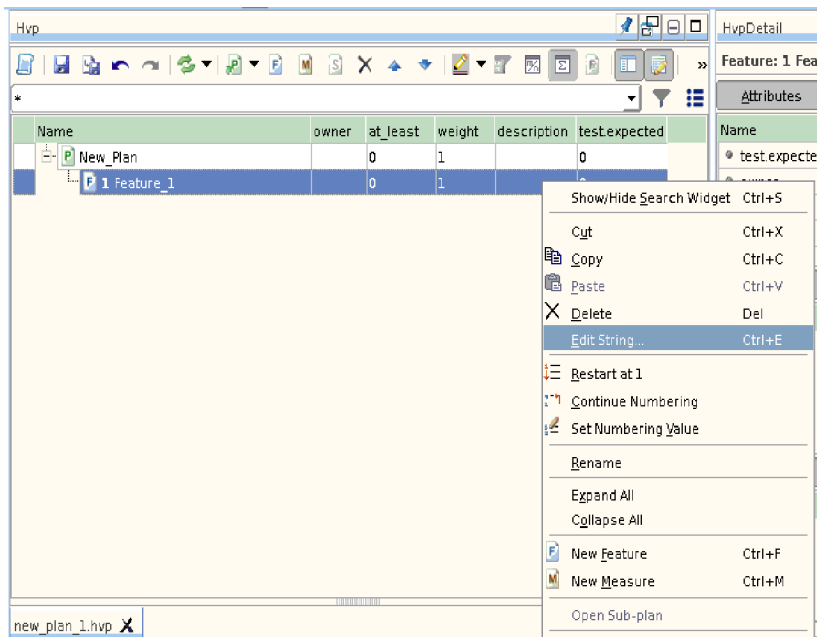


Figure 56 Editing Strings



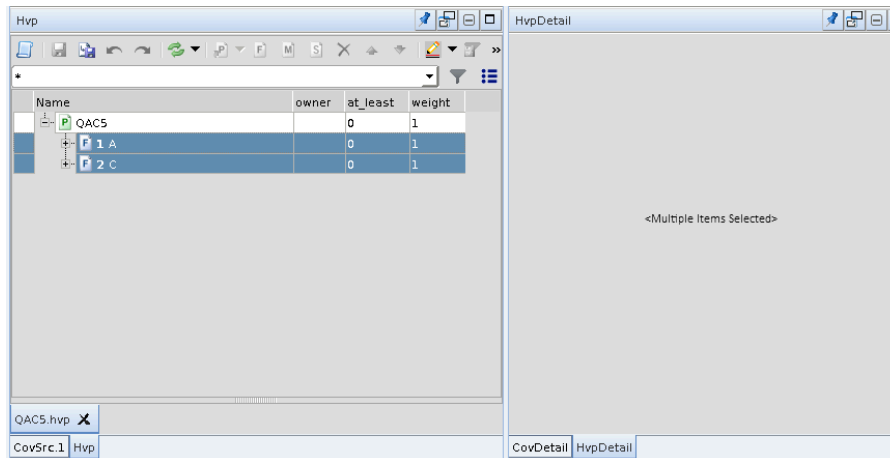
The **Planner** editor allows you to select multiple items of a plan by using the following operations as shown in Figure 57:

- Ctrl + Mouse Click
- Shift + Mouse Click
- Ctrl + A
- Drag mouse to select

The following operations are supported with multiple selections:

Operation	Behavior
Copy/Cut	Limited to items of the same type. Items in lower levels of the same path are ignored.
Delete	Items in lower levels of the same path are ignored.
Delete Review Remind	Deleting multiple review reminders.
Unlink issue	Unlinking multiple issues.
Drag & Drop (Move)	Limited to items of the same type. Items in lower levels of the same path are ignored.

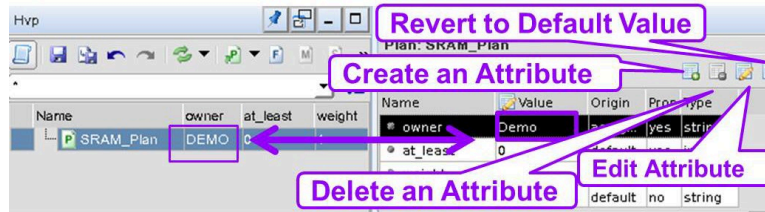
Figure 57 Selecting Multiple Features



4. Change the values for the new plan attributes in the *HvpDetail* pane. You can double-click the **Value** field or click the  icon to change the specific attribute value. For example, as shown in Figure 58, you can see the owner is changed to *DEMO*

both in the *HvpDetail* pane and the *Hvp* pane. The **Value** field is also editable and you can change the value in the **Attribute** pane.

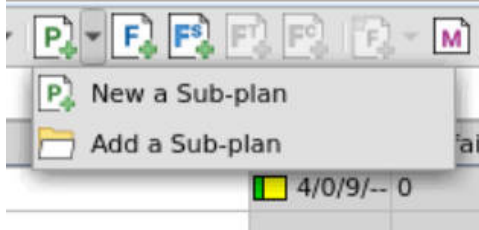
Figure 58 Edit Attributes in the Attributes pane



Creating a Sub-Plan

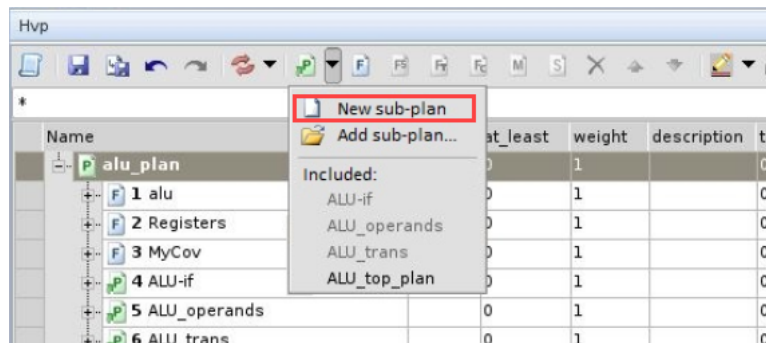
You can add sub-plans for features, top-plans, or other sub-plans using the HVP planner in the *Hvp* pane. Sub-plans can be newly created or they can be added to the current HVP from a list of previously created sub-plans. To create a sub-plan, perform the following steps:

1. Select a feature, top-plan, or sub-plan.
2. Add a sub-plan belonging to the selected object with the help of the menu as shown in the following screenshot:



Creating a New Sub-Plan

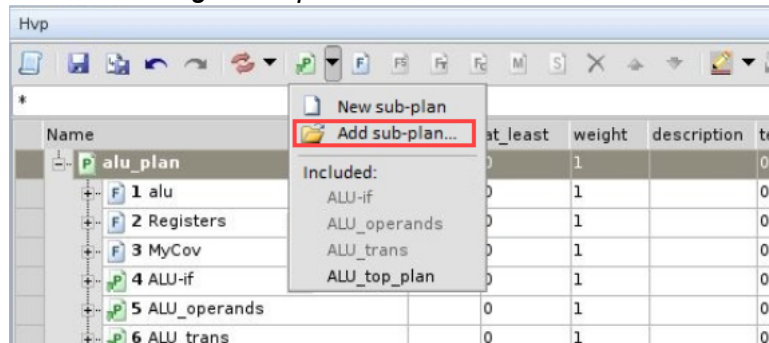
Open the menu as shown in the following screenshot and choose **New sub-plan** to create an empty sub-plan as a placeholder for the currently selected plan or feature in the HVP.



Adding an Existing Sub-Plan

Open the menu as shown in the following screenshot and choose **Add a Sub-Plan** to specify a sub-plan from a list of previously added sub-plans to the current HVP.

Figure 59 Adding a sub-plan



Auto-Saving Option for HVP Editor

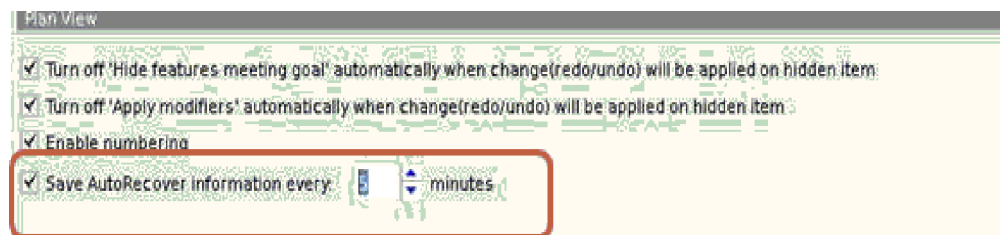
Verdi Coverage provides an **Autosave** feature with the HVP editor that helps prevent data loss if the tool crashes for some reason. Verdi performs an autosave in the background every 5 minutes. However, the original `.hvp` and `.spc` files does not change until you click **Plan -> Save**.

Consider an example of a plan named `MyPlan.hvp`. If you choose the **Recover** option, the tool loads the HVP data from `.MyPlan.hvp` and `.MyPlan.hvp.spc`. Verdi Coverage does NOT modify the original `MyPlan.hvp` until you perform **Plan -> Save**.

Preference Setting for AutoSave

You can use the preference setting shown in Figure [Autosave Preferences](#) to enable or disable **Autosave** and also modify the **Autosave** interval.

Figure 60 Autosave Preferences

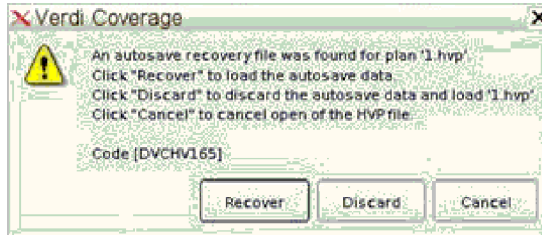


Recovering/Discarding Data From an Autosave

Consider an example where you are editing `MyPlan.hvp`. If you exit *Coverage* without saving it or if there is a crash, Verdi created the files `.MyPlan.hvp` and `.Myplan.hvp.spc`

automatically in the background. When you open this plan the next time, Verdi checks whether the *.MyPlan.hvp* file exists; if the file exists and the modified date of the hidden file is later than the file *MyPlan.hvp*, the following warning message is shown:

Figure 61 Recover/Discard/Cancel Autosaved Data



In this scenario, you can:

- Click the **Recover** button to load the Autosave data.
- Click the **Discard** button to discard the Autosave data and load *MyPlan.hvp*.
- Click the **Cancel** button to cancel the opening of the HVP file.

After you select an option and confirmed, the *.MyPlan.hvp* file is deleted.

Creating Features for Planner

You can add features in a planner from the compiled result in the Verdi platform, from the coverage database or directly from design to strengthen the current coverage. When the database is loaded into Verdi Coverage, the status of current coverage is displayed in the Coverage mode. Verdi Coverage provides a method to directly add the weak sources to a planner.

You can also link to spec-based PDF files for features in the HVP planner. This is a way to populate the plan in the *Hvp* pane with a requirement specification already written by the designer and then link the specification with the feature, so that the requirements in the design specification are not forgotten or lost and are incorporated in the verification plan. After the feature setting in the *Hvp* pane is complete, you can link a requirements specification for the feature using the spec-linking feature of Verdi Coverage.

Additionally, the explicit structure of *Hvp* hierarchy is also provided. After you add a feature or sub-plan, the numbering of added item is automatically numbered.

The following sections describe these methods to add features:

- [Adding Features From Coverage Database](#)
- [Adding Features From the Design](#)

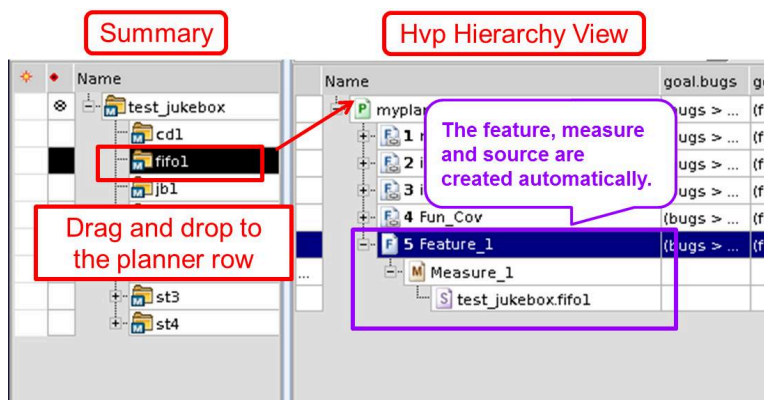
- Specification Linking in Planner
- Running a Specification Review
- Configuring the Linked Feature

Adding Features From Coverage Database

When the coverage database is loaded into Verdi Coverage, you can use the following method to add features:

- Drag an item (source) from the tabs in the *Summary* pane and drop to the planner row in the *Hvp* pane. As shown in Figure [Create a New Feature from Summary](#), a new feature is created with its numbering, the measure and source.

Figure 62 *Create a New Feature from Summary*

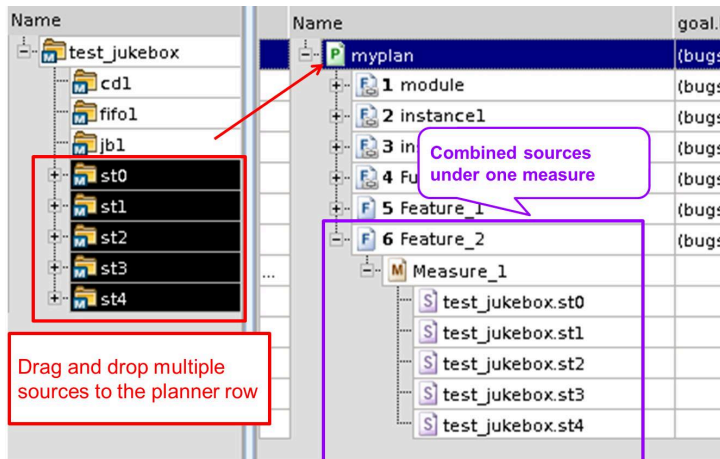


- Select the plan in the *Hvp* pane. Select a source in the *Modules*, *Groups*, or *Asserts* tabs of the *Summary* pane and use the **Add to Plan** command in the right-click command menu.
- Select the plan in the *Hvp* pane. Select a source in the *Hierarchy* pages of the *Summary* pane and use the **Add Instance to Plan** or **Add Tree to Plan** commands in the right-click command menu to add the instance or the entire hierarchy of the selected source to the plan.
- Drag a named object from the *Line*, *Toggle*, *FSM*, and *Condition* tabs of the *CovDetail* pane and drop to the row of the plan in the *Hvp* pane. For example, you can drag a named line from the *Line* page or a variable from the *Toggle* page and drop it to the selected plan in the *Hvp* pane. Alternatively, use the **Add Tree to Plan** right-click command to add the selected object to the Hvp planner. For the UPF power plan, you can drag a Covergroup or a Coverpoint from the *CovDetail* pane and drop to the row of the plan in the *Hvp* pane.

If the source is a group or assert, *Group* or *Assert* is automatically specified as the metric of the new measure. If the source comes from the *Hierarchy* tab, the *Modules* tab or the Verdi platform, all code coverage metrics included in the database are automatically specified for the measure.

When multiple sources are dragged and then dropped to a planner, sources are combined under the measure (refer to Figure [Select Multiple Sources](#)).

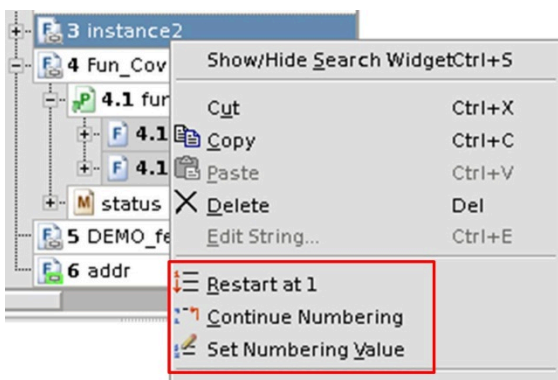
Figure 63 Select Multiple Sources



You can use the **Rename** command in the right-click menu of the *Hvp* pane to rename the plan, feature, measure, or source names. Additionally, the numbering of each feature can be modified using the commands in the right-click menu (see Figure [Numbering Setting](#)):

- **Restart at 1:** Set the numbering starting from 1.
- **Continue Numbering:** Reset the numbering to automatic setting.
- **Set Numbering Value:** Specify a numbering.

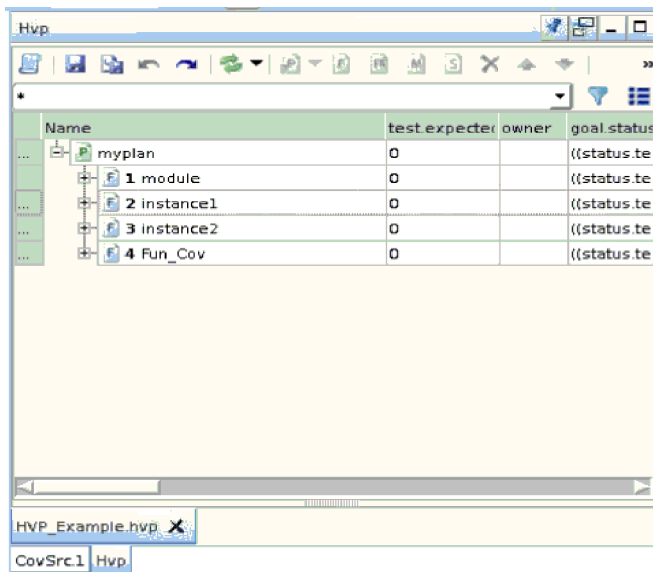
Figure 64 Numbering Setting



Inserting a Sibling Feature in a Plan

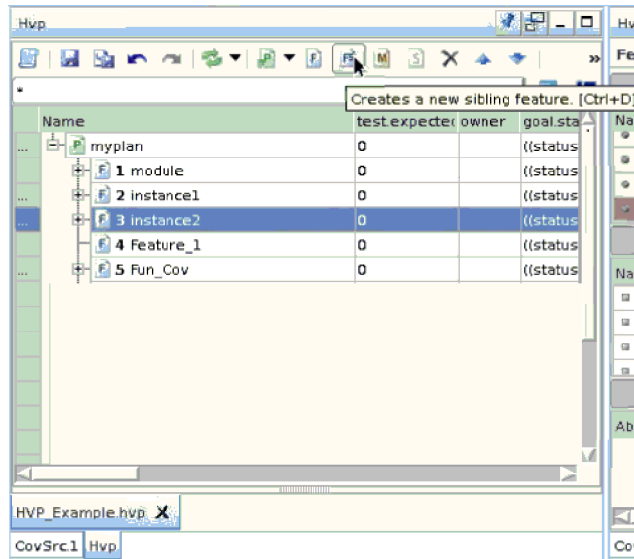
Verdi allows you to add or delete features in a plan. If you add a feature using the **Create a New Feature** icon, Verdi inserts the feature as the last child feature item of the selected parent item (plan/feature). However, this approach requires you to reselect the parent feature for each sibling feature created. With the new **Create Sibling Feature** icon in the *Hvp* pane, there is no need to reselect the parent feature for each new sibling feature. The icon allows you to rapidly create a set of features beneath a common parent feature. Creating a set of features at the same level provides ease of usability and enhances user experience.

Consider a plan `myplan` that consists of multiple features, such as `module`, `instance1`, `instance2`, and `Fun_Cov` (see the figure below).

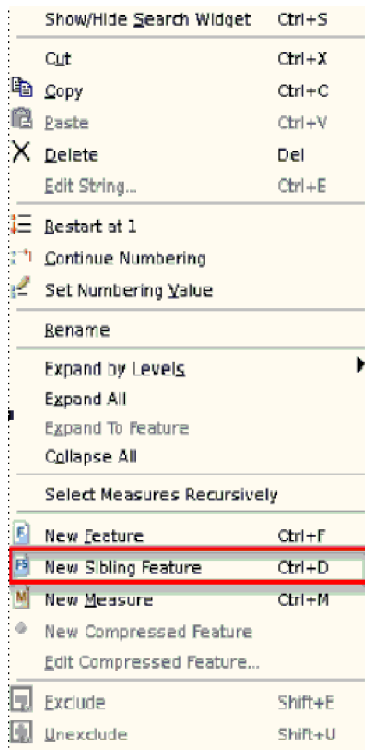


If you add a feature in `myplan` using the **Create a New Feature**  icon in the *Hvp* pane, Verdi inserts the feature at the end of `Fun_Cov`. To insert the feature after `instance2`, perform one of the following methods:

- Select `instance2`, click the **Create a New Sibling Feature**  icon in the *Hvp* pane (see the figure below).



- Right-click `instance2` and select **New Sibling Feature** (see the figure below).



- Select `instance2` and use keyboard shortcut: **CTRL + D**.

Adding Features From the Design

You can also create features and add them to a plan directly from instances, modules, groups, and asserts directly from the design. You can select an instance sub-tree, create features in a plan based on that sub-tree and then map them.

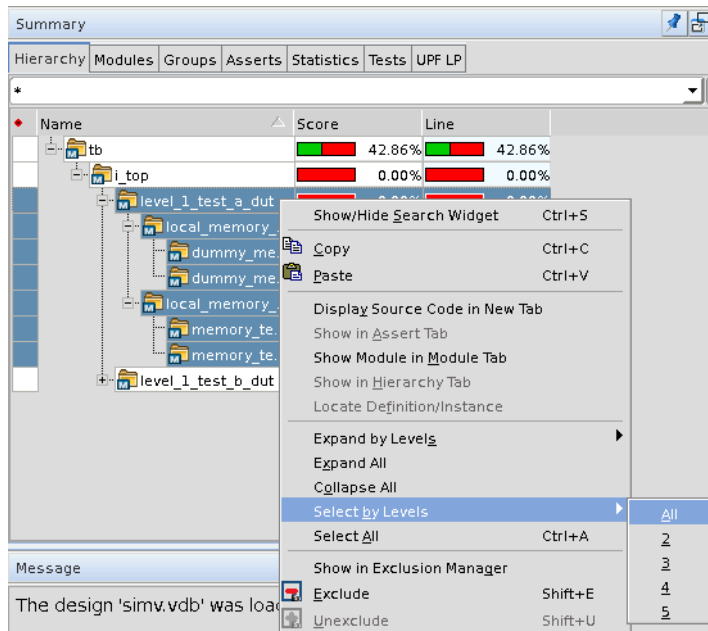
Creating a New Feature and Sibling Features

You can create a feature tree mapping of a selected leaf instance or instances included in the tree under the leaf instance, using the following steps. The order of selection in summary table and plan does not affect the result.

1. Select an instance for which you want to create a feature in the plan, in the **Hierarchy** tab and right-click on it as shown in Figure [Selecting the Level of Sub-Instances](#). If you want to select all the instances included in the tree under the leaf instance, then select the menu item **Select by Levels -> All**.

You can also select multiple instances using the **Ctrl** or **Shift** keys.

Figure 65 Selecting the Level of Sub-Instances

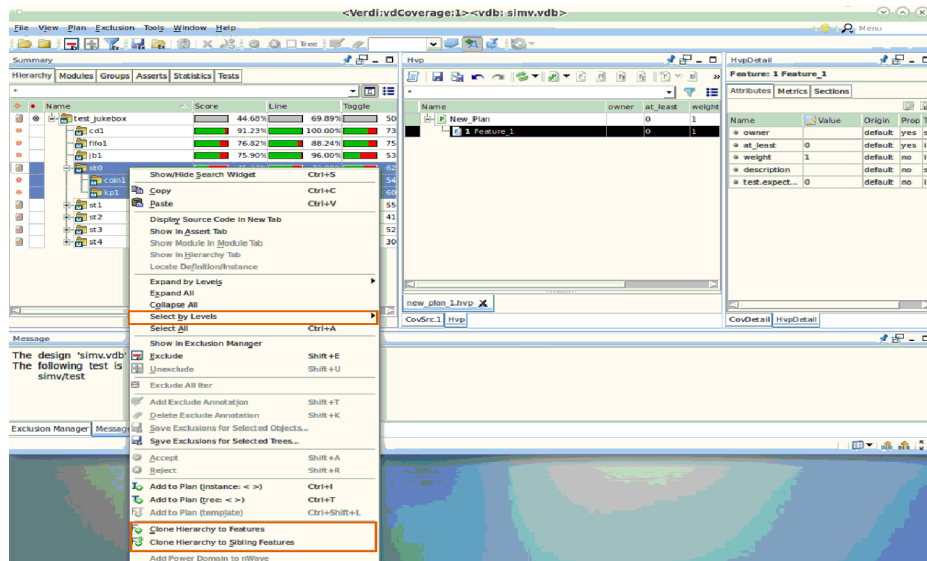


2. After you have selected the instance(s) in the *Hierarchy* tab, select the plan into which these features should be added in the *Hvp* tab. In this case, the features should be added to the plan named *New_Plan*.
3. Once both the plan and the instances are selected, right-click on the selected instance(s) and select either the **Clone Hierarchy to Features** or **Clone Hierarchy to Sibling Features** menu options.

The **Clone Hierarchy to Features** menu option creates a new feature under the plan *New_Plan* that measures selected coverage items. The **Clone Hierarchy to Sibling Features** menu option creates a new sibling feature of the selected feature that measures selected coverage items.

Figure [Selecting Instances for Feature Creation](#) shows these two menu options:

Figure 66 *Selecting Instances for Feature Creation*



The following table lists out the scenarios in which the **Clone Hierarchy to Features** or **Clone Hierarchy to Sibling Features** menu options are enabled or disabled:

Selection	Action		
Selection in Summary Table	Selection in Plan	Clone Hierarchy to Features	Clone Hierarchy to Sibling Features
Hierarchy: Single Instance	Plan	Enabled	Disabled
	Feature	Enabled	Enabled
	Sub-plan	Disabled	Enabled
Hierarchy: Multiple Instances (connected as a subtree, called S)	Plan	Enabled	Disabled
	Feature	Enabled	Enabled
	Sub-plan	Disabled	Enabled
Hierarchy: Multiple Instances (disconnected)	PlanFeatureS ub-plan	Disabled	Disabled

Other Tabs (Module/Group/Assert): Single/Multiple Items (such as module, group, and so on)	Plan	Enabled	Disabled
	Feature	Enabled	Enabled
	Sub-plan	Disabled	Enabled
None	Any	Disabled	Disabled
Any	None	Disabled	Disabled
	Measure		
	Source		
	In Sub-plan		
	Multiple Items		

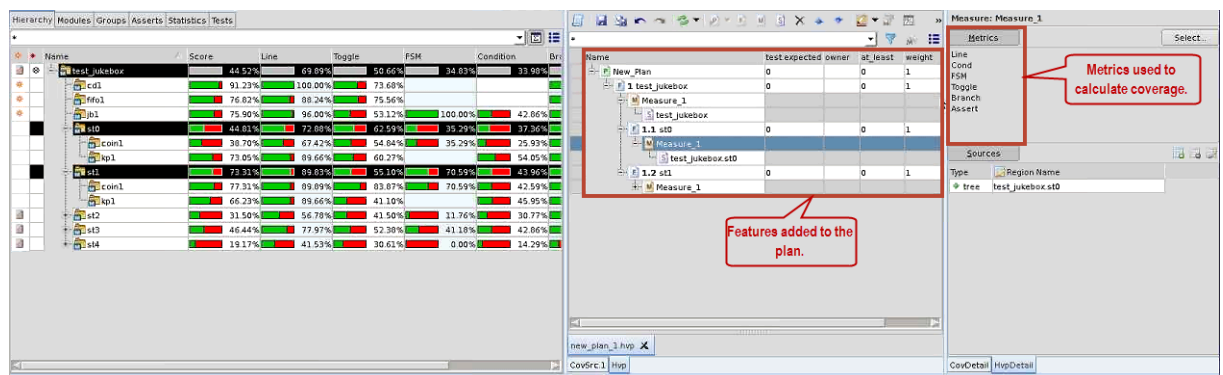
4. The newly created features are displayed in the *Hvp* tab under *New_Plan* with the leaf instance and the instances under the tree of the selected leaf instance annotated. Coverage is calculated for all the metrics, that is, Line, Cond, FSM, Toggle, Branch, and Assert.

Note:

To choose the level of annotation or the metrics for which coverage needs to be calculated, see section *Choosing Annotation Level and Metrics for Coverage Calculation* in the *Verdi LCA Feature Guide*.

5. These metrics are displayed in the *HvpDetail* tab as shown in the following figure:

Figure 67 New Features Created Under Plan



The following actions are performed when creating features from a single and multiple instances. Different selections in the summary table result in different actions.

Action Selection	Clone Hierarchy to Features	Clone Hierarchy to Sibling Features
Hierarchy: Single Instance	Creates a new feature under the selected feature/plan that measures selected coverage items.	Creates a new sibling feature of the selected feature that measures selected coverage items.
Hierarchy: Multiple Instances (connected as a subtree called S)	Creates a new feature subtree under selected feature/plan that mirrors the structure of S, measuring each instance of S.	Create a new sibling feature subtree to selected feature that mirrors the structure of S, measuring each instance of S.
Hierarchy: Multiple Selected Instances (NOT connected as a subtree)	Disallowed	Disallowed
Other Tabs (Module/Group/Assert): Single/Multiple Coverage Items (such as module, group, and so on)	Creates new features under selected feature/plan that measures each selected coverage items.	Creates new sibling features to selected feature that measures each selected coverage items.

The following sections describe the position, order, and naming of the features created:

Position of the New Feature(s)

- **Create Features:** The newly created features are the last child of the parent (selected item).
- **Create Sibling Features:** The newly created sibling features are the closest sibling of the selected item. This is also applicable for multiple selection support.

Order of the New Feature(s)

The order of new features is the same as the order in which you select the instances using the **Ctrl** key. When you select multiple instances using the **Shift** key, the resultant features are ordered top-to-bottom.

Naming of New Features

By default, the name of a new feature is the name of the instance from the design hierarchy. However, if the name is a keyword in the HVP language such as `test` or `feature`, it is renamed by adding a suffix `_renamed`. For example, if a feature is created from an instance named `feature`, the feature that is created from it is named `feature_renamed`. You are notified of this change with warning message similar to the following one, in the message window.

Verdi Planner enables you to track a verification project and ensures that all its verification goals are achieved. It allows you to create a hierarchical verification plan to measure what is important for each project. A linked and executable verification plan ties every part of the original written specification and test plan documents to their corresponding verification goals. The following figure shows how the verification specifications, plans, and data come together to track progress and status of the verification project:

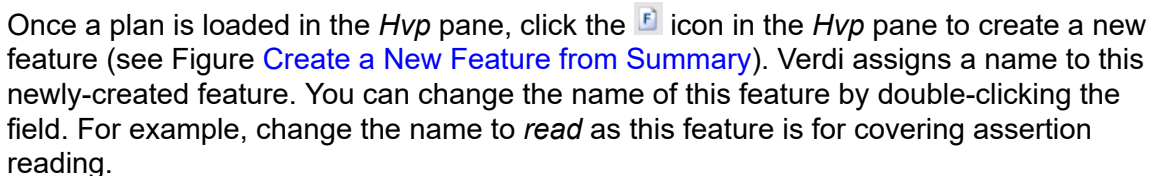
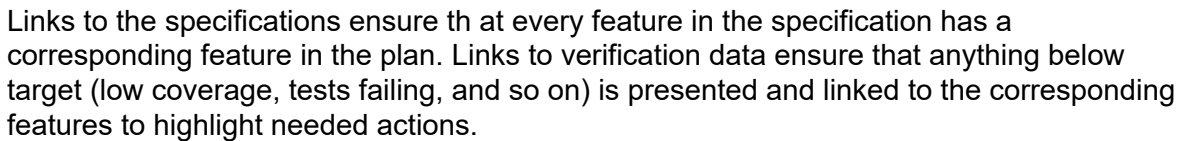
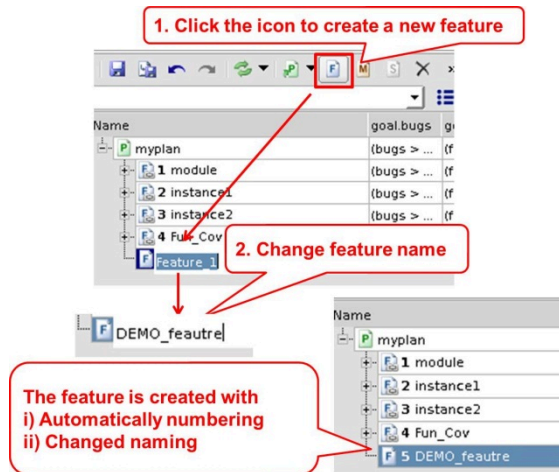
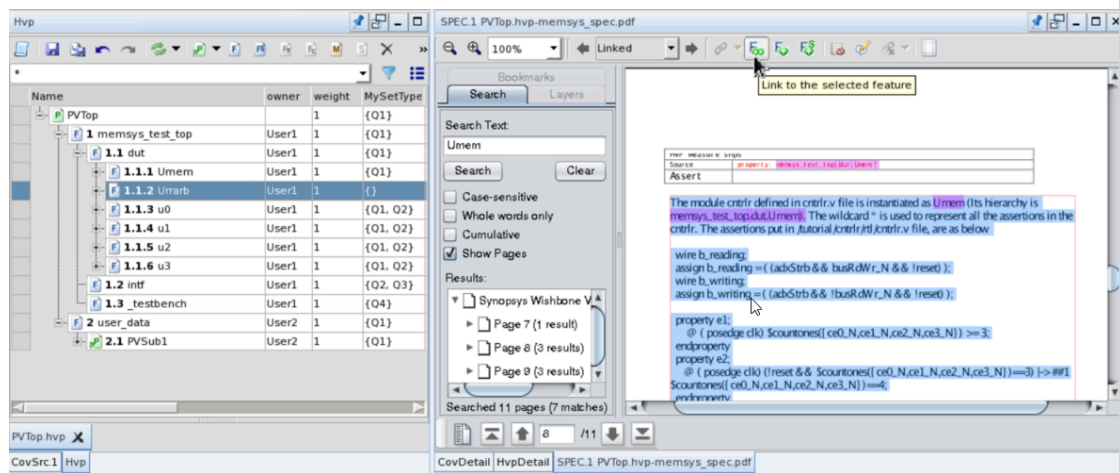


Figure 68 Create a New Feature

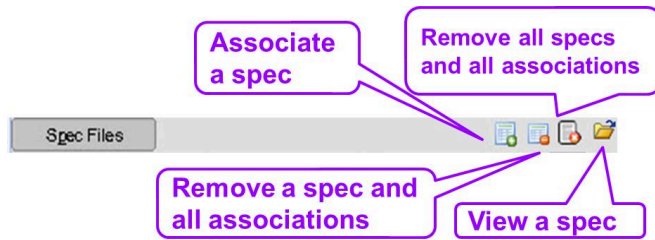


The Undo or Redo icons are provided in the *Hvp* pane to undo or redo an edited action during the edition. For example, in the figure below, the text related to *Umem* is linked with the *Urrarb* feature instead of the *Umem* feature, you can undo the action using the Undo icon.



As shown in Figure [Icons in Spec Files Pane](#), you can use the icons to manage specs and associations in the *Spec Files* pane. Click the icon to open the *Choose a spec file* form and select a specification file to be linked. You can also use the **Plan -> Open Spec** command to open a specification in the HVP planner.

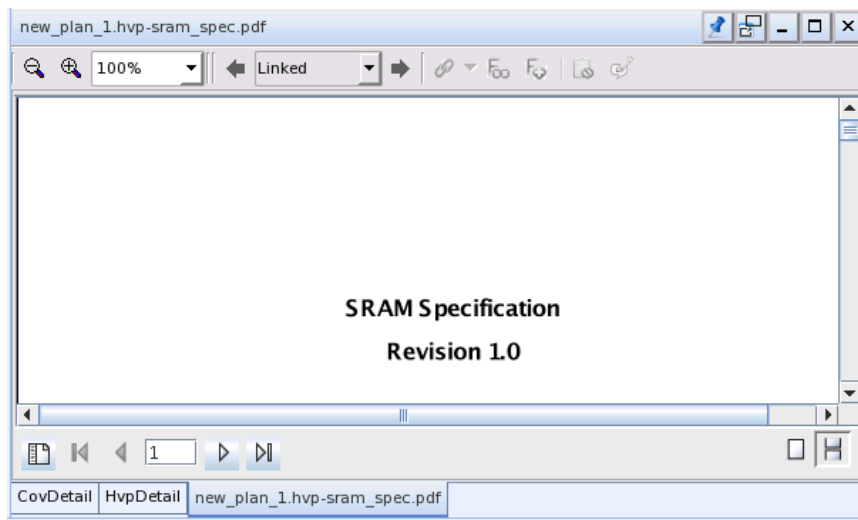
Figure 69 Icons in Spec Files Pane



After a specification file is opened, you can double-click the specification row or use the  icon to view the contents and link a section of the specification to a feature in the HVP planner.

The specification file's contents are displayed in its own tab. As shown in Figure [Tab for the Content of Spec](#), the requirement SRAM specification is displayed in a new `new_plan_1.hvp-sram_spec.pdf` tab.

Figure 70 Tab for the Content of Spec

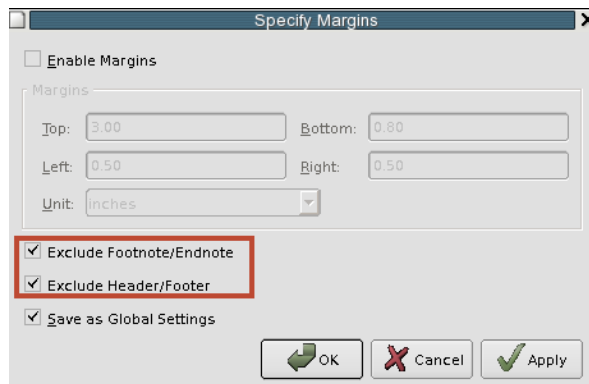


To link a section of the text, perform the following steps:

1. Drag to highlight the section of text to be linked.
2. Select the feature in the *Hvp* pane with which the section is linked.
3. Click the  icon to link the section to the selected feature.

To ensure that the text in the header, footer, or endnotes are excluded from spec-linking, you can select the **Exclude Footnote/Endnote** and **Exclude Header/Footer** options in the **Specify Margins** form as shown in the following figure:

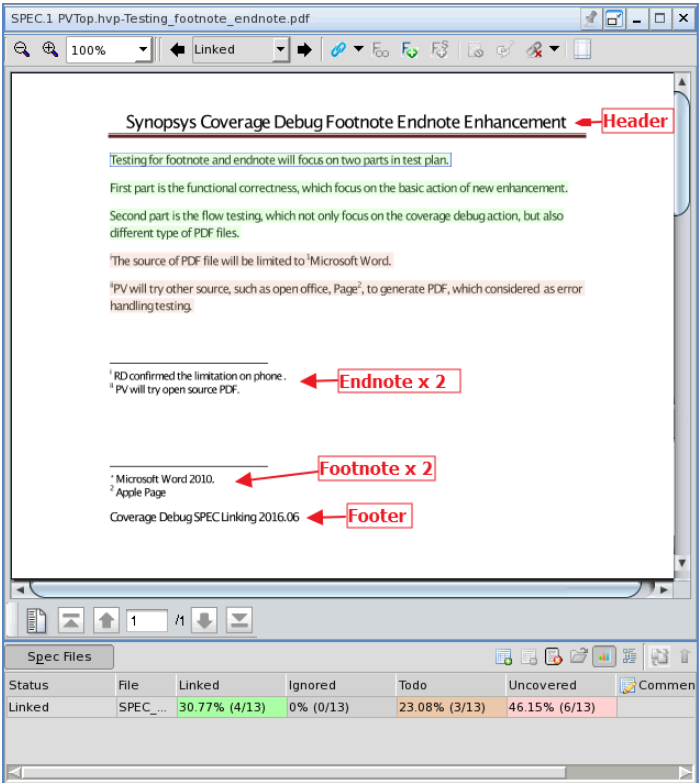
Figure 71 *Excluding Headers and Footnotes/Endnotes*



The image shows a 'Specify Margins' dialog box. At the top, there is a checkbox for 'Enable Margins' which is unchecked. Below this is a 'Margins' section with four input fields: 'Top' (3.00), 'Bottom' (0.80), 'Left' (0.50), and 'Right' (0.50). Below these fields is a 'Unit' dropdown menu set to 'inches'. At the bottom of the dialog, there are three checkboxes: 'Exclude Footnote/Endnote' (checked), 'Exclude Header/Footer' (checked), and 'Save as Global Settings' (checked). The first two checkboxes are enclosed in a red rectangular box. At the very bottom are three buttons: 'OK' (with a green arrow icon), 'Cancel' (with a red X icon), and 'Apply' (with a green checkmark icon).

If you select the **Exclude Footnote/Endnote** option, all foot notes or end notes present in the text are excluded from spec-linking. Similarly, if you select the **Exclude Header/Footer** option, all the headers and footers are excluded.

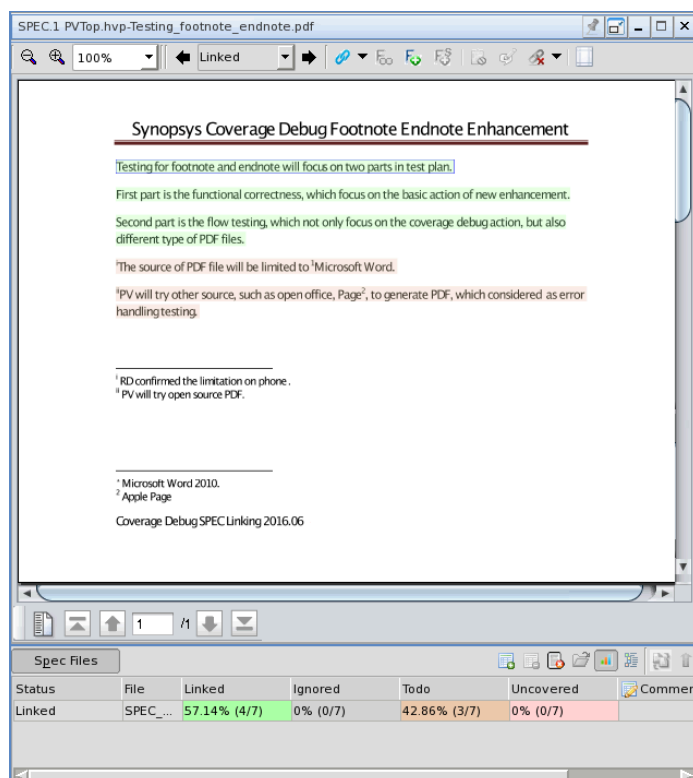
For example, consider the `Testing_footnote_endnote.pdf` file in the following figure that shows total lines as 13 including header, footer, endnotes, and footnotes:



If you exclude header, footnote, endnote, and footer [header x 1 + footnote x 2 + endnote x 2 + footer x 1 = 6] from spec-linking, the planner shows total lines as 7 as shown in the following figure:

Chapter 5: Using Verification Planner

Creating Features for Planner



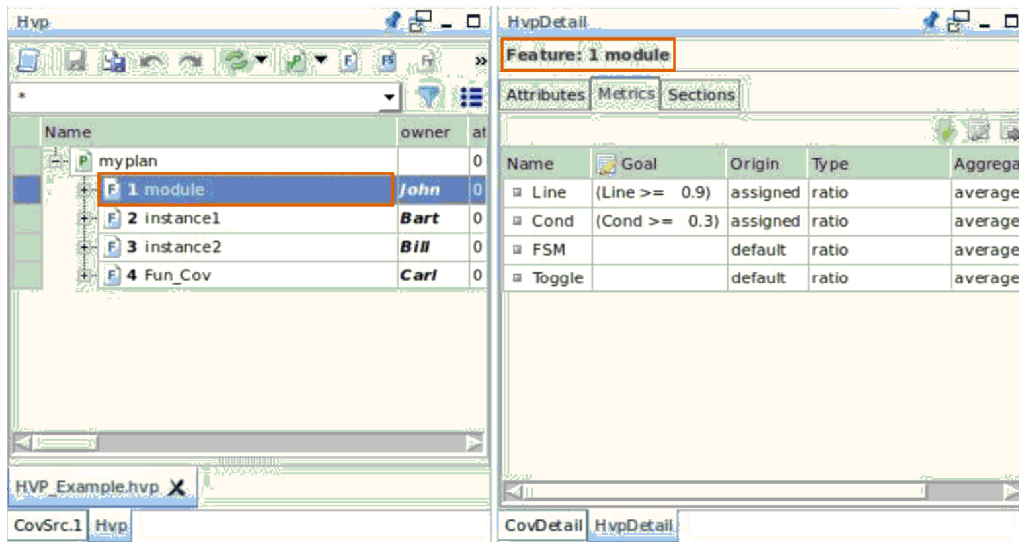
For more information about excluding sections of the specification document from linking, see the [Excluding Ignore Sections From the Specification Coverage Computation](#) section.

Note:

The exclusion of header/footer and footnote/endnote is only supported with PDFs generated using Microsoft office. PDFs generated using any other tools do not fully support this feature.

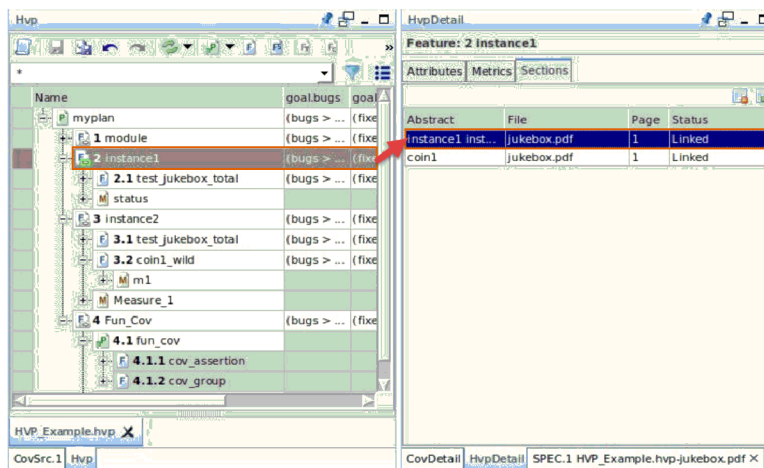
You can select the feature in the *Hvp* pane to show the corresponding *Feature* page in the *HvpDetail* pane for the selected feature details. As shown in Figure [Feature Details](#), the *Attributes*, *Metrics*, and *Section* panes are shown in the *HvpDetail* pane for the **Feature: 1 module** in the *HvpDetail* pane.

Figure 72 Feature Details



You can also link a feature to multiple sections. The sections are listed in the *Sections* tab of the *HvpDetail* pane (see Figure [Multiple Sections Linked to One Feature](#)).

Figure 73 Multiple Sections Linked to One Feature

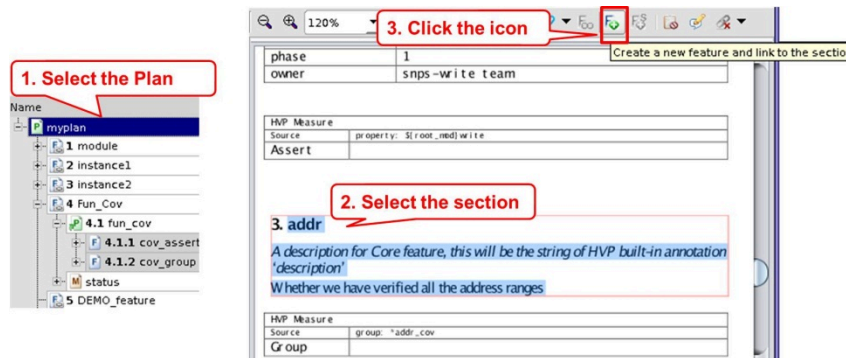


To create and link a new feature in the *Hvp* pane, perform the following steps (see Figure [Create and Link a New Feature](#)):

1. Select the plan in the *Hvp* pane.
2. Drag to highlight the section of text to be linked in the specification.

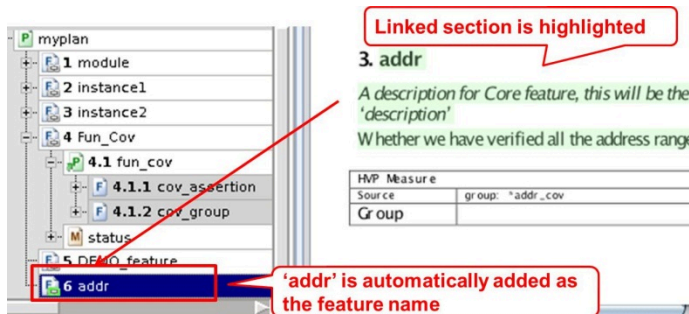
3. Click the **Create a new feature and link to the section**  icon to create the feature and link it to the section in the specification.

Figure 74 Create and Link a New Feature



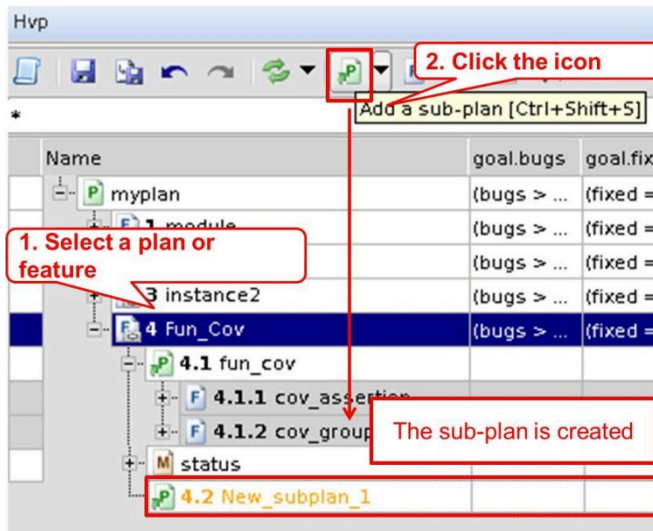
Linking the specification to a new feature automatically copies the first row of words in the highlighted text and name the feature (refer to Figure [Created and Linked Feature](#)). You can also use the  icon to create a new sibling feature of the selected one and link it to the section in the specification.

Figure 75 Created and Linked Feature



After a plan or a feature is created, you can use the  icon to add a sub-plan under a plan or feature (refer to Figure [Create a Sub-Plan](#)).

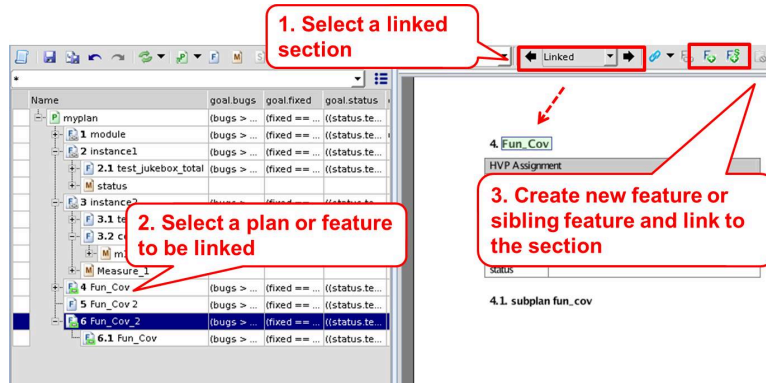
Figure 76 Create a Sub-Plan



A section in the specification can be linked to many features. You can use the following steps to link the same section to more than one feature (see Figure [Multi-Features Link to One Section](#)):

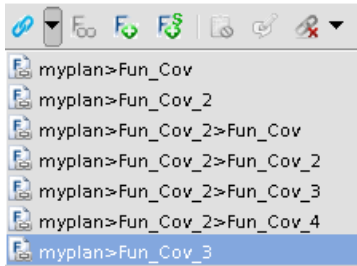
1. Click the ◀ or ▶ icons in the specification toolbar to select a section which is linked to another feature.
2. Select a plan or a feature in the *Hvp* pane.
3. Click the icon to create a new feature and link to the selected section.
4. Click the icon to create a new sibling feature and link to the selected section.

Figure 77 Multi-Features Link to One Section



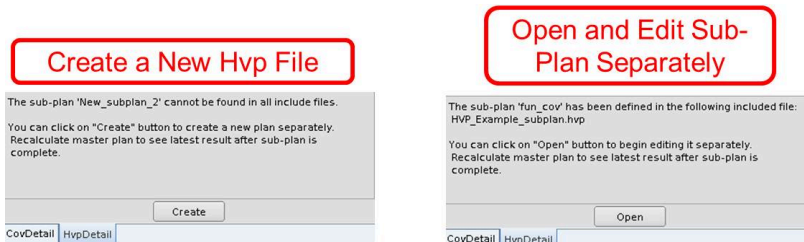
You can select a linked feature by clicking the  icon to see all the linked features for one section (see Figure [Linked Features List](#)). After selecting one feature in the list, the corresponding feature is highlighted in the *Hvp* pane.

Figure 78 *Linked Features List*



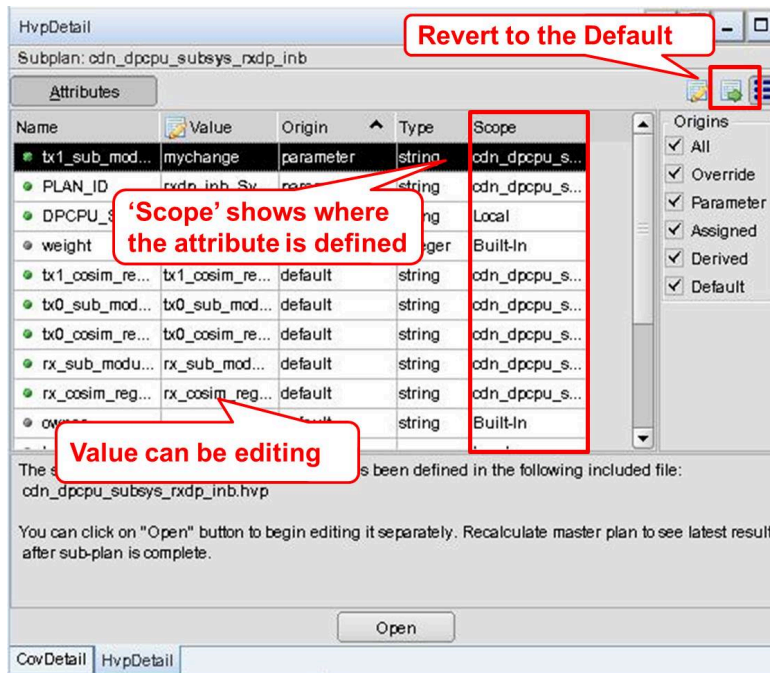
If a new sub-plan is selected in the *Hvp* pane, you can click the **Create** button in the *HvpDetail* pane to create a new Hvp file or reuse an existed Hvp file for the sub-plan. If a sub-plan having corresponding Hvp file is selected in the *Hvp* pane, you can click the **Open** button in the *HvpDetail* pane to edit the sub-plan in a separate tab in the *Hvp* pane (see Figure [Create and Edit Sub-Plan](#)).

Figure 79 *Create and Edit Sub-Plan*



After editing the sub-plan, all attributes (derived from main plan and defined in sub-plan itself), are listed in the *Attribute* pane in the *HvpDetail* pane (see Figure [Attributes Pane of Sub-Plan](#)).

Figure 80 Attributes Pane of Sub-Plan



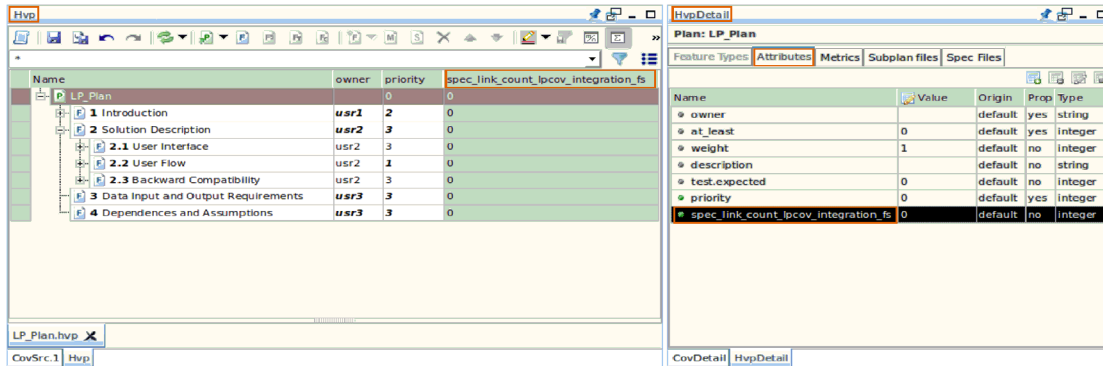
You can directly edit the attribute values, which automatically create corresponding parameters for the sub-plan instance. You can also click the  icon to revert the changed to the default value. The scope where the attribute is defined is shown in the **Scope** field. If the attribute is defined in both main plan and sub-plan, it shows the main plan in the **Scope** field. When a sub-plan is opened in another tab, the numbering of the sub-plan is renumbered, starting from 1.

Specification Linking Count

Verdi Coverage provides a set of link count attributes with the `spec_link_count` prefix that identifies all linked specifications with a plan. These attributes show the number of links to a feature or sub-feature from a given specification document. These attributes are automatically created when you associate specifications with the plan. You can link the plan with more than one specification; the number of the associated specifications determines the number of `spec_link_count` attributes in the plan. Each specification generates only one `spec_link_count` attribute. The initial value of this attribute is zero and it gets updated when the feature of the plan gets linked to or unlinked from the specification. You cannot edit this attribute; it is automatically populated when the specification is linked with its associated text.

When you associate a specification file with a plan, Verdi Coverage generates the `spec_link_count_<filename>` attribute to count the number of links in the specification from the plan. The type of this attribute is *integer* and its propagation is set to *no* so that

the value does not propagate to the child node. The attribute is available in the *Hvp* pane and in the *Attributes* section of the *HvpDetail* pane as shown in the following figure:



You can use the `spec_link_count` attribute for quick filtering and for creating modifiers, like any other integer-valued attribute. To use the `spec_link_count` attribute, filter only the features that link to a specific specification. For example, consider the following filter statement:

```
remove feature where spec_link_count_my_doc == 0;
```

In the example, the statement creates a filtered plan with only features that link to the `my_doc` specification.

For details on filters and modifiers, see the [Using Modifiers](#) section.

PDF Reader Bookmark Support

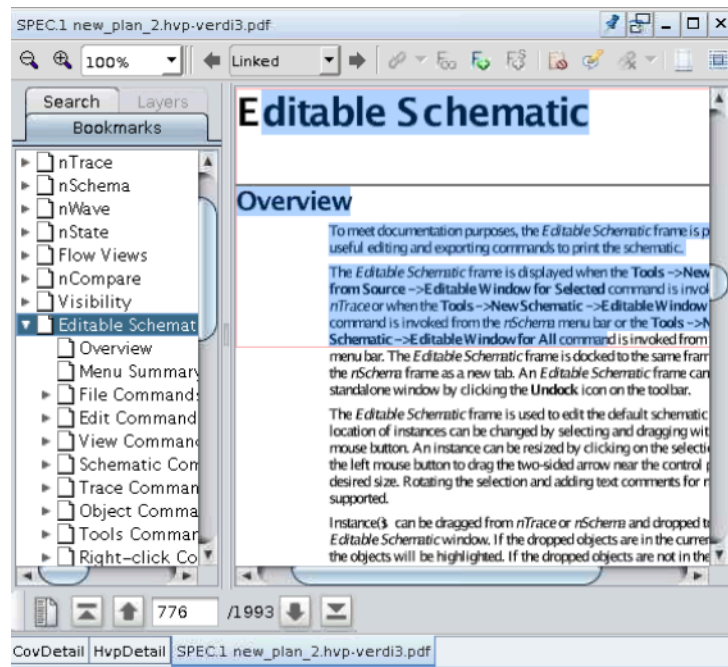
Verdi enables you to use bookmarks to perform specification linking quickly. You can select a bookmark in the specification document to directly move to the required section and then link the section to the plan.

When you open the specification to create the link with the plan, it does not show bookmarks by default.

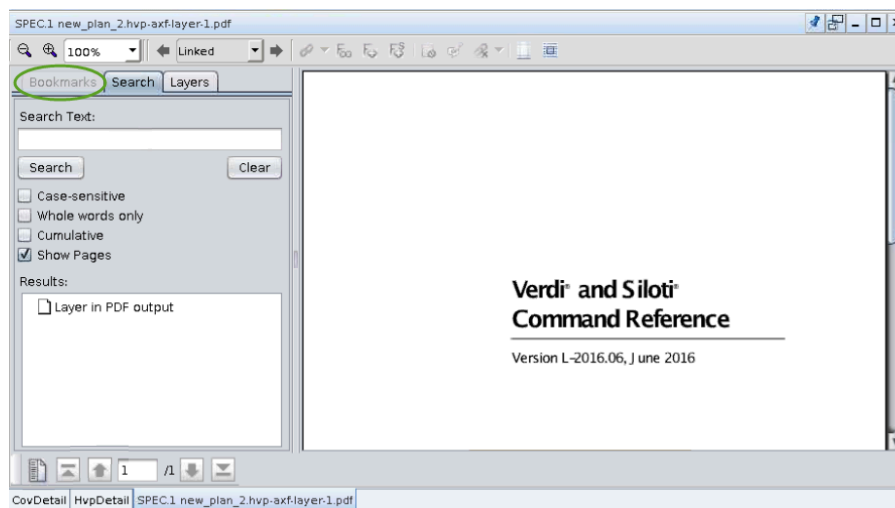
To perform specification linking using bookmarks, perform the following steps:

1. In the *HvpDetail* pane, select **Spec Files**.
2. Click the **Associate a Spec** icon to open the **Choose a spec file** form.
3. Select the specification file and click **Open**. Then, double-click the specification file to open it.
4. Click the **Show/Hide Utility Pane** icon in the **Specification** pane to display bookmarks.
5. Select a bookmark from the **Bookmarks** tab to show its corresponding text in the specification file. Select the text to create a link and click **Create a new feature and**

link to the section  icon to link the section to the plan as shown in the following figure:



If a PDF does not contain any bookmarks, then this feature is not enabled. For example, see the figure below that shows that the *Bookmarks* tab is disabled.



Excluding Ignore Sections From the Specification Coverage Computation

Computing specification coverage for all features in the specification requires them to be verified against the corresponding features in the verification plans. Normally for SoC-level testing, it is not required to verify IP-level features; therefore, these features of the specification are not relevant for calculating specification coverage. Using specification linking, you can search for these irrelevant sections in the entire specification and mark them as ignored.

Previously, specification total lines were used as a denominator for calculating specification coverage as follows:

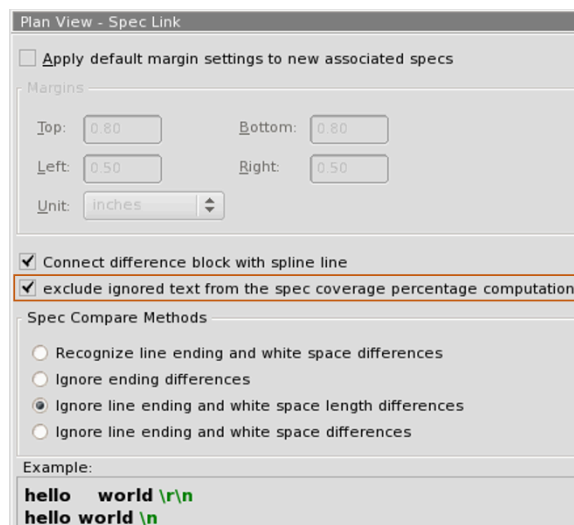
```
Denominator = Total Lines  
Uncovered Lines = Total Lines - Linked Lines - Todo Lines - Ignored Lines
```

Now, specification linking can exclude ignored sections from the specification total lines and use its result as a denominator for calculating coverage as follows:

```
Denominator = Total Lines - Ignored Lines  
Uncovered Lines = Total Lines - Linked Lines - Todo Lines - Ignored Lines
```

To exclude ignored sections from the specification total computation, select the **Exclude ignored text from spec coverage** option from the **Plan View - Spec Link** dialog box as shown in the following figure:

Figure 81 Plan View - Spec Link



For example, consider the `v1.pdf` specification that has total lines as 276. Previously, if 21 lines were marked as linked, 32 lines as todo, and 77 lines as ignored, then the denominator and uncovered lines were calculated as follows:

```
Denominator = 276
Linked = 21/276
Todo = 32/276
Ignored = 77/276
Uncovered = 276 - 21 - 32 - 77 = 146/276
```

The following figure shows how denominator was calculated previously:

Status	File	Linked	Ignored	Todo	Uncovered
Linked	v1.pdf	7.61% (21/276)	31.16% (86/276)	10.14% (28/276)	82.25% (227/276)

Now, if 21 lines are marked as linked, 32 lines as todo, and 77 lines as ignored, then the denominator and uncovered lines are calculated as follows:

```
Denominator = 276 - 77 = 199
Linked = 21/199
Todo = 32/199
Ignored = 77/276 (For ignored, total lines are used as a denominator)
Uncovered = 276 - 21 - 32 - 77 = 146/199
```

The following figure shows how denominator is calculated now:

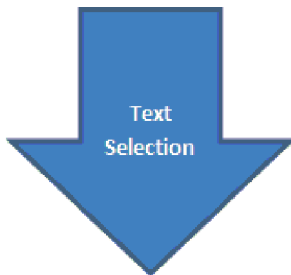
Subplan Files					
Spec Files					
Status	File	Linked	Ignored	Todo	Uncovered
Linked	v1.pdf	11.05% (21/190)	31.16% (86/276)	14.74% (28/190)	74.21% (141/190)

CovDetail HvpDetail SPEC.1 link.hvp-v1.pdf

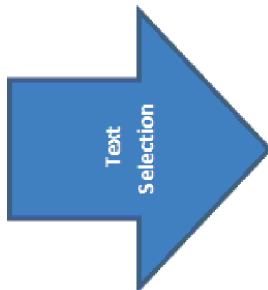
Limitations

The specification-linking feature has the following limitation:

- Rotated text is not well supported, you might not be able to select it. This also impacts line coverage. For example, in the following figure, the text selected covers two lines.



When the text is rotated as shown in the following figure, the text selection covers many lines instead of two lines as shown in the previous figure. This might impact line coverage.



Running a Specification Review

When you have an updated specification, you can update and review the linked specification in the *Spec Review Window*. There might be some mismatched and missing sections of the linking features between the new and linked specifications. In these cases, unresolved sections have to be reviewed.

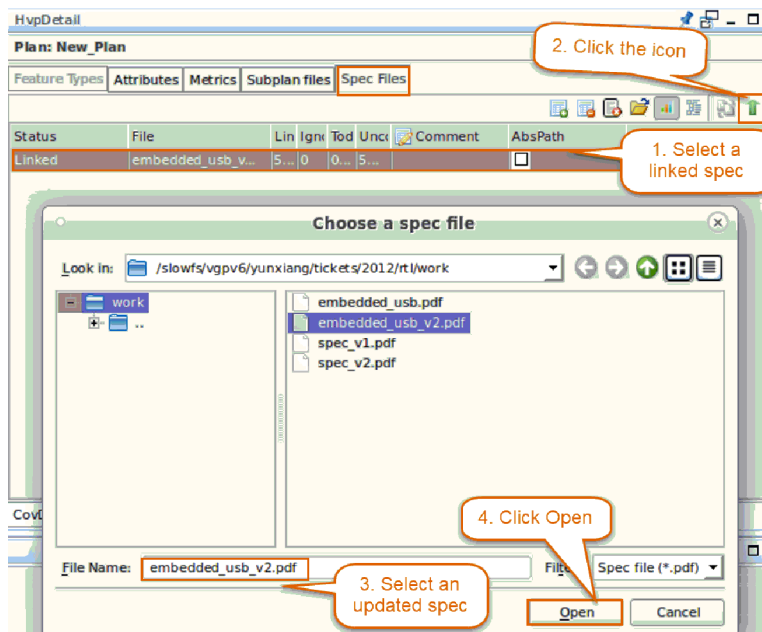
The following sections describe how you can use this feature to perform the specification review.

Using Specification Review Flow

To start reviewing a specification, you need to select a plan in the *HVP* tab and then, perform the following steps:

1. Select the linked specification you want to inspect and review it in the *Spec Files* tab of the *HvpDetail* pane (see Figure [Select a New Specification](#)).
2. Click the  icon and select the specification file you want to compare in the invoked *Choose a spec file* dialog box.
3. Click the **Open** button.

Figure 82 Select a New Specification



4. The *Spec Review Window* form is opened with the comparison between the current and the new specifications, as shown in Figure [Spec Review Window](#). The status field of the selected current specification is changed to *Updating*.

Figure 83 Spec Review Window

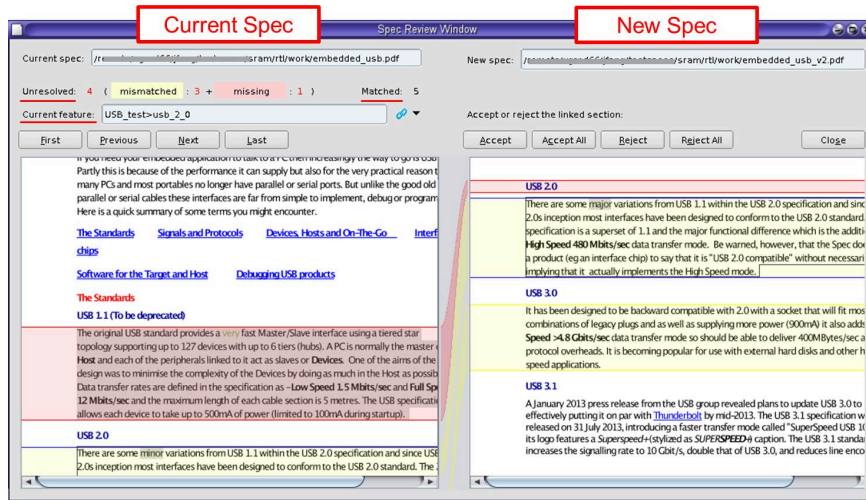


Figure 84 the Updating Status



The **Current spec** and **New spec** fields show the paths of the current and new specifications. The **Review Summary** field (as shown in Figure Review Summary) in the *Spec Review Window* shows how many unresolved sections need to be reviewed, including mismatched and missing ones.

Figure 85 Review Summary

Unresolved: 4 (mismatched : 3 + missing : 1) Matched: 5

The matched, unmatched, new, or missing sections or lines are identified. The current specification is annotated for the unresolved section links and the new specification is annotated with all mapped section links. The missing linked sections that cannot be mapped 100% accurately (the content and the relative location of the section link are not changed) are highlighted in red. The mismatched linked sections that contain mismatched wording are highlighted in yellow. A matched section is highlighted in green.

Reviewing Specifications

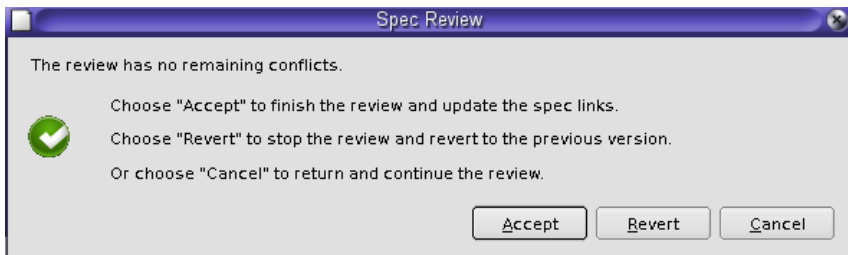
In the *Spec Review Window*, you can click the **Previous** or **Next** button to find out and check the previous or next unresolved section. After checking it, you can use the **Accept** or **Reject** buttons to accept or reject the mappings of the mismatched section links from

the current specification to the new one. For missing sections, you can only use the **Reject** button to reject it. You can also use the **Accept All** and **Reject All** buttons to accept and reject all unresolved sections. After you accept or reject a mapped section, the **Review Summary** field is also updated accordingly to show one less unresolved section.

After all the unresolved sections are reviewed and either accepted or rejected, the **Matched** number in the **Review Summary** field displays 0 and disappears to signify that there are no more updated sections to review.

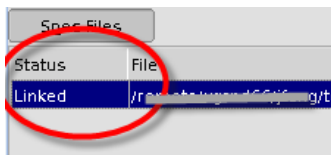
You can click the **Close** button to stop review in the invoked *Spec Review* form.

When your review has no remaining conflicts, the following form is opened.

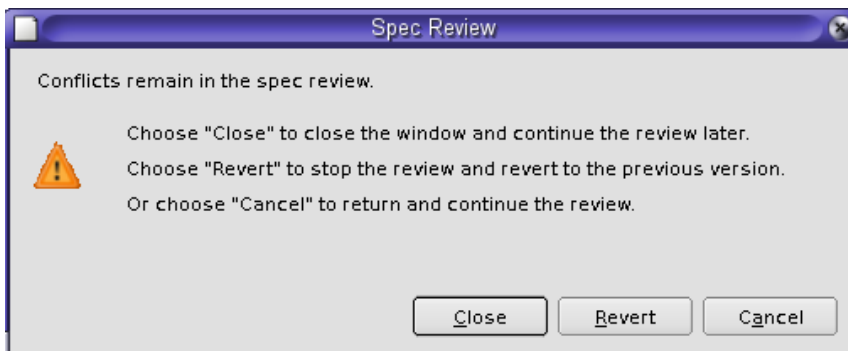


If you want to accept the updates during the review, click the **Accept** button. The linked specification in the HVP planner is updated to the new one. If you want to cancel updating the specification, you can click the **Revert** button to cancel all decisions (accepted or rejected) made in this reviewing. The linked specification remains the original one. The status is changed to 'Linked' after clicking either **Accept** or **Revert** the buttons.

Figure 86 the Linked Status



When a review still has some unresolved sections, the following form is opened.



Click the **Close** button to close the review first and click the  icon to continue the review later. Click the **Revert** button to stop the review and revert the linked specification to the previous version. You can also click the **Cancel** button to back to the current review.

You can click the  undo icon in the *Hvp* pane to undo the following actions during the review:

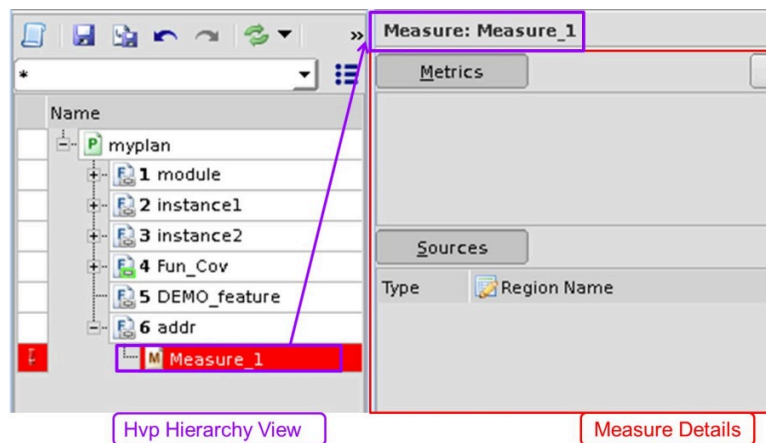
- Start a specification review with a new specification.
- Review all unresolved sections.
- Complete the review and the old specification is replaced by the new one.

Configuring the Linked Feature

You can specify coverage metrics in the *Hvp* pane by clicking the  icon or using the **New Measure** command in the right-click menu.

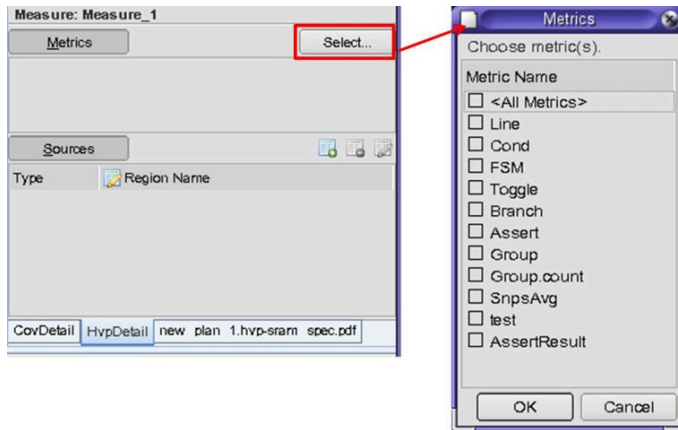
After a measure is created and selected in the *Hvp* pane, the *Metrics* and *Section* panes are shown in the *HvpDetail* pane for the selected measure. As shown in Figure [Measure Detail](#), you can configure the selected measure in the *HvpDetail* pane.

Figure 87 Measure Detail



Select a measure in the *Hvp* pane and click the **Select** button in the *Metrics* pane of the *HvpDetail* pane to set the coverage metric for the selected measure. As shown in Figure [Select Metrics](#), the metrics are listed in the *Metrics* form.

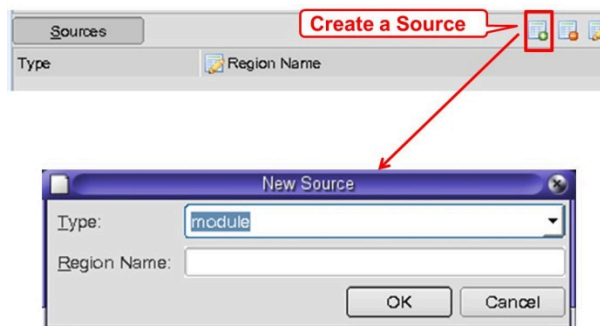
Figure 88 Select Metrics



For example, if your requirement is to check only assertion metrics in the design, select the "Assert" metric and click the **OK** button in the *Metrics* form.

The HVP planner also supports correlating the selected source code information with a metric. You can click the  icon in the *Sources* pane of the *HvpDetail* pane to directly assign the source code type and region name in the *New Source* form.

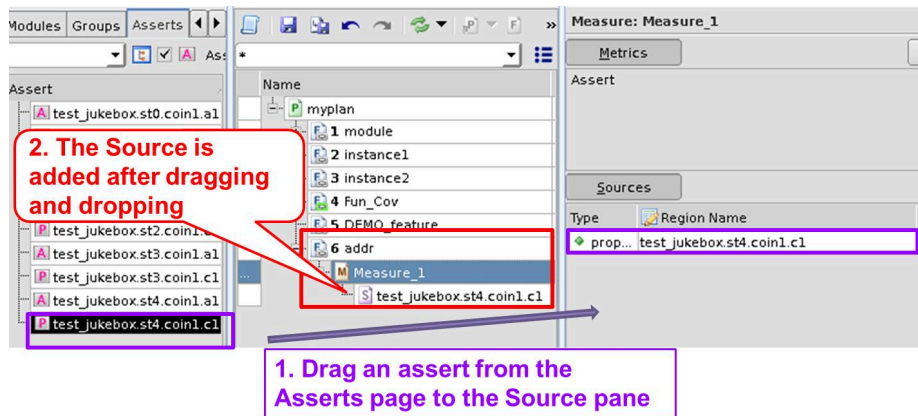
Figure 89 Assign New Source



Alternatively, you can drag an item from the *Hierarchy*, *Modules*, *Groups*, or *Asserts* pages in the *Summary* pane and drop to the *Sources* tab of the *HvpDetail* pane to correlate a metric and source code type. For example, drag an assert from the *Asserts* page in the *Summary* pane and drop it to the *Sources* pane of the *HvpDetail* pane.

As shown in Figure [Assign New Source by Dragging and Dropping](#), you can see a property is added in the *Measure:Measure_1* details page of the *HvpDetail* pane and the source is added under the *Measure_1* measure in the *Hvp* pane.

Figure 90 Assign New Source by Dragging and Dropping



You can also drag a source from the tabs in the *Summary* pane and drop to the feature row in the *Hvp* pane. A new measure is created and the metrics are automatically specified for the source (refer to the [Adding Features From Coverage Database](#) section for details).

After the source information has been populated, you can recalculate the metrics in this plan by clicking the  icon in the *Hvp* pane. After calculating the plan, the average percentage scores of all the built-in metrics mapped to a feature are shown in the *Hvp* pane.

You can use the **Plan -> Save Plan As** command to save the plan in *hvp* format. The saved plan can be reloaded in the future.

You can also mask features of a verification plan that have met all of their goals.

Click the  icon in the *Hvp* pane toolbar to enable the **Hide features meeting goals** option. After this option is enabled, only the features that have not met their goals are kept in the *Hvp* pane.

Measure Concealment in Reports

Currently, HVP provides the flexibility to create multiple measures and metric sources within features. Though this flexibility is useful for many HVP use models, it might produce complex reports for simple models. HVP is now enhanced to simplify usage and reporting for portions of the plan that follow a simplified pattern of a single source for each feature. This is referred to as a 1:1:1 feature because one feature contains one measure and one source.

You can conceal a measure in a report if a feature qualifies the 1:1:1 compression. Verdi can infer from the structure of the feature whether it is qualified for the 1:1:1 compression. To qualify for the compression, the feature should meet the following criteria:

1. The feature is a leaf feature, that is, it does not contain any subfeature.
2. The feature contains exactly one measure.
3. The measure contains exactly one source.
4. The source name should not contain wildcard and regular expressions.

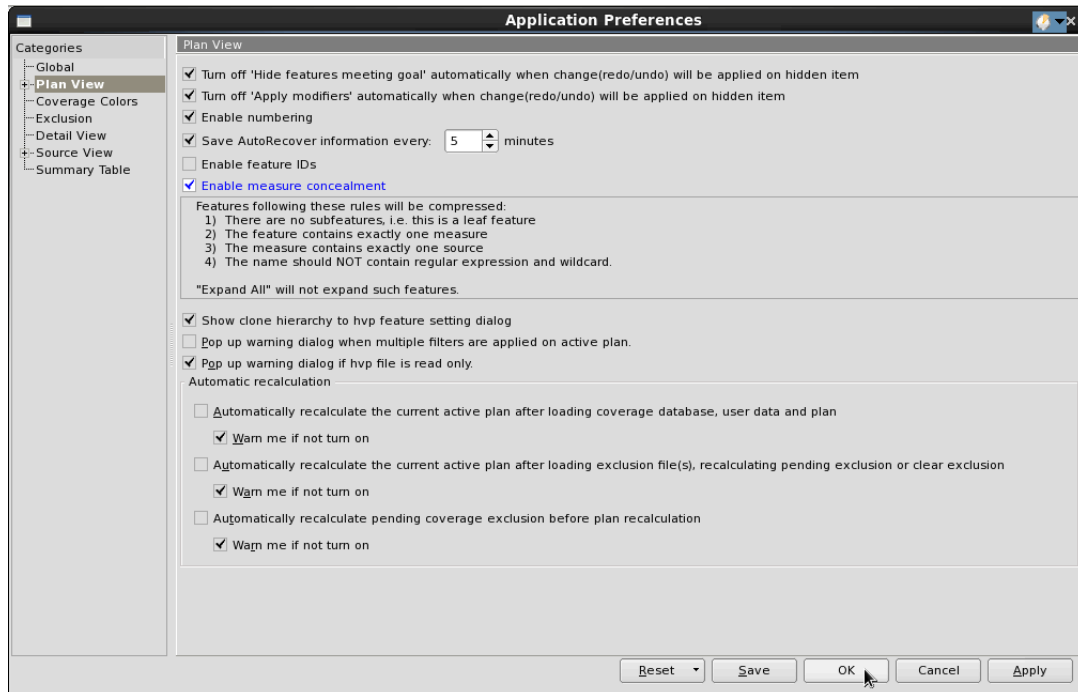
For example, the following feature is qualified for the 1:1:1 compression:

```
feature cd;  
  Metrics = "Line, Cond";  
  measure test Measure_1;  
    source = "cd";  
  endmeasure  
endfeature
```

Use Model

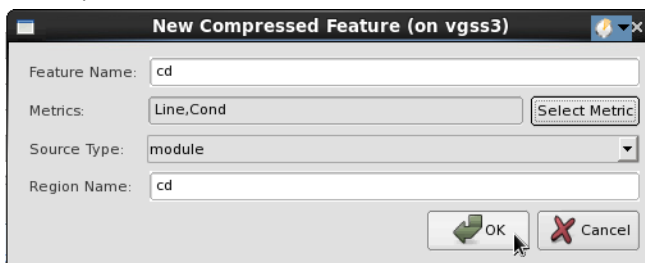
To conceal measures for features that have one measure and one source, perform the following steps in the Verdi GUI:

1. Select **Tools > Preferences** to open the **Application Preferences** dialog box and select **Enable measure concealment** (see the figure below) in **Plan View**.

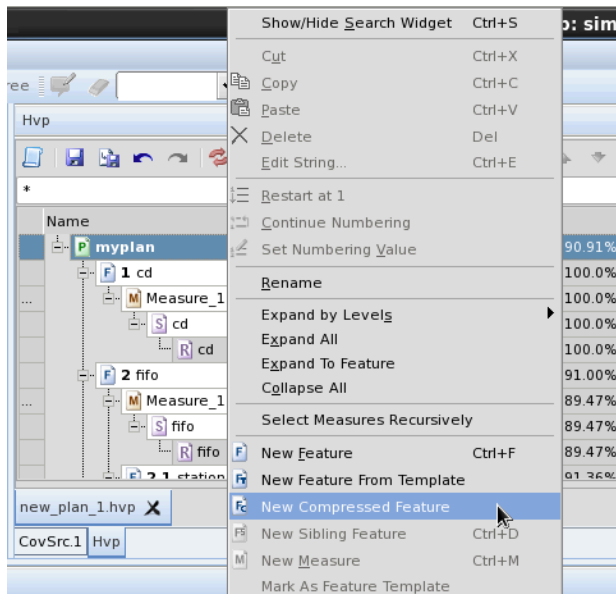


2. Click **OK** to enable measure concealment that allows you to create a compressed feature and a compressed feature template.

To create a compressed feature in a plan, click the **Creates a New Compressed Feature** icon in the *Hvp* pane. The icon displays the **New Compressed Feature** dialog box to specify the feature name, its metrics, its source type, and its region name (see the figure below):



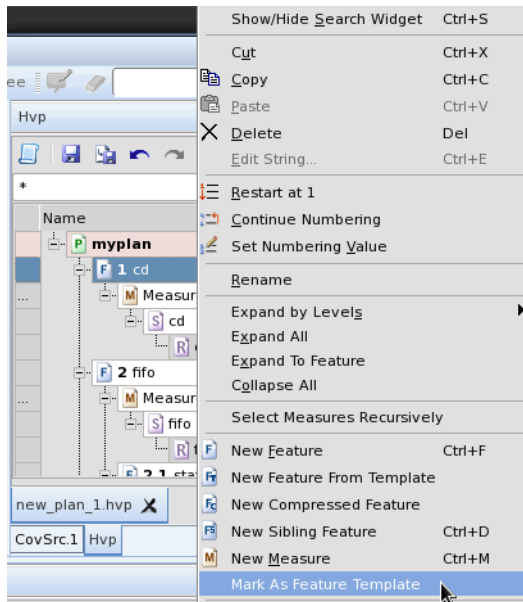
Alternatively, you can right-click the plan and select **New Compressed Feature** to create a compressed feature (see the figure below):



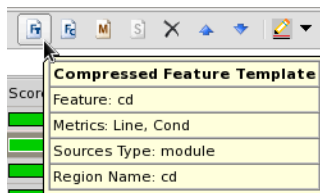
For example, a new compressed feature, `cd` with metrics as `Line` and `Condition`, gets added to the plan, `myplan` as follows:

Name	Score	Line	Cond
myplan	69.72%	90.91%	60.78%
1 cd	100.0%	100.0%	100.0%
Measure_1	100.0%	100.0%	100.0%
cd	100.0%	100.0%	100.0%
cd	100.0%	100.0%	100.0%

To create a compressed feature template, use the **Compressed Feature Template** icon in the *Hvp* pane. Alternatively, you can right-click a feature and select **Mark As Feature Template** to make the compressed feature template (see the figure below).



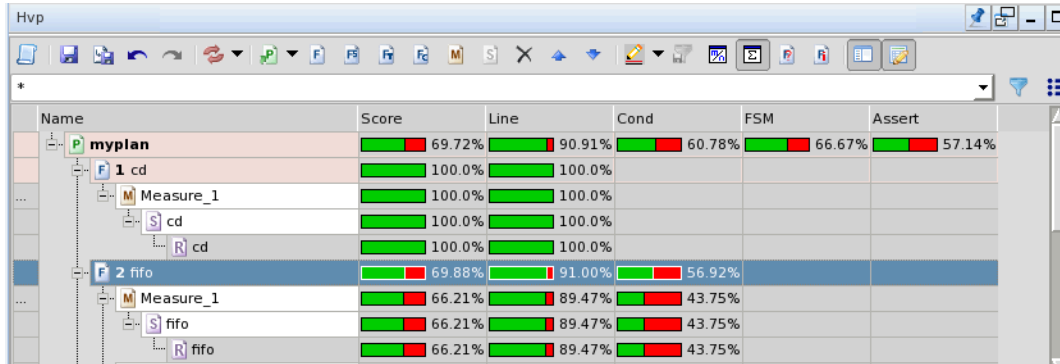
Once a compressed feature becomes a template, the **Compressed Feature Template** icon shows the details of the compressed feature template. For example, if you mark `cd` as a feature template and select the **Compressed Feature Template** icon, the icon shows the details - such as its name, metrics, source type, and region name - of the compressed feature in the tool tip as follows:



You can also mark an existing feature in the *Modules* tab or in the *Groups* tab as a compressed feature. For this, right-click the existing feature and select **Add to Plan (template)**. For example, consider the `fifo` feature in the *Modules* tab of the *Summary* pane that has metrics as Line, Toggle, Condition, and Branch (see the figure below):

Summary								
Hierarchy Modules Groups Asserts Statistics Tests								
Name	Score	Line	Toggle	FSM	Condition	Branch		
cd	92.98%	100.00%	78.95%			100.00%		
coin_fsm	79.94%	91.36%	92.86%	72.22%	61.22%	82.05%		
fifo	66.21%	89.47%	64.95%		43.75%	66.67%		
test_jukebox.fifo1	66.21%	89.47%	64.95%		43.75%	66.67%		
jukebox	89.90%	96.00%	76.00%	100.00%	90.00%	87.50%		
test_jukebox.jb1	89.90%	96.00%	76.00%	100.00%	90.00%	87.50%		
kp_fsm	73.15%	90.91%	75.81%	66.67%	69.44%	78.95%		
station	79.07%		79.07%					
test_jukebox	85.04%	99.02%	68.60%			87.50%		

If you mark `fifo` as a compressed feature by selecting **Add to Plan (template)**, the feature gets added to the plan, becomes compressed, and shows only Line and Condition metrics details (see the figure below).



Name	Score	Line	Cond	FSM	Assert
myplan	69.72%	90.91%	60.78%	66.67%	57.14%
1 cd	100.0%	100.0%			
Measure_1	100.0%	100.0%			
cd	100.0%	100.0%			
R cd	100.0%	100.0%			
2 fifo	69.88%	91.00%	56.92%		
Measure_1	66.21%	89.47%	43.75%		
fifo	66.21%	89.47%	43.75%		
R fifo	66.21%	89.47%	43.75%		

Loading a Predefined HVP File

If a predefined plan has already been created, a plan can be loaded into the HVP planner. The *hvp* format file can be created in the coverage GUI with the method described in the [Configuring the Linked Feature](#) section or created ahead of time in *hvp* format. The default HVP file with *.hvp* extension name can be loaded using the **Plan -> Open Plan** command.

After a plan is loaded into the *Hvp* pane, you can recalculate the coverage metrics per this plan by clicking the  icon.

The HVP planner cannot store items that it does not recognize. The following are the two ways to generate XML files which can be loaded to Verdi:

- Use the command `Save As` to save an XML file without the skipped contents. See [Saving the Plan](#).
- Use the command `hvp genxls -plan <plan_file>` to generate an XML file without the skipped contents.

An existing plan can also be loaded in the XML Spreadsheet 2003 format using the **Plan -> Load Excel XML** command.

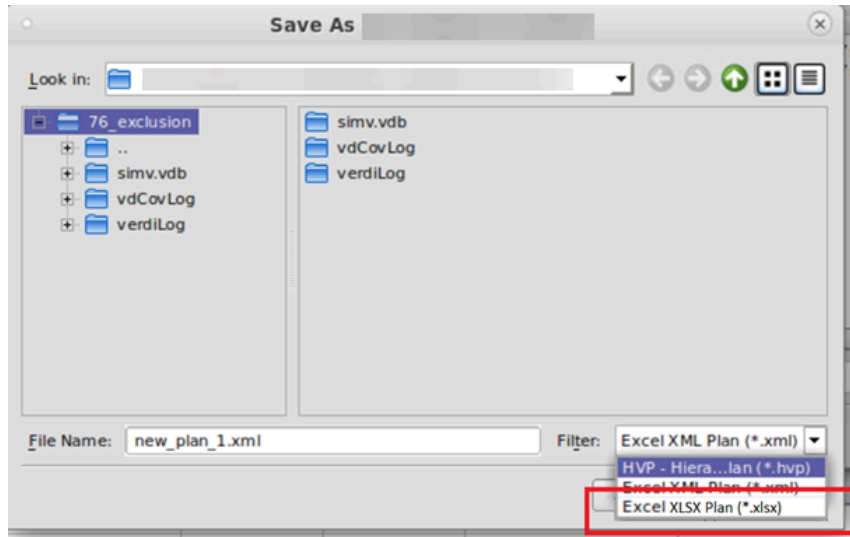
You can also generate a report in the XML Spreadsheet 2003 format from an existing HVP file by using the **Plan -> Report** command. The generated report is for viewing purposes only, it cannot be loaded into Verdi Coverage as an HVP file.

Saving the Plan

Click the  icon to save the changes made in a current plan.

To save the plan in XML format:

1. Choose **Plan > Save Plan As**.
2. Select **Excel XLSX Plan** from the **Filter** list in the **Save As** dialog box.



You can also open the Save As dialog box by clicking the  icon.

User Data As a Spreadsheet Worksheet

Verdi Planner provides user data in a single spreadsheet in the same XML plan document that contains test data. For this, you can enter both data in separate tabs called *test* and *user data* respectively in the same spreadsheet. To specify test data and user data, use `hvp test` and `hvp userdata` as table tags respectively.

The figure below shows a single XML spreadsheet with tabs: *test* and *userdata*. In the *userdata* tab, the `hvp userdata` tag is used to specify user data. The user metric is in tag row *C1:my_int* and *D1:status*. Empty cells in C indicates that there is no data for its corresponding metric, *my_int*, and empty cells in D indicate that there is no data for the given metric status.

	A	B	C	D
1	hvp userdata	name	my_int	status
2		pat2001-01	1	reviewed
3		pat2001-02	2	not_reviewed
4		pat2002	3	reviewed
5		pat2003		not_reviewed
6		pat2005		reviewed
7		NO_PATTERN	6	not_reviewed
8		pat2006	7	
9		pat2007	8	
10		pat2008	9	
11				
12				

For user data, Verdi Planner generates a user data file with the following naming convention:

```
<plan_filename>.data
```

where, `data` is the suffix for HVP User Data files.

In the above figure, Verdi Planner generates the user data file with the name as `spec.xml.data`.

This section consists of the following subsections:

- [Use Model](#)
- [Backward Compatibility](#)

Use Model

To annotate user data text files against an XML spreadsheet plan that results in scores being annotated to the `.xml` report, use either of the following user interfaces:

- [Command Line](#)
- [Verdi GUI](#)

Command Line

To annotate user data text files against an XML spreadsheet plan, use the following command:

```
hvp annotate -plan <plan_file>
```

where,

- `-plan <plan_file>` is a spreadsheet or doc XML file with user data for your verification plan.

For example:

```
hvp annotate -plan test.xml
```

To import the user data XML spreadsheet plan, use the following command:

```
-userdata [path_userdata]
```

```
hvp annotate -plan -userdata [path_userdata]
```

where,

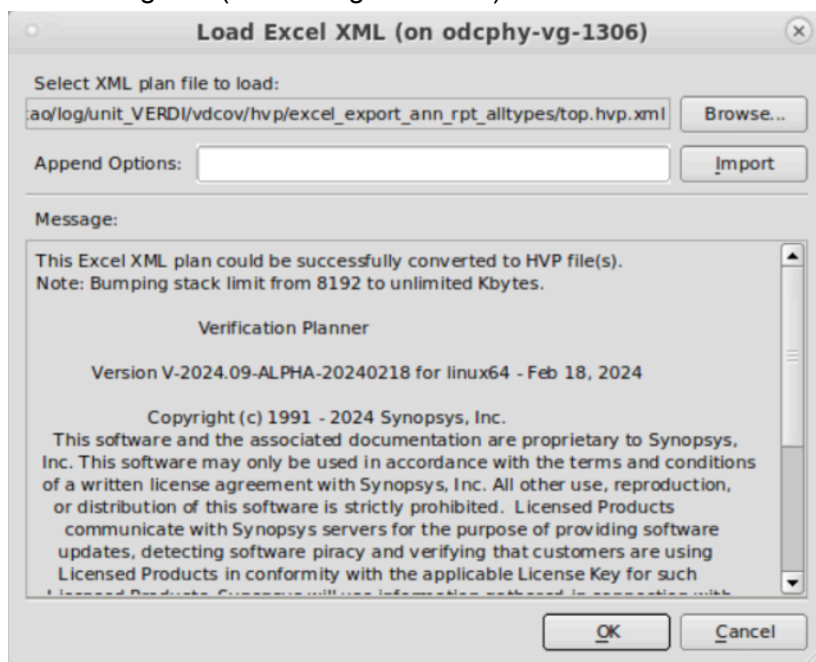
- `-userdata [path_userdata]` is the user data path to import the XML file.

Verdi GUI

To annotate user data text files against an XML spreadsheet plan using the Verdi GUI, perform the following steps:

1. Load the Verdi Coverage GUI using the following command:

```
verdi -cov -covdir <coverage_database.vdb>
```
2. Change the mode of the window to Plan mode using **Windows -> Plan Mode**.
3. Load an XML Spreadsheet using **Plan -> Load Excel XML** to open the **Load Excel XML** dialog box (see the figure below):



4. Select an XML plan file having user data using the **Browse** button.

5. Import XML plan file using the **Import** button.
 6. Click **OK** to automatically generate `<plan_filename>.data` along with `<plan_filename>.hvp`.
-

Backward Compatibility

To generate an XML spreadsheet plan with user datasheets, you need the latest VCS and Verdi Coverage version, that is above L-2016.06-1. For previous versions, Verdi Planner does not annotate user data to the plan even if the XML spreadsheet contains user data.

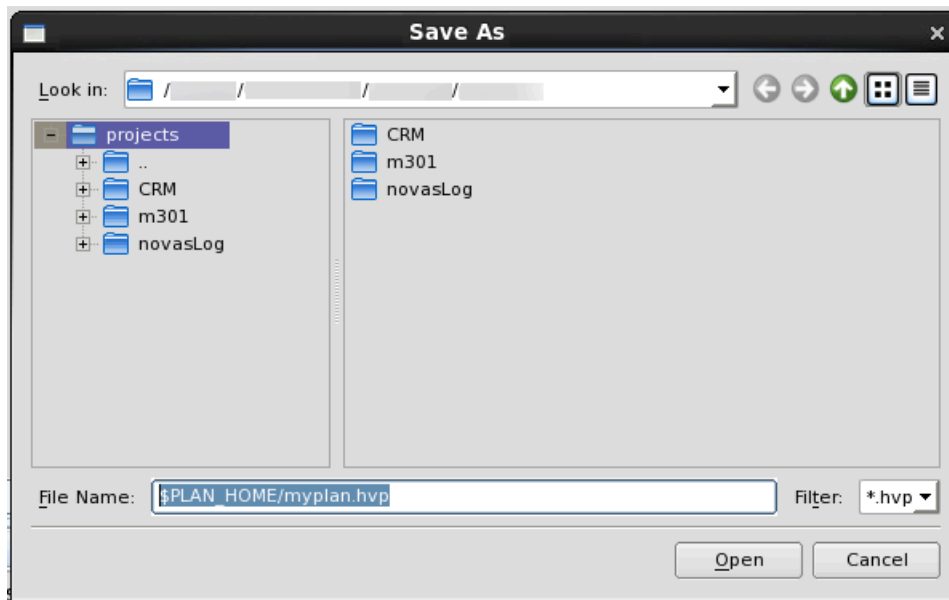
Loading and Saving a Verification Plan Using Environment Variables

In previous releases, you used to browse through different directories to load and save a verification plan. Verdi Planner is now enhanced to support environment variables for loading and saving a plan. Using environment variables, you just need to specify the file name instead of specifying the long path of the file. You might use environment variables to specify the path of a project, plan, CRM, and design root in the **File Name** field of the **Load** or **Save** dialog box.

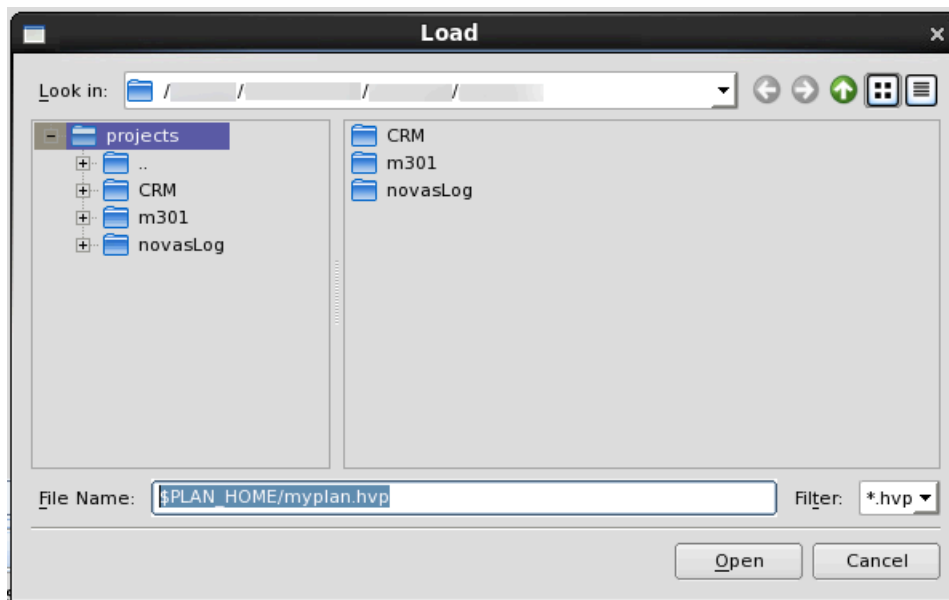
To set the environment variables for the design root, project, plan, and CRM, use the following commands:

```
setenv DESIGNROOT <designroot_path>
setenv PROJ_HOME <project_path>
setenv PLAN_HOME $PROJ_HOME/<plan_path>
setenv CRM_HOME $PROJ_HOME/<crm_file_path>
```

An example to use an environment variable in the **File Name** field of the **Save As** dialog box in the Verdi GUI is shown in the following figure:



An example to use an environment variable in the **File Name** field of the **Load** dialog box in the Verdi GUI is shown in the following figure:



To load a verification plan, you can also provide environment variables on the Verdi command line as follows:

```
verdi -cov -covdir $DESIGNROOT/<designroot_path> -plan  
$PLAN_HOME/<plan_name> -userdata $CRM_HOME/<crm_file_path>
```

For example, the following command loads `myplan.hvp` using `DESIGNROOT`, `PLAN_HOME`, and `CRM_HOME` environment variables:

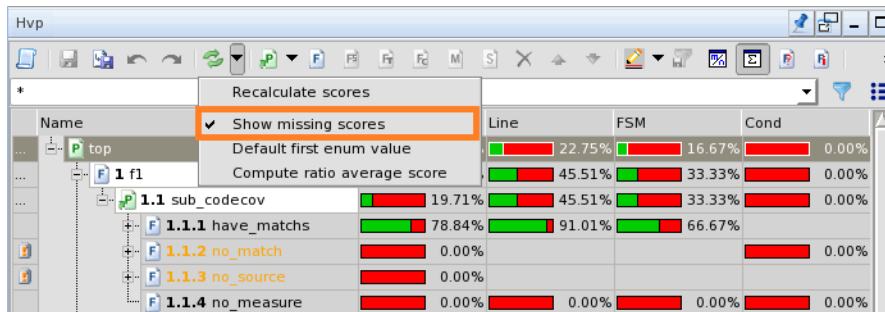
```
verdi -cov -covdir $DESIGNROOT/simv.vdb -plan $PLAN_HOME/myplan.hvp  
-userdata $CRM_HOME/bugs.txt
```

Showing Missing Scores

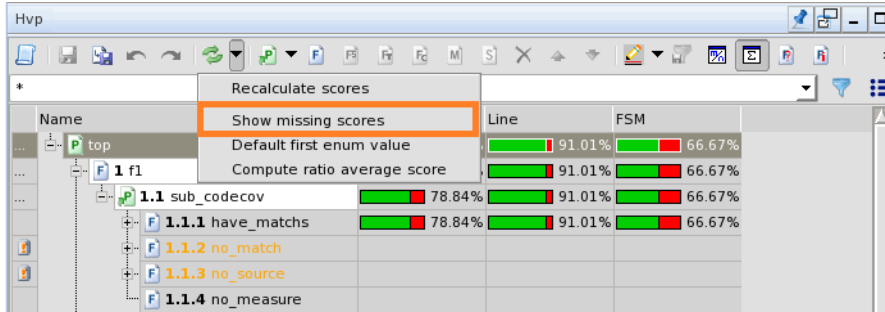
Verdi enables unmatched coverage sources to affect plan scores using the **Show Missing Scores** option in the Verdi GUI. These unmatched coverage sources are scored as zero, by default. You might deselect this option to produce no scores for the unmatched coverage scores. The option produces no score in the following scenarios:

- The sources do not match any object in the `.vdb` database.
- The measure statements have no sources.
- The sources match an object in the `.vdb` database but when no metric is measured.

Consider the `top` plan that has a feature `f1` with has four sub-features, `have_matches`, `no_match`, `no_source`, and `no_measure`. The coverage score of `no_match`, `no_source`, and `no_measure` is zero as shown in the following figure:



In the figure above, even unmatched coverage features — such as `no_match`, `no_source`, and `no_measure` — are considered for coverage computation as the **Show Missing Scores** option is selected. If you deselect this option, then these unmatched coverage features are not included for coverage computation as shown in the following figure:



The `-hvp_no_score_missing` option disables unmatched coverage sources to affect plan scores. There is no score for unmatched coverage source.

Graphical Traceability Support

Verdi Planner provides a new report that can be generated from an annotated verification plan that exports data in the FreeMind format and allows you to see relationships between features and enum metric values. The most common usage is to chart the relationship between features and their test pass/fail results. This allows you to use the graphical report to trace from a feature to its test results and vice versa.

This section consists of the following subsections:

- [Use Model](#)
- [Limitations](#)

Use Model

Verdi Planner exports an annotated plan to the FreeMind XML format that enables you to view HVP enum results using the FreeMind tool. The FreeMind node structure is similar to the HVP hierarchy tree. Only the naming conventions are different, such as HVP test enum result is leaf node in the FreeMind tool, HVP plan or feature is fork in the FreeMind tool, and HVP region is the FreeMind bubble.

To export an HVP to the FreeMind XML format, the **FreeMind XML Format** option is added to the Verdi GUI. This section consists of the following subsections:

- [User Interface](#)
- [User Flow](#)

User Interface

To export an HVP to the FreeMind XML format, perform the following steps:

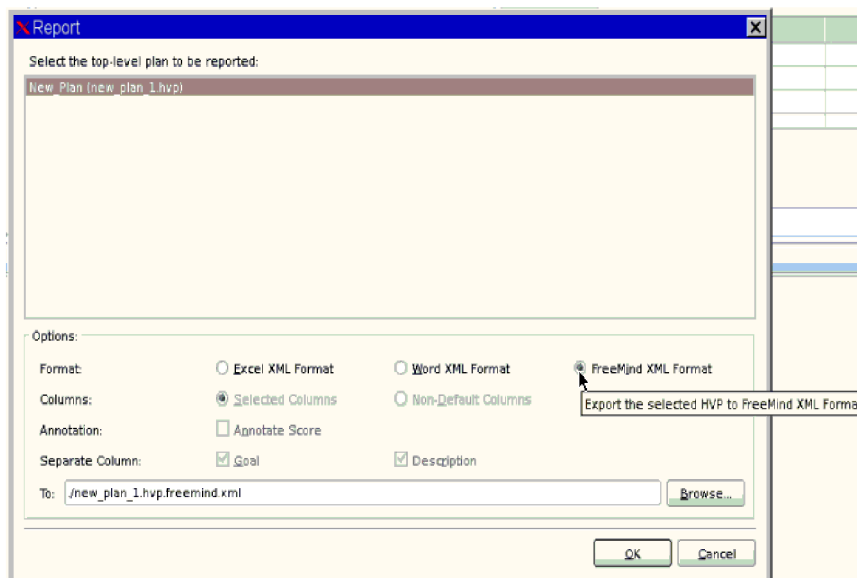
1. Load the Verdi Coverage GUI using the following command:

```
verdi -cov -plan <plan_file> -userdata <userdata_files>
```

For example:

```
verdi -cov -plan test.hvp -userdata test.hud
```

2. To select the top-level plan to export, select **Plan -> Report**. It opens the **Report** dialog box (see the figure below).



3. Click **FreeMind XML Format** from the **Options** group box to export the plan to the FreeMind XML format. When this format is selected, all other configurations such as **Columns**, **Annotation**, and **Separate Column** become disabled and the text box of the exported XML file is updated to `<plan_file>.freemind.xml`.

User Flow

Consider the following example to see the flow of `test.hud` and `test.hvp`:

test.hud

```
HVP metric = test
/cfg/block1/test11 = warn
/cfg/block1/test12 = fail
/cfg/block1/test13 = pass
/cfg/block1/test14 = pass
/cfg/block1/test15 = fail
```

```
/cfg/block1/test21 = pass
/cfg/block1/test22 = fail
/cfg/block1/test23 = unknown
/cfg/block1/test24 = warn
/cfg/block1/test25 = assert
```

```
/cfg/block2/test1 = assert
/cfg/block2/test2 = pass
/cfg/block2/test3 = fail
/cfg/block2/test4 = pass
/cfg/block2/dummy1 = warn
```

test.hvp

```
plan top;
  feature common;
    feature block1;
      test = ((test >= 10) && (test.fail <= 1));
      feature Feature_1;
        measure test Measure_1;
        source = "/cfg/block1/test11_";
      endmeasure
    endfeature
    measure test, test.percent.fail m1;
    source = "/cfg/block1/test*";
    endmeasure
    measure Line m_code;
    source = "tree: test_jukebox.st1";
    endmeasure
    measure test Measure_1;
    source = "/cfg/block1/test11", "/cfg/block2/
      dummy1";
    endmeasure
  endfeature
  feature block2;
    test = ((test >= 10) && (test.fail <= 1));
    measure test, test.percent.fail m2;
    source = "/cfg/block1/test11";
    endmeasure
  endfeature
endfeature
endplan
```

The Verdi GUI shows the following format of test.hud and test.hvp:

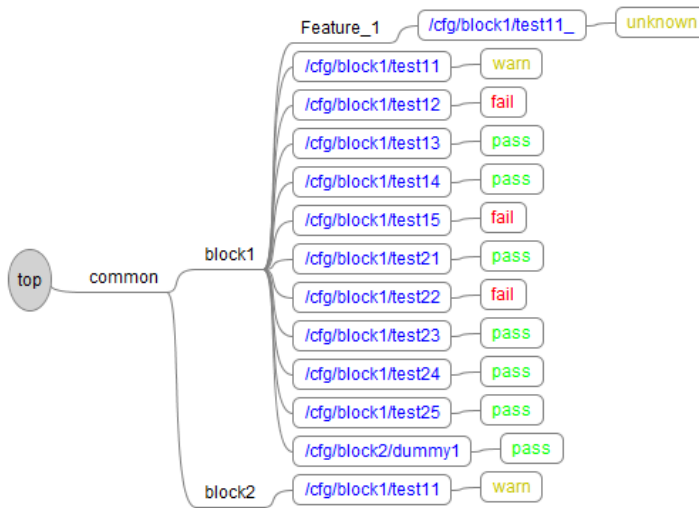
test.hud

	Test Name	Type	User Test Name	Status
	/cfg/block1/test11	Test Record		warn
	/cfg/block1/test12	Test Record		fail
	/cfg/block1/test13	Test Record		pass
	/cfg/block1/test14	Test Record		pass
	/cfg/block1/test15	Test Record		fail
	/cfg/block1/test21	Test Record		pass
	/cfg/block1/test22	Test Record		fail
	/cfg/block1/test23	Test Record		unknown
	/cfg/block1/test24	Test Record		warn
	/cfg/block1/test25	Test Record		assert
	/cfg/block2/dummy1	Test Record		warn
*	/cfg/block2/test1	Test Record		assert
*	/cfg/block2/test2	Test Record		pass
*	/cfg/block2/test3	Test Record		fail
*	/cfg/block2/test4	Test Record		pass

test.hvp

Name	owner	at_least	testpass	testfail	testwarn	testunknown	testassert
top		0	3	3	5	2	1
1 common		0	3	3	5	2	1
1.1 block1		0	3	3	4	2	1
1.1.1 Feature_1		0	0	0	0	1	0
Measure_1			0	0	0	1	0
/cfg/block1/test11		0	0	0	0	1	0
no matching region			0	0	0	1	0
m1		3	3	3	2	1	1
/cfg/block1/test*		3	3	3	2	1	1
/cfg/block1/test11		0	0	0	1	0	0
/cfg/block1/test12		0	0	1	0	0	0
/cfg/block1/test13		1	0	0	0	0	0
/cfg/block1/test14		1	0	0	0	0	0
/cfg/block1/test15		0	1	0	0	0	0
/cfg/block1/test21		1	0	0	0	0	0
/cfg/block1/test22		0	1	0	0	0	0
/cfg/block1/test23		0	0	0	0	1	0
/cfg/block1/test24		0	0	0	1	0	0
/cfg/block1/test25		0	0	0	0	0	1
m_code							
test_juicebox.t1							
no matching region							
Measure_1		0	0	0	2	0	0
/cfg/block1/test11		0	0	0	1	0	0
/cfg/block1/test11		0	0	0	1	0	0
/cfg/block2/dummy1		0	0	0	1	0	0
/cfg/block2/dummy1		0	0	0	1	0	0
1.2 block2		0	0	0	1	0	0
m2		0	0	0	1	0	0
/cfg/block1/test11		0	0	0	1	0	0
/cfg/block1/test11		0	0	0	1	0	0

The FreeMind XML format is as follows:



Limitations

The limitations of this feature are as follows:

- Only the HVP plan, feature, or region that has matched enum value is dumped, whereas others are skipped. Also, HVP measure/source is skipped.
- During exporting, the existing opened HVP handle is used. If HVP is not refreshed, the exported plan is not the latest one.

To calculate the specification coverage for both main plan and subplan, Verdi Coverage uses the *Merge Specification Coverage* algorithm. The algorithm provides priority to calculate each plan, which is categorized as "Linked", "Ignored", and "Todo". The priority sequence of these categories is as follows:

Linked > Ignored > Todo

The following table shows the priority sequence as per the Merge Specification Coverage algorithm:

Main Plan	Subplan	Merge Specification Coverage Algorithm
Linked	Ignored	Linked
Unlinked	Todo	Todo
Todo	Ignored	Ignored

Main Plan	Subplan	Merge Specification Coverage Algorithm
Todo	Linked	Linked

Note:

Main plans and subplans should share the same margin for merging the specification coverage, otherwise, coverage merge cannot happen among plans.

Using Derived Test Metrics

Verdi Planner supports the following additional built-in derived test metrics to track verification progress:

- [Percent Metric for Test Metric](#)
- [test.completion Metric](#)

Using these derived metrics, you can gather the percentage of tests with different status and test completion percentage in a verification plan.

Percent Metric for Test Metric

In previous releases, the built-in test metric of **enum** type reports integer count for sub-metrics **test.pass**, **test.fail**, **test.warn**, **test.unknown**, and **test.assert**. For example,

```
test = 6;  
test.pass = 3;  
test.fail = 3;
```

In previous releases, the test metric reports the basic number of tests with status **pass**, **fail**, **warn**, **unknown**, and **assert** in the HVP annotated report. It does not display the percentage of these scores.

Verification Planner for Verdi reports percentage score for each **enum** value of the test metric. This enables you to get the percentage of tests with status **pass**, **fail**, **warning**, **unknown**, or **assert**. The following is the list of the derived percent sub-metrics:

- **test.percent.pass**
- **test.percent.fail**
- **test.percent.warn**

- **test.percent.unknown**
- **test.percent.assert**

The score of a percent metric is the percentage of each enum's value of the test metric against the total number of tests in that feature.

For example, if a is the integer count of the built-in test metric **enum** value and b is the integer count of the total number of tests, then the percentage metric is computed as follows:

$$(a / b) * 100$$

For example, if

```
test.pass = 8
test.fail = 2
test = 10
```

then the value of **test.percent.pass = 8/10** and is reported as 80%.

The following is the example for the usage of percent metrics:

```
feature f1;
  test.percent.pass = test.percent.pass > 60%;
  measure test, test.percent.pass m1;
    source = "testname*";
  endmeasure
endfeature
```

Consider the following example:

```
feature parent:
  test:6[pass:3 fail:3 warn:0 assert:0 unknown:0 ]
  test.percent.pass:50.00%
  feature block1
    test:5[pass:3 fail:2 warn:0 assert:0 unknown:0 ]
    test.percent.pass:60.00%
  feature block2
    test:1[pass:0 fail:1 warn:0 assert:0 unknown:0 ]
    test.percent.pass: 0.00%
```

In this example, the **test.percent.pass** score in **feature parent** is calculated from the test score of:

```
feature parent test:6[pass:3 fail:3 warn:0 assert:0 unknown:0] 50%
```

which is equal to 3/6. The total test score of **feature parent** is the sum of the test scores of **feature block1** and **feature block2**.

The following figure shows the derived sub-metrics displayed by Verdi coverage:

Figure 91 Viewing Percent Metric in Verdi Coverage

	goal.test	test pass/fail/mis	test.fail	pass%	fail%
top		3/3/0 3	3	50.00%	50.00%
1 parent		3/3/0 3	3	50.00%	50.00%
1.1 block1	((test >= 10)...	3/2/1 2	2	60.00%	40.00%
M m1		3/2/1 2	2	60.00%	40.00%
S /cfg/block1/test*		3/2/1 2	2	60.00%	40.00%
1.2 block2		0/1/1 1	1	0.00%	100.0%
M m1		0/1/1 1	1	0.00%	100.0%
S /cfg/block2/test*		0/1/1 1	1	0.00%	100.0%

test.completion Metric

The **test.completion** metric allows you to measure the progress of tests being written against the number of tests expected to be written.

The score of the **test.completion** metric is the ratio of total integer count of all sub-metrics of the test metric (**test.pass**, **test.fail**, **test.warn**, **test.assert**, **test.unknown**) and the expected number of tests in a plan/feature. The expected number of tests should be set using the **test.expected** attribute.

Where, **test.completion**=**total_test_count**/**test.expected** and
total_test_count=**test.pass**+**test.fail**+**test.warn**+**test.assert**+**test.unknown**

Test.expected Attribute

You can use the **test.expected** built-in attribute assignment, as shown below, to set the expected number of tests in feature/plan scope.

test.expected = <number>;

The expected number is like an annotation value kept in feature/plan node. The default value of **test.expected** is 0.

Figure 92 Viewing test.expected in Verdi Coverage

Name	test.expected	goal.test	test pass/fail/n	test.pass
New_Plan	0		29/4/3	15
1 Goal_pass	2		1/0/1	1
2 Goal_fail	2		0/1/1	0
3 Goal_warn	4		3/0/1	0
4 Goal_assert	0		5/0/0	1
5 Goal_unknown	0		3/0/0	1
6 Goal_completion	5		4/1/0	2
7 Goal_complex	0	((test > ...	13/2/0	10

The **test.completion** metric can be calculated only after setting the **test.expected** attribute.

The following example illustrates the usage of the **test.completion** metric:

```
feature Goal_pass;
test.expected = 2;
test.percent.pass = (test.percent.pass >= 0.5);
measure test, test.percent.pass, test.completion Measure_1;
source = "test3";
endmeasure
endfeature
```

Figure 93 Viewing test.completion Metric in Verdi Coverage

Name	Line	test pass/fail/mis	pass%	test completion
 top	 73.03%	 12/4/19	 68.75%	 45.71%
 1 parent	 73.03%	 12/4/19	 68.75%	 45.71%
 1.1 block1	 89.09%	 7/3/10	 60.00%	 50.00%
 m1		 7/3/10	 60.00%	
 m_code	 89.09%			
 1.2 block2	 53.64%	 4/0/10		
 m1		 4/0/10		
 m_code	 53.64%			
 1.3 block3	 76.36%	 1/0/9		 10.00%
 1.3.1 block3_sub		 1/0/9		 10.00%
 m1		 1/0/10		
 m_code	 76.36%			
 m1		 0/1/10		

Using Test Name as HVP Source

Verdi Coverage lets you add a test as the source for an HVP plan from the following sources:

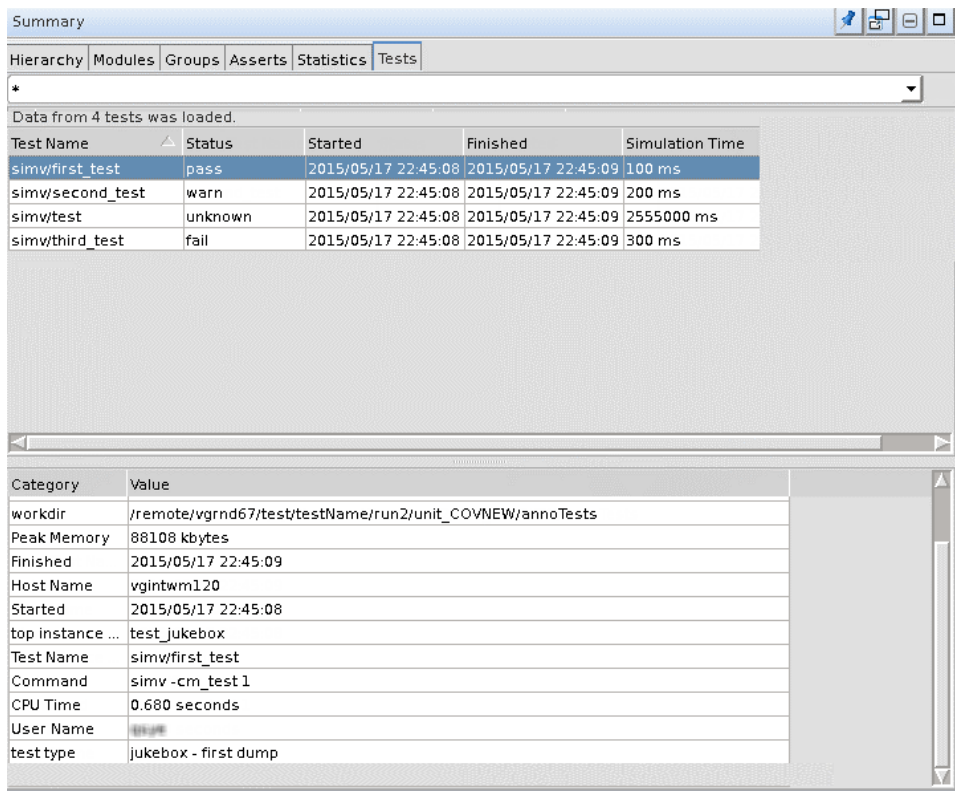
- User Data File
- Loaded Test List

The following sections describe how you can use this feature:

Adding a Test Source

Verdi Coverage lets you add a test from the *Test* Pane via drag-drop and Copy/Paste operations. If you add a source using this method, Verdi automatically selects the corresponding test metric.

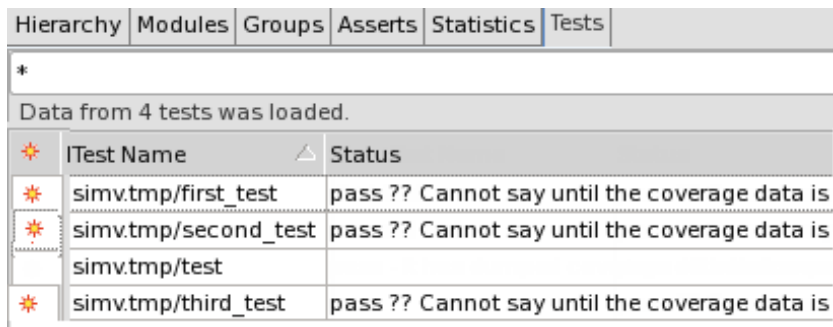
Figure 94 Test Details



Used/Unused Marker for Test Pane

Verdi Coverage marks used/unused tests in the Test pane as shown the following figure:

Figure 95 Used/Unused Marker



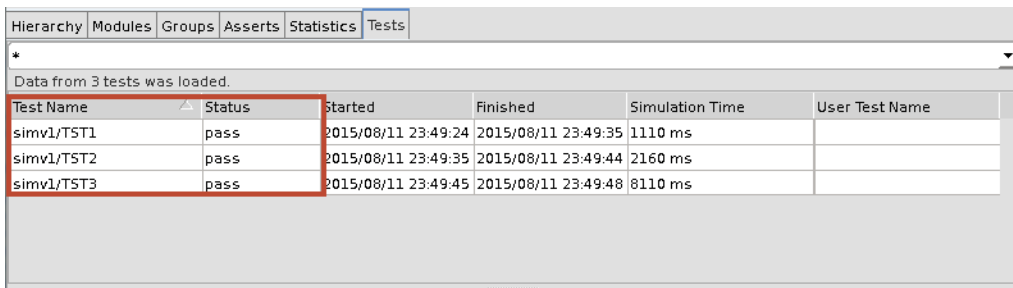
Conflict Resolution

When Verdi Coverage detects a conflict, it resolves it using the following rules:

Conflict	Take Priority
VDB and Execution Manager DB	Execution Manager session
VDB directory and user data	User data
VDB and VDB have the same user test name	If there is more than one test run with the same user name, Verdi matches them all like a wildcard.

For example, consider a scenario where a test passes for the loaded VDB as shown in the following figure:

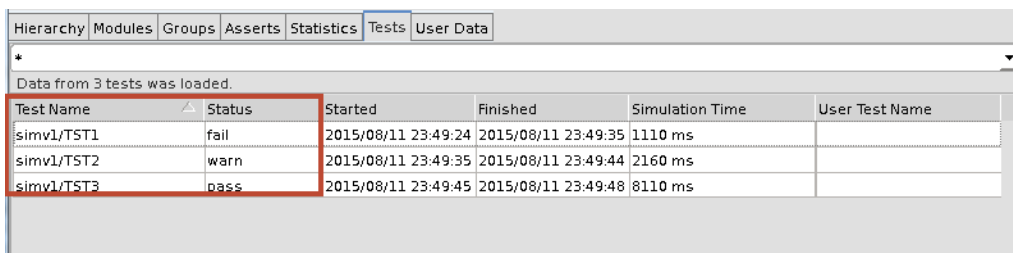
Figure 96 Tests Run With VDB



Test Name	Status	Started	Finished	Simulation Time	User Test Name
simv1/TST1	pass	2015/08/11 23:49:24	2015/08/11 23:49:35	1110 ms	
simv1/TST2	pass	2015/08/11 23:49:35	2015/08/11 23:49:44	2160 ms	
simv1/TST3	pass	2015/08/11 23:49:45	2015/08/11 23:49:48	8110 ms	

Now, consider that you also load user data; the test status changes to the following:

Figure 97 Tests Run With User Data



Test Name	Status	Started	Finished	Simulation Time	User Test Name
simv1/TST1	fail	2015/08/11 23:49:24	2015/08/11 23:49:35	1110 ms	
simv1/TST2	warn	2015/08/11 23:49:35	2015/08/11 23:49:44	2160 ms	
simv1/TST3	pass	2015/08/11 23:49:45	2015/08/11 23:49:48	8110 ms	

As user data results takes precedence over VDB test results as user data gets higher precedence in a conflict situation.

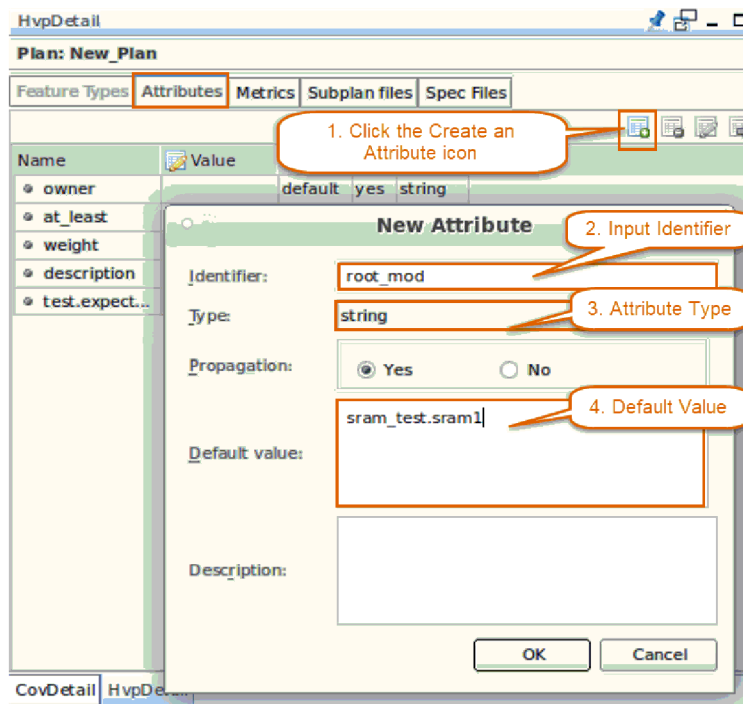
Using User-Defined Attributes and Metrics

Verdi Coverage supports creation of specialized attributes and metrics that are customizable to the design in the planner.

Creating Attributes

Create an attribute for the selected plan by clicking the  icon in the *Attribute* pane of the *HvpDetail* pane as shown in Figure [Add a Customized Attribute](#). You can specify the identifier, type, and default value in the *New Attribute* form.

Figure 98 Add a Customized Attribute



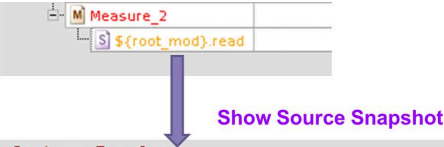
As shown in Figure [Added root_mode Customized Attribute](#), the *root_mode* attribute is an attribute of the *myplan* plan.

Figure 99 Added root_mode Customized Attribute

Plan: myplan				
Attributes				
Name	Value	Origin	Prop	Type
owner		default	yes	string
at_least	0	default	yes	integer
weight	1	default	no	integer
description		default	no	string
root_mod	sram_test...	default	yes	string

The new attribute can be used as a variable as part of the property to denote the source information. For example, add a source region under the *Measure_2* measure and rename the source region with `${root_mod}.read` (see Figure [Show Source Snapshot](#)). You can click the  icon to show the source snapshot in a new tab after the plan is modified. The source text shows the definition of the *root_mod* user-defined metric.

Figure 100 Show Source Snapshot



```

269 feature Fun_Cov;
270     owner = "Carl";
271     measure bugs, fixed, status status;
272     source = "/cfg/data/Function_coverage";
273
274     endmeasure
275     measure Measure_2;
276     source = "tree: ${root_mod}.read";
277     endmeasure
278     subplan fun_cov;
279 endfeature
  
```

Enum Type Attribute

Verification Planner provides an attribute type named `enum`.

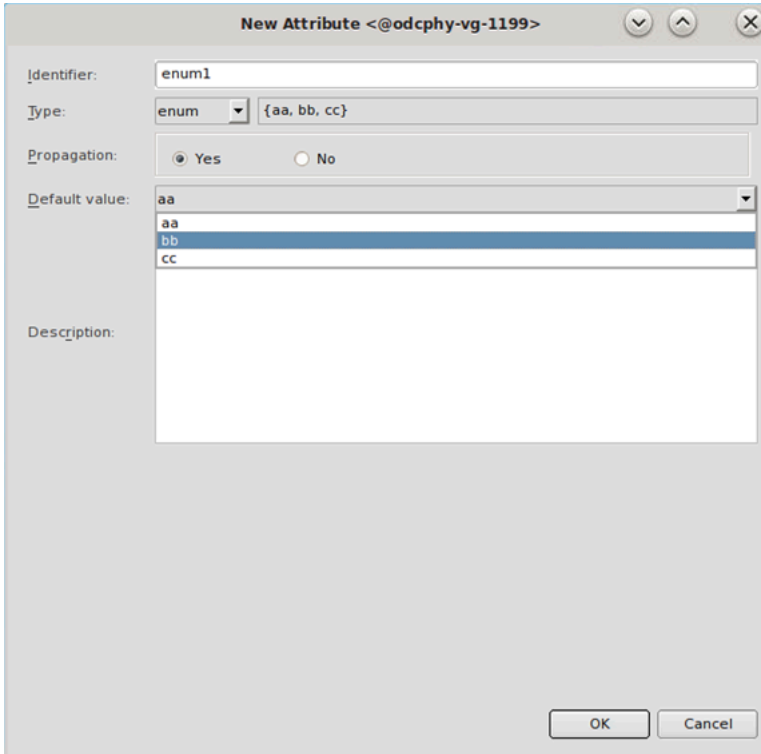
The following section describes how to create and edit an attribute of enum type in the **Attribute** pane of the **HvpDetail** pane of the Verdi GUI:

- [Adding an Enum Type Attribute using the Verdi GUI](#)
- [Editing an Enum Type Attribute using the Verdi GUI](#)

Adding an Enum Type Attribute using the Verdi GUI

To create an `enum` type attribute in Verdi, perform the following steps:

1. Click the **Create an attribute**  icon to display the **New Attribute** dialog box.
2. In the **New Attribute** dialog box, specify the **Identifier** as the name of the attribute (say, `enum1`). To make an attribute of type `enum`, select **Type** as **enum** and specify its values as follows:



Enclose the specified values in square brackets and separate them by commas. For example, `{aa, bb, cc}`.

3. To add the values of this attribute in the sub-plans along with the main plan, mark **Propagation** to **Yes**.
4. To specify the default value of this attribute, use the **Default value** field. If you do not specify the default value of this attribute, the default value of this attribute is considered as empty.
5. Click **OK** to add the `enum1` attribute in the *Attributes* tab of the **HvpDetail** window. You can see its corresponding plan in the **Hvp** window in the Verdi GUI.

Editing an Enum Type Attribute using the Verdi GUI

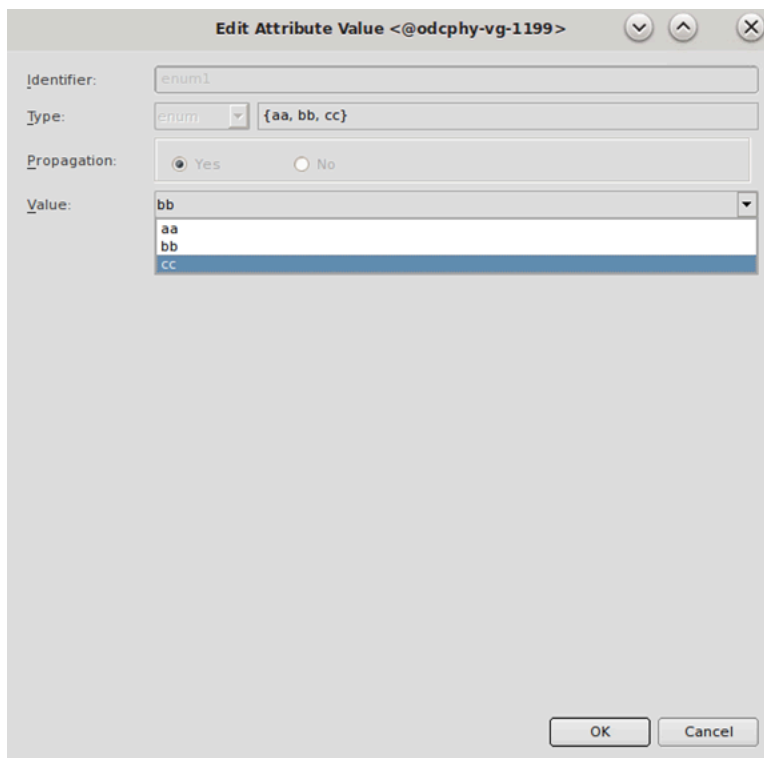
The **Edit Attribute Value** dialog box allows you to edit the values of enum attributes.

The **Edit Attribute Value** dialog box can be displayed by double-clicking the value of an enum attribute cell. You can also edit an attribute value by doing any of the following:

- Clicking on the **Edit Attribute** button  in the **HvpDetail** pane with a single item selected in the **Hvp** pane and a specified attribute selected in the **HvpDetail** pane.
- Double-clicking the specified value cell in the **HvpDetail** pane.

When multiple items are selected in the **Hvp** pane, the value change of the **Edit Attribute Value** dialog box (triggered by double-clicking the specified attribute value) takes effect on all values of the selected items.

You can select a value from the **Value** list of the valid enum values as follows:



Set Type Attributes

Verification Planner provides a new attribute type named `set`. A set is a group of discrete objects or numbers, considered as an entity. Each object or number in a set is known as a member or an element of the set. When the elements of a set is listed, these elements are enclosed in curly brackets. For example, a set (say, A) of natural numbers from 1 to 5 is denoted as follows:

$A = \{1, 2, 3, 4, 5\}$

The order in which the elements of a set appear does not matter. For example, consider another set (say, B) that consists of natural numbers from 1 to 5 as follows:

$B = \{4, 3, 2, 1, 5\}$

Set A and B are considered as equal as they have the same elements in different order.

If two sets have same elements but the elements appear multiple times in one set, even then these sets are considered as same. For example, consider the following set C:

$C = \{4, 2, 2, 3, 1, 1, 5, 5\}$

Set A, B, and C are considered as same irrespective of multiple elements in different order.

If every member of set X is also a member of set Y, then X is a subset of Y. For example, X {1, 3} is a subset of Y {1, 2, 3}. However, Z{1, 4} is not a subset of Y as element 4 does not exist in Y {1, 2, 3}.

The `set` type enables you to specify multiple values for an attribute. To declare an attribute of the `set` type in the HVP, use the following syntax:

```
attribute set{<element_list>} identifier = <default_set_value>;  
identifier = <set_value>;
```

where `<set_value>` is an identifier. Identifiers in `<set_value>` must start with an alphabetic character, but otherwise can use alphanumeric or the underscore character in their names.

For example, the following code shows the definition of the `my_set` attribute of type `set` with values as `S0`, `S1`, `S2`, and `S_other`:

```
attribute set{S0, S1, S2, S_other} my_set = {};  
my_set = {S0, S2};
```

Attributes of type `set` in a plan can be propagated to sub-plans by setting their propagation value to `Yes`. The attributes propagated from a parent plan cannot be modified in a sub-plan. The attributes defined in a sub-plan can be used within the sub-plan but are not visible in the parent plan.

The `set` attribute type allows you to perform filtering. The `set` type can be filtered using the `==` and `!=` operators. Operators such as `>`, `<`, `>=`, `<=`, `+`, `-`, `*` and `/` are not supported.

For example, consider the following filter that shows the usage of `==` and `!=` operators:

```
filter eg_filter;  
  //# == and != is an exact match,  
  //# {S0} == {S1,S2} returns false  
  //# {S0,S1,S2} != {S0,S1} returns true  
  remove feature where my_set == {S0,S1};  
  keep feature where my_set != {};  
endfilter
```

In the example above, the `my_set == {S0, S1}` filter removes the features where `my_set` is exactly `{S0, S1}`. However, the `my_set != {}` filter restores any feature that has a null `my_set`.

Verification Planner also provides the `contains` function for filtering with the `set` type attribute:

```
contains(<set_attribute_identifier>, <set_val>)
```

The `contains` function returns `true` when `<set_val>` is equal to or a subset of the value of the `set` type attribute specified by `<set_attribute_identifier>`. Also, consider the following `contains` statement that always returns `true` because the null set is a subset of every set:

```
contains(any_attribute, {})
```

Consider the following filter:

```
filter eg_filter;
  //# returns true when my_set contains both S0 and S1
  //# {S0,S1,S2} != {S0,S1} returns true
  remove feature where contains (my_set, {S0,S1});
  keep feature where my_set == {S0, S1};
  remove feature where contains (my_set,{S0}) &&(contains(my_set,{S1}));
endfilter
```

In this example, the filter removes all features that contain `S0`, `S1`, and some other set member. It does not remove any feature that has `my_set` defined exactly as `{S0, S1}`.

Note:

The `contains(my_set, {S0, S1})` criteria is exactly the same as the `"contains(my_set, {S0}) && contains(my_set, {S1})"` criteria.

This section consists of the following subsections:

- [HVP](#)
- [Verdi GUI](#)

HVP

Consider the following HVP file, `spec.hvp`, which shows the definition of the `set` type attribute, `my_set` and six features (`F_1`, `F_2`, `F_3`, `F_4`, `F_5`, `F_sub1`, and `F_sub2`) that contains different values of the attribute:

```
plan New_Plan;
  attribute set{S0,S1,S2,S_other} my_set = {S0,S2};
  feature F_1;
    my_set = {S0,S1};
  endfeature
  feature F_2;
```

```

    my_set = {S0,S2};
endfeature
feature F_3;
    my_set = {S1,S2};
endfeature
feature F_4;
    my_set = {};
endfeature
feature F_5;
    my_set = {S0,S1,S2,S_other};
endfeature
feature F_sub1;
    my_set = {S0,S2};
endfeature
feature F_sub2;
    my_set = {S0,S2};
endfeature
endplan

```

The New_Plan HVP plan is represented in a spreadsheet XML plan as follows:

	A	B	C
1	hvp plan New_Plan	feature	\$my_set
2		plan	
3		F_1	{S0, S1}
4		F_2	{S0, S2}
5		F_3	{S1, S2}
6		F_4	{}
7		F_5	{S0, S1, S2, S_other}
8		F_sub1	{S0, S2}
9		F_sub2	{S0, S2}

The attribute definitions of my_set in the spreadsheet plan appears as follows:

	A	B	C	D	E
1	hvp attribute New_Plan	type	propagate	name	default
2		set{S0, S1, S2, S_other}	yes	my_set	{S0, S2}
3					

In a Word XML plan, you can add the attribute definitions of my_set as follows:

HVP Attribute my_set	
type	set{S0, S1, S2, S_other}
propagate	yes
name	my_set
default	{S0, S2}

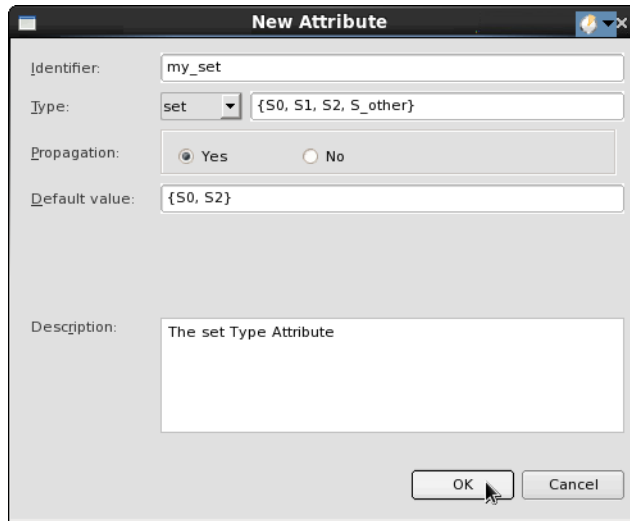
Verdi GUI

- [Adding a Set Type Attribute by Using the Verdi GUI](#)
- [Editing a Set Type Attribute by Using the Verdi GUI](#)

Adding a Set Type Attribute by Using the Verdi GUI

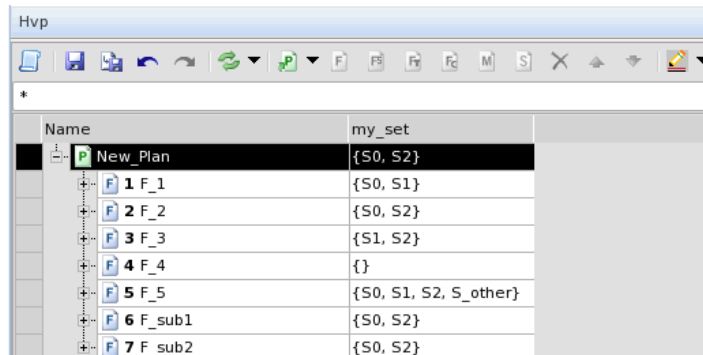
To create a `set` type attribute in Verdi, perform the following steps:

1. Click the **Create an attribute**  icon to display the **New Attribute** dialog box.
2. In the **New Attribute** dialog box, specify **Identifier** as the name of the attribute (say, `my_set`). To make an attribute of type `set`, select **Type** as **set** and specify its values as follows:



3. To add the values of this attribute in the sub-plans along with the main plan, mark **Propagation** to **Yes**.
4. To specify the default value of this attribute, use the **Default value** field. If you do not specify the default value of this attribute, the default value of this attribute is considered as an empty set, that is `{ }`.
5. Add the description of the attribute in the **Description** field.

6. Click **OK** to add the `my_set` attribute in the *Attributes* tab of the **HvpDetail** window. You can see its corresponding plan in the **Hvp** window in the Verdi GUI, which is as follows:

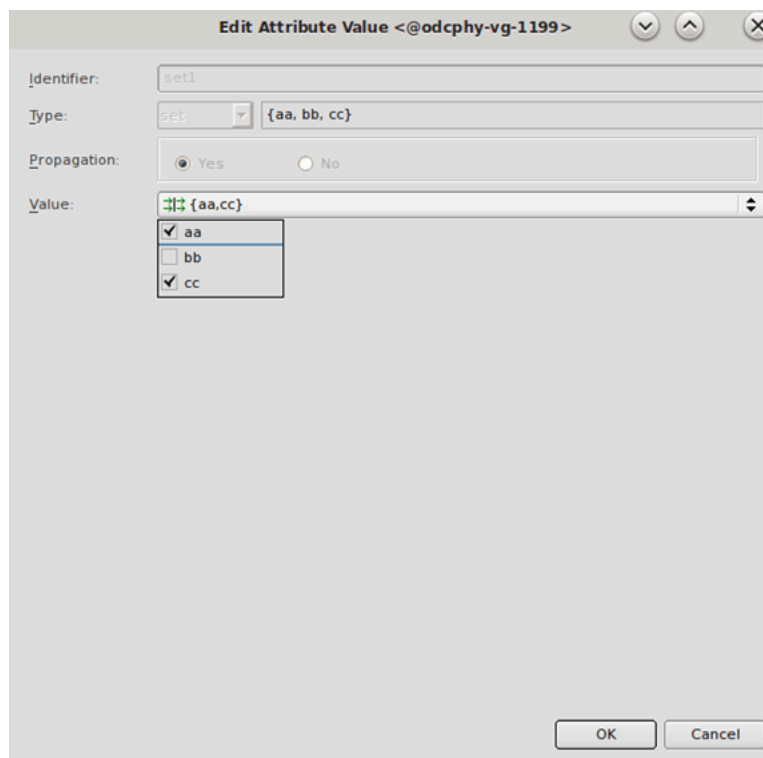


The screenshot shows the 'Hvp' window with a table of plans. The table has two columns: 'Name' and 'my_set'. The plans are listed as follows:

Name	my_set
New_Plan	{S0, S2}
1 F_1	{S0, S1}
2 F_2	{S0, S2}
3 F_3	{S1, S2}
4 F_4	{}
5 F_5	{S0, S1, S2, S_other}
6 F_sub1	{S0, S2}
7 F_sub2	{S0, S2}

Editing a Set Type Attribute by Using the Verdi GUI

The **Edit Attribute Value** dialog box allows users to edit the values of set attributes. You can edit values by selecting them from the drop-down list of valid set values as follows:



The **Edit Attribute Value** dialog can be displayed by double-clicking the value of the set attribute cell. You can also edit an attribute value by doing any of the following:

- Clicking on the **Edit Attribute** button  in the **HvpDetail** pane with a single item selected in the **Hvp** pane and a specified attribute selected in the **HvpDetail** pane
- Double-clicking the specified value cell in the **HvpDetail** pane.

When multiple items are selected in the **Hvp** pane, the value change of the **Edit Attribute Value** dialog (triggered by double-clicking the specified attribute value) takes effect on all values of the selected items.

Filters on a Plan

You can apply filters on the features of the plan using the following methods:

- [Load Modifiers](#)
- [Quick Filter](#)

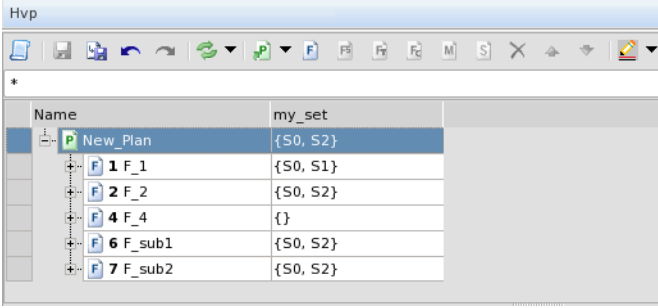
Load Modifiers

To apply filters on the plan using the `.hvpmod` file, click the **Load modifiers**  icon in the **Hvp** window. The icon enables you to load the modifiers in the Verdi GUI.

To load a specific modifier, select its file name from the **Hvp** window. Loading the modifier implements the criteria specified in the `.hvpmod` file. For example, consider the `my_filter.hvpmod` file that contains the following criteria:

```
filter my_filter;  
  remove feature where contains (my_set, {S1, S2});  
endfilter
```

When the `my_filter.hvpmod` file is loaded, it implements the above criteria on the `New_Plan` plan and removes the `F_3` feature and the `F_5` feature as they match the criteria (see the figure below):

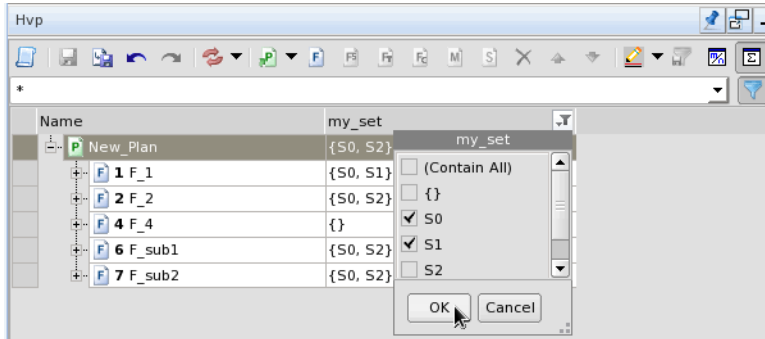


Name	my_set
New_Plan	{S0, S2}
1 F_1	{S0, S1}
2 F_2	{S0, S2}
4 F_4	{}
6 F_sub1	{S0, S2}
7 F_sub2	{S0, S2}

Quick Filter

You can also apply filters using the **Quick Filter**  icon in the **Hvp** pane. The icon uses the `contains` operator for extracting the filtered results instead of the `==` operator. To

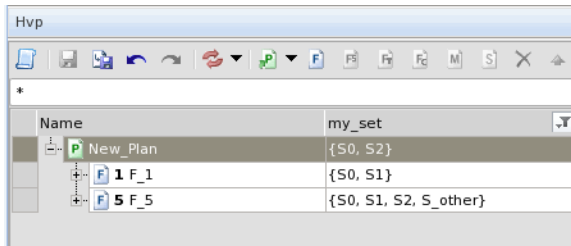
select the criteria for quick filtering, click  adjacent to the `set` type variable column. For example, click  adjacent to the `my_set` variable to display the list of criteria that can be used for filtering (see the figure below).



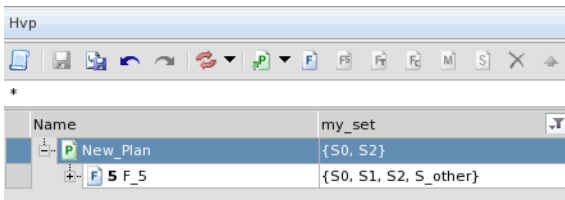
where, the **Contain All** option selects all elements of the defined set. This option selects all other check boxes except {}.

A set can have any number of elements starting from none to many. The number of elements in a set is called the cardinality. A set with no element is known as the empty set (or the null set) and is represented by {}. The cardinality of the null set is zero. A null set is always a member of every set including the null set itself. Therefore, evaluating whether {} is contained within any set should always resolve to "true". The {} option in the **Hvp** pane selects all sets as an empty set is contained in all sets. This option deselects all other check boxes.

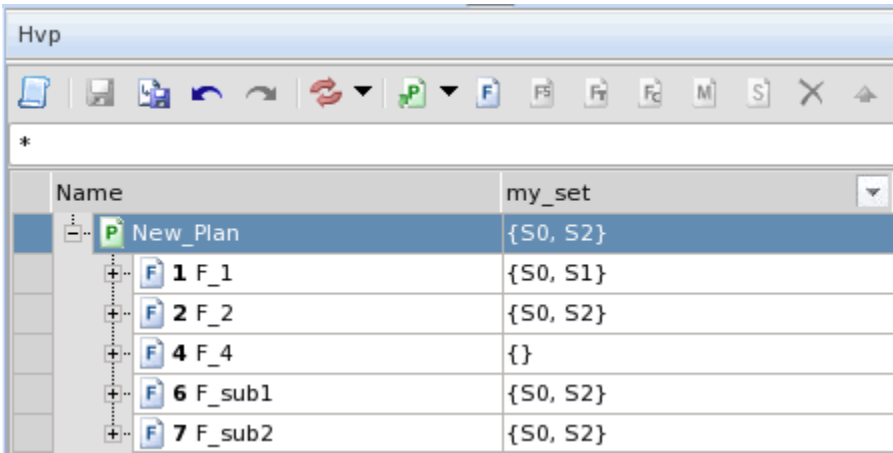
For example, if you select `S0` and `S1` as criteria for filtering, Verdi returns the features that contain {`S0`, `S1`} instead of {`S0`} or {`S1`} (see the figure below).



In the example, if you select **Contain All**, Verdi returns the features that contain {`S0`, `S1`, `S2`, `S_other`} (see the figure below).



In the example, if you select {}, Verdi returns all features (see the figure below).



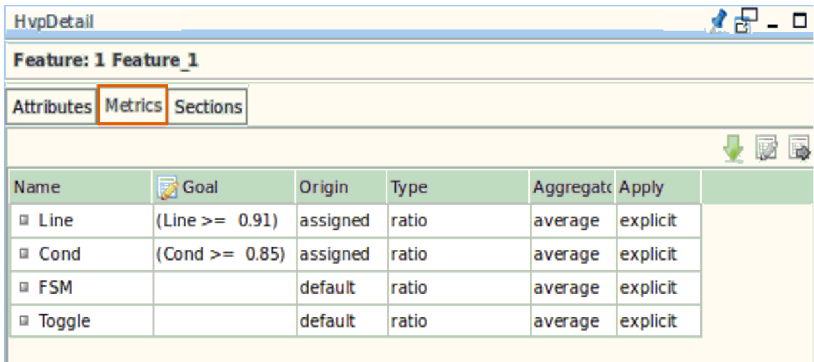
The screenshot shows the 'Hvp' pane with a table of metrics. The table has two columns: 'Name' and 'my_set'. The 'Name' column contains a tree view with a root 'New_Plan' and several sub-items: '1 F_1', '2 F_2', '4 F_4', '6 F_sub1', and '7 F_sub2'. The 'my_set' column contains the corresponding metric sets for each item.

Name	my_set
New_Plan	{S0, S2}
1 F_1	{S0, S1}
2 F_2	{S0, S2}
4 F_4	{}
6 F_sub1	{S0, S2}
7 F_sub2	{S0, S2}

Creating Metrics

After a plan is created or loaded in the *Hvp* pane, the default metrics are shown in the *Metrics* tab of the *HvpDetail* pane with the  icon. The tool only displays the metrics in the pane selected for the measure in the feature view as shown in [Figure Default Metrics](#). You can add your own metrics and recalculate the coverage for the plan.

Figure 101 Default Metrics

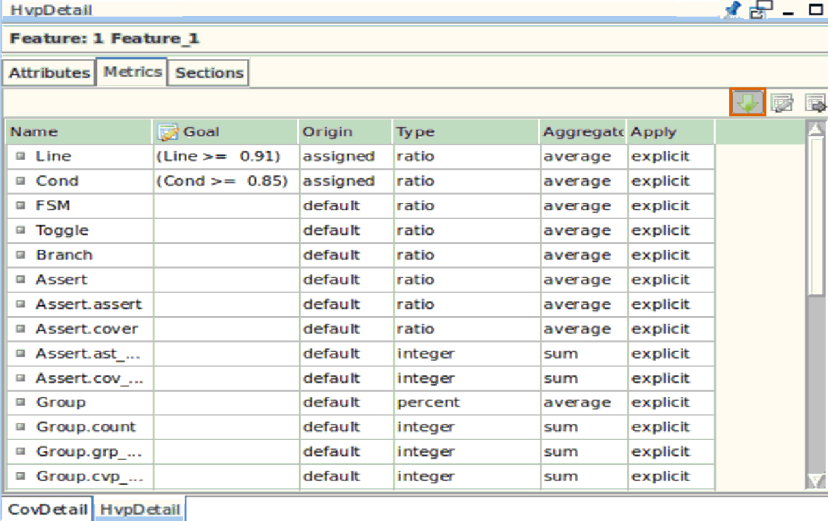


The screenshot shows the 'HvpDetail' pane with the 'Metrics' tab selected. The tab has a sub-header 'Feature: 1 Feature_1' and a table of metrics. The table has columns: 'Name', 'Goal', 'Origin', 'Type', 'Aggregate', and 'Apply'. The 'Name' column has checkboxes next to each item. The 'Goal' column contains the goal expression for each item. The 'Origin' column contains the origin of the metric. The 'Type' column contains the type of the metric. The 'Aggregate' column contains the aggregate function. The 'Apply' column contains the apply method.

Name	Goal	Origin	Type	Aggregate	Apply
☒ Line	(Line >= 0.91)	assigned	ratio	average	explicit
☒ Cond	(Cond >= 0.85)	assigned	ratio	average	explicit
☒ FSM		default	ratio	average	explicit
☒ Toggle		default	ratio	average	explicit

You can toggle the button to display or hide the metrics as shown in [Figure Toggling to Display All Metrics](#).

Figure 102 Toggling to Display All Metrics



Name	Goal	Origin	Type	Aggregate	Apply
Line	(Line >= 0.91)	assigned	ratio	average	explicit
Cond	(Cond >= 0.85)	assigned	ratio	average	explicit
FSM		default	ratio	average	explicit
Toggle		default	ratio	average	explicit
Branch		default	ratio	average	explicit
Assert		default	ratio	average	explicit
Assert.assert		default	ratio	average	explicit
Assert.cover		default	ratio	average	explicit
Assert.ast_...		default	integer	sum	explicit
Assert.cov_...		default	integer	sum	explicit
Group		default	percent	average	explicit
Group.count		default	integer	sum	explicit
Group.grp_...		default	integer	sum	explicit
Group.cvp_...		default	integer	sum	explicit

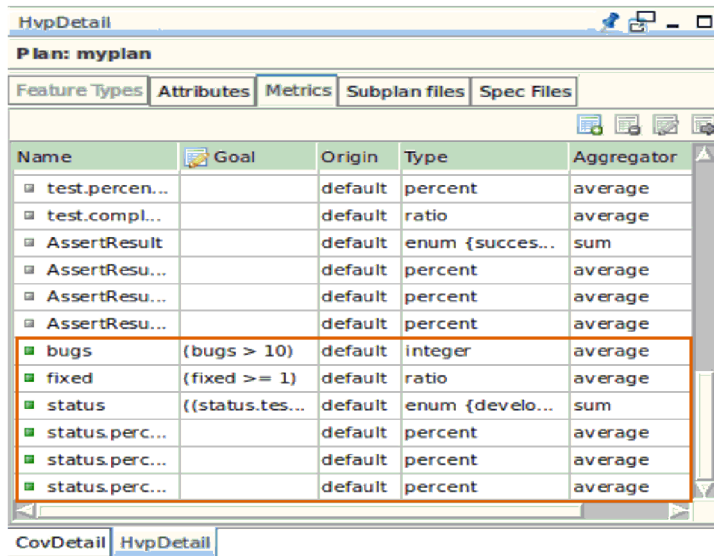
If your data which contain the metrics information is loaded by the **Plan -> Manage User Data** command, the metrics in the loaded data are also shown in the *Metrics* tab with the  green icon. For example, Figure [Metrics Defined in the User Data](#) shows the metrics that is defined in the loaded user data as follows:

```
HVP metric = bugs
/cfg/data/instance_1 = 15
/cfg/data/instance_2 = 23
/cfg/data/Function_coverage = 0

HVP metric = fixed
/cfg/data/instance_1 = 11/12
/cfg/data/instance_2 = 23/23
/cfg/data/Function_coverage = 0/0

HVP metric = status
/cfg/data/instance_1 = testing
/cfg/data/instance_2 = done
/cfg/data/Function_coverage = developing
```

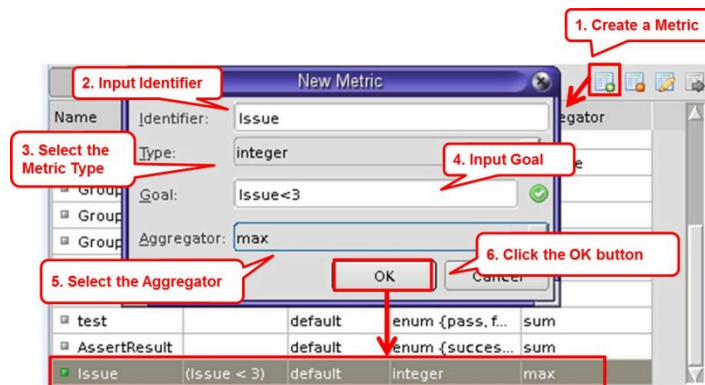
Figure 103 Metrics Defined in the User Data



Name	Goal	Origin	Type	Aggregator
test.percen...		default	percent	average
test.compl...		default	ratio	average
AssertResult		default	enum {succes...	sum
AssertResu...		default	percent	average
AssertResu...		default	percent	average
AssertResu...		default	percent	average
bugs	(bugs > 10)	default	integer	average
fixed	(fixed >= 1)	default	ratio	average
status	((status.tes...	default	enum {develo...	sum
status.perc...		default	percent	average
status.perc...		default	percent	average
status.perc...		default	percent	average

You can also create a metric for the selected plan by clicking the  icon in the *Metrics* pane of the *HvpDetail* pane. As shown in Figure [Add a New Metric](#), after setting done for the new metric and clicking the **OK** button, the new added metric is shown in the *Metrics* pane with the  green icon.

Figure 104 Add a New Metric



1. Create a Metric

2. Input Identifier

3. Select the Metric Type

4. Input Goal

5. Select the Aggregator

6. Click the OK button

Name	Identifier	Type	Goal	Aggregator
test		default	enum {pass, f...	sum
AssertResult		default	enum {succes...	sum
Issue	(Issue < 3)	default	integer	max

Adjusting Weight in a User-Defined Aggregate Metric

Verdi enables you to define weight for each base metric of a user-defined aggregate metric. Weight for each base metric is set as 1, by default. You can specify the weight using the following interfaces:

- [Using Verdi Planner](#)
- [Using HVP](#)

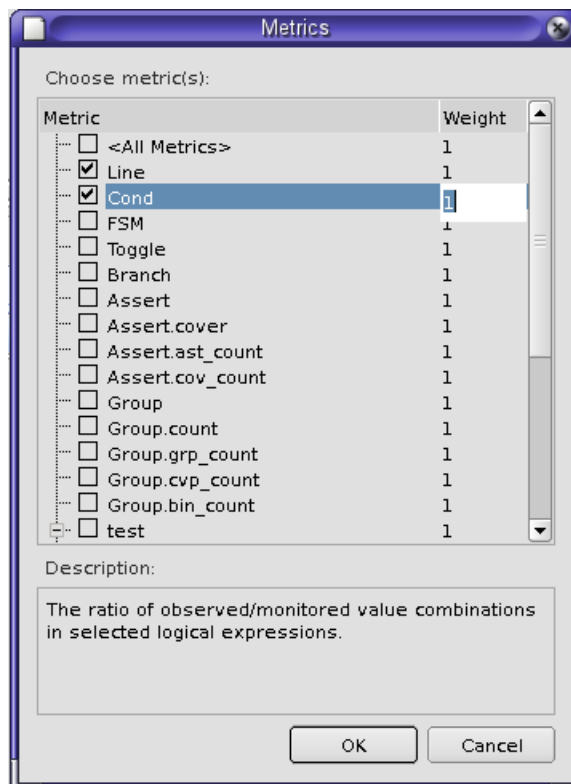
Using Verdi Planner

Verdi Planner enables you to specify weight in a user-defined aggregate metric while creating a new metric or while editing an existing user-defined metric.

To specify the weight while creating a new user-defined aggregate metric, perform the following steps:

1. Click the  icon in the *Metrics* pane of the *HvpDetail* pane to open the **New Metric** dialog box.
2. Specify the name of the metric in the **Identifier** text box, say *MyAggr_Line_Cond*, and select **Type** as **aggregate**. Then, click **Define** to open the **Metrics** dialog box to edit **Weight** of the selected metric:

Figure 105 Metrics Dialog Box



Note:

It is recommended that the value of Weight should be more than 0. Otherwise, Verdi Planner displays a warning message.

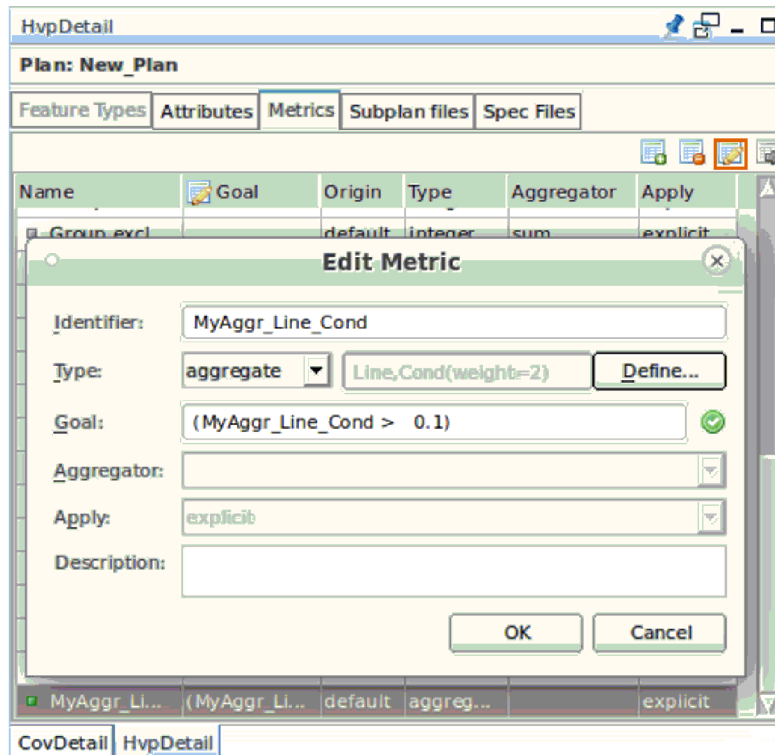
3. Click **OK**.

You can calculate the aggregate score using the **Recalculate**  icon in the *Hvp* pane.

To specify the weight while editing a user-defined aggregate metric, perform the following steps:

1. Select the user-defined metric, say *MyAggr_Line_Cond*, and click the **Edit Metric**  icon in the *Metrics* pane of the *HvpDetail* pane to display the **Edit Metric** dialog box:

Figure 106 Edit a Metric



2. Click **Define** to open the Metrics dialog box to edit **Weight** of the selected metric.
3. Click **OK**.

You can calculate score using the **Recalculate**  icon in the *Hvp* pane.

Using HVP

To specify weight in a user-defined aggregate metric, use the following HVP syntax:

```
metric metric-type identifier;  
  goal = goal-specification;  
  aggregator = aggregator-specification;  
endmetric
```


where,

metric-type: Specify metric type as **aggregate** and its syntax is as follows:

```
metric aggregate {metric_reference(weight=<value>)}
```

Consider the following example in which `MyAgg_Line_Cond` is a user-defined aggregate metric that calculates average of Line and Condition metrics:

```
metric aggregate {Line(weight=1.0), Cond(weight=2.0)} MyAgg_Line_Cond;  
  apply = explicit;  
  goal = (MyAgg_Line_Cond > 0.1);  
endmetric
```

You can calculate the aggregate score of the metric using the following:

$$\text{Score of the user-defined metric} = (\text{metric1} * \text{metric1_weight} + \text{metric2} * \text{metric2_weight}) / (\text{metric1_weight} + \text{metric2_weight})$$

Therefore, you can calculate the score of `MyAgg_Line_Cond` using the following:

$$(\text{Line} * 1 + \text{Cond} * 2) / (1 + 2)$$

Descriptions for Attribute and Metric Definitions

Verification Planner provides two types of attributes and metrics: *Built-in* and *User-defined*. Built-in attributes and metrics are implicitly computed during annotation for every verification plan and you cannot customize them. On the other hand, you can create your own user-defined attributes and metrics based on your requirements. For details on creating user-defined attributes and metrics in Verdi, see [Creating Attributes](#) and [Creating Metrics](#).

Verification Planner provides a new feature, which allows you to add a description to your user-defined attributes and metrics. The feature works as follows:

- You can add descriptions to all attribute types, that is, `integer`, `real`, `string`, `enum`, and `set`. Also, this option is applicable to all metric types, that is, `integer`, `enum`, `real`, and `percent`.
- You might specify more than one line as a description.
- You might use special and escape characters in the description.

This section consists of explaining how the description feature works with the following interfaces:

- [HVP](#)
- [Verdi GUI](#)

This section also consists of the following subsection:

- [Limitation](#)

HVP

To add a description to an attribute definition in a plan, use the following syntax:

```
<att_name>.description = <enquoted_description_string>
```

For example, to add a description to the definition of a user-defined attribute `Attr_Integer`, use the following in the plan:

```
attribute integer Attr_Integer = 1;  
Attr_Integer.description = "This is an integer-type attribute. \n This is  
a user-defined attribute.";
```

To add a description to a metric definition in your plan, use the following syntax:

```
metric metric-type identifier;  
  goal = goal-specification;  
  aggregator = aggregator-specification;  
  description = <enquoted_description_string>;  
endmetric
```

For example, to add a description to a user-defined metric `color`, use the following in the plan:

```
metric enum {red, yellow, green, purple, black} color;  
  goal = (color.green >=4);  
  aggregator = sum;  
  description = "This is a color metric. \nThis is a user-defined  
metric.";  
endmetric
```

To add a description to a user-defined attribute or metric in the spreadsheet XML plan, add a new column for the description in its attribute and metric definition sheet as follows:

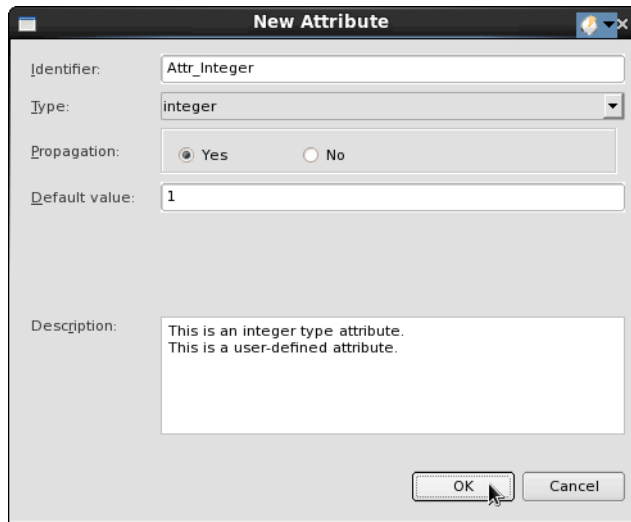
	A	B	C	D	E	F
1	hvp attribute top	type	propagate	name	default	description
2		integer	yes	Attr_Integer	1	This is an integer-type attribute. This is a user-defined attribute.

	A	B	C	D	E	F
1	hvp metric top	name	type	aggregator	goal	description
2		color	enum	sum	(color.green >=4)	This is a color metric. This is a user-defined metric.

Verdi GUI

To add a description to an attribute definition in a plan, perform the following steps:

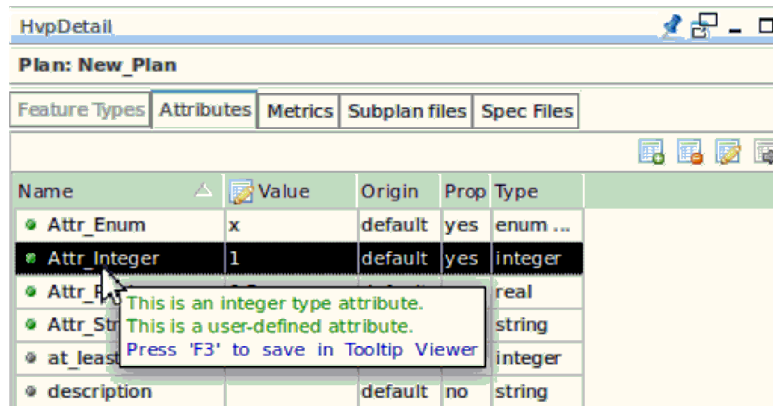
- 1. Click the **Create an attribute**  icon to display the **New Attribute** dialog box.
- 2. In the **New Attribute** dialog box, specify **Identifier** as the name of the attribute (say, `Attr_Integer`). To make an attribute of type `integer`, select **Type** as **integer**.
- 3. To add the values of this attribute in the sub-plans along with the main plan, mark **Propagation** to **Yes**.
- 4. To specify the default value of this attribute, use the **Default value** field.
- 5. To add a description to the attribute, use the **Description** field as follows:



The 'New Attribute' dialog box contains the following fields:

- Identifier:** Attr_Integer
- Type:** integer (selected from a dropdown)
- Propagation:** Yes (selected with a radio button), No
- Default value:** 1
- Description:** This is an integer type attribute.
This is a user-defined attribute.
- Buttons:** OK, Cancel

- Click **OK** to add the attribute in the *Attributes* tab of the *HvpDetail* pane. To see the description of this attribute, hover the pointer over the attribute in the *Attributes* tab. For example, see the figure below for the description of `Attr_Integer`:



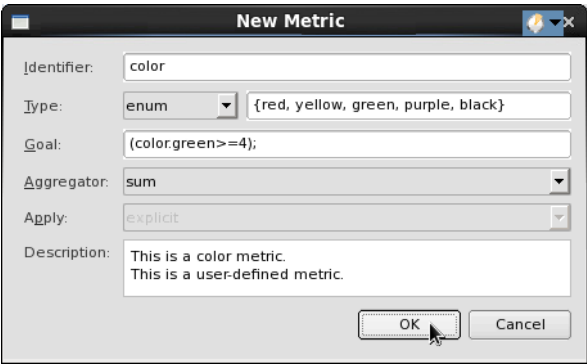
The *HvpDetail* pane shows the *Attributes* tab. A tooltip is displayed over the `Attr_Integer` row, showing its description.

Name	Value	Origin	Prop	Type
Attr_Enum	x	default	yes	enum ...
Attr_Integer	1	default	yes	integer
Attr_Real				real
Attr_String				string
at_least				integer
description		default	no	string

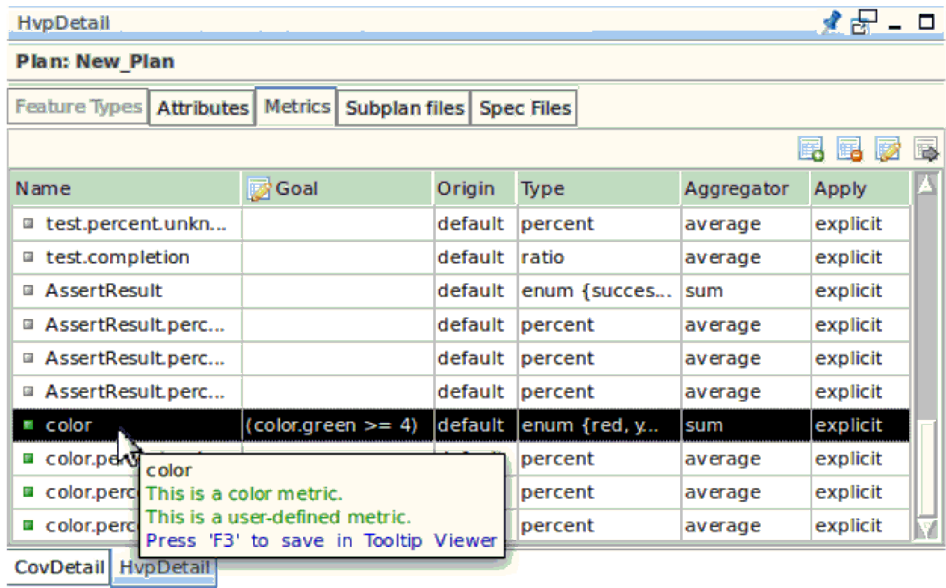
Tooltip for `Attr_Integer`:
This is an integer type attribute.
This is a user-defined attribute.
Press 'F3' to save in Tooltip Viewer

To add a description to a metric definition in your plan, perform the following steps:

- Click the **Create a metric** icon to display the **New Metric** dialog box.
- In the **New Metric** dialog box, specify **Identifier** as the name of the metric (say, `color`). To make a metric of type `enum`, select **Type** as `enum` and specify its values as `{red, yellow, green, purple, black}`.
- You can specify the coverage goal of the metric in the **Goal** field. For example, specify `(color.green >=4)`; in the **Goal** field.
- Select **Aggregator** as `sum` or `uniquesum` based on your requirement.
- To add a description to the metric, use the **Description** field (see the figure below):



6. Click **OK** to add the metric in the *Metrics* tab of the *HvpDetail* pane. To see the description of this metric, hover the pointer over the metric in the *Metrics* tab. For example, see the figure below for the description of `color`:



Limitation

The feature has the following limitation:

- If you add a description for an attribute in the main plan, the planner does not add the description to its sub-plans. Similarly, if you add the description in a sub-plan, the planner does not add the description to the attribute/metric in the main plan.

Using Modifiers

You can load modifiers to modify a plan hierarchy outside of *Hvp* planner definition files using plan modifiers: override and filter.

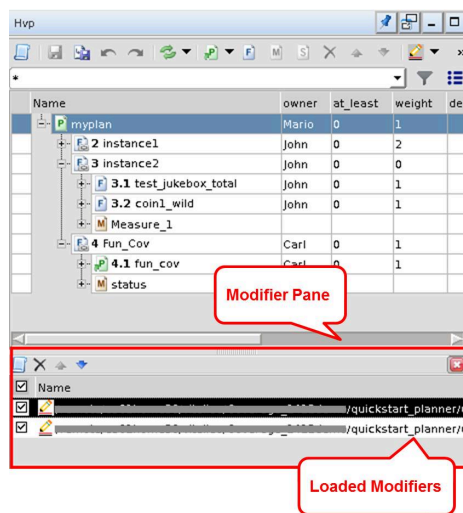
You can use one of the following methods to load modifiers:

- Click the  icon and select the modifier file(s) in the invoked *Load Modifier* form.
- Add the **-mod <modifier_file>** option on the command line when you invoke Verdi Coverage. For example,

```
%> verdi -cov -plan <plan>.hvp -mod <modifier_file>.hvpmod
```

After the modifier is loaded into *Hvp* planner, the planner is modified accordingly and the *Modifier* pane is shown with the modifier.

Figure 107 Load Modifier



You can recover the modification by disabling the  check-box in the beginning of the row. The first listed modifier (in the top) modifies the plan first. You can also use the  or  icons to change the order of the modifiers.

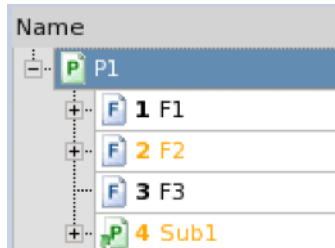
Hiding Complete Features Using the Toggle Filter

Verdi enables you to toggle between complete and incomplete features of a plan. A feature is an incomplete feature if it satisfies the following rules:

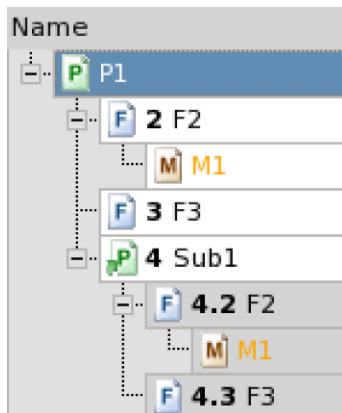
- A feature is a leaf feature if it does not have any sub-feature or sub-plan.
- A feature has no measures or a feature has measures but these measures have no source.

To toggle between complete and incomplete features, Verdi provides the **Hide Complete Feature**  icon in the **Hvp** pane. The icon hides all complete features of a plan to focus mainly on its incomplete features.

Consider the P_1 plan in the following figure that consists of a complete feature, F_1 , and incomplete features, F_2 , F_3 , and Sub_1 :



To hide complete features of P_1 , click the **Hide Complete Feature**  icon. The following figure shows only the incomplete features, such as F_2 , F_3 , and Sub_1 :



where, F_2 is an incomplete feature as its measure M_1 does not have any source.

F_3 is a leaf feature as it does not have any sub-feature or sub-plan.

Propagation of Values

Starting with this release, Verdi Planner highlights attribute values in a plan or sub-plan that are modified. If you edit an attribute field in Verdi Planner, its **Origin** field becomes *assigned*. In addition, Verdi provides you an option to revert to the default values of the fields. Once the values are reverted, their **Origin** field is assigned as *default* or *derived* based on the propagation property of the field.

This section consists of the following subsections:

- [Highlighting Values](#)
- [Reverting to Default Values](#)

Highlighting Values

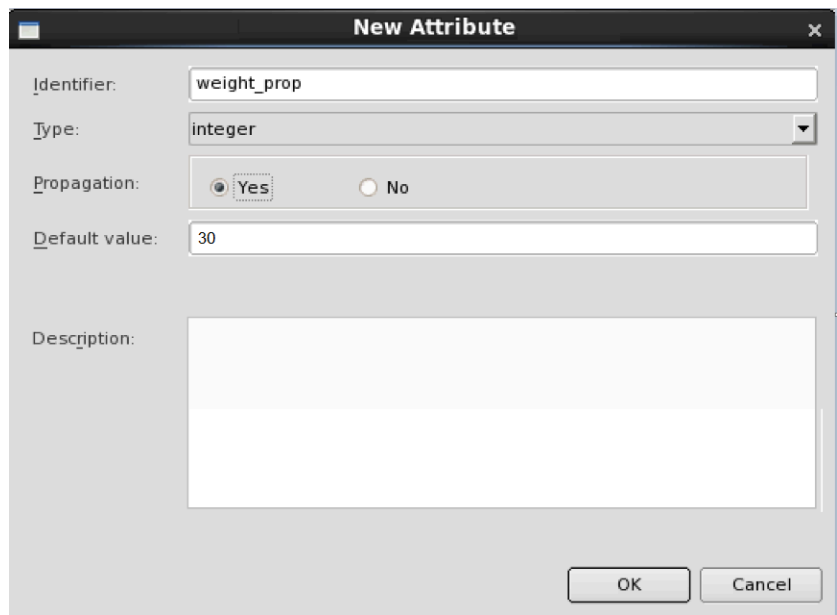
Consider the following figure that shows the `P1` plan, the `owner` column and the `weight` column:

Name	owner	weight
 P1	Ustr00	1
 1 F1	Ustr00	1
 1.1 F11	Ustr00	1
 1.2 F12	Ustr00	1
 1.2.1 F121	Ustr00	1
 2 F2	Ustr00	1
 2.1 F21	Ustr00	1
 2.2 F22	Ustr00	1
 3 F3	Ustr00	1

If you change the weight value of `F1` from `1` to `5`, Verdi Planner highlights this updated value in bold and italics format as shown in the following figure:

Name	owner	weight
 P1	Ustr00	1
 1 F1	Ustr00	<i>5</i>
 1.1 F11	Ustr00	1
 1.2 F12	Ustr00	1
 1.2.1 F121	Ustr00	1
 2 F2	Ustr00	1
 2.1 F21	Ustr00	1
 2.2 F22	Ustr00	1
 3 F3	Ustr00	1

You might propagate a value of an attribute from a feature to its sub-features or a plan to its sub-plan. For this, specify the **Propagation** field of the attribute to **Yes**. For example, the following figure shows an attribute, `weight_prop`, whose **Propagation** field is marked as **Yes**:



The 'New Attribute' dialog box is shown with the following fields:

- Identifier:
- Type:
- Propagation: ☒ Yes ☐ No
- Default value:
- Description:

Buttons: OK, Cancel

Once you click **Ok** in the **New Attribute** dialog box, the default value of this attribute is applied to all features and their sub-features as shown in the following figure:

Name	owner	weight	weight_prop
P1	Usr00	1	30
1 F1	Usr00	5	30
1.1 F11	Usr00	1	30
1.2 F12	Usr00	1	30
1.2.1 F121	Usr00	1	30
2 F2	Usr00	1	30
2.1 F21	Usr00	1	30
2.2 F22	Usr00	1	30
3 F3	Usr00	1	30

If you change the `weight_prop` value of `F1` to `100`, this value gets propagated to the sub-features of `F1`. The following figure shows that the `weight_prop` value of `F11`, `F12`, and `F121` is `100`:

Name	owner	weight	weight_prop
P1	Ustr00	1	30
1 F1	Ustr00	5	100
1.1 F11	Ustr00	1	100
1.2 F12	Ustr00	1	100
1.2.1 F121	Ustr00	1	100
2 F2	Ustr00	1	30
2.1 F21	Ustr00	1	30
2.2 F22	Ustr00	1	30
3 F3	Ustr00	1	30

Though the values of F11, F12, and F121 are propagated, you might change the values as per your requirement. For example, you might update the value of F11, F12, and F121 to 101, 102, and 103 respectively. Once the values are updated, Verdi Planner highlights these updated values as shown in the following figure:

Name	owner	weight	weight_prop
P1	Ustr00	1	30
1 F1	Ustr00	5	100
1.1 F11	Ustr00	1	101
1.2 F12	Ustr00	1	102
1.2.1 F121	Ustr00	1	103
2 F2	Ustr00	1	30
2.1 F21	Ustr00	1	30
2.2 F22	Ustr00	1	30
3 F3	Ustr00	1	30

Calculating the Count of Covergroup Bins

Verdi enables you to view the covered, uncovered, excluded, and total unweighted count of covergroup bins under a feature in an HVP plan. Verdi provides the following *Group.*_count* built-in submetrics in the *HVP* pane:

- **Group.covered_count:** Enable this submetric to view the number of covered bins for each feature in a plan.
- **Group.uncovered_count:** Enable this submetric to view the number of uncovered bins for each feature in a plan.
- **Group.excluded_count:** Enable this submetric to view the number of excluded bins for each feature in a plan.
- **Group.bin_count:** Enable this submetric to view the total number of covered and uncovered bins for each feature in a plan.

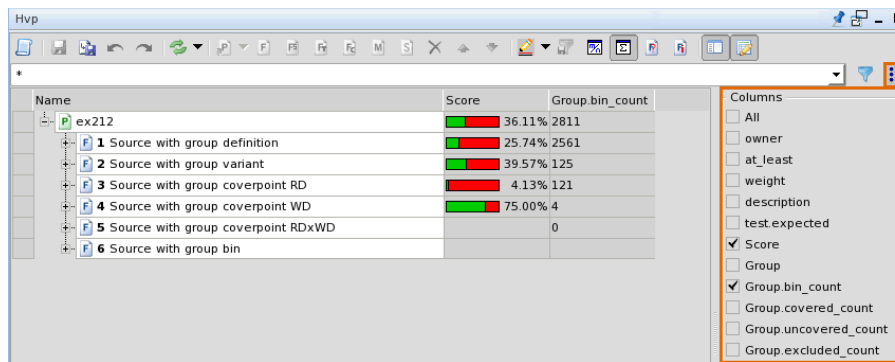
For details on *Group.*_count* submetrics, see the following sections:

- [Enabling Group.*_count Submetrics](#)
- [Generating the URG Report to View Group.*_count Submetrics](#)

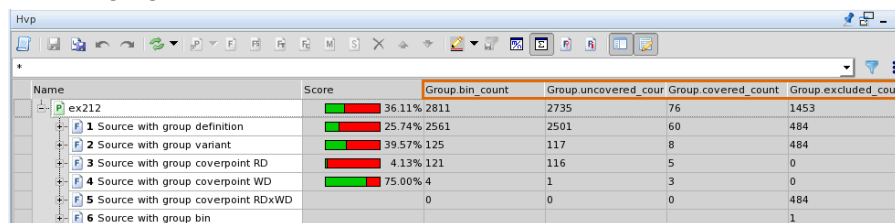
Enabling Group.*_count Submetrics

To enable the covered, uncovered, and excluded count of covergroup bins, perform the following steps:

1. Click the  *Configure Columns* icon in the *HVP* pane to display built-in and user-defined submetrics. The following figure shows the **Score** and **Group.bin_count** columns in the *HVP* pane as their check boxes are selected under **Columns**:



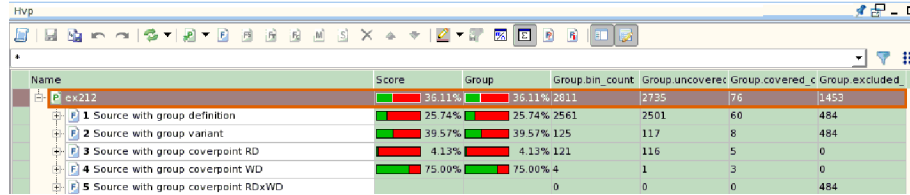
2. Select **Group.covered_count**, **Group.uncovered_count**, and **Group.excluded_count** to display the number of covered, uncovered, and excluded bins respectively for all features in a plan. For example, the *Group.*_count* columns are highlighted in the following figure:



Name	Score	Group.bin_count	Group.uncovered_count	Group.covered_count	Group.excluded_count
ex212	36.11% 2811	2735	76	1453	
1 Source with group definition	25.74% 2561	2501	60	484	
2 Source with group variant	39.57% 125	117	8	484	
3 Source with group coverpoint RD	4.13% 121	116	5	0	
4 Source with group coverpoint WD	75.00% 4	1	3	0	
5 Source with group coverpoint RDxWD	0	0	0	484	
6 Source with group bin				1	

In the figure, note that the *Score aggregation mode: self + children*  icon is clicked. This icon calculates the aggregate of each *Group.*_count* submetric of a plan. For example, Verdi calculates the aggregate of **Group.bin_count**, **Group.covered_count**,

Group.uncovered_count, and **Group.excluded_count** of the `ex212` plan when  is clicked and displays their aggregate in the top row as highlighted in the following figure:



Name	Score	Group	Group.bin_count	Group.uncovered	Group.covered_c	Group.excluded
ex212	36.11%	36.11% 2011	2735	76	1453	
1 Source with group definition	25.74%	25.74% 2561	2501	60	484	
2 Source with group variant	39.57%	39.57% 125	117	8	484	
3 Source with group coverpoint RD	4.13%	4.13% 121	116	5	0	
4 Source with group coverpoint WD	75.00%	75.00% 4	1	3	0	
5 Source with group coverpoint RDxWD			0	0	0	484

Defining Source Targets

For the features having groups, group instances, or bins as source targets, Verdi returns the covered, uncovered, excluded, and total count of matched groups, instances, or bins respectively in a plan. The syntax of a source target in a measure in an HVP plan is as follows:

```
measure Group, Group.bin_count, Group.covered_count,
  Group.uncovered_count, Group.excluded_count measure_1;
  source = "group/group instance/group bin/
    group instance bin: <group object name>";
endmeasure
```

Example of Source Targets

Consider the following covergroup definition, variant, instance, coverpoint, and cross coverpoint:

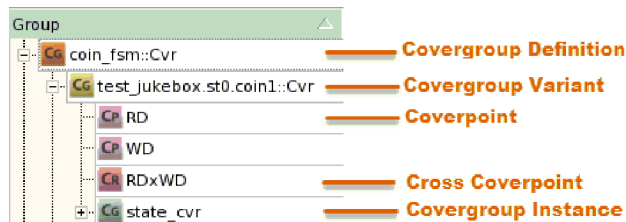


Table [Source Targets Examples](#) shows the examples of source targets:

Table 3 Source Targets Examples

Source Targets	Examples
Covergroup Definition	source = "group: coin_fsm::Cvr";
Covergroup Variant	source = "group: test_jukebox.st0.coin1::Cvr";
Coverpoint	source = "group: test_jukebox.st0.coin1::Cvr.RD";
Cross Coverpoint	source = "group: test_jukebox.st0.coin1::Cvr.RDxWD";

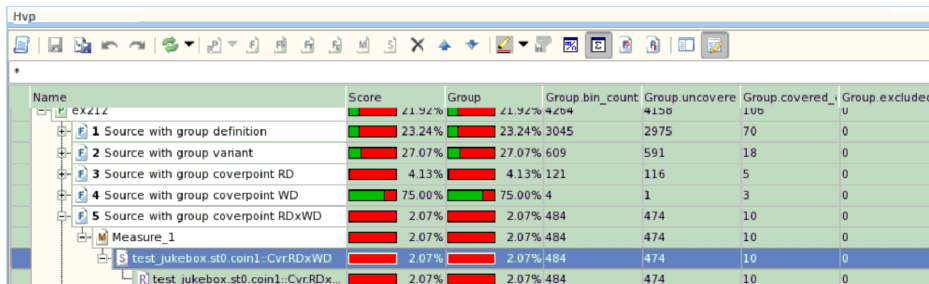
Table 3 Source Targets Examples (Continued)

Source Targets	Examples
Group Bin	source = "group bin: test_jukebox.st0.coin1::Cvr.RDxWD.HIGH-MED";
Group Instance Bin	source = "group instance bin: test_jukebox.st0.coin1.state_cvr.RDxWD.HIGH-MED";
Group Instance	source = "group instance: test_jukebox.st0.coin1.state_cvr";
Group Instance Coverpoints	source = "group instance: test_jukebox.st0.coin1.state_cvr.*";

Managing Exclusions

If a group, group instance, or bin is excluded from coverage computation, the **Group.excluded_count** submetric shows the count of excluded bins. Other **Group.*_count** submetrics show 0.

For example, consider the following figure that shows the count of **Group.bin_count**, **Group.uncovered_count**, and **Group.covered_count** submetrics of the cross coverpoint, `test_jukebox.st0.coin1::Cvr.RDxWD` as 484, 474, and 10 respectively:



In this figure, note that **Group.excluded_count** of this cross coverpoint is 0.

Consider the following figure in which `test_jukebox.st0.coin1::Cvr.RDxWD` is excluded:

Name	Score	Group	Group.bin_count	Group.uncovered	Group.covered_count	Group.excluded_count
ex212	36.11%	36.11%	2811	2735	76	1453
1 Source with group definition	25.74%	25.74%	2561	2501	60	484
2 Source with group variant	39.57%	39.57%	125	117	8	484
3 Source with group coverpoint RD	4.13%	4.13%	121	116	5	0
4 Source with group coverpoint WD	75.00%	75.00%	4	1	3	0
5 Source with group coverpoint RDXWD			0	0	0	484
Measure_1			0	0	0	484
test_jukebox.st0.coin1::Cvr.RDXWD			0	0	0	484

In this figure, **Group.excluded_count** shows the count as 484 and other **Group.*_count** submetrics show 0 because `test_jukebox.st0.coin1::Cvr.RDXWD` is excluded.

Generating the URG Report to View Group.*_count Submetrics

To generate a URG report in Verdi, use **Tools > Generate URG Report**. For generating a URG report for **Group.*_count** submetrics, Verdi uses the following URG command:

```
urg -plan <filename.hvp> -dir <simv.vdb> -elfile <filename.el> -format
text -show ratios
```

For example, if a source target is a covergroup definition, `coin_fsm::Cvr`, Verdi generates the following URG report:

```
Measure Measure_1:
Metrics: Group, Group.bin count, Group.covered_count, Group.uncovered_count, Group.excluded_count
SCORE GROUP Group.bin_count Group.covered_count Group.excluded_count Group.uncovered_count
2.34 2.34 60/2561 2561 60 484 2561

Sources:
group: coin_fsm::Cvr
SCORE GROUP Group.bin_count Group.covered_count Group.excluded_count Group.uncovered_count NAME
6.40 6.40 8/125 125 8 484 117 test_jukebox.st0.coin1::Cvr(x)
2.96 2.96 18/609 609 18 0 591 test_jukebox.st1.coin1::Cvr
1.81 1.81 11/609 609 11 0 598 test_jukebox.st2.coin1::Cvr
2.96 2.96 18/609 609 18 0 591 test_jukebox.st3.coin1::Cvr
0.82 0.82 5/609 609 5 0 604 test_jukebox.st4.coin1::Cvr
```

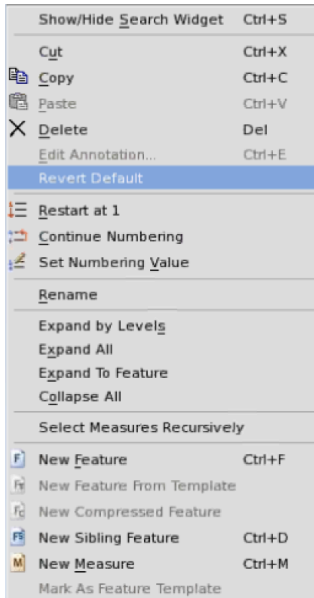
Note:

If you use **Verdi > Tools > Generate URG Report** to generate a URG report from Verdi before deleting VDB, the coverage database gets copied into the `/tmp` directory path.

You can see the `/tmp` directory path when you run the URG command. However, if the URG report is not generated, the database is not saved anywhere else. You might need to regenerate the database. The **Copy Database Fullname** option is used to save the database name only.

Reverting to Default Values

To revert the updated values to their default values, right-click the values and select the **Revert Default** option (see the figure below).



Consider the example in the [Highlighting Values](#) section. To revert the `weight_prop` value of F1 to its default value, select the **Revert Default** right-click option. The option reverts only the `weight_prop` value of F1 and not the `weight_prop` value of its sub-features, F11, F12, and F121 (see the figure below):

Name	owner	weight	weight_prop
P1	Usr00	1	30
1 F1	Usr00	5	30
1.1 F11	Usr00	1	100
1.2 F12	Usr00	1	100
1.2.1 F121	Usr00	1	100
2 F2	Usr00	1	30
2.1 F21	Usr00	1	30
2.2 F22	Usr00	1	30
3 F3	Usr00	1	30

Shipped Example for User Reference

You can refer to a coverage example and plan that is shipped for your reference. To do so, perform the following steps:

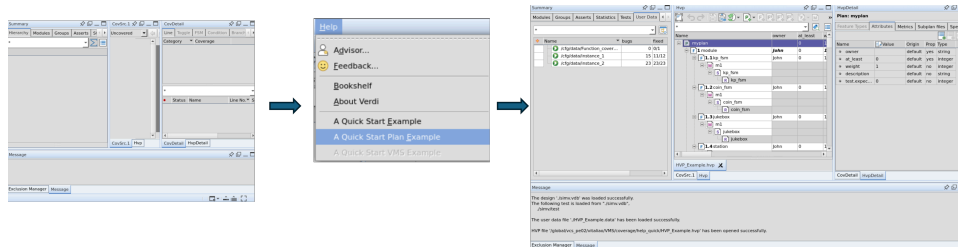
1. Set up your environment with Verdi and VCS settings.
2. Use the following command to open an empty coverage GUI:

```
%> verdi -cov
```

3. From the **Help** menu, select **A Quick Start Plan Example**.

Result

- The example shipped for coverage and plan is copied to your local directory.
- The simulation results, including coverage data, are generated in your local directory through an automatic compilation and simulation process.
- The `simv.vdb` and `.hvp` files are loaded automatically into the Verdi Coverage GUI.



- The coverage data is automatically annotated to open the `.hvp` files including User Data.

Name	bugs	fixed
/cfg/data/Function_cover...	0	0/1
/cfg/data/instance_1	15	11/1
/cfg/data/instance_2	23	23/2

test_jukebox.st2...					
test_jukebox.st3...					
test_jukebox.st4...					
Measure_1	23	100.0%	1		
/cfg/data/instance_2	23	100.0%	1		
/cfg/data/instance_2	23	100.0%	1		
4 Fun_Cov	100.0%	23.24%	0	0.00%	1

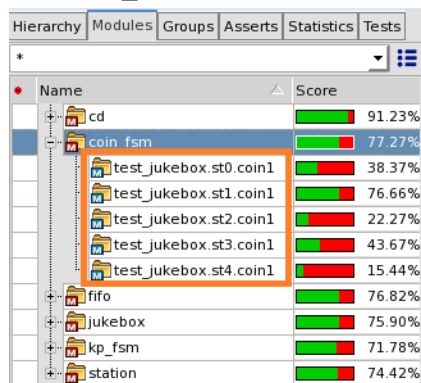
6

Using the Exclusion Manager

Verdi Coverage provides the Exclusion Manager feature to increase coverage by taking out the exclusions that are not needed for the design. When a design has a lot of code or combinations that never occur or there are parts of the design that designer does not care about, then the exclusion features are used to exclude the above coverage targets from the coverage score. Thus, you can focus on the coverage results from the design that can happen.

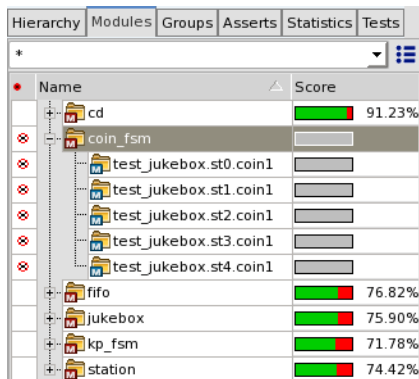
When you identify a coverage target that can be ignored, or is impossible by design, you can mark it for exclusion directly in the GUI and add an annotation explaining why it is excluded. You can then save those exclusions and annotations to an exclude file that can be reused in the next and future sessions. Excluded objects are removed from the coverage score calculation and appear disabled in reports.

You might exclude an object either at a module level or an instance level. An exclusion on a module excludes the corresponding object in all of its instances. For example, consider the `coin_fsm` module that has five instances as shown in the following figure:



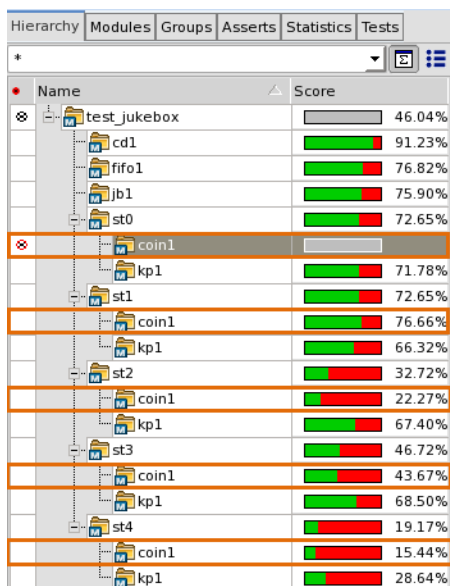
Name	Score
cd	91.23%
coin_fsm	77.27%
test_jukebox.st0.coin1	38.37%
test_jukebox.st1.coin1	76.66%
test_jukebox.st2.coin1	22.27%
test_jukebox.st3.coin1	43.67%
test_jukebox.st4.coin1	15.44%
fifo	76.82%
jukebox	75.90%
kp_fsm	71.78%
station	74.42%

If `coin_fsm` is excluded, all the instances under it are automatically excluded as shown in the following figure:



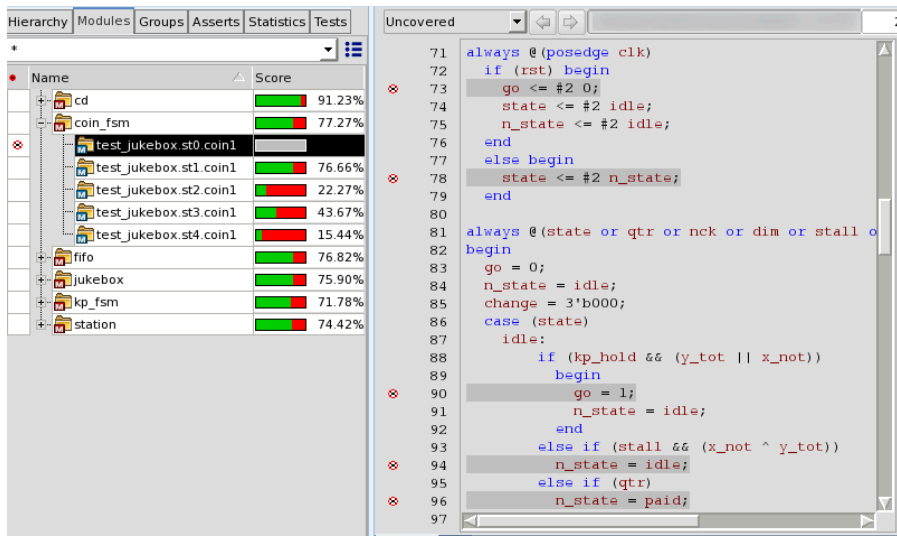
Name	Score
cd	91.23%
coin_fsm	
test_jukebox.st0.coin1	
test_jukebox.st1.coin1	
test_jukebox.st2.coin1	
test_jukebox.st3.coin1	
test_jukebox.st4.coin1	
fifo	76.82%
jukebox	75.90%
kp_fsm	71.78%
station	74.42%

If a new instance is created for the excluded module, the new instance is automatically excluded. On the other hand, if you exclude an object at the instance level, then its other instances are not excluded automatically. For example, if you exclude `test_jukebox.st0.coin1` at the instance level, then its other instances in `test_jukebox.st1`, `test_jukebox.st2`, `test_jukebox.st3`, and `test_jukebox.st4` are not excluded as shown in the figure below:

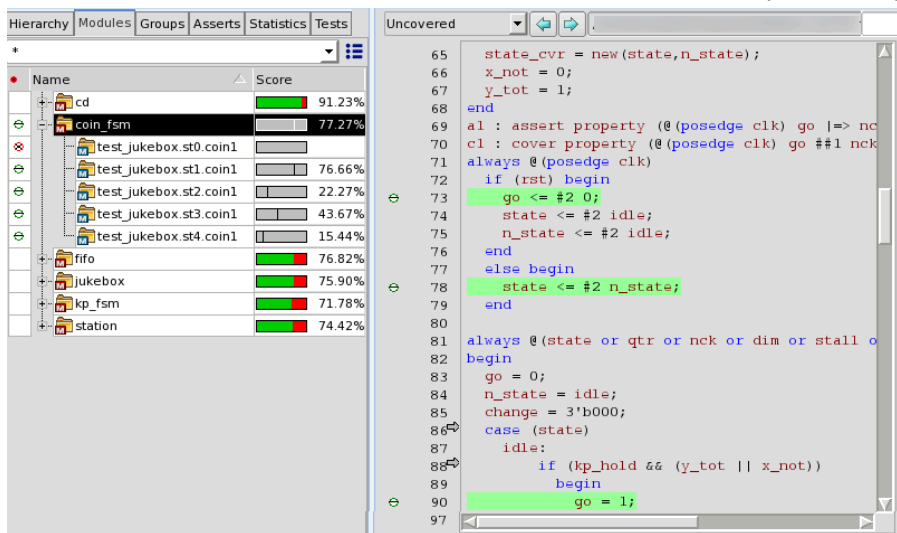


Name	Score
test_jukebox	46.04%
cd1	91.23%
fifo1	76.82%
jb1	75.90%
st0	72.65%
coin1	
kp1	71.78%
st1	72.65%
coin1	76.66%
kp1	66.32%
st2	32.72%
coin1	22.27%
kp1	67.40%
st3	46.72%
coin1	43.67%
kp1	68.50%
st4	19.17%
coin1	15.44%
kp1	28.64%

Instance exclusions do not affect the coverage score of a module, even if all of its instances are excluded. For example, if you exclude `test_jukebox.st0.coin1`, Line 73, Line 78, Line 90, and so on are excluded as shown in the following figure:



Note that these lines are not excluded at the module level (see the figure below).



If you intend to exclude an object for all instances of a module, ensure that you are in the module exclusion mode and not the instance exclusion mode. These two modes look similar. The main difference is that for module exclusions, a module scope is selected in the *Modules* tab of the *Summary* pane. For instance-based exclusions, you select an instance scope in the *Hierarchy* tab of the *Summary* pane. As it is not possible to convert instance exclusions to module exclusions, if you intend to create module exclusions confirm that you are in the module exclusion mode before excluding objects.

The *Exclusion Manager* pane of Verdi Coverage combines all the exclusions from the different Verdi Coverage panes together.

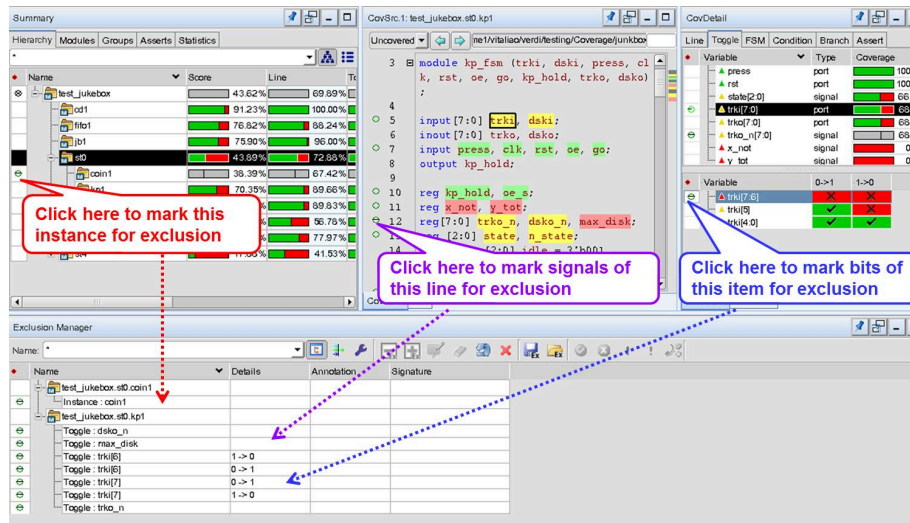
This chapter discusses the following topics:

- [Using the Exclusion Manager Pane](#)
- [Using the Toolbar to Manage Exclusions](#)
- [Reviewing Exclusions](#)
- [Filter Support for Annotations and Elfile](#)
- [Displaying Multiple Annotated Exclusions](#)
- [Exclusions With Checksum Mismatch](#)
- [Adding Annotations to an Excluded Object](#)
- [Handling Duplicate Annotations of an Excluded Object](#)
- [Saving Exclusions to a File](#)
- [Seed and Propagated Exclusions](#)
- [Exclusions on a Topology Folder](#)
- [Exclusion Propagation to Connections](#)
- [Exclusion Propagation From Line Coverage](#)

Using the Exclusion Manager Pane

The *Exclusion Manager* pane is displayed by default. As shown in Figure [Add Exclusion Through Exclusion Column](#), you can mark an instance for exclusion by clicking an item in the *Hierarchy* page of the *Summary* pane.

Figure 108 Add Exclusion Through Exclusion Column



The following table shows the icons marked at the beginning of the row when the exclusion is set or executed.

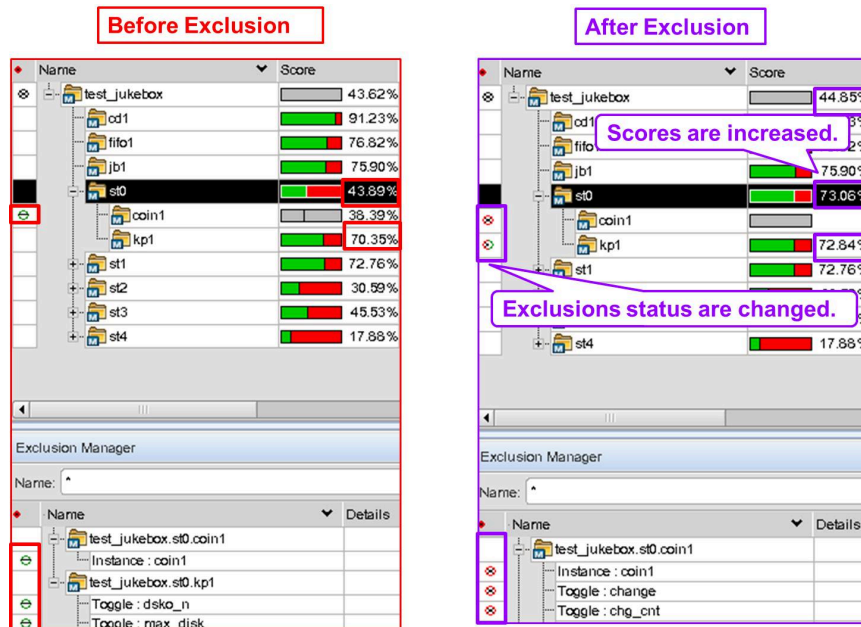
Icon	Status
	The item can be marked to exclude.
	The item is marked to be excluded.
	The previously excluded item is marked to be included.
	The item is excluded.
	The item is partially excluded.
	The item is marked to be partial excluded in the <i>Coverage Source Code</i> pane.
	The item is excluded as propagated from other exclusions.

After being marked, the item is added into the *Exclusion Manager* pane. You can also mark some signals of a line in the *CovSrc* pane per different coverage types or some bits of a signal in the *Toggle* page of the pane on demand.

When the exclusions are set and automatically located in the *Exclusion Manager* pane, use the **Recalculate** command in the right-click menu or the icon to apply the exclusions and update the coverage scores.

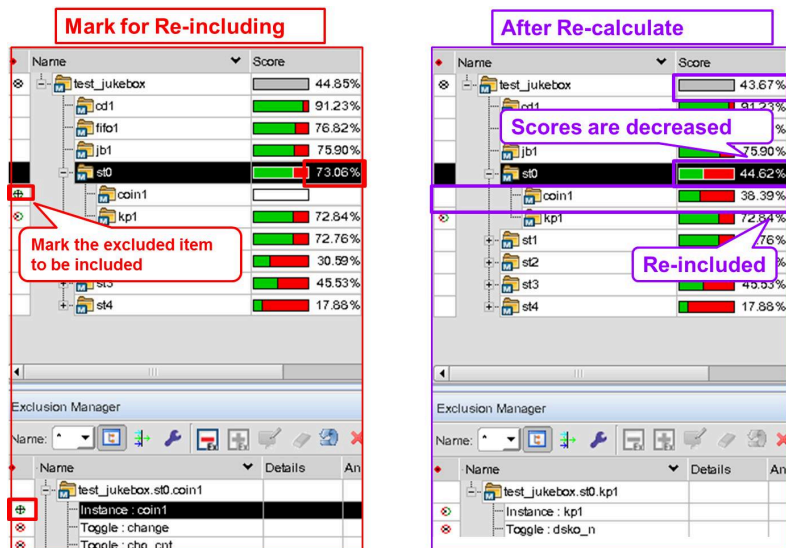
After the coverage scores are updated, the icons in the **Exclusion** column change to show the exclusions that are excluded from the coverage score. As shown in [Figure Recalculate With Exclusions](#), the  icon in the **Exclusion** column and in the **coin1** row shows the **coin1** instance is excluded at this calculation.

Figure 109 Recalculate With Exclusions



You can also include the previously excluded items again by clicking the  icon to change it to the  icon in the exclusions summary of the *Exclusion Manager* pane or in the **Exclusion** column of the *Summary*, *CovSrc*, or panes. For example, as shown in [Figure Recalculate for Re-included Items](#), you can see the previously excluded item that was marked to be included is now reincluded in the coverage score. You can also exclude/include a line in the expanded macro.

Figure 110 Recalculate for Re-included Items



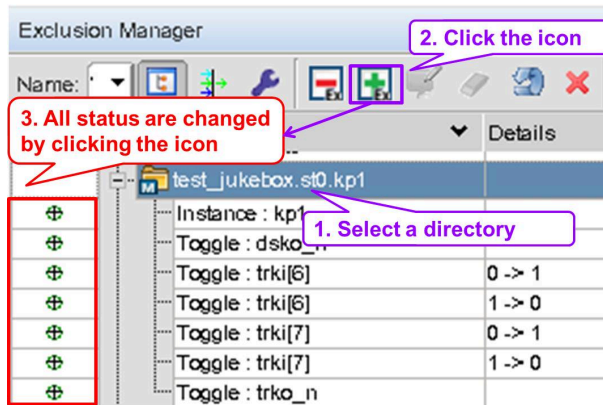
Using the Toolbar to Manage Exclusions

You can select items in the *Exclusion Manager* pane and use the toolbar icons of the *Exclusion Manager* pane to change the exclusion status of the selected items, and to save and load exclusions. The rows in the *Summary* pane can also be selected.

Icon	Description
	To exclude marked items
	To re-include the selected items
	To delete all exclusions in the <i>Exclusion Manager</i> window.
	To save exclusions
	To load exclusions

If you select a directory in the *Exclusion Manager* pane, you can use the icon or icon to exclude or re-include all items belonging to this directory. As shown in the Figure [Apply to All Items by Using Toolbar Icon](#), the exclusion status of all items that belong to the selected directory is changed.

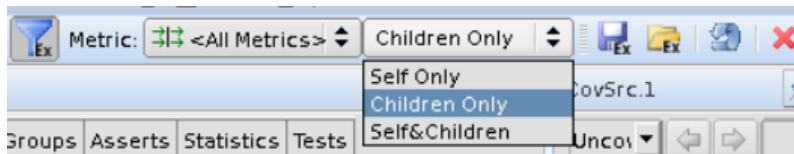
Figure 111 Apply to All Items by Using Toolbar Icon



Using Advanced Exclusion Options

You can enable the advanced exclusion options by toggling the  button in the toolbar as shown in Figure [Advanced Exclusion Options](#):

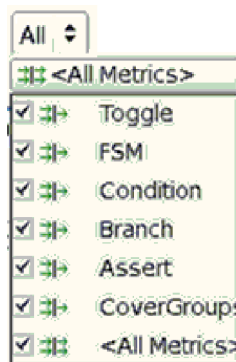
Figure 112 Advanced Exclusion Options



Options for Metrics

Figure [Metric Options](#) lists the metrics available; you can choose one of the metrics or can choose the last option <All Metrics>.

Figure 113 Metric Options



The metric combos are enabled /disabled using the rules listed in the following table:

Item Selected	Enable/Disable
Detail object	Disabled
Module/Instance/Tree item	Covergroup is disabled; all other loaded metrics are enabled.
Covergroup	Disabled

Options for Targets

Figure [Target Options](#) lists the targets you can choose from:

Figure 114 Target Options



It provides the following options:

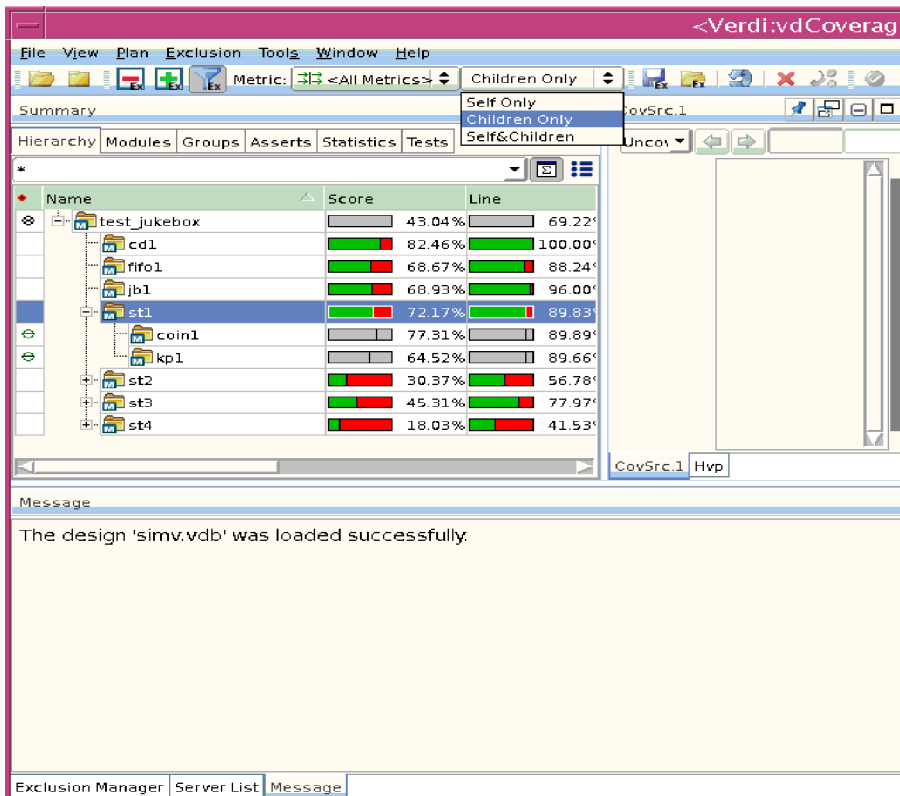
- **Self Only:** The exclusion action is applied to the selected scope itself.
- **Children Only:** The exclusion action is applied to the selected scope's children only.
- **Self & Children:** It is complete tree exclusion. The option is enabled only when an instance is selected, otherwise it is disabled.

For example, if you want to exclude the child instances of an instance, use the following steps:

1. Turn on **Advance Exclusion Options** from the toolbar.
2. Select an instance on the **Hierarchy** summary.
3. Select **Children Only** from the toolbar.
4. Click the **Exclude Marked Items** button from the toolbar.

Figure [Excluding Child Instances](#) shows how you can exclude child instances.

Figure 115 Excluding Child Instances



Following are the three ways to save exclusions:

- **Save all exclusions:** Saves all exclusions in the *Exclusion Manager* pane into a specified file. To save all exclusions in an *.el exclusion file, click **File-> Save all exclusions**.
- **Save exclusions not saved:** Saves exclusions that are not yet saved into a specified file including the exclusions that are already existing in the file.

To save new exclusions for entire design, click the **Save Exclusions Not Saved** icon from the toolbar. Alternatively, click **File> Save Exclusions Not Saved**.

- **Save selected exclusions:** Saves exclusions for selected instances, trees, modules, and groups into a specified file including the exclusions that are already existing in the file. To save exclusions for the selected instances, trees, modules, and groups, click **Exclusion->Save Exclusions for Selected Objects/Trees**.

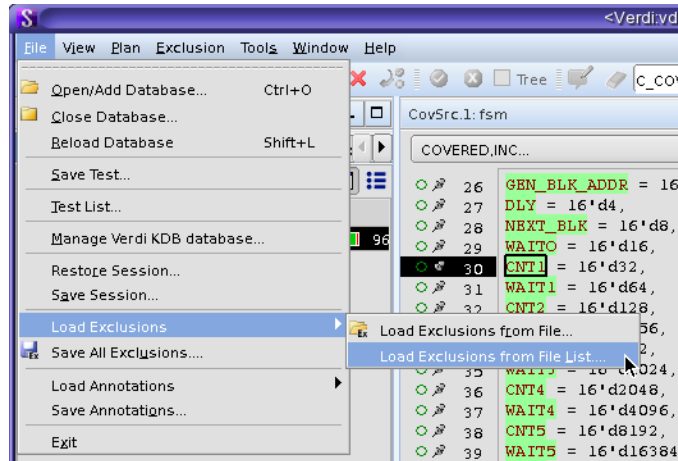
To save the exclusion file with the selected metrics, enable the **Save propagated exclusions from module of covergroup variant Metrics** option.

Loading Exclusion Files

You can click the  icon to load an *.el exclusion file to the *Exclusion Manager* pane.

You can load exclusion files (.el files) using the **File** menu commands **Load Exclusions from File** or **Load Exclusions from File List** as shown in the following figure:

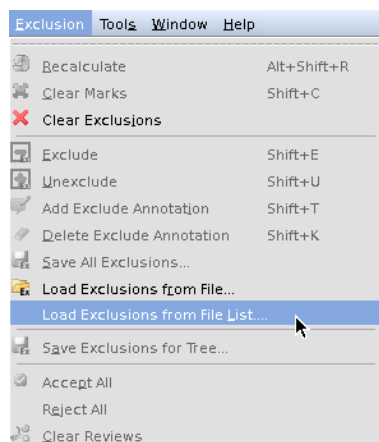
Figure 116 Loading Exclusion Files From the File Menu



You can specify the .el file that needs to be loaded using the **Load Exclusions from File** option. You can also load a file which contains a list of .el files using the **Load Exclusions from File List** option.

These options are also available in the **Exclusion** menu as shown in the following figure:

Figure 117 Loading Exclusion Files From the Exclusion Menu



The command corresponding to the **Load Exclusions from File** option is `vdCov -elfile <ElFile>`. The command corresponding to the **Load Exclusions from File List** option is `vdCov -elfilelist <ElFileList>`.

Here, the `elfile` file path specified in the `<ElFileList>` file can be the absolute path or relative path. By default, the relative path in the `<ElFileList>` is relative to the `<ElFileList>` itself. Therefore, there is no effect on the path even if you change the current working directory. However, if you want `<ElFileList>` to consider the relative path to the working directory, you can use the `-relative_to_workingdir` option while specifying the `elfile`.

You can also specify the file path using a user-defined environment variable added in the `<ElFileList>` file. For example, run the command `setenv ANY_ENV_VAR_NAME /remote/exclude_dir` in the console, then add `$ANY_ENV_VAR_NAME/exclusion1.el` in the `<ElFileList>`.

Note:

When there is an existing excluded item in the `.el` file loaded using `-elfilelist`, there might be a conflict while trying manually performing exclusion operations on the existing excluded item/scope.

Adding Annotations for Exclusions

The       icons are provided in the Coverage toolbar to add the annotation for an added exclusion as listed in table [Icons for Annotation](#). Before using the  icon to add an annotation to the selected exclusion, you need to use the  icon to create annotation(s) first. [Icons for Annotation](#) lists the descriptions for each icon.

Table 4 *Icons for Annotation*

Icon	Description
	Add an annotation selected in the  field to the selected exclusion.
	Delete the annotation from the selected exclusion.
	Edit annotations in the invoked <i>Edit Annotation</i> form.
	Add annotation automatically when excluding.

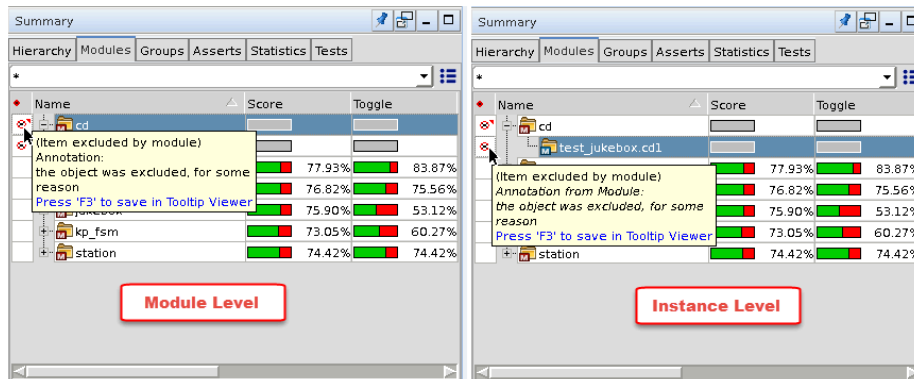
Table 4 Icons for Annotation (Continued)

	Specify the advanced options in the invoked Metric:  <All Metrics> Self Only field for the added annotation.
---	---

Module Exclusion Annotations at the Instance Level

When an object is excluded at the module level, it is also excluded in all of the module's instances. Also, when you add exclusion annotations to the object at the module level, these annotations are added to the object at the instance level. These annotations are visible at the instance level in the tooltips of the object. The tooltips display the exclusion annotations from the module level. The scope of these annotations is applicable to *Hierarchy* and *Modules* tabs in the *Summary* pane.

For example, in the figure below, the exclusion annotation is added for `cd` in the *Modules* tab. Now, hover the pointer over its object at the instance level, that is `test_jukebox.cd1`. It shows the exclusion annotation of `cd` in its tooltip at the instance level.



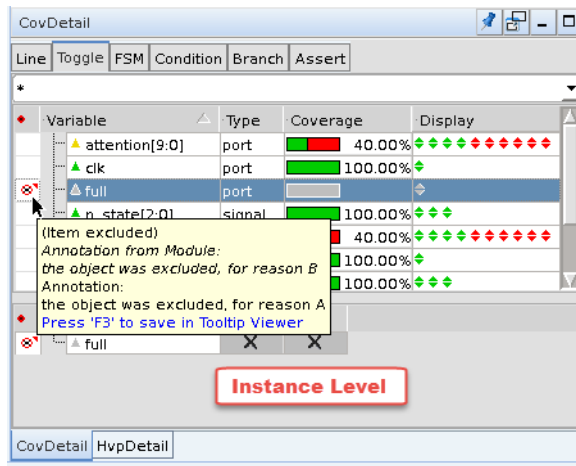
An instance might also contain its own annotations. In this scenario, the tooltip of the instance shows both annotations, that is, from the instance level and from the module level. To identify the module-level annotations, see the exclusion annotations in *italics* in **Annotation from Module** in the tooltip. In the figure below, the annotations from the module and the instance are as follows:

Module

the object was excluded, for reason B

Instance

the object was excluded, for reason A

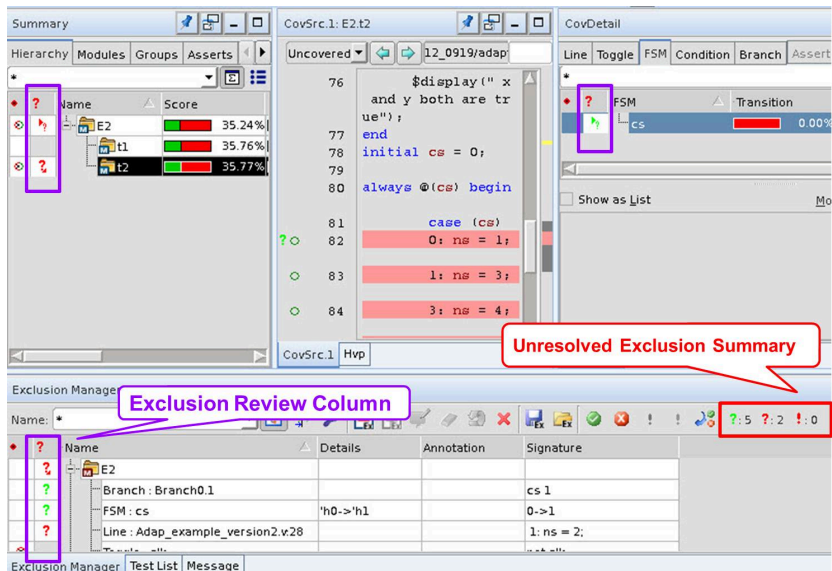


As described in the [Using the Toolbar to Manage Exclusions](#) section, you can load a saved exclusion file into the *Exclusion Manager* pane. You might have an exclusion file that was created recently, in the latest format, but the design has evolved. There might be some conflicting exclusions when you load an exclusion file with old format into the latest database. In these cases, unresolved exclusions need to be reviewed after loading the exclusion file.

The following sections describe how to use the features for the exclusion review.

After loading an exclusion file, you can find unresolved icons in the *Exclusion Review* columns if some unresolved exclusions exist in the loaded exclusion file (see figures [Load Exclusion File With Unresolved Exclusions](#)). You have to inspect and review the unresolved exclusions.

Figure 118 Load Exclusion File With Unresolved Exclusions



The **?:5 ? :2 !:0 Unresolved Exclusion Summary** field in the toolbar of the *Exclusion Manager* pane shows how many unresolved exclusions are needed to be reviewed. Table [Unresolved Exclusion Status](#) shows the icons marked at the *Exclusion Review* columns when the exclusion is unresolved or needs to be reviewed.

Table 5 Unresolved Exclusion Status

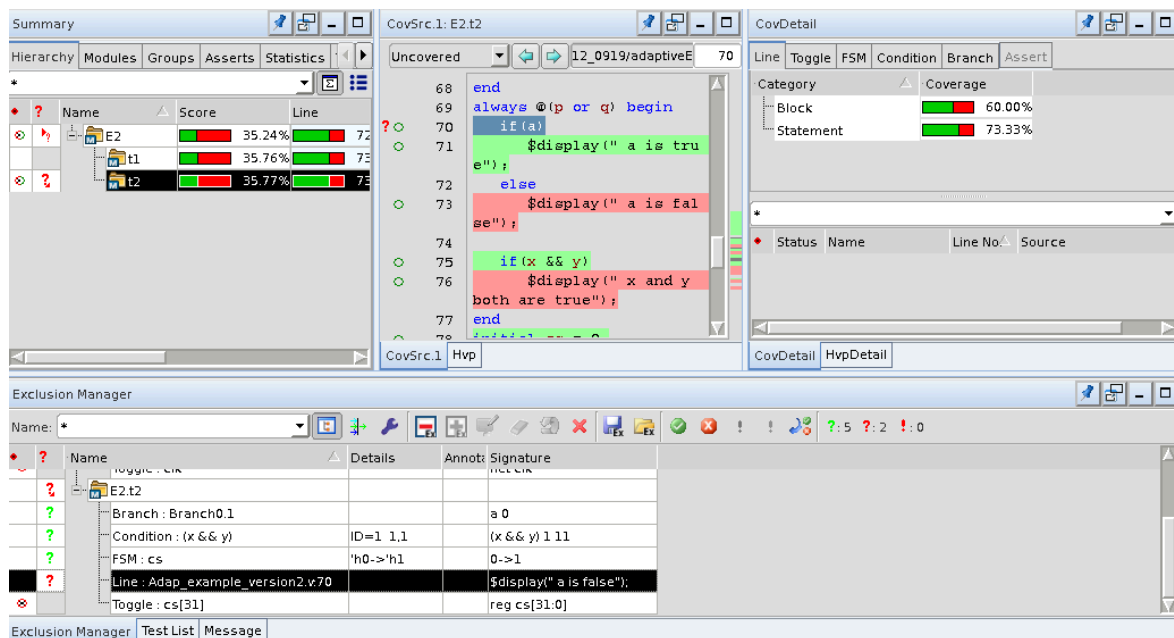
Unresolved Icon	Status
?	Unresolved mappable exclusions with signatures matching
?	Unresolved mappable exclusions with signatures mismatching
!	Unresolved un-mappable exclusions
?	Both reviewed and un-reviewed exclusions under the hierarchy
!	Exclusions are rejected. They include un-reviewed and reviewed exclusions
✓	Exclusions under the hierarchy have been reviewed and accepted

Reviewing Line Coverage Exclusions

To review the exclusion in the *Exclusion Manager* pane, double-click on the exclusion row (with  or ) . The corresponding source code of the unresolved exclusion in the *CovSrc* pane is shown. You can check the highlighted source code to see whether it maps the adaptive exclusion shown in the **Signature** field in the *Exclusion Manager* pane or not.

You can also search the adaptive exclusion in the source code. Use the “Ctrl” and “S” hotkey in the Coverage Source Code to bring up the **Search** field and click the  or  icons to find the previous or next matched string in the source code.

When you review and discover the signature matched, you can click in the circle icon in *CovSrc* to both accept the exclusion and exclude the item in the *Exclusion Manager* pane. The *Exclusion Manager* pane is updated with another row of the same line with exclusion icon.



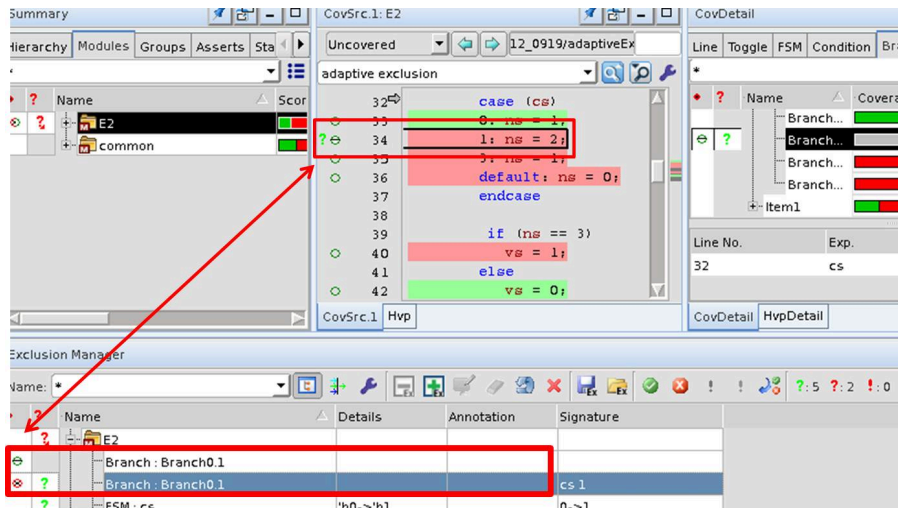
The screenshot displays the Verdi Coverage tool interface with three main panes:

- Summary Pane (Left):** Shows a hierarchy of modules and groups. The 'E2' module is selected, showing a score of 35.24% and a line count of 72. Below it, 't1' and 't2' are listed with scores of 35.76% and 35.77% respectively.
- CovSrc Pane (Middle):** Displays the source code for 'E2.t2'. The code is shown with line numbers 68 to 77. Line 70 is highlighted, showing an 'if' statement. The code is as follows:


```

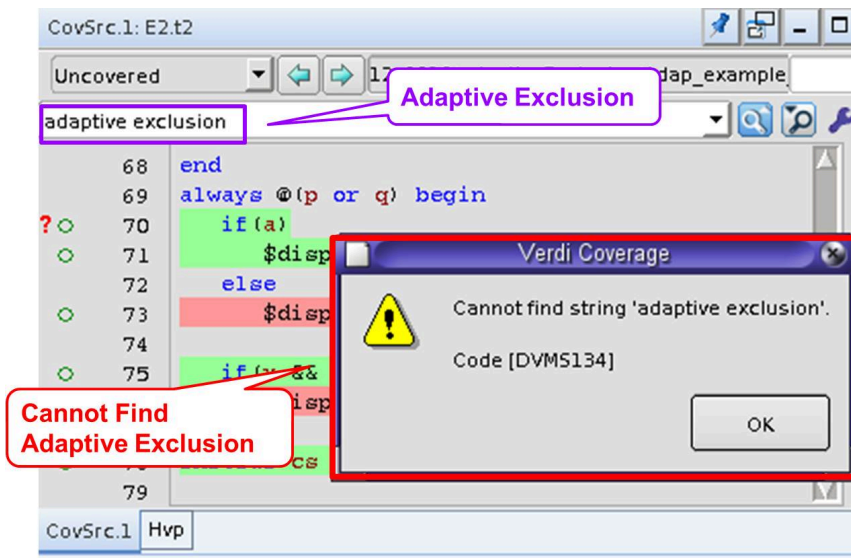
68 end
69 always @(p or q) begin
70   if (a)
71     $display(" a is tru
72   e");
73   else
74     $display(" a is fal
75   se");
76   if (x && y)
77     $display(" x and y
78     both are true");
79 end

```
- CovDetail Pane (Right):** Shows coverage details for the selected line. It includes a table with columns: Line, Toggle, FSM, Condition, Branch, Assert, and Coverage. The table shows coverage for 'Block' (60.00%) and 'Statement' (73.33%).
- Exclusion Manager Pane (Bottom):** Shows a list of exclusions. The 'Line : Adap_example_version2.v70' exclusion is selected, showing its signature as '\$display(" a is false");' and its toggle as 'reg cs[31:0]'. The exclusion is marked with a red question mark icon.



If you cannot find the mapped signature in the source code when using the search field to find the adaptive exclusion as shown, a pop-up window is displayed with an error code.

Figure 119 Search Adaptive Exclusion in Source Code



After reviewing the line, for the exclusion with the **?** icon, you can mark the line as reviewed and accept it by using the **Accept** right-click menu item command or click the **?** icon in the **Exclusion Review** column to cycle to the **✓ reviewed and accept** icon. You can also reject the exclusion by using the **Reject** right-click menu item or selecting the **✗ reviewed and reject** icon.

For the exclusion with the  icon, you can mark the line as reviewed by using the **Mark Unmap Reviewed** right-click menu item or click the  icon in the column to cycle to the  reviewed icon.

After you set a line as reviewed, the **Unresolved Exclusion Summary** field is also updated accordingly to show one less exclusion for reviewing.

After all the unresolved exclusions are reviewed and either accepted or rejected, the Summary Review displays 0 and disappears to signify that there are no more exclusions to review. Then, you can click the  icon to update the coverage scores with the new exclusions.

You can clean out the reviewed exclusions after all the exclusions are reviewed. Then, you can click the  icon to save the latest exclusion since the exclusions are updated to the current reviewed exclusions.

Attributes on Coverage Exclusions

Verdi Coverage enables you to define attributes in an exclusion project file, attach them to exclusion objects, and use filtering expressions to select which exclusions are applied.

This section consists of the following subsections:

- [Defining and Creating Attributes](#)
- [Editing Attribute Values](#)
- [Exclusion Filter](#)

To load an exclusion project and exclusion file while invoking Verdi, use the following command:

```
verdi -cov -covdir simv.vdb -excl_project excl.proj -elfile 1.el
```

where, `excl.proj` is the name of an exclusion project and `1.el` is the name of an exclusion file.

Defining and Creating Attributes

Verdi Coverage supports a built-in attribute, `Annotation`, but also allows you to define your own integer, real, string, date, enum, or set-type attributes. A couple of examples are shown below:

```
attribute integer phase = 0;  
attribute string assignee = "";  
attribute enum {new,in_progress,approved,rejected} status = new;
```

To manage exclusions/annotation attributes, the following dialog box is available from Annotation Manager:

When exclusions are saved to a file, the values for any attributes that are used on those exclusions are saved to the same file.

If a coverable object has multiple exclusions map to it, multiple attributes might be shown next to that object in the report. For example, if the signal a has two exclusion objects, one with Phase = phase1 and other with Phase = phase2, both exclusion objects are displayed with their respective attribute values as shown in the figure:

Name	Details	Annotation
Toggle : a		
Toggle : a		phase1
Toggle : a		phase2

The figure shows that the attributes are applied to exclusion objects and not to coverable objects. Coverable objects do not have attributes in this flow.

Editing Attribute Values

Verdi Coverage allows you to edit the value of the built-in attribute, `Annotation` as well as supported attributes, such as `integer`, `string`, and `real`, `enum` and `date` directly in the Exclusion Manager window.

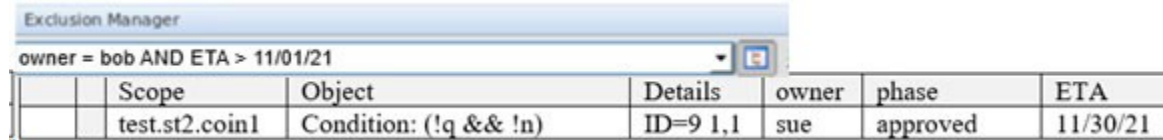
Exclusion Filter

Verdi Coverage provides a filter that determines which exclusions are applied to the design. It allows conditions based on user-defined attributes instead of only on built-in `Annotation` and `Elfile` fields. `Annotation` is the one built-in attribute type that is used for filtering. Also, if the `Elfile` attribute is enabled, it appears in the *Annotation Manager* table and can be used for filtering.

Here are few examples of filter expressions using user-defined attributes:

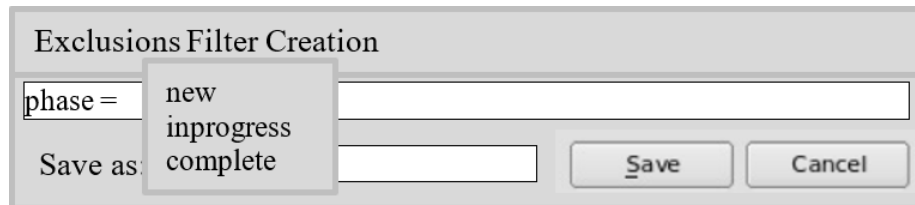
```
owner="Bob"
phase>1 OR status=inprogress
status=approved AND datecreated<06/01/2021
```

The following figure shows the group of exclusions with a filter (owner = bob AND ETA>11/01/21). Only one exclusion matched the criteria and it is the only one that is applied to the design:



Scope	Object	Details	owner	phase	ETA
test.st2.coin1	Condition: (!q && !n)	ID=9 1,1	sue	approved	11/30/21

When an attribute of type `enum` or `set` is used, the GUI shows a drop-down that allows you to pick the value or values to be used in the filter as shown below:



Exclusions Filter Creation

phase = new
inprogress
complete

Save as:

Filter expressions can be named and saved for later use. For example, a complex filter such as, can be named saved:

```
priority = 1 AND owner = "Lucy" AND status = approved
```

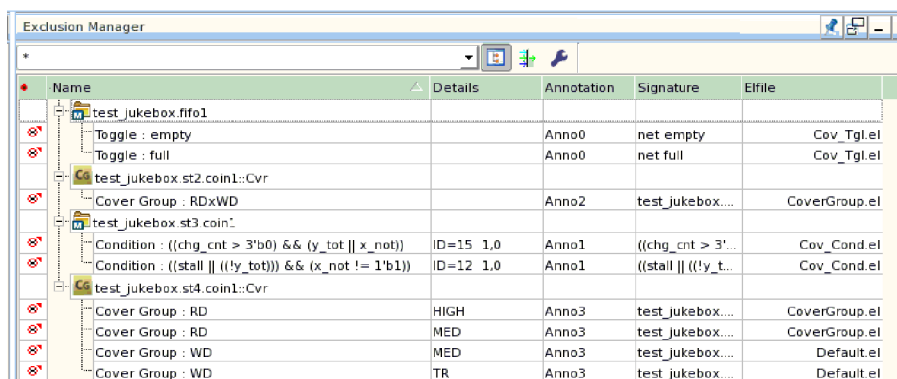
To load the exclude file, `excl1.el` and use the saved filter, say `cov_filter`, which is defined in it, use the following command:

```
% urg -elfile excl1.el -elfilter cov_filter -dir  
simv.vdb ... (EQ 1)
```

Filter Support for Annotations and Elfile

The *Exclusion Manager* pane enables you to filter exclusion items based on **Name**, **Annotations**, and **Elfile** columns. Filtering based on annotations enables you to view the annotations of searched items. Filtering based on exclusion file names provides you the list of searched results having the required exclusion file name.

Consider the following example that shows **Name**, **Annotations**, and **Elfile** columns for the *Exclusion Manager* pane:

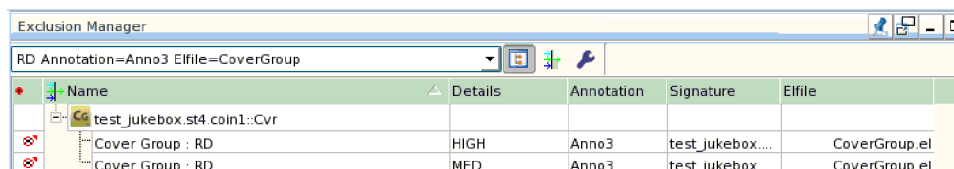


Name	Details	Annotation	Signature	Elfile
test_jukebox.fifo1				
Toggle : empty		Anno0	net empty	Cov_Tgl.el
Toggle : full		Anno0	net full	Cov_Tgl.el
test_jukebox.st2.coin1::Cvr				
Cover Group : RDxWD		Anno2	test_jukebox...	CoverGroup.el
test_jukebox.st3.coin1				
Condition : ((chg_cnt > 3'b0) && (y_tot x_not))	ID=15 1.0	Anno1	((chg_cnt > 3'...	Cov_Cond.el
Condition : ((stall (!y_tot)) && (x_not != 1'b1))	ID=12 1.0	Anno1	((stall (!y_t...	Cov_Cond.el
test_jukebox.st4.coin1::Cvr				
Cover Group : RD	HIGH	Anno3	test_jukebox...	CoverGroup.el
Cover Group : RD	MED	Anno3	test_jukebox...	CoverGroup.el
Cover Group : WD	MED	Anno3	test_jukebox...	Default.el
Cover Group : WD	TR	Anno3	test_jukebox...	Default.el

To search for the excluded items whose name is RD, Annotation is Anno3, and file name contains CoverGroup, use the following filter string in the Search field of the *Exclusion Manager* pane:

RD Annotation=Anno3 Elfile=CoverGroup

This filter string provides the following search results:



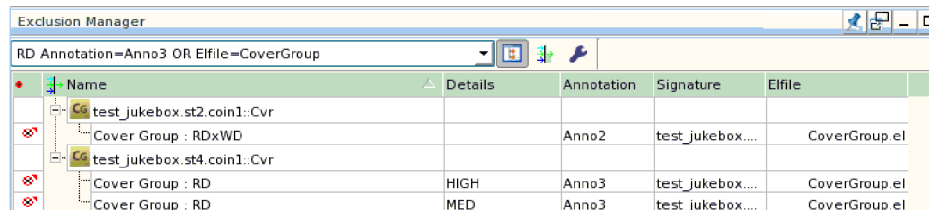
Name	Details	Annotation	Signature	Elfile
test_jukebox.st4.coin1::Cvr				
Cover Group : RD	HIGH	Anno3	test_jukebox....	CoverGroup.el
Cover Group : RD	MED	Anno3	test_jukebox....	CoverGroup.el

The *Exclusion Manager* pane allows you to use AND and OR operators. If no operator is specified between the column names in the filter string, the *Exclusion Manager* pane considers it as AND.

Consider a scenario in which you need to search the excluded items for the name as RD with either annotation as Anno3 or file name as CoverGroup. For this, use the following filter string:

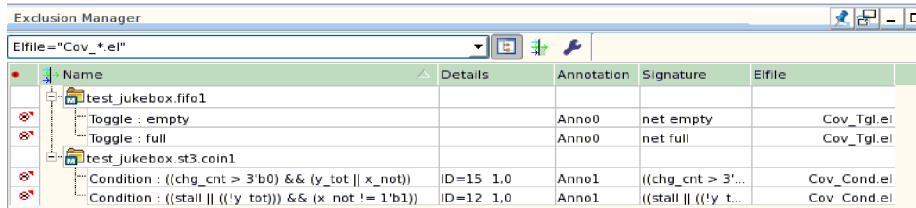
RD AND Annotation=Anno3 OR Elfile=CoverGroup

This filter string provides the following search results:



Name	Details	Annotation	Signature	Elfile
test_jukebox.st2.coin1::Cvr				
Cover Group : RDxWD		Anno2	test_jukebox....	CoverGroup.el
test_jukebox.st4.coin1::Cvr				
Cover Group : RD	HIGH	Anno3	test_jukebox....	CoverGroup.el
Cover Group : RD	MED	Anno3	test_jukebox....	CoverGroup.el

The *Exclusion Manager* pane also supports wildcard and regular expressions as filter string. For example, if you input the filter string, Elfile="Cov_*.el", the *Exclusion Manager* pane shows the following search results:



Name	Details	Annotation	Signature	Elfile
test_jukebox.fifo1				
Toggle : empty		Anno0	net empty	Cov_Tgl.ei
Toggle : full		Anno0	net full	Cov_Tgl.ei
test_jukebox.st3.coin1				
Condition : ((chg_cnt > 3'b0) && (y_tot x_not))	ID=15 1,0	Anno1	((chg_cnt > 3'...	Cov_Cond.ei
Condition : ((stall ((!y_tot))) && (x_not != 1'b1))	ID=12 1,0	Anno1	((stall (!y t...	Cov_Cond.ei

Note:

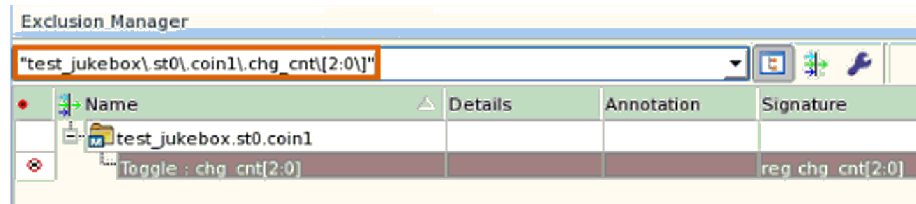
All filter strings are automatically saved in the `novas.rc` file. When Verdi Coverage is invoked from the directory in which `novas.rc` is saved, the filter string list is restored.

You can drag-and-drop a scope or an object from the *Summary* pane or the *CovDetail* pane to the Search field of Exclusion Manager to find exclusions. In this case, you do not need to type a string in the Search field. Note that this drag-and-drop method works properly in the default *Wildcards* mode. In the *Regular Expression* mode, you need to add the escape character `"\"` to the special characters of a regular expression to display search results.

Note:

To change the Filter mode from *Wildcards* to *Regular Expression*, click **Tools -> Preferences -> Global -> View Filters**.

For example, consider the `chg_cnt[2:0]` signal in the `test_jukebox.st0.coin1` scope. If you drag-and-drop this signal to the Search field of Exclusion Manager, you get the search string as `test_jukebox.st0.coin1.chg_cnt[2:0]`. This search string does not give any search result in the *Regular Expression* mode as it contains special characters `"."`, `"["`, and `"]"`. In this case, you need to add the escape character `"\"` to these special characters to display search results as shown in the following figure:



Name	Details	Annotation	Signature
test_jukebox.st0.coin1			
Toggle : chg_cnt[2:0]			reg chg_cnt[2:0]

Displaying Multiple Annotated Exclusions

Coverable objects in Verdi are marked as excluded when they have one or more exclusions applied to them. Exclusions can be loaded from one or more exclusion files or can be applied interactively in Verdi.

By default, if an object is excluded due to exclusions loaded from multiple exclusion files, and two or more of those exclusions have annotations, the exclusion file that is

loaded last overwrites the annotation from any previous exclusion file. If Verdi Coverage is invoked with the `-excl_append_annotation` switch, Verdi appends annotations from all exclusions to the excluded coverable object. For example, there are exclusions of `cd1` in both `1.el` and `2.el`, and each of those files contains an annotation on its exclusion, then the switch concatenates both in the *Exclusion Manager* pane for `cd1`, as follows:

	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1		anno1 anno2		1.el 2.el

The `-excl_append_annotation` switch is now removed. If more than one annotation exists for an excluded object, then all annotations are displayed as separate rows (see the figure below):

	Name	Annotation	Elfile
	test_jukebox.cd1		
	Instance : cd1		
	Instance : cd1	anno1	1.el
	Instance : cd1	anno2	2.el

Exclusions With Checksum Mismatch

When the design changes and exclusions that were created with a previous version are loaded, Verdi may require that you review some of the exclusions before it will apply them to remove corresponding coverable objects from coverage scoring. An exclusion is flagged when the checksum saved for it in the elfile does not match the checksum saved in the coverage database. Flagged exclusions can be accepted, rejected, or left unreviewed.

If a reviewed exclusion is accepted, it is saved to its associated elfile with a new checksum. If a reviewed exclusion is rejected, it gets removed from Verdi. If an exclusion is unreviewed, it is saved to its associated elfile with its original checksum.

Adding Annotations to an Excluded Object

You may add an annotation to an excluded coverable object that is propagated to its exclusion objects. The following figure shows a coverable object, `cd1` and its exclusions:

Name	Annotation	Elfile
test_jukebox.cd1		
Instance : cd1		
Instance : cd1	anno1	1.el
Instance : cd1	anno2	2.el

To add an annotation to an exclusion, perform the following steps:

1. Double-click the object to enter the edit mode.
2. Specify an annotation in its **Annotation** column. The specified annotation is propagated to the **Annotation** column of all of its contained excluded objects.

Note:

For more details on how to add annotations to an exclusion, see the [Adding Annotations for Exclusions](#) section.

Handling Duplicate Annotations of an Excluded Object

If the annotation of an excluded object is changed and is similar to another annotation of the same excluded object, then only one entry of that exclusion is retained in *Exclusion Manager*. For example, consider the following two exclusions with annotations `anno1` and `anno2` in the 1.el file:

Exclusion Manager					
*					
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		1.el

Consider a scenario in which `anno2` is renamed to `anno1`. Now, there are two excluded objects with same annotations, `anno1`, in the exclusion file, 1.el. Verdi retains only one of the excluded objects from the exclusion file and removes the other excluded objects, as shown in the following figure:

Exclusion Manager						
*						

There are certain scenarios related to saving exclusions to a file. These scenarios are explained in the following subsections:

- Saving All Exclusions for Selected Objects
- Saving Selected Exclusions for an Object

Saving All Exclusions for Selected Objects

To save all exclusions for selected objects to a file, perform the following steps:

1. In *Exclusion Manager*, select an object whose all exclusions you want to save. For example, select the `Instance:cd1` object, as shown in the following figure:

Exclusion Manager					
*					
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		2.el
	Instance : cd1		anno3		3.el

2. Right-click the selected object in *Exclusion Manager* and select **Save Exclusions for Selected Objects** to save all its exclusions to a file, say 4.el. Once saved, these exclusions appear in *Exclusion Manager*, as shown in the following figure:

Exclusion Manager					
*					
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		2.el
	Instance : cd1		anno3		3.el
	Instance : cd1		anno1		4.el
	Instance : cd1		anno2		4.el
	Instance : cd1		anno3		4.el

Saving Selected Exclusions for an Object

To save selected exclusions for an object to a file, perform the following steps:

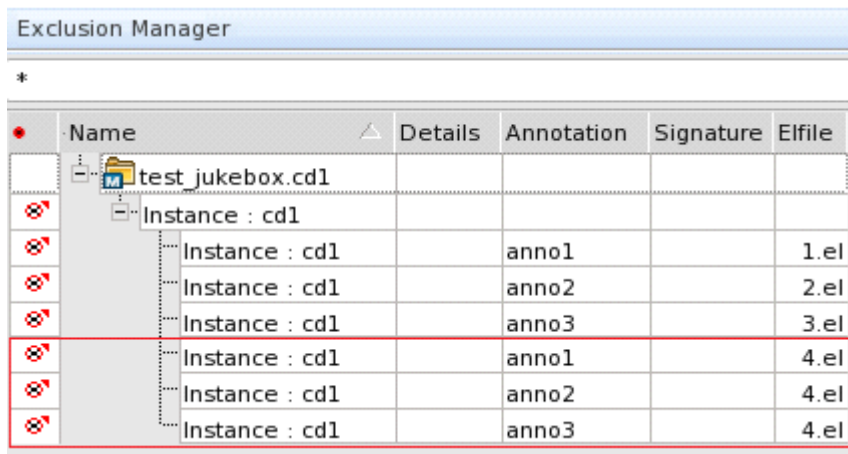
1. In *Exclusion Manager*, select the exclusions for an object that you want to save.
For example, select exclusions with annotations `anno1` and `anno2`, as shown in the following figure:

Exclusion Manager					
*					
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		2.el
	Instance : cd1		anno3		3.el

2. Right-click the selected exclusions and select **Save Exclusions for Selected Objects** to save the selected exclusions to a file, say 4.el. Once saved, these exclusions appear in *Exclusion Manager*, as shown in the following figure:

Exclusion Manager					
*					
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		2.el
	Instance : cd1		anno3		3.el
	Instance : cd1		anno1		4.el
	Instance : cd1		anno2		4.el

Note that duplicate exclusions are saved only once. For example, when the `Instance:cd1` object and its two exclusions with annotations `anno1` and `anno2` are saved to 4.el, exclusions with `anno1` and `anno2` become duplicates in 4.el. These duplicate exclusions are saved only once, as shown in the following figure:



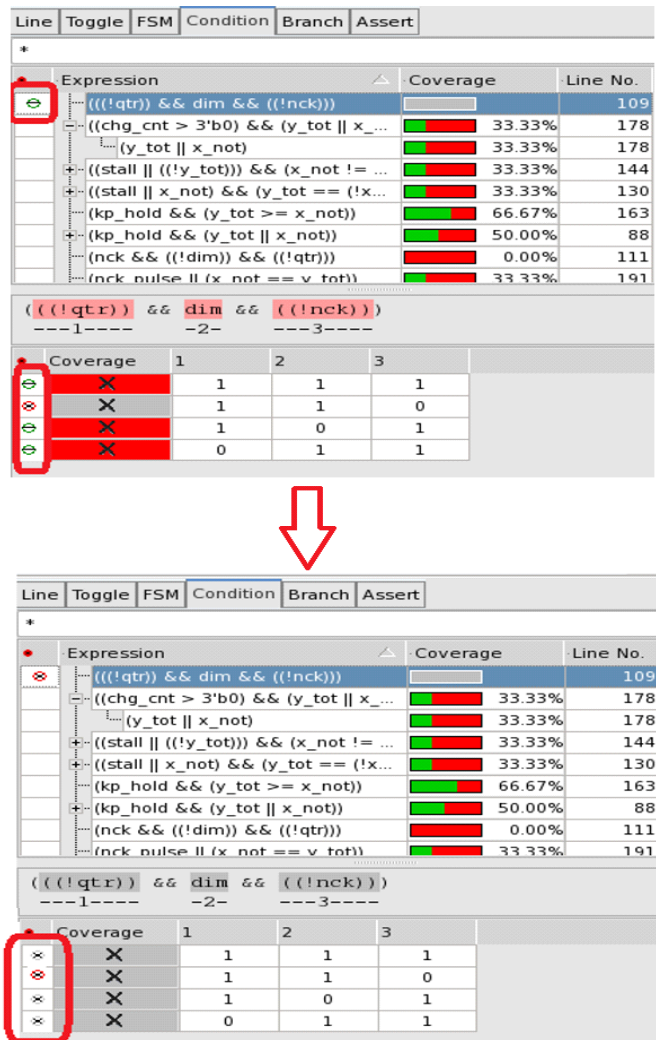
	Name	Details	Annotation	Signature	Elfile
	test_jukebox.cd1				
	Instance : cd1				
	Instance : cd1		anno1		1.el
	Instance : cd1		anno2		2.el
	Instance : cd1		anno3		3.el
	Instance : cd1		anno1		4.el
	Instance : cd1		anno2		4.el
	Instance : cd1		anno3		4.el

Seed and Propagated Exclusions

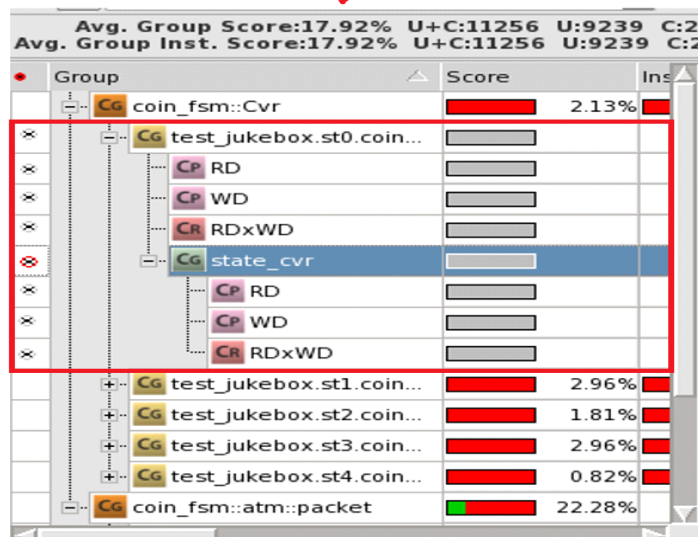
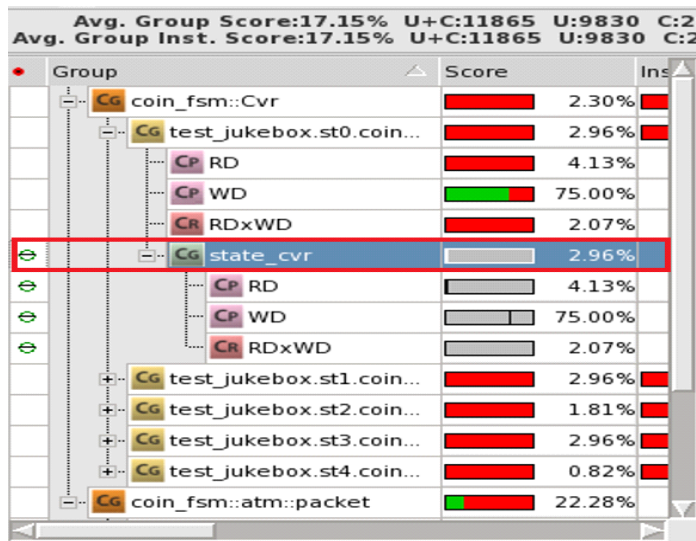
Seed exclusions are excluded objects that are excluded directly either manually or via an exclusion file. Propagated exclusions are the exclusions that are propagated by seed exclusions and are not excluded directly.

Here are certain scenarios for seed and propagated exclusions:

- If a parent object is excluded, its child exclusions are excluded as propagated exclusions, as shown in the following figure:



- If all instance FSM groups are excluded, then its variant is also excluded, as shown in the following figure:



- If all child exclusions of a parent object are excluded, then the parent object is excluded as a propagated exclusion, as shown in the following figure:

Line	Toggle	FSM	Condition	Branch	Assert
•					
Expression	Coverage	Line No.			
(((qtr) && dim && (!!nck)))	0.00%	109			
((chg_cnt > 3'b0) && (y_tot x_...	33.33%	178			
(y_tot x_not)	33.33%	178			
((stall (!!y_tot))) && (x_not != ...	33.33%	144			
((stall x_not) && (y_tot == (!x_...	33.33%	130			
(kp_hold && (y_tot >= x_not))	66.67%	163			
(kp_hold && (y_tot x_not))	50.00%	88			
(nck && (!!dim)) && (!!qtr)))	0.00%	111			
(nck_pulse (x_not == v_tot))	33.33%	191			
(((qtr) && dim && (!!nck)))					
---1-----2-----3-----					
Coverage	1	2	3		
	1	1	1		
	1	1	0		
	1	0	1		
	0	1	1		



Line	Toggle	FSM	Condition	Branch	Assert
*					
Expression	Coverage	Line No.			
☒ (((!qtr)) && dim && (!!nck)))		109			
☐ ((chg_cnt > 3'b0) && (y_tot x_...	33.33%	178			
☐ (y_tot x_not)	33.33%	178			
☐ ((stall (!!y_tot))) && (x_not != ...	33.33%	144			
☐ ((stall x_not) && (y_tot == (!x_...	33.33%	130			
☐ (kp_hold && (y_tot >= x_not)	66.67%	163			
☐ (kp_hold && (y_tot x_not)	50.00%	88			
☐ (nck && (!!dim)) && (!!qtr)))	0.00%	111			
☐ (nck_pulse (x_not == v_tot))	33.33%	191			
(((!qtr)) && dim && (!!nck)))					
---1-----2-----3-----					
Coverage	1	2	3		
☒ X	1	1	1		
☒ X	1	1	0		
☒ X	1	0	1		
☒ X	0	1	1		

Exclusions on a Topology Folder

Verdi now enables you to apply exclusions on all coverable objects inside a topology folder with one action. A topology folder can be a module, class, generate block, parameterized module, program block, or UVM hierarchy. Excluding one of these folders now excludes all coverable objects inside it.

This section consists of the following subsections:

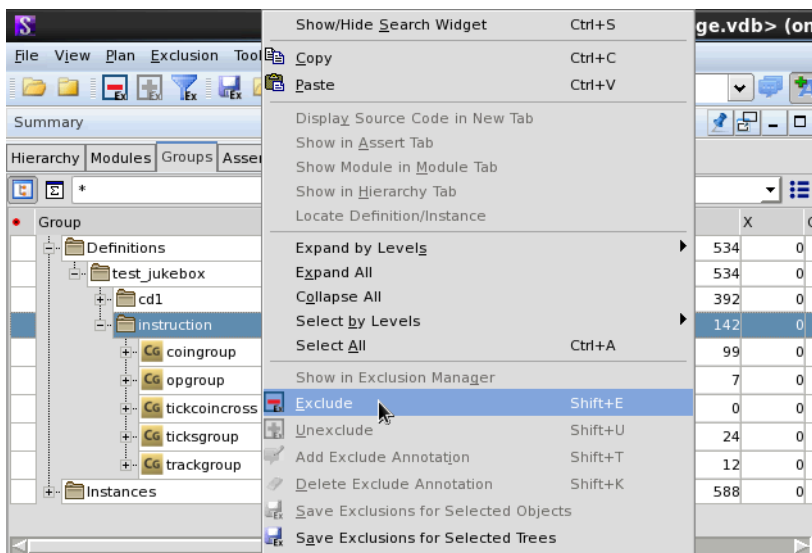
- [Marking Exclusions](#)
- [Tree Exclusions](#)
- [Saving Exclusions](#)
- [Adding Exclusion Annotations](#)

Marking Exclusions

You can mark exclusions on a topology folder using the **Exclude**  icon. The icon applies exclusions on all coverable objects in the topology folder. The icon is available in the main menu, the Exclusion toolbar, and in the right-click menu.

For example, consider a topology folder, `instruction` that contains coverable objects as `trackgroup`, `ticksgroup`, `tickcoincross`, `opgroup`, and `coingroup`. To mark this folder as excluded, perform the following steps:

1. Click the **Display Covergroup Instance tree**  icon in the *Summary* pane.
2. In the *Groups* tab, select the topology folder, `instruction`. When you select the topology folder, the **Exclude** icon becomes enabled.
3. To exclude the `instruction` folder and all the coverable objects under it, click the  icon in the main menu or in the **Exclusion** menu. Alternatively, right-click the folder to select **Exclude** as shown in the following figure:



4. After the exclusions are applied on the folder, exclusion markers are applied on the coverable objects. For exclusion markers, see the highlighted icons in the following figure:

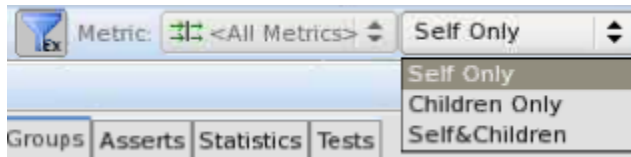
Group	Score	Definition	U+C	U	C	X
Definitions	28.40%		398	6	392	2518
test jukebox	28.40%		398	6	392	2518
cd1	99.42%		398	6	392	0
instruction	0.00%					2518
coingroup						99
opgroup						7
tickcoincross						2376
ticksgroup						24
trackgroup						12

Note:

No exclusion marker is displayed next to the topology folder (see `instruction` in the above figure).

Tree Exclusions

You can apply exclusions to either a topology folder or its sub-folders, or a tree (that is, both a topology folder and its sub-folders). When you select a topology folder, Verdi enables the **Advanced Exclusion Options** icon in the main toolbar. When you click this icon, Verdi displays the following exclusion toolbar:



You can decide the scope of applying exclusions using any of the following options from the exclusion toolbar:

- **Self Only:**

If you select this option, Verdi applies exclusions to the coverable objects that exist in a topology folder. The exclusions are not applied on the coverable objects that are present inside the sub-folders of the topology folder.

For example, consider the following figure that shows a topology folder, `cd1`, its direct coverable object, `disc`, and its sub-folder, `honker`. The `honker` sub-folder contains one coverable object, `song`.

Group	Score
Definitions	84.99%
test_jukebox	84.99%
cd1	97.47%
disc	97.47%
honker	97.47%
song	97.47%
instruction	80.00%

If you select **Self Only**, Verdi applies exclusions only to its direct coverable object, which is `disc`.

- **Children Only**

If you select this option, Verdi applies exclusions to the coverable objects that exist in the sub-folders of a topology folder. The exclusions are not applied on the coverable objects that are present directly in the topology folder.

In the example, if you select **Children Only**, Verdi applies exclusions only to the coverable objects inside the `honker` folder, which is `song`.

- **Self&Children**

If you select this option, Verdi applies exclusions to all coverable objects inside the selected topology folder. In the example, if you select **Self&Children**, Verdi applies exclusions to both `song` and `disc`.

Saving Exclusions

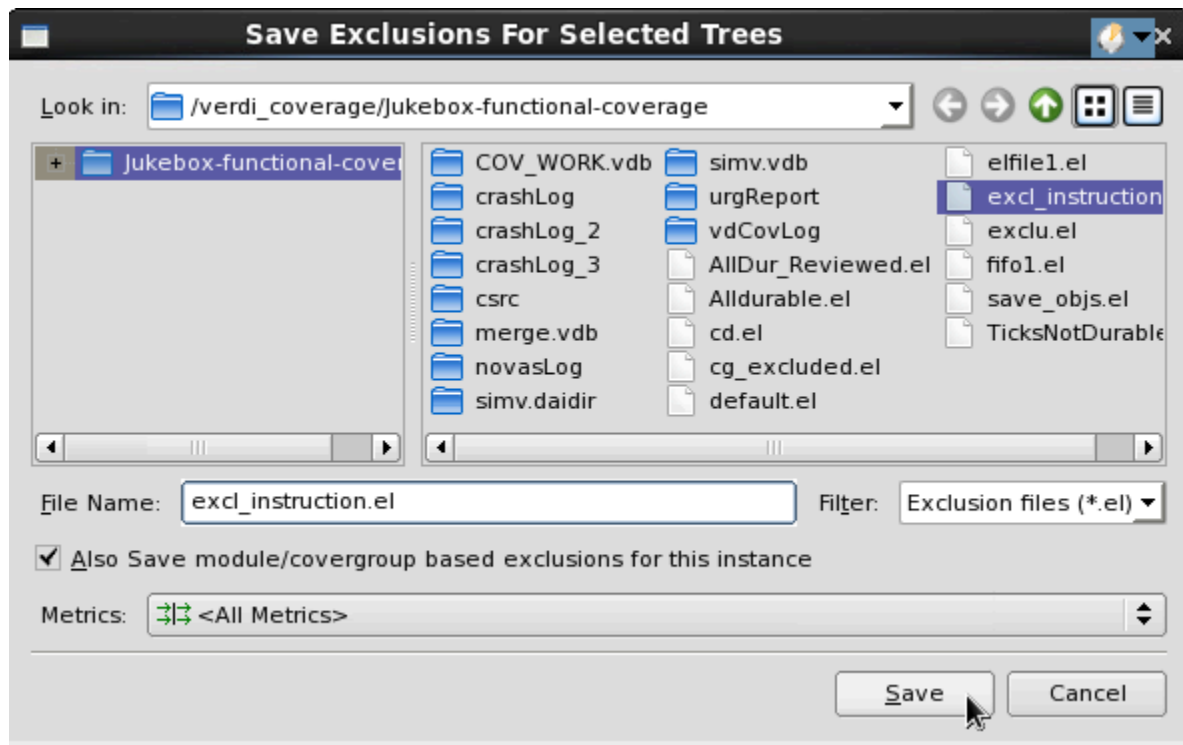
Verdi allows you to save exclusions for coverable objects that you want to exclude. Note that exclusions are not saved for a topology folder but for its coverable objects. For example, consider a folder, `cd1`, which contains coverable objects `cg1`, `cg2`, and `cg3`. If you save exclusions for `cd1`, exclusions are saved for `cg1`, `cg2`, and `cg3`, and not for `cd1`. Therefore, if a new coverable object, `cg4`, is added to the `cd1` folder and it needs to be excluded, you need to explicitly exclude `cg4`.

To save exclusions, Verdi provides the following two options:

- **Save Exclusions for Selected Objects:** This option saves all exclusions applied to only direct coverable objects in the topology folder. This option does not save exclusions applied to the coverable objects in its sub-folders.
- **Save Exclusions for Selected Trees:** This option saves all exclusions applied to coverable objects in the topology folder and its sub-folders.

To save exclusions, perform the following steps:

1. Right-click the folder on which the exclusions are already applied and select **Save Exclusions for Selected Objects** or **Save Exclusions for Selected Trees**.
2. Specify the exclusion file name, `excl_instruction.el`, in the **Save Exclusions for Selected Objects** or **Save Exclusions for Selected Trees** dialog box and click **Save**.



In the figure above, the **Also, Save module/covergroup based exclusions for this instance** option enables you to save exclusions that are applied on modules and covergroups. This option is selected by default.

Note:

The **Also, Save module/covergroup based exclusions for this instance** option is removed from P-2019.06 and later releases.

Adding Exclusion Annotations

Verdi enables you to add annotations for exclusions on a topology folder or its sub-folders or a tree. There are two ways to add annotations:

1. Exclude a coverable object and perform the following steps to add annotations:
 - a. Select the coverable object for which the annotation needs to be added.
 - b. Specify the annotation in the **Annotations**  box in the main toolbar.
 - c. Click the **Add an annotation to the selected items**  button from the main toolbar to add the annotation to the selected coverable object.
2. Add an annotation for a coverable object before excluding it. For this, perform the following steps:
 - a. Select the coverable object for which the annotation needs to be added.
 - b. Click the **Adding Annotations Automatically When Excluding**  icon from the main toolbar. This icon is enabled by default.
 - c. Specify the annotation in the **Annotations**  box in the main toolbar.
 - d. Exclude the coverable object and note that the specified annotation is automatically attached to the object.

You can also add exclusion annotations by right-clicking a topology folder and selecting **Add Exclude Annotation**. It opens the **Add Exclusion Annotation** dialog box (as shown below):



In the **Add Exclusion Annotation** dialog box, click **Add** to display a field to add an exclusion annotation. Then, click **OK**.

You can add exclusion annotations for either a topology folder or its sub-folders, or a tree (that is, both a topology folder and its sub-folders) using the **Advanced Exclusion Options**  icon in the main toolbar as explained in the [Tree Exclusions](#) section.

To delete an annotation either click the **Delete the selected annotation from the selected items**  button in the main toolbar or click **Delete Exclusion Annotation** from the right-click menu.

Exclusion Propagation to Connections

Verdi Coverage supports toggle exclusion propagation to signal connections and can automatically exclude the related exclusion signals for a given signal.

As an example, consider the following design:

```
module Add(a,b b, c);
    input a;
    input b;
    output c;
    assign c = a+b;
endmodule

module Add2(a2,b2,c2);
    input a2;
    input b2;
    output c2;
    Add inst2(a2,b2,c2);
endmodule

module TOP;
    wire w1;
    reg [3:0] w2;
    wire w3;
    Add2 inst(w1,w2[3],w3);
endmodule

TOP.w3 connected ports are:
TOP.inst.c2
TOP.inst.inst2.c
```

With this enhancement, if you exclude the `TOP.w3` signal for `vdCov`, the connected ports, `TOP.inst.c2` and `TOP.inst.inst2.c` also get excluded automatically.

This also works for vector/bus signals as follows:

For `Top.inst.b2`, the connected ports are as follows:

```
TOP.inst.inst2.b
TOP.w2[3]
```

Verdi Coverage recognizes the related signals connected to the user-selected signal and excludes them automatically. This feature is discussed in more detail in the subsequent sections.

Prerequisite

As Verdi Coverage needs the Elaboration database to retrieve the connected signals, you need to load the corresponding Elaboration database. It is recommended to use the Unified Front end flow to generate the Elaboration database.

Use Model

To automatically exclude the connected signals for the selected signals in toggle exclusion, perform the following steps:

1. After importing a design, you can select the signal or multiple signals in the toggle list view for which you want to exclude connected signals. For this, use the new CSM **Exclude Connected Signals** offered by Verdi Coverage.

Figure 120 Excluding Selected Signals

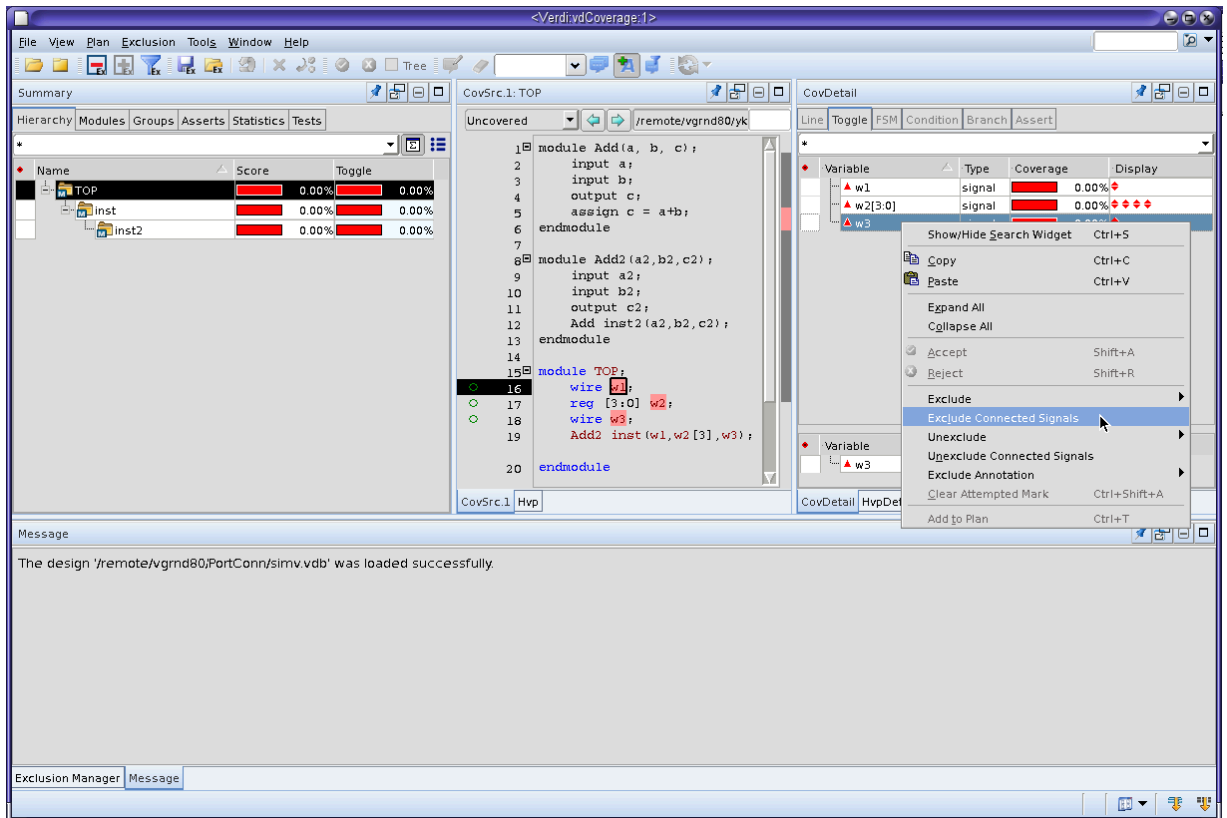
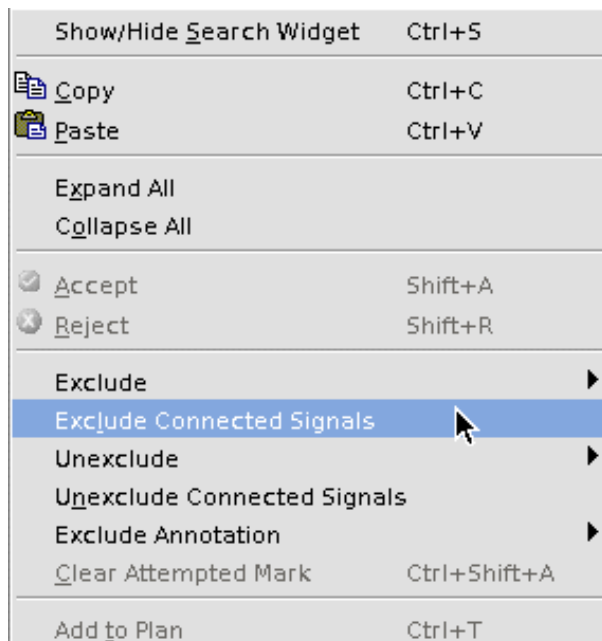
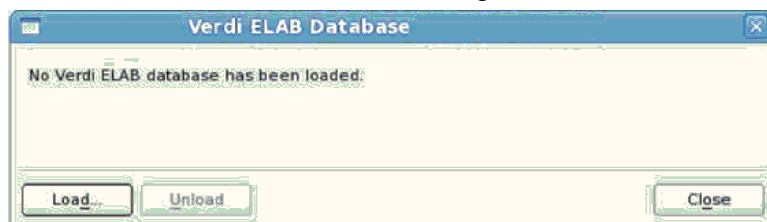


Figure 121 New Option - Exclude Connected Signals



2. Verdi Coverage needs the Elaboration database to be loaded to get the connected signals, so you must load the Elaboration database into vdCov.

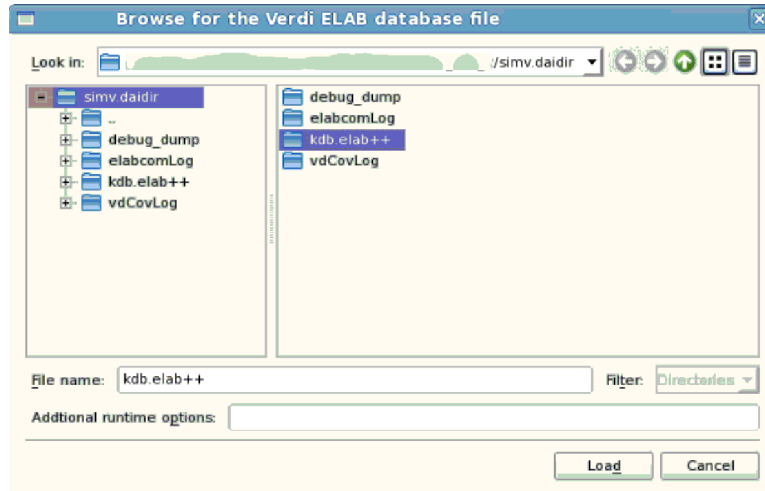
Figure 122 Verdi ELAB Database Dialog Box



If you have not already loaded the Elaboration database, the **Verdi ELAB Database** dialog box is displayed. Specify the Elaboration database that needs to be loaded and

click the **Load** button. This opens the following dialog box that lets you browse for and load the Elaboration database:

Figure 123 Browse for the Verdi ELAB Database File



After you have selected the Elaboration database that needs to be loaded, Verdi Coverage loads it.

Note:

It is recommended to use the Elaboration database from the Unified Front end flow.

3. After loading the Elaboration database, the **Connected Signals** dialog box is shown with the loaded Elaboration database path. The signals connected to the selected signal(s) are also shown in the *Connected Signals List View* pane along its direction (input, output, or inout). The items are grouped by the selected signals in the *Connected Signals List View* pane.

In the following figure, if you select one signal, multiple signals connected to the signal are selected in the *Connected Signals List View* pane:

Figure 124 Signals Connected to User Selected Signal(s)

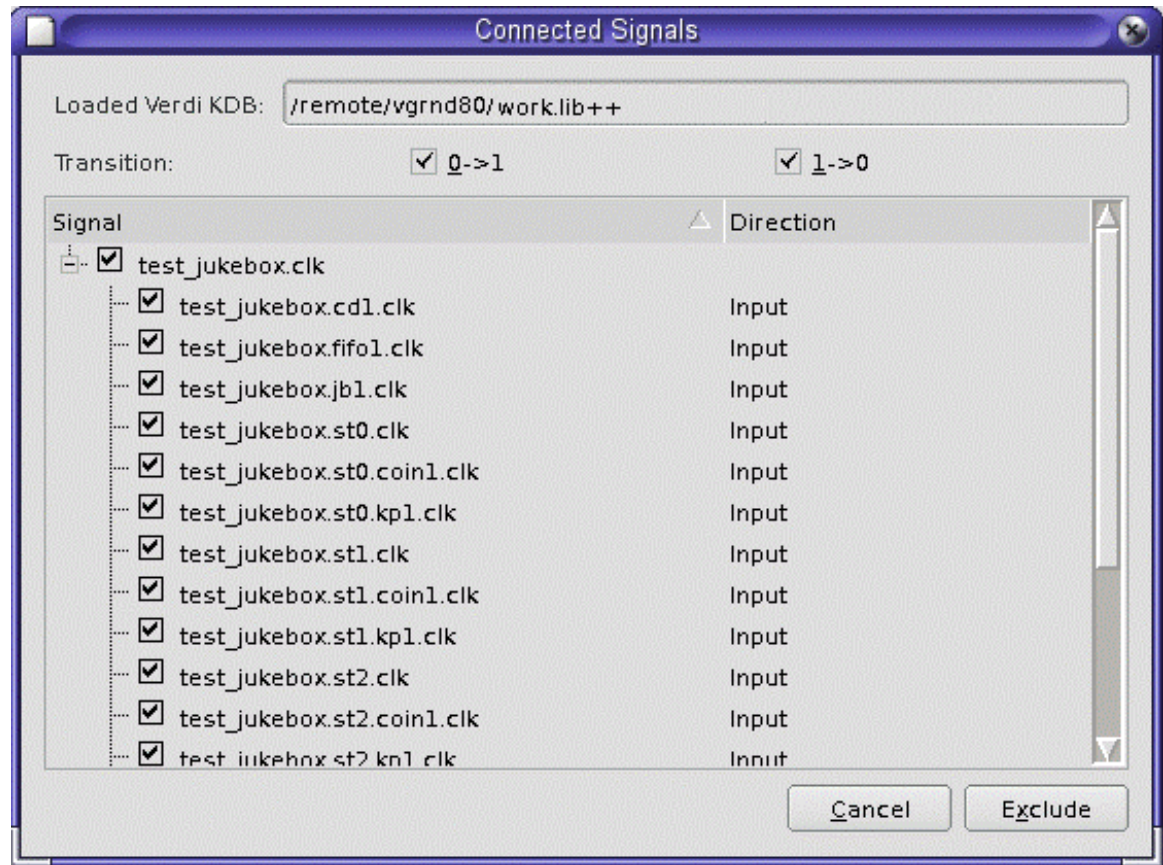
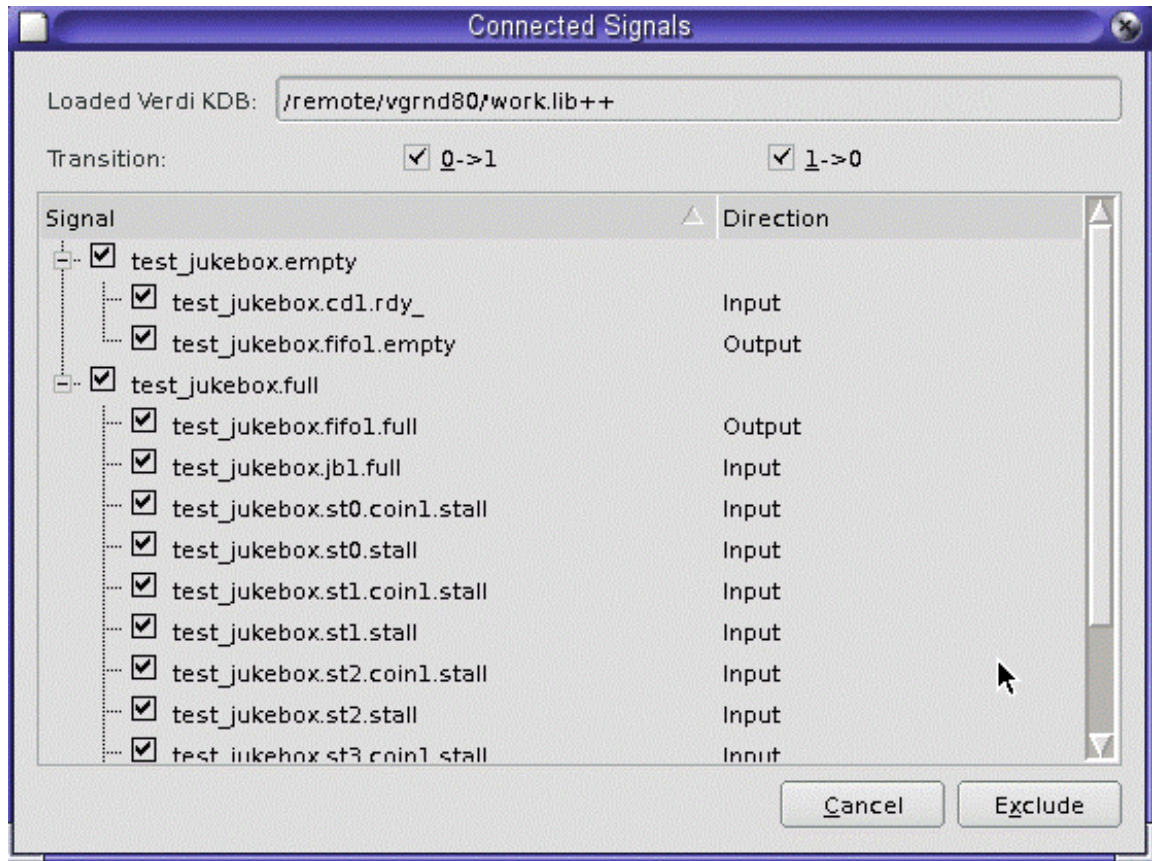


Figure 125 Signals Connected to User Selected Signal(s)



You can select/deselect the signals to exclude and can check the transition 0->1 and 1->0 to exclude a signal. When the dialog box is displayed, you can double-click a signal on the dialog box to view its source.

4. Once you have selected the signals to be excluded, click the *Exclude* button to exclude the signal in the *Connected Signals List View* pane and its associated signals. These signals are displayed in Exclusion Manager.

Managing the Verdi ELAB Database

As the Elaboration database is already loaded, the next time you want to exclude a signal (and its connected signals), you can get the connected signals automatically.

Verdi Coverage offers a new menu item **Manage Verdi ELAB Database** in the *File* main menu. Clicking this menu item to display the **Verdi ELAB Database** dialog box

that indicates whether there is already an ELAB database loaded along with its path, as follows:

Figure 126 No Verdi ELAB Loaded

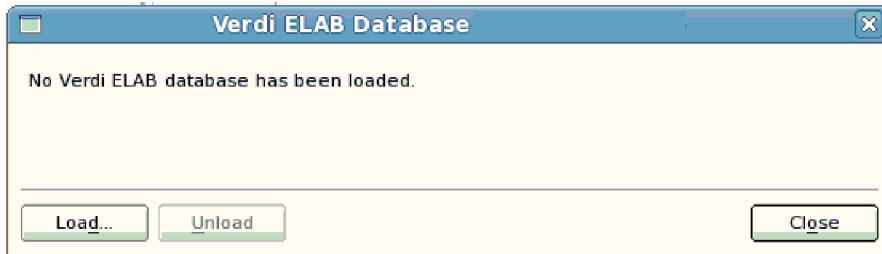
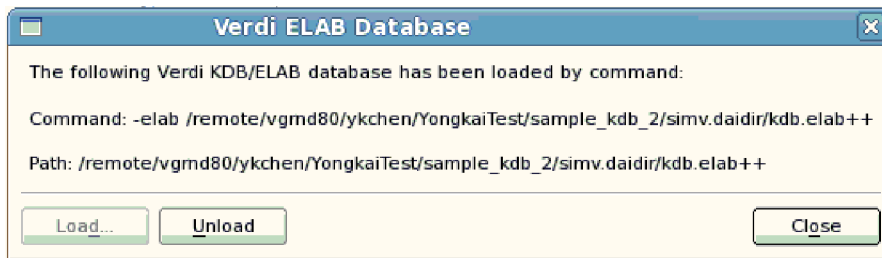


Figure 127 Loaded the ELAB Database With Its Path



New/Updated Tcl Commands

The following Tcl commands are added/updated with this release:

- `gui_load_kdb`: Loads the KDB library file.
`gui_load_kdb -path <KDB_library_path>`
- `gui_unload_kdb`: Unloads the existing loaded KDB library file.
- `gui_dump_connected_signals`: Dumps all the connected signals' names.
`gui_dump_connected_signals -name <signalFullPath>`
- If there is an MDA element, add the `-mda` parameter as follows:
`gui_dump_connected_signals -name top.inst.mda[3][4][1:5] -mda`
- `gui_exclude_signal`: Excludes the specified signal, even though the signal is actively shown in the current UI.
`gui_exclude_signal -name <signalFullName>`
- If you want to specify the transition, add a parameter: `-transition 0to1` or `-transition 1to0`.

- If you want to un-exclude (include) the signal, add the `-include` parameter.
- If you want to add exclusion annotations, add the `-annotation <anno>` parameter.

Exclusion Propagation From Line Coverage

Verdi allows you to automatically propagate line exclusions to condition and branch coverage metrics. This propagation reduces manual effort in creating the same exclusions again for condition and branch metrics.

This section consists of the following subsections:

- [Use Model](#)
- [Example](#)
- [Limitations](#)

Use Model

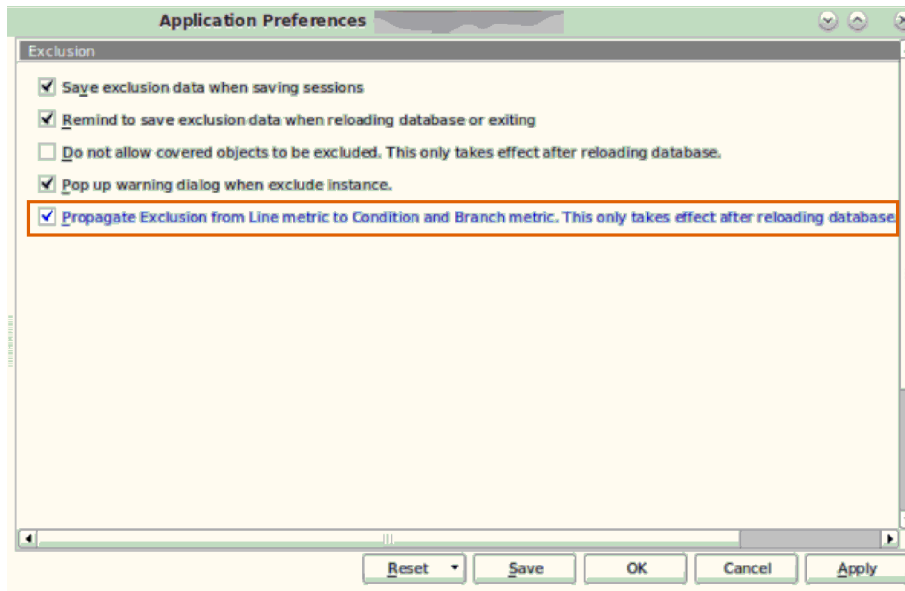
To propagate line coverage exclusions to condition and branch metric, use the following command while invoking Verdi:

```
verdi -cov -excl_propagation
```

You can also enable exclusion propagation from inside Verdi. For this, perform the following steps:

1. Select Tools -> Preferences to display the **Application Preferences** dialog box (Figure [Application Preferences Dialog Box](#)).
2. In the **Application Preferences** dialog box, select the **Exclusion** category.
3. In the **Exclusion** category, select the following option:
Propagate Exclusion from Line metric to Condition and Branch metric. This only takes effect after reloading the database.
4. Reload the database.
5. Right-click a signal in the *Hierarchy* pane and select **Exclude** to propagate line coverage exclusions to condition and branch metric.

Figure 128 Application Preferences Dialog Box

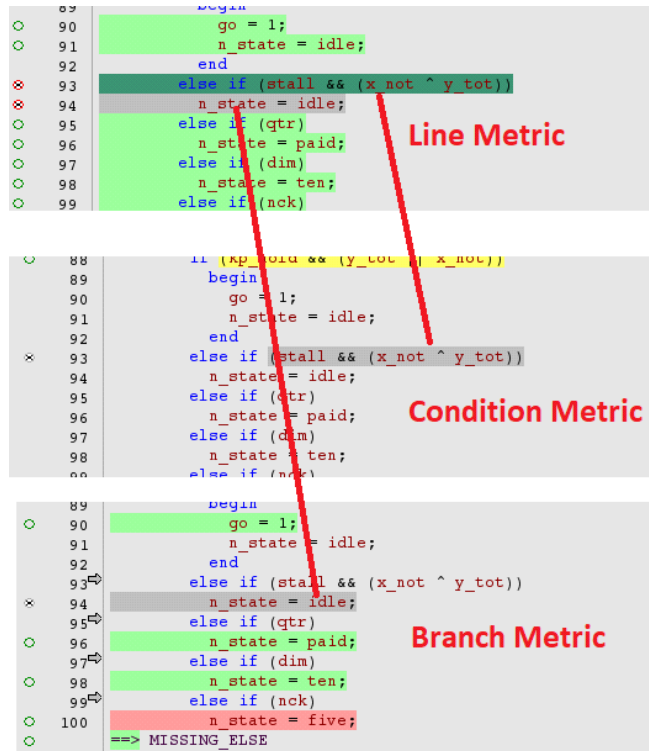


Example

Consider the following code snippet in which Line 93 and Line 94 are marked as excluded:

```
93 else if (stall && (x not ^ y_tot))
94   n_state = idle;
95 else if (qtr)
96   n_state = paid;
97 else if (qtr)
98   n_state = paid;
```


When `verdi -cov -excl_propagation` is used while invoking Verdi, the exclusions at Line 93 and Line 94 are propagated to condition and branch coverage metrics, as shown here:



Limitations

This feature has the following limitations:

- Propagating condition and branch exclusions is not supported.
- Exclusions are not propagated for `MISSING_ELSE` and `MISSING_DEFAULT`.

7

Loading and Saving Sessions

Verdi Coverage provides save and load session features to save the current process or restore a previously saved session to continue the unfinished process.

The analyzed coverage, Exclusion Manager and HVP planner status can be recorded using the **File -> Save Session** command. Specify a name and path for the session file in the Save Session form and click the **Save** button. The following information is saved in the session file:

- General status
- Preferences settings
- Docked panes layout
- Coverage status in the *Summary* pane
- Each Coverage Source Code and corresponding Coverage details status
- Exclusion Manager status
- HVP Planner and corresponding status in the *HvpDetail* pane

You can use the **File -> Restore Session** command to restore the saved session, and the Verdi Coverage session is restored.

Note:

To save the session including additional changes in the *HVP Planner* and *Exclusion Manager* panes, the changes must first be saved in these separate panes.

8

Interacting With Verdi Debug

Verdi Coverage is interlinked with the Verdi platform using which you can locate covered and uncovered situations in your design. You can also check whether a signal, module, or instance found in the Verdi platform is covered or uncovered by using this ability.

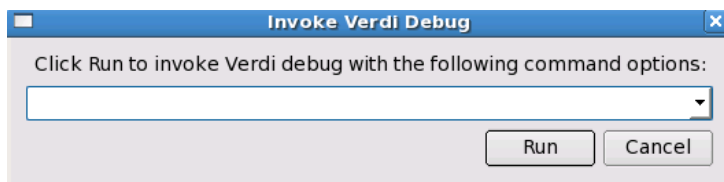
The following sections describes how you can invoke the Verdi platform and use the Interactive Debug capabilities between Verdi Coverage and the Verdi platform:

- [Invoking the Verdi Platform](#)
- [Using Interactive Debug Between Verdi and Verdi Coverage](#)

Invoking the Verdi Platform

You can use the **Tools -> Debug** command to invoke the main window of the Verdi platform from Verdi Coverage. The *Invoke Verdi Debug* form is opened (see Figure [Invoke the Verdi Platform for Debugging](#)) and you can input the options for invoking the Verdi platform. For example, you can load the `work` library into the Verdi platform while invoking the Verdi platform by entering `-lib work` in the blank field and then clicking the **Run** button in the *Invoke Verdi Debug* form.

Figure 129 Invoke the Verdi Platform for Debugging



The contents from the last five executions are provided in the drop-down menu.

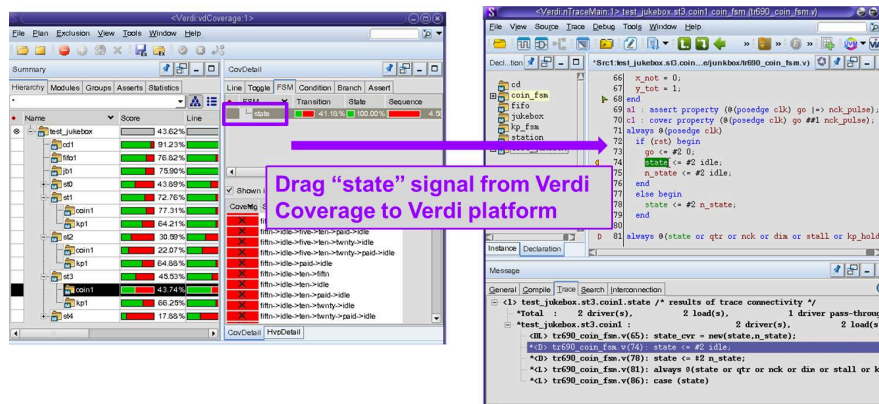
Note:

The invoked Verdi platform does not automatically load the design. You need to manually load your design and ensure the designs are the same between Verdi Coverage and the Verdi platform.

Using Interactive Debug Between Verdi and Verdi Coverage

After loading the design in the Verdi platform, you can drag and drop a signal, module or instance from Verdi Coverage to the Verdi platform. As shown in Figure [Drag and Drop Signal to Verdi Platform](#), drag the *state* signal from the *FSM* tab of the *CovDetail* pane and drop it to the Verdi platform. The *Hierarchy Browser* pane highlights the module that the *state* signal belongs to and this signal is highlighted in the source code pane. Also, the *Trace* page of the *Message* pane shows the tracing results of the *state* signal.

Figure 130 Drag and Drop Signal to Verdi Platform



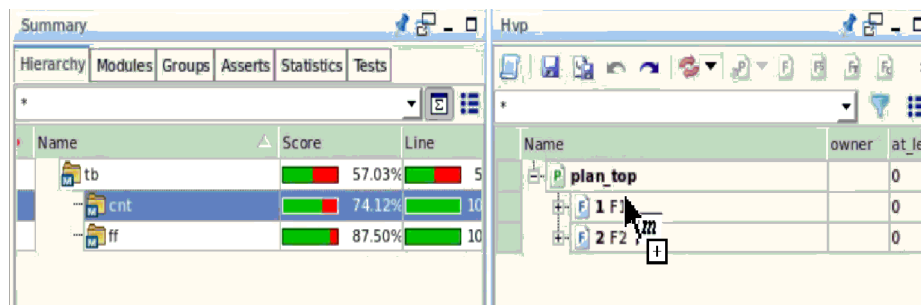
The table [Drag and Drop to Verdi Platform Behaviour](#) describes the drag and drop behavior from Verdi Coverage to the Verdi platform.

Note:

Dragging the line, condition, and branch expressions is not supported.

Verdi drag and drop location depends on the top-left corner of the icon, which is present on the right-side of the cursor.

Figure 131 Drag and Drop location



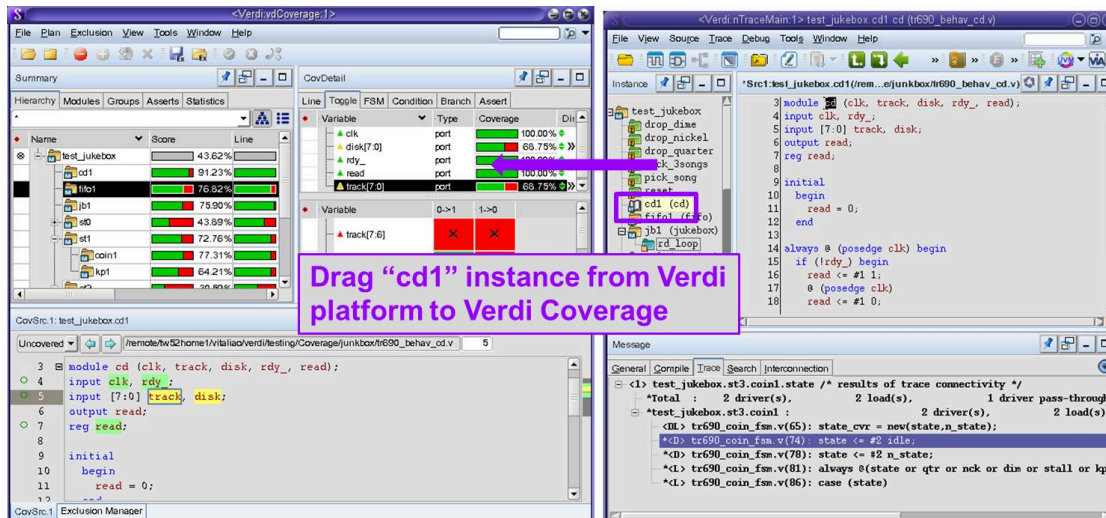
For example, in the above image, the drop location is on F1, as the top-left corner of the *m* icon is present on F1.

Table 6 Drag and Drop to Verdi Platform Behaviour

Source Item / Destination Window/Pane	Module/Instance	Signal	Assert	Cover-group Instance	Cover-point/Cross	FSM State Signal	FSM Transition/Sequence
nWave	Show signals of scope	Show signal	Show signal	NA	NA	Show state signal	Show state signal
Source code	Show in source	Show in source	Show in source	Show in source	Show the parent in source	Show in source	Show in source
Hierarchy Browser	Locate in Hierarchy Browser tree and sync with the source code pane	NA	Locate in Hierarchy Browser tree and sync with the source code pane	Locate in <i>decl_tree</i> page	Show the parent in <i>decl_tree</i> page	NA	NA
nState	NA	NA	NA	NA	NA	Show in nState with corresponding FSM	Show in nState with corresponding FSM
Assertion	Show assertions of scope	NA	Show assertion	NA	NA	NA	NA
nSchema	Show schematic	Show schematic	NA	NA	NA	Show in schematic	Show in schematic

You can also drag and drop a signal, module or instance from the Verdi platform to Verdi Coverage. As shown in Figure [Drag and Drop Signal to Verdi Coverage](#), the `cd1` instance is dragged from the *Hierarchy Browser* pane in the Verdi platform and dropped into the *Toggle* tab of the *CovDetail* pane in Verdi Coverage. The signals of `cd1` are listed and the toggle coverage is shown in the *Toggle* page. The source code of the `cd1` instanced module in the *CovSrc* pane is also shown.

Figure 132 Drag and Drop Signal to Verdi Coverage



The table [Drag and Drop to Verdi Coverage Behavior](#) describes the drag and drop behavior from the Verdi platform to Verdi Coverage.

Table 7 Drag and Drop to Verdi Coverage Behavior

Source Item / Destination Pane	Module/Instance	Signal	Assert	Covergroup	nState
Summary	Show in the Summary pane	NA	Show in the Summary pane; Switch to the Asserts tab	Show in the Summary pane; Switch to the Groups tab	NA
	Change to the scope	Show in the Toggle tab of the CovDetail pane	Show in the Assert tab of the CovDetail pane	Show in the Covergroup tab of the CovDetail pane	Show in the FSM tab of the CovDetail pane
CovSrc	Show in the CovSrc pane	Show in the CovSrc pane	Show Assert in the CovSrc pane	Show covergroup in the CovSrc pane	Show covergroup in the CovSrc pane
HVP pane	Add scope source into HVP hierarchy	NA	Add assert source into HVP hierarchy	Add covergroup source into HVP hierarchy	NA

9

Using VC Execution Manager

VC Execution Manager ("ExecMan") is a versatile, modular system for managing compilation, regression, and tracking of the design verification process. The execution engine automatically merges VCS coverage data from each test run.

ExecMan's main functionality is to run a set of regression simulations, collect results, and generate interactive reports for exploring those results. ExecMan stores regression results in a relational database. A web-based user interface is provided to enable browsing of regression results as well as files generated by verification tools. In addition, results can be viewed from within Verdi Coverage.

To aid in quick analysis of simulation failures, ExecMan can rerun the fastest running examples of each unique failure category, and automatically prepare detailed debug results for analysis.

As regressions involve running many tests on a large set of hardware, ExecMan includes features to overcome common system failures. Disk access interruptions and failed hosts are tolerated, for example.

ExecMan is preconfigured to work with VCS and UVM methodologies but can be configured to run any verification tool. It works in conjunction with grid managers, such as LSF or Sun Grid, as well as directly executing simulation jobs on local hosts.

ExecMan requires two license keys: `VC-EXECMAN-SERVER` and `VC-EXECMAN-CLIENT`.

You can run VC Execution Manager and view run results from Verdi Coverage. To invoke ExecMan from the Verdi Coverage GUI, use either of the following methods:

- **Tools > Open Tests Browser**
- **Tools > Run Tests**

For more details on ExecMan, see the *VC Execution Manager User Guide* on SolvNetPlus.